



# Pathology associated with kidney failure and anesthesia

Maria I Dalamagka

## ► To cite this version:

Maria I Dalamagka. Pathology associated with kidney failure and anesthesia. Magna Scientia Advanced Research and Reviews, 2024, 12, pp.106 - 109. 10.30574/msarr.2024.12.1.0149 . hal-05432456

**HAL Id: hal-05432456**

**<https://hal.science/view/index/docid/5432456>**

Submitted on 26 Dec 2025

**HAL** is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



Distributed under a Creative Commons Attribution 4.0 International License

## Pathology associated with kidney failure and anesthesia

Maria I. Dalamagka \*

*Department of Anesthesia, General Hospital of Larisa, Greece.*

Magna Scientia Advanced Research and Reviews, 2024, 12(01), 106–109

Publication history: Received on 10 August 2024; revised on 26 September 2024; accepted on 28 September 2024

Article DOI: <https://doi.org/10.30574/msarr.2024.12.1.0149>

### Abstract

Chronic kidney disease (CKD) is defined as either a glomerular filtration rate (GFR) of  $<60 \text{ ml min}^{-1} 1.73 \text{ m}^{-2}$  for 3 months or more, irrespective of cause, or kidney damage leading to a decrease in GFR, present for 3 months or more. The damage may manifest as abnormalities in the composition of blood or urine, on radiological imaging, or in histology. CKD is classified into five stages depending on GFR, ranging from Stage 1 - normal GFR to Stage 5 - established renal failure. Patients with kidney disease undergoing surgery and anesthesia are at high risk for increased adverse events that include, cardiovascular complications and mortality, and further deterioration of renal function and development of acute kidney injury (AKI) due to perioperative injuries that are caused by hemodynamic instability, hypovolemia, or drug toxicity. Special consideration should be placed on preventing further deterioration of renal function as well as protection of existing renal function in patients with moderate to severe impairment from the effects of anesthetics and pain medications. Analgesics such as nonsteroidal anti-inflammatory drugs (NSAIDs) can contribute to a reduction of the residual renal function in CKD and should be avoided. Renal IR injury is a risk factor for acute renal failure and delayed graft function. Pathogenic factors for renal IR include, but are not limited to, the following: oxidative stress, inflammation, cellular necrosis, and apoptosis. Further clinical studies are required to address the optimal medication regimen that can be used for postoperative pain management in the more severe stages of CKD, including hemodialysis.

**Keywords:** Chronic kidney disease; GFR; Acute kidney injury; Renal function.

### 1. Introduction

More than 1.8 million people in the UK have a diagnosis of chronic kidney disease- CKD with around a further million thought to be undiagnosed. The estimated disease prevalence in the UK of CKD stages 3–5 is thought to be between 4.3% and 8.5% and this appears to be increasing. The global prevalence of CKD is 9.1%. In 2017, CKD was recognised as the 12th leading cause of death, responsible for 1.2 million deaths each year worldwide. Deaths attributable to CKD are set to increase further, with predictive modelling suggesting that by 2040 2.2 million people will die annually because of CKD. Chronic kidney disease is defined by the Renal Association as an abnormality of kidney structure or function that lasts more than 3 months. It only becomes evident when fewer than 40% of nephrons are functioning. Tests which enable diagnosis include electrolyte abnormalities due to tubular disorders, proteinuria (albumin to creatinine ratio (ACR)  $> 3 \text{ mg/mmol}$ ), haematuria of renal origin, histological or radiological abnormalities in structure and abnormal function with resultant raised creatinine and/or cystatin C (eGFR  $<60 \text{ ml/min/1.73 m}^2$ ) on more than two occasions 90 days apart. Glomerular filtration rate (GFR) is the internationally accepted measure to express renal function. However, GFR is not routinely measured because it is a complex procedure that involves measurement of plasma or urinary clearance of an exogenous marker such as inulin. More commonly, eGFR is used: this uses serum creatinine, cystatin, or both, age, and sex to mathematically derive the eGFR. According to the NIDDK, more than 30 million American adults may have CKD. Safe anesthetic management requires an understanding of CKD pathophysiology to prevent aggravation of pre-existing disease. Patients who are dialysis-dependent are at particularly high risk of requiring emergency medical and surgical interventions due to frequent presence of comorbidities such as acute

\* Corresponding author: Maria I. Dalamagka

cardiovascular events, heart failure, vascular-access-related infections, endocarditis, bowel ischemia/bleeding, limb ischemia/necrosis, bone fractures, and cancer. Chronic kidney disease is associated with an increased risk of ischaemic heart disease. Most patients do not progress to end-stage renal failure but die as a result of fatal cardiovascular complications. Left ventricular hypertrophy can occur because of chronic volume overload, pressure overload, or both. Patients undergoing renal replacement therapy (RRT) may exhibit calciphylaxis, which describes accumulation of calcium in small blood vessels. Patients with end-stage renal failure were considered hypocoagulable. Acquired uraemic platelet dysfunction and thromboasthenia decrease platelet adhesion and increase vessel wall fragility, which contribute to bleeding diatheses [1-9].

### 1.1. Anesthetic management of patients in CKD

Depending on the patient's status and the surgical procedure, the preoperative evaluation may require close communication between the primary care physician, nephrologist, surgeon, and anesthesiologist to determine if a patient is optimized for surgery. The following assessments are recommended for patients with CKD: comorbid conditions, severity of CKD assessed by level of kidney function, complications related to level of kidney function, risk for loss of kidney function, and risk for cardiovascular disease. The risk for cardiovascular complications should be promptly evaluated. The patient should undergo a routine electrocardiogram. All present preoperative abnormalities, such as anemia, hyperkalemia, and metabolic acidosis, should be preoperatively corrected. A hemoglobin value of 10 g/dl is strongly recommended. Calcium chloride, insulin and dextrose, sodium bicarbonate, and resins can be used to correct hyperkalemia. If the patient is under dialysis treatment, the final dialysis prior to surgery should be scheduled 12-24 hours before surgery. The anesthetic management of patients suffering from CKD is complex. Due to delayed gastric emptying and neuropathy, there is risk of gastric acid aspiration. Gastric aspiration prophylaxis can be managed using sodium citrate, metoclopramide, anti-H<sub>2</sub> drugs, and rapid induction. Short-acting anesthetic drugs are recommended (propofol, remifentanyl, cisatracurium, vecuronium). Sevoflurane can deteriorate renal function by fluoride ion and compound A production, so isoflurane remains the preferred anesthetic inhalator agent. When selecting a neuromuscular blocking agent (NMBA) for use in patients with CKD, the anesthesiologist should consider the impact of renal impairment on the elimination of the drug, the potential for drug accumulation with incremental doses, and the production of active metabolites. To prevent postoperative residual curarization (PORC), long-acting NMBAs should be avoided. Opioids may be used, as they have no direct toxic effects on the kidney. They do, however, have an antidiuretic effect, and they may cause urinary retention. Patients with Stage 5 CKD who have undergone renal transplantation require immunosuppression. These drugs are usually given by the oral route. If enteral administration is inappropriate, then IV administration with dose adjustment will be required. Desflurane and isoflurane are metabolised to a minimal extent and there are no concerns regarding their use in CKD. Atracurium is the neuromuscular blocking agent (NMBA) of choice in patients with CKD. Its metabolism is unique. Atracurium undergoes ester hydrolysis and Hofmann degradation, both of which are independent of renal function. Cisatracurium can also be used in patients with CKD. It is predominantly eliminated by Hofmann degradation. Sugammadex has been used successfully in clinical research to reverse blockade from aminosteroid NMBAs in patients with severe renal impairment. Opioids have no direct nephrotoxic effects. They do have an antidiuretic effect, which may manifest clinically as urinary retention. Remifentanyl is not dependent on renal function for its elimination. An appropriate dose reduction of morphine and Fentanyl should be considered. Regional anaesthesia is a useful way of avoiding systemic drug accumulation and can be used as the primary anaesthetic technique, depending on the type of surgery. Central neuraxial blockade is not absolutely contraindicated, but consequent hypotension must be attended to in order to prevent worsening of renal perfusion. The ability of the kidneys to excrete water and sodium is impaired, causing difficulty in handling large fluid loads. As glomerular filtration decreases, sodium and water are retained and because of the increased hydrostatic pressure, fluid moves into the extravascular space leading to the clinical manifestations of fluid overload such as generalised and pulmonary oedema. Loss of nephron function results in renal retention of potassium. As GFR declines, the remaining nephrons adapt by increasing renal excretion and this works as a compensatory mechanism when the GFR is >15 ml/min. The tendency towards hyperkalaemia is ameliorated by instituting a low potassium diet and avoiding drugs known to cause hyperkalaemia [10-22].

## 2. Results and Discussion

Renal ischemia/reperfusion (IR) injury is a leading cause of preoperative acute kidney injury (AKI), which frequently complicates major vascular, cardiac, transplant, and liver surgeries. AKI has been shown to occur after some major surgeries. Novel interventions that protect against IR injury are needed to improve early graft function after kidney transplantation. Available general and local anesthetics, including third generation inhaled anesthetics, propofol, and amide-class local anesthetics, are effective and safe with a low incidence of side effects. Propofol, a widely used anesthetic, has shown potential as a novel organ-protective agent through its efficient membrane-targeted and cytoprotective effects. This is not surprising since lipid emulsions are major components of propofol, which

accumulating data show provide organ protective effects against IR injury in many organs, such as the heart, kidney, liver, and intestines. Sevoflurane and enflurane, did not cause deterioration of postoperative renal function in patients with preexisting renal issues. The metabolism of enflurane to inorganic fluoride during and after surgery did not cause a clinically significant level of renal disease or dysfunction. There is significant clinical evidence for volatile anesthetic-mediated organ protection. Sevoflurane preconditioning significantly lessened the postoperative rise of transaminase levels in patients undergoing liver resection. A short period of sevoflurane preconditioning in patients undergoing coronary artery bypass graft surgery was shown to significantly decrease both the release of a myocardial contractile dysfunction marker and the levels of plasma cystatin C concentrations. Ketamine ameliorate the upregulation of inflammatory pathways and reduction of metabolism caused by hypoxia. Thiopental pretreatment reduced renal IR injury induced by free radicals. Opioids may be used, as they have no direct toxic effects on the kidney. They do, however, have an antidiuretic effect. Patients with Stage 5 CKD who have undergone renal transplantation require immunosuppression. Nonsteroidal anti-inflammatory drugs (NSAIDs) can contribute to a reduction of the residual renal function in CKD and should be avoided.

Chronic kidney disease may result in a wide array of neurological manifestations including myoclonus, asterixis, chorea, uraemic encephalopathy and seizures. Autonomic neuropathy is common in patients with CKD because of decreased baroreceptor sensitivity, sympathetic overactivity and parasympathetic dysfunction. In haemodialysis, dialysate is pumped in a counter-current direction to blood flow and solutes equilibrate after diffusion across a semipermeable membrane. This too has associated complications, which include those related to vascular access, hypotension, arrhythmias and dysequilibration syndrome. Disease-specific treatment should be continued throughout the perioperative period where it is safe and practicable to do so. Certain medications such as ACE inhibitors remain controversial in the preoperative setting with no consensus regarding whether preoperative ACE inhibitor therapy is beneficial or harmful. Therefore, preoperative antihypertensive management should be considered on an individual basis and remains at the discretion of the anaesthetist. Should a patient require dialysis before surgery, liaise with the patient's specialist team.

---

### 3. Conclusion

The majority of research studies have focused on the non-anesthetic effects of propofol, such as its antioxidant, immunomodulatory, analgesic, and neuroprotective effects. It was found that propofol and ketamine were the 2 anesthetic agents that produced the least renal oxidative stress. AKI incidence, morbidity, and ICU stay were all found to be significantly lower for patients who received dexmedetomidine. Bupivacaine's ability to decreased cell damage suggests that it may have some protective effects against renal IR injury. Further clinical studies are required to address the optimal medication regimen that can be used for postoperative pain management in the more severe stages of CKD, including hemodialysis.

---

### References

- [1] Livio M, Gotti E, Marchesi D. Uraemic bleeding: role of anaemia and beneficial effect of red cell transfusions. *Lancet*. 1982; 2:1013-1015.
- [2] Ocak G, Lijfering W.M, Verduijn M. Risk of venous thrombosis in patients with chronic kidney disease: identification of high-risk groups *J Thromb Haemost*. 2013; 11:627-633.
- [3] Títóff V, Moury H.N, Títóff I.B. Seizures, antiepileptic drugs, and CKDAm *J Kid Dis*. 2019; 73:90-101.
- [4] Craig R.G, Hunter J.M. Recent developments in the perioperative management of adult patients with chronic kidney disease *Br J Anaesth*. 2008; 101:296-310.
- [5] National Institute for Health and Care Excellence. Chronic kidney disease in adults: assessment and management, clinical guideline CG182 NICE, London, 2015.
- [6] Wright M, Southcott E, MacLaughlin H. Clinical practice guideline on undernutrition in chronic kidney disease. *BMC Nephrol*. 2019; 1620:370.
- [7] Gemmell L, Docking R.D, Black E. Renal replacement therapy in critical care *BJA Educ*. 2017; 17:88-93.
- [8] Bradley T, Teare T, Milner Q. Anaesthetic management of patients requiring vascular access surgery for renal dialysis *BJA Educ*. 2017; 17:269-274.
- [9] Lea-Henry T.N, Carland J.E, Stocker S.L. Clinical pharmacokinetics in kidney disease *CJASN*. 2018; 13:1085-1095.

- [10] Fukazawa Kyota, Lee H. Thomas. Volatile anesthetics and AKI: risks, mechanisms, and a potential therapeutic window JASN. 2014; 25:884-892.
- [11] R. G. Craig, J. M. Hunter. Recent developments in the perioperative management of adult patients with chronic kidney disease, BJA: British Journal of Anaesthesia, Volume 101, Issue 3, September 2008, Pages 296–310, <https://doi.org/10.1093/bja/aen203>.
- [12] Centers for Disease Control and Prevention. National Chronic Kidney Disease Fact Sheet, 2017. National Chronic Kidney Disease Fact Sheet, 2017.
- [13] Krishnan M. Preoperative care of patients with kidney disease. Am Fam Physician. 2002 Oct 15;66(8):1471-6, 1379. PubMed PMID: 12408421.
- [14] Domi R, Huti G, Sula H, Baftiu N, Kaci M, et al. From Pre-Existing Renal Failure to Perioperative Renal Protection: The Anesthesiologist's Dilemmas, Anesth Pain Med. 2016; 6(3):e61545. doi: 10.5812/aapm.32386.
- [15] Sellbrant I, Brattwall M, Jildenstål P, Warren-Stomberg M, Forsberg S, Jakobsson JG. Anaesthetics and analgesics; neurocognitive effects, organ protection and cancer reoccurrence an update. Int J Surg 2016; 34: 41–46.
- [16] Andrews DT, Royse C, Royse AG. The mitochondrial permeability transition pore and its role in anaesthesia-triggered cellular protection during ischaemia-reperfusion injury. Anaesth Intensive Care 2012; 40: 46–70.
- [17] L, Ruffenach G, Kararigas G, et al. Intralipid protects the heart in late pregnancy against ischemia/reperfusion injury via Caveolin2/STAT3/GSK-3 $\beta$  pathway. J Mol Cell Cardiol 2017; 102: 108–116.
- [18] Li J, Motayagheni N, Barakati N, Eghbali M. Intralipid protects the heart in late pregnancy against ischemia/reperfusion injury by reducing cardiomyocyte apoptosis via Mir122 induction. Circ Res 2017; 119:A442.
- [19] Motayagheni N, Eghbali M. Complete reversal of xylazine-induced bradycardia with intralipid in female mice. Circ Res 2016; 119:A253.
- [20] Motayagheni N, Phan S, Eshraghi C, Eghbali M. Inhibition of leptin receptor abolishes intralipid-induced cardioprotection against ischemia-reperfusion injury. Cardiology 2016; 134: 241.
- [21] Motayagheni N, Eghbali M. Reversal of xylazine-induced bradycardia with intralipid. Cardiology 2016; 134: 431.
- [22] Motayagheni N. From bupivacaine to intralipid: Leading edge. J Anesth Clin Res 2016; 4: 00164.



# Remifentanil as an anesthetic agent in children

Maria I Dalamagka

## ► To cite this version:

Maria I Dalamagka. Remifentanil as an anesthetic agent in children. GSC Advanced Research and Reviews , 2024, 20, pp.001 - 004. 10.30574/gscarr.2024.20.1.0238 . hal-05432506

**HAL Id: hal-05432506**

**<https://hal.science/view/index/docid/5432506>**

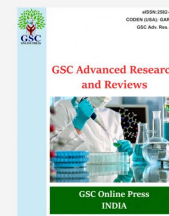
Submitted on 26 Dec 2025

**HAL** is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



Distributed under a Creative Commons Attribution 4.0 International License



## Remifentanyl as an anesthetic agent in children

Maria I Dalamagka \*

*Anesthesiology Department, General Hospital of Larisa, Greece.*

GSC Advanced Research and Reviews, 2024, 20(01), 001–004

Publication history: Received on 19 May 2024; revised on 29 June 2024; accepted on 02 July 2024

Article DOI: <https://doi.org/10.30574/gscarr.2024.20.1.0238>

### Abstract

Remifentanyl hydrochloride is an ultra-short-acting opioid that undergoes rapid metabolism by tissue and plasma esterases. Remifentanyl is increasingly used as an adjuvant to general anaesthesia in neonates undergoing surgery because of its favourable haemodynamic and respiratory effects compared with other agents. The pharmacokinetic profile of remifentanyl appears well suited for use in the sedation of children for short, day case procedures and remifentanyl has been demonstrated to be an attractive drug for this purpose. Owing to higher clearances, younger children will require higher infusion rates of remifentanyl than older children and adults to achieve equivalent plasma concentrations. Propofol and remifentanyl appear to be gaining popularity for short procedures in children. Together they provide sedation, anxiolysis and analgesia, possibly best provided by separate continuous infusions. They also promote haemodynamic stability, minimal respiratory depression, with a rapid recovery profile. The use of remifentanyl in neonates and young infants is increasing despite the difficulties in obtaining high standard objective evidence in this age group. Remifentanyl's pharmacokinetic and pharmacodynamic profile in this age group is similar to that of older children, which may have several theoretical advantages. Reports of a rapid development of  $\mu$ -receptor tolerance with remifentanyl are in conflict; activity at  $\delta$ -opioid receptors may contribute. Remifentanyl produce a fall in blood pressure and cardiac index, mainly as a result of a fall in heart rate. Although atropine is able to reduce the fall in heart rate, is not able to prevent completely the reduction in cardiac index. Remifentanyl has been described as a titratable opioid. It has been reported to be ideally suited to the sedation of selected neurosurgical patients who require frequent neurological examinations. It may also provide suitable analgesia based sedation in mechanically ventilated children postoperatively.

**Keywords:** Remifentanyl; Anaesthesia; Sedation; Children; Surgery

### 1. Introduction

Remifentanyl has a strong affinity for  $\mu$  opioid receptors although less for  $\kappa$  and  $\delta$ . The former is antagonised by naloxone. Remifentanyl is a potent  $\mu$ -agonist that retains all the pharmacodynamic characteristics of its class regarding analgesia, respiratory depression, muscle rigidity, nausea and vomiting, pruritus. Remifentanyl is a potent respiratory depressant and this effect has been widely studied in children. It offers a unique titratability when its effects need to be quickly achieved or suppressed. Remifentanyl is an ultra-short acting synthetic opioid; it shares some similarities with other piperidine derivatives such as alfentanil and fentanyl, but with a potency 26–65 times that of alfentanil. Remifentanyl is a synthetic opioid, first introduced into clinical practice in 1996. Its unique pharmacokinetic profile has resulted in a gradual increase in its popularity in paediatric anaesthesia. It is an opioid of high potency and rapid clearance, consequently lacking problems of accumulation. These characteristics give it a high degree of predictability and it has become an attractive choice for a wide variety of anaesthetic challenges, from premature neonates to the elderly. Neonates and infants have a higher clearance than older children and as a result, remifentanyl has additional benefits in this group. Remifentanyl showed an extremely rapid elimination similar to that in adults. The fast clearance rates observed in neonates and infants, as well as the lack of age related changes in half life, are in sharp contrast to the

\* Corresponding author: Maria I Dalamagka

pharmacokinetic profile of other opioids. Remifentanyl in a bolus dose of 5 microg/kg may cause hypotension in anesthetized children [1-12].

---

## 2. Total intravenous anesthesia

Propofol administered in conjunction with an opioid such as remifentanyl is used to provide total intravenous anesthesia for children. Recommended target concentrations for both propofol and remifentanyl depend on the type of surgery, the degree of surgical stimulation, the use of local anesthetic blocks, and the ventilatory status of the patient. Total intravenous anesthesia has been widely practiced in adult anesthesia since the introduction of propofol into routine clinical practice in 1982. Although there were reports of the use of total intravenous anesthesia in children as early as 1989, it is only in recent years that the practice in children has become more common. The main benefits of using a total intravenous anesthesia technique include reduced postoperative nausea and vomiting with propofol use, quicker recovery, improved quality of recovery. Results for both propofol and remifentanyl in children, required a larger loading dose, expressed as per kilogram, than healthy adults. Total intravenous anesthesia techniques use propofol as the principal drug for induction and maintenance of anesthesia. The commonly used medications include the opioids remifentanyl, alfentanil, and sufentanil. Ketamine is occasionally used, but it has a long context-sensitive half-time with consequent delayed awakening after prolonged exposure. Coadministration of midazolam reduces the propofol dose for induction and the amnesic effect. The high potency, fast onset, and short context-sensitive half-time of 2 to 3 min means that remifentanyl can be effectively used by infusion for prolonged periods of time. There is additivity of anesthetic effect when used with propofol described in adults and in children 1 to 11 year, although effects such as apnea appear synergistic. A pharmacodynamics model describing the propofol and remifentanyl additive interaction for anesthesia in children 1 to 12 year using Bispectral Index as an effect measure is similar to that reported in adults. Maintenance infusion requirements for propofol in neonates differ substantively from those in older infants and children. These may be attributed mostly to differences in pharmacokinetics. Delayed awakening, hypotension, and an increased incidence of bradycardia have been reported in neonates and infants. The initial loading dose of remifentanyl may cause hypotension and bradycardia in children, prompting some to target the plasma rather than effect-site concentration when initiating an infusion. Remifentanyl clearance is increased and volume of distribution also increased, care should be taken using bolus doses larger than 0.25 mcg/kg in infants less than 6 months in whom blood pressure decreases can cause a lowering of cerebral perfusion. An increase in remifentanyl infusion rate is a better option. The anesthetic requirement in children with cognitive impairment may be reduced. Processed electroencephalogram use may better dictate dose. Propofol metabolism is reduced in neonates and matures over the first six months of life. Remifentanyl volume of distribution and clearance are increased in neonates compared to children and adults. Sensitivity to the both sedative effects and analgesic effects is increased. Adverse effects such as hypotension are increased in neonates given propofol. These effects can lead to delayed recovery after anesthesia [13-25].

---

## 3. Emergence agitation

Emergence agitation (EA) is one of the main concerns in the field of pediatric anesthesia using short acting volatile anesthetics, such as sevoflurane. This postanesthetic behavioral disturbance is frequently developed in ophthalmological surgery, and EA may cause harm to the child and distress to caregivers, leading to dissatisfaction with the anesthetic. Fentanyl and remifentanyl are short-acting opioids that are widely used in pediatric patients. Both drugs have been used as sedatives and analgesics, which can affect the hemodynamic response of blunt surgical stimulation as well as tracheal intubation. The clinical usefulness of these drugs has been suggested to reduce EA occurring post sevoflurane anesthesia. However, opioids still have some adverse effects on respiratory depression, postoperative vomiting, and opioid-induced hyperalgesia during the postoperative period. EA is commonly seen in children during the early post anesthesia period following general anesthesia. Several factors can affect the incidence of EA, including age, type of anesthetics, type of surgery, preoperative anxiety, and postoperative pain. Among these factors, EA is more common at a young age, and when sevoflurane is used during general anesthesia.

---

## 4. Outcomes

Remifentanyl is an attractive drug for use in younger children and may be feasible for use in situations where spontaneous ventilation is advantageous. Remifentanyl provides titratable analgesia with haemodynamic stability and prompt recovery allowing early neurological assessment which are the fundamental goals of neuroanaesthetic management. With regard to the neurocognitive effect of remifentanyl, evaluated as an index of potential neuroprotection, recent studies have demonstrated that morphine and fentanyl, but not remifentanyl, disrupt cholinergic neurotransmission. This disruption is known to cause postoperative delirium and impaired memory function. The addition of remifentanyl to propofol reduce mean arterial pressure and heart rate without affecting CBFV

(cerebral blood flow velocity). This implies that CBF (cerebral blood flow) autoregulation is preserved during remifentanyl/propofol anaesthesia. Remifentanyl/propofol anaesthesia increases the autoregulatory index with the suggestion of a widened autoregulatory plateau despite the lower flow status of this anaesthetic combination. Combination of remifentanyl and midazolam infusions have also proved satisfactory for sedation in children requiring ventilation after cardiac surgery.

---

## 5. Discussion

Fentanyl is a short acting opioid that possesses synthetic  $\mu$  receptor stimulating properties and is widely used in children as an intraoperative sedative and analgesic. Fentanyl has a short peak time and action duration after intravenous administration. Like other opioids, it has potential negative effects through central nervous system actions. Earlier administration of fentanyl, during intubation, based on hemodynamic stability during tracheal intubation, intraoperative analgesia, and postoperative complications. Remifentanyl has an ester linkage and these unique pharmacokinetics allow for more rapid onset and offset. In addition, a short context sensitive half life of remifentanyl allowed for intraoperative continuous infusion. Postoperative pain might develop with the use of remifentanyl owing to its faster elimination properties. Remifentanyl has been shown to be a useful agent for sedation during cardiac catheterisation in children owing to its favorable pharmacokinetic properties and haemodynamic stability during stressful events. Remifentanyl has also been shown to provide satisfactory peri-operative control of stress response to cardiac surgery. Use of remifentanyl has been associated with N<sub>2</sub>O and volatile agents for general anesthesia and with propofol for total intravenous anesthesia (TIVA). It seems very useful for sedation inside and outside the operating room and in intensive care for both short painful procedures and synchronization with mechanical ventilation.

---

## 6. Conclusion

Remifentanyl representing the most predictable opioid in paediatric anesthesia. Remifentanyl is an opioid of high potency and rapid clearance, lacking problems of accumulation. These characteristics give it a high degree of predictability and it has become an attractive choice for a wide variety of anaesthetic challenges, from premature neonates to the elderly. Neonates and infants have a higher clearance than older children and as a result, remifentanyl has additional benefits in this group. It is well known that continuous opioid infusions can cause tolerance. Therefore, there is evidence that remifentanyl infusions may result in increased postoperative pain and increased opioid requirements.

---

## Compliance with ethical standards

### *Disclosure of conflict of interest*

No conflict of interest to be disclosed.

---

## References

- [1] Bürkle H, Dunbar S, Van Aken H. Remifentanyl: a novel, short acting,  $\mu$  opioid. *Anesthesia and Analgesia* 1996; 83: 646–51.
- [2] Glass PSA, Gan TJ, Howell S. A review of the pharmacokinetics and pharmacodynamics of remifentanyl. *Anesthesia and Analgesia* 1999; 89: S7–14.
- [3] Reyntjens K, Foubert L, De Wolf D, Vanlerberghe G, Mortier E. Glycopyrrolate during sevoflurane–remifentanyl-based anaesthesia for cardiac catheterization of children with congenital heart disease. *British Journal of Anaesthesia* 2005; 95: 680–4.
- [4] Ross AK, Davis PJ, Dear G deL, et al. Pharmacokinetics of remifentanyl in anesthetized pediatric patients undergoing elective surgery or diagnostic procedures. *Anesthesia and Analgesia* 2001; 93: 1393–401.
- [5] Davis PJ, Wilson AS, Siewers RD, Pigula FA, Landsman IS. The effects of cardiopulmonary bypass on remifentanyl kinetics in children undergoing atrial septal defect repair. *Anesthesia and Analgesia* 1999; 89: 904–8.
- [6] Rigby-Jones AE, Priston MJ, Sneyd JR, et al. Remifentanyl–midazolam sedation for paediatric patients receiving mechanical ventilation after cardiac surgery. *British Journal of Anaesthesia* 2007; 99: 252–61.
- [7] Davis PJ, Cladis FP. The use of ultra-short-acting opioids in paediatric anaesthesia. The role of remifentanyl. *Clinical Pharmacokinetics* 2005; 44: 787–96.

- [8] Reid JE, Mirakhur RK. Bradycardia after administration of remifentanyl. *British Journal of Anaesthesia* 2000; 84: 422–3.
- [9] Wang J, Winship S, Russell G. Induction of anaesthesia with sevoflurane and low-dose remifentanyl: asystole following laryngoscopy. *Anesthesia and Analgesia* 1998; 81: 994–5.
- [10] Elliott P, O'Hare R, Moyna Bill K, Phillips AS, Gibson FM, Mirakhur RK. Severe cardiovascular depression with remifentanyl. *Anesthesia and Analgesia* 2000; 91: 58–61.
- [11] DeSouza G, Lewis MC, TerRiet MF. Severe bradycardia after remifentanyl. *Anesthesiology* 1997; 87: 1019–20.
- [12] Briassoulis G, Spanaki AM, Vassilaki E, Fytrolaki D, Michaeloudi E. Potentially life-threatening bradycardia after remifentanyl infusion in a child. *Acta Anaesthesiologica Scandinavica* 2007; 51: 1130–1.
- [13] Weale NK, Rogers CA, Cooper R, Nolan J, Wolf AR. Effect of remifentanyl infusion rate on stress response to the pre-bypass phase of paediatric cardiac surgery. *British Journal of Anaesthesia* 2004; 92: 187–94.
- [14] Bakan M, Cakmakkaya S, Onsel I, Kaya G. Inadvertent bolus administration of high-dose remifentanyl during anesthesia in a 6-year-old girl. *Pediatric Anesthesia* 2006; 16: 1202.
- [15] Robinson DN, O'Brien K, Kumar R, Morton NS. Tracheal intubation without neuromuscular blockade in children: a comparison of propofol combined either with alfentanil or remifentanyl. *Paediatric Anaesthesia* 1998; 8: 467–71.
- [16] Klemola U-M, Hiller A. Tracheal intubation after induction of anesthesia in children with propofol–remifentanyl or propofol–rocuronium. *Canadian Journal of Anesthesia* 2000; 47: 854–9.
- [17] Weber F, Füssel U, Gruber M, Hobbhahn J. The use of remifentanyl for intubation in paediatric patients during sevoflurane anaesthesia guided by bispectral index (BIS) monitoring. *Anaesthesia* 2003; 58: 749–55.
- [18] Batra YK, Al Qattan AR, Ali SS, Qureshi MI, Kuriakose D, Migahed A. Assessment of tracheal intubating conditions in children using remifentanyl and propofol without muscle relaxant. *Pediatric Anesthesia* 2004; 14: 452–6.
- [19] Blair JM, Hill DA, Wilson CM, Fee JPH. Assessment of tracheal intubation in children after induction with propofol and different doses of remifentanyl. *Anaesthesia* 2004; 59: 27–33.
- [20] Crawford MW, Hayes J, Tan JM. Dose-response of remifentanyl for tracheal intubation in infants. *Anesthesia and Analgesia* 2005; 100: 1599–604.
- [21] Morgan JM, Barker I, Peacock JE, Eissa A. A comparison of intubating conditions in children following induction of anaesthesia with propofol and suxamethonium or propofol and remifentanyl. *Anaesthesia* 2007; 62: 135–9.
- [22] Foubert L, Reyntjens K, De Wolf D, Suys B, Moerman A, Mortier E. Remifentanyl infusion for cardiac catheterization in children with congenital heart disease. *Acta Anaesthesiologica Scandinavica* 2002; 46: 355–60.
- [23] Tirel O, Chanavaz C, Bansard JY, et al. Effect of remifentanyl with and without atropine on heart rate variability and RR interval in children. *Anaesthesia* 2005; 60: 982–89.
- [24] Chanavaz C, Tirel O, Wodey E, et al. Haemodynamic effects of remifentanyl in children with and without intravenous atropine. An echocardiographic study. *British Journal of Anaesthesia* 2005; 94: 74–9.
- [25] Scott LJ, Perry CM. Remifentanyl: a review of its use during the induction and maintenance of general anaesthesia. *Drugs* 2005; 65: 1793–823.
- [26] Shinohara K, Aono H, Unruh GK, Kindscher JD, Goto H. Suppressive effects of remifentanyl on hemodynamics in baro-denervated rabbits. *Canadian Journal of Anesthesia* 2000; 47: 361–6.



# Obesity in children and general anesthesia

Maria I Dalamagka

## ► To cite this version:

Maria I Dalamagka. Obesity in children and general anesthesia. GSC Biological and Pharmaceutical Sciences, 2022, 19, pp.308 - 309. 10.30574/gscbps.2022.19.2.0206 . hal-05432493

**HAL Id: hal-05432493**

**<https://hal.science/view/index/docid/5432493>**

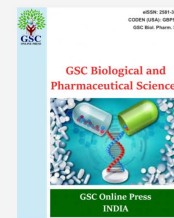
Submitted on 26 Dec 2025

**HAL** is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



Distributed under a Creative Commons Attribution 4.0 International License



(CASE REPORT)



## Obesity in children and general anesthesia

Maria I Dalamagka \*

*Department of Anaesthesia, General Hospital of Larisa, Larisa, Thessaly, Greece.*

GSC Biological and Pharmaceutical Sciences, 2022, 19(02), 308–309

Publication history: Received on 25 April 2022; revised on 27 May 2022; accepted on 29 May 2022

Article DOI: <https://doi.org/10.30574/gscbps.2022.19.2.0206>

### Abstract

Childhood obesity has reached epidemic levels in developed as well as in developing countries. Overweight and obesity in childhood are known to have significant impact on both physical and psychological health. Obese 10-year-old boy undergoes general anesthesia for emergency peritonitis surgery. The purpose of this study was to study the intraoperative behavior of obese children as they are accompanied by respiratory and metabolic disorders.

**Keywords:** Obesity; Anesthesia; Allergia; Children; Bridion; Astma

### 1. Introduction

The World Health Organization (WHO) has identified childhood obesity as “one of the most serious public health challenges of the 21st century.” In 2016, WHO found that globally, the number of overweight children under age 5 topped 41 million, and the number between ages 5 and 19 was 340 million. Socio-cultural factors have also been found to influence the development of obesity. Our society tends to use food as a reward, as a means to control others, and as part of socializing. Nearly 30% of obese children suffer from asthma. Obese children are more likely to suffer from respiratory infections which can necessitate a change in timing of elective surgery. Patients with bronchial asthma have an increased risk of developing intra-operative bronchospasm, especially if they are receiving general anesthesia with an endotracheal tube. Bronchospasm occurs because as bronchoconstriction becomes more severe it takes a progressively longer time for the alveoli to empty their carbon dioxide during exhalation [1-5].

### 2. Case Study

A child aged 10 years and 60 kg came with an acute abdomen image for appendectomy. The child had allergic bronchial asthma, with a known allergic predisposition, known egg allergy and frequent asthma attacks two years ago. Preoperative administration of Solu medrol 120 mg, Dexatone 4 mg and Ondansetron 4 mg was performed. Introduction to anesthesia was performed with Dormicum 1 mg, Ketamine 10 mg, Fentanyl 0.1 mg, Propofol 18 mg, Esmeron 50 mg. After intubation, entered to mechanical ventilation, with a volume control ventilation model and a Primus Dräger ventilator. Maintenance of anesthesia was achieved with Sevoflurane 2.5%, additional Fentanyl 0.1 mg and morphine 3 mg were given. The duration of the surgery was one hour. At the end of the surgery, Bridion of 200 mg was given with pure inhaled oxygen and the awakening was delayed for half an hour.

### 3. Management and Outcome

The concept of cross-reactivity between drugs used in the perioperative setting and food is often quoted, but usually not supported by evidence. Pre-existing allergic diseases and other confounding non-allergic but related clinical entities need to be carefully considered during the preoperative evaluation, as they may result in perioperative life-threatening

\* Corresponding author: Maria I Dalamagka

Department of Anaesthesia, General Hospital of Larisa, Larisa, Thessaly, Greece.

conditions. For this reason it was given as a hypnotic drug Dormicum, Ketamine and Hypnomidate, instead of Propofol, although studies have shown that it can be used. The awakening was delayed due to the recirculation of medicines and drugs are not calculated based on ideal body weight because the child had acute pain and anxiety as it would lead to complications in the induction of anesthesia. Also given intravenous Solu Medrol and Dexaton removing sensitization.

---

#### 4. Discussion

Expiratory reserve volume is decreased in obesity. Total lung capacity is usually preserved except with morbid obesity, excessive central adiposity or with obesity- hypoventilation syndrome. Obese patient have decreased FRC which may lead to atelectasis. Atelectasis means incomplete expansion. Obesity leads to an increased metabolic rate, increased oxygen demand, increased CO<sub>2</sub> production and increased alveolar ventilation. These is decreased chest wall compliance due to adipose tissue that lies over the thorax. Lung volumes are decreased as the increased abdominal mass forces the diaphragm cephalad. For this reason Bridion was given to the awakening, as the remaining muscle relaxation would lead to life-threatening situations for the patient.

---

#### 5. Conclusion

Obese children have special needs for anesthesia in emergency surgeries and should take into account the metabolic and respiratory disorders that may occur as a result of obesity.

---

#### Compliance with ethical standards

##### *Acknowledgments*

General Hospital of Larisa.

##### *Statement of ethical approval*

The present research work does not contain any studies performed on animals/humans subjects by any of the authors'.

##### *Statement of informed consent*

Written informed consent was obtained from the patient for publication of this case report. A copy of the written consent is available.

---

#### References

- [1] Popkin BM, Doak CM. The obesity epidemic is a worldwide phenomenon. *Nutr Rev.* 1998; 56: 106–14.
- [2] Bhave S, Bavdekar A, Otiv M. IAP National Task Force for Childhood, Prevention of Adult Diseases: Childhood Obesity. IAP National Task Force for Childhood Prevention of Adult Diseases: Childhood Obesity. *Indian Pediatr.* 2004; 41: 559–75.
- [3] Raj M, Sundaram KR, Paul M, Deepa AS, Kumar RK. Obesity in Indian children: Time trends and relationship with hypertension. *Natl Med J India.* 2007; 20: 288–93.
- [4] Brenn BR. Anesthesia for Pediatric Obesity. *Anesthesiol Clin North America.* 1 DEC 2005; 23(4): 745-64
- [5] Blaho K, Winbery S, Merigian K. Pharmacological considerations for the pediatric patient. *Optom Clin.* 1996; 5: 61-90.



## Emergency Medicine

Maria I Dalamagka

### ► To cite this version:

Maria I Dalamagka. Emergency Medicine. World Journal of Advanced Research and Reviews, 2024, 23, pp.394 - 401. 10.30574/wjarr.2024.23.2.2389 . hal-05432486

**HAL Id: hal-05432486**

**<https://hal.science/view/index/docid/5432486>**

Submitted on 26 Dec 2025

**HAL** is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



Distributed under a Creative Commons Attribution 4.0 International License

## Emergency Medicine

Maria I. Dalamagka \*

*Anesthesiology Department, General Hospital of Larisa, Greece.*

World Journal of Advanced Research and Reviews, 2024, 23(02), 394–401

Publication history: Received on 25 June 2024; revised on 04 August 2024; accepted on 06 August 2024

Article DOI: <https://doi.org/10.30574/wjarr.2024.23.2.2389>

### Abstract

Emergency defines the potential provision of rapid and effective Medical Nursing Care, in situations where life is threatened. Emergency medicine is defined as the branch of Medicine that operates within an organized EMS. Emergency Medicine System (EMS) is defined as the organized set of provision of immediate pre-hospital, in-hospital and inter-hospital care. Objectives of Emergency Medicine are the provision of an integrated system of pre-hospital, in-hospital and inter-hospital emergency care; improving the quality of emergency medical care; reduction of mortality, morbidity and disability related to injury and sudden illness. Collection of epidemiological data related to accident prevention and health promotion.

**Keywords:** Emergency Medicine; Trauma; Airway; CPR.

### 1. Introduction

ABC and its variations are the basic steps used by medical professionals in emergency medicine. In its original form it represents the following: airway, breathing and circulation. The protocol was originally developed as a memory aid for rescuers performing CPR, and its most widely known use is in the care of the unconscious patient, although it is also used as a reminder of patient assessment and treatment priorities in many acute medical conditions. Breathing and circulation are essential to life. ATLS (Advanced Trauma Life Support) program of the American College of Surgeons. ABCDE' S : mnemonic rule for approaching multiple injuries and labeling life-threatening injuries, based on the order in which they are life-threatening. Systematically checked: A(Airway) ensuring the patency of the upper airways, while simultaneously protecting the cervical spine; B(Breathing) ensuring breathing and ventilation; C(Circulation) ensuring circulation and stopping any bleeding; D(Disability): Neurologic (status) assessment of the neurological condition; E(Exposure/Environmental control) full exposure of the patient with simultaneous protection from hypothermia. Check of responsiveness, neck and head injury (fall from a height, car accident); assessment of the situation. A tap on the sternum will determine that the patient is indeed unconscious. Not performance of CPR on a patient, who is not fully awake, but is breathing and has circulation. Close approach if it's breathing (See - Hear - Feel). Airway control; the victim must be in the correct position. If there is any exudate (vomit, blood, etc.) in or around the person's mouth must be removed to clear the airway. It must be on a surface as flat as possible. Ensuring that hands of patient are under the sides and his back and legs are straight; open of the airway, head and trachea must be properly aligned; tilt head back towards the sky. There may be situations where the airway is blocked. This can be from a foreign object, the victim's tongue, or vomit or other bodily fluids. Quickly turn the victim's head to one side to aid dislodgement. If the tongue is blocking the airway; try the mandibular traction method; Lift the chin slightly towards the sky without moving the rest of the head. This helps the tongue to drop to the floor of the jaw, despite the arrangement of the airway. Breath control: obvious signs of breathing; there are some obvious signs, that the victim is breathing; look for the rise and fall of the diaphragm. If the patient shows great difficulty in breathing, the circulation is checked. Placement of the hand near the nose and mouth of the patient to see if it's breathing. Normal breathing, there is no need for CPR. Opening the airway may be enough to start the patient breathing again. Traffic control; the index and middle finger, just below the jaw, to the right or left of

\* Corresponding author: Maria I. Dalamagka

the larynx or on the Adam's apple. This is the carotid artery and should provide a strong pulse if the blood is circulating well. If the pulse is weak or if there is no pulse, perform CPR; is not breathing and his pulse is weak or absent, you should perform CPR (30 compressions at a rhythm. We give two full inhalations lasting 1-1.5 sec each)

## 2. Wound

Treatment of the injured follows the practice of collection and rapid transport or the practice of on-site stabilization. Its goal is to identify and deal with situations that put his life in immediate danger. This phase requires recognition and treatment of life-threatening conditions such as tension pneumothorax, massive hemothorax, airway obstruction, cardiac tamponade, and hemorrhagic shock. Time works negatively in the case of the injured. The primary assessment should be completed in the shortest possible time. Primary assessment consists of a structured rapid assessment and treatment of situations, which put the injured person's life in immediate danger. The assessment follows the following steps: A- Airway (Airway) and cervical spine stabilization; B- Ventilation (Breathing); C- Circulation; D- Neurological Assessment (Disability); E-Exposure. Air duct and stabilization of cervical spine: assessment of airway patency must be done quickly because it is essential. The first approach will give us important information. If the victim is talking, then there is an adequate airway and some degree of ventilation. Absence of breathing, reduced chest expansion or the presence of pathological sounds such as snoring or wheezing, are signs of complete or even partial obstruction of the airway, which require immediate treatment. On the other hand, its opening airway in the multi-injured patient always encounters the obstacle of possible damage to the cervical spine. For this reason cervical spine should be immobilized. The basic means of securing the airway are manual opening manipulations, auxiliary means (airways and suction), while specialized manipulations include tracheal intubation and the use of supraglottic ventilation devices. Stabilization of cervical spine: It is a process that already begins with the control of the airway. Stabilization of cervical spine is achieved by the use of a hard collar. The presence of neurological deficit or pain as well as the mechanism of injury play an important role. Also cervical spine stabilizes at a disturbed level of consciousness and substance use. Ventilation (Breathing): assessment of ventilation by speech and pulse oximetry. A respiratory rate of 12-20/min is considered to be in the normal range. Also, the symmetry and expansion of the hemithoraxes give some evidence for the adequacy of ventilation, as does the auscultation of the ribcage. In hypoxemia, which is not corrected with supplemental oxygen administration, selective tracheal intubation and mechanical ventilation are indicated. Regarding respiratory support, the available pre-hospital means are the self-expanding bag (ambu) and portable ventilators (choice of ventilation model, administration of high FiO<sub>2</sub>). Circulation: if the patient responds to verbal stimuli, then he has an elementary circulation, capable of maintaining his brain function. The existence of confusion, agitation and the reduced level of consciousness can perhaps be attributed to hemodynamic instability. Palpation of the pulse in peripheral vessels, such as the carotid artery, indicates the presence of blood pressure  $\approx$  80 mmHg, while simultaneously counting the number of pulses and its quality. The skin provides equally important information. Cold clammy skin indicates hypovolemia. Capillary refill speed ( $\approx$  2 min) and pulse oximetry are altered by cold and the state of the vascular network. Neurological assessment (Disability): rapid neurological examination is concerned with the assessment of the level of consciousness, the assessment of the size of the pupils and the presence of mobility in the extremities. The basic tool for assessing the level of consciousness is the Glasgow scale. The Glasgow scale measured in the first few minutes often changes rapidly after the first resuscitation efforts. However, the presence of a non-improving Glasgow score  $<9$  is an indication for sedation, tracheal intubation and mechanical ventilation. A second point of attention is the pupils of the eyes. The existence of anisocoria, mydriasis without reaction to light or myoses, probably indicate serious intracranial pathology. Exposure: removal of casualty's clothing and rapid assessment of injuries is essential in finding injuries. As hypothermia is a major problem during transport of the multi-casualty, exposure must be limited to the shortest possible time. Secondary assessment: secondary assessment begins after primary assessment has been completed and immediately life-threatening situations have been addressed. In no case is the transfer delayed to clinical assessment is completed at this stage. The head is examined for signs of external bleeding, abrasions, bruises or hematomas, fractures. The cervix is explored for hematoma or subcutaneous emphysema. Grinding in the area of the larynx, in combination with the larynx and subcutaneous emphysema are signs of a fracture of the larynx. At the same time, the carotid arteries are palpated and listened for a murmur. Examination of the chest is of particular importance, since it may conceal potentially life-threatening injuries, which at first assessment did not give any signs. Thus screening for decreased or paradoxical tracheal mobility or displacement, the presence of strangulation, the presence of percussive bluntness or tympani and finally listening to abnormal sounds may reveal the presence of hemothorax or pneumothorax. If this is a titratable injury, whether the course involves the heart and great vessels or the lung parenchyma should be investigated. Examination of the abdomen likewise includes, in principle, episcopal investigation of possible signs of injury. Of particular importance, however, is the palpation of the abdomen for the presence of tenderness. This examination in a relatively stable patient is done during the secondary assessment. But since we have to deal with an unstable patient, who requires sedation, intubation and mechanical ventilation, a quick assessment of the abdomen for the existence of tenderness should be done already from the primary assessment. The pelvis may be a site of extensive blood loss. In addition to exploring for hematomas, external bleeding, abrasions, or bruising, the pelvis should be explored with

anterior-posterior pressure of the pubic symphysis and bilateral pressure on the iliac crests for pain or instability. Finally, the limbs are examined for deformity, reduced mobility, disturbed blood supply. Trauma involving the chest, abdomen, pelvis, and retroperitoneum is often accompanied by significant blood loss. Fractures of long bones with hematomas may have a blood loss of 2-4 units. The most common cause of shock in injured people is bleeding. Other forms of shock such as cardiogenic shock (cardiac tamponade, tension pneumothorax, myocardial contusion), neurogenic shock (spinal cord injury) and septic shock (usually occurring in the second year) may be present but are uncommon. The American College of Surgeons, in an effort to evaluate blood loss and establish a uniform strategy in the treatment of the injured, has established a classification of blood loss according to the clinical picture of the injured and suggests the corresponding treatment. Treatment of hypoglycemia is by administration of crystalloid solutions in sufficient quantities. Blood and blood products are not a priority. Of course, the main condition for dealing with hemorrhagic shock is the control of the bleeding. We should not underestimate blood loss from external wounds, bleeding that is easily controlled with external pressure. Casualty response to rapid administration (bolus) of 2000 ml R/L (recommended by ATLS) may give us information about the amount of bleeding, the need for a transfusion and the possibility of surgery.

---

### 3. Traumatic Brain Injury (TBI)

- GCS = 13 - 15. Loss of consciousness for a short period; very good prognosis; mortality <1%
- GCS = 9 - 12. Good prognosis; mortality <5%
- Severe GCS <8. Mortality >40% high rate of residual damage.

Primary lesions include vascular rupture, hematoma formation, early ischemia and neuronal necrosis. Secondary lesions occur gradually in a second year. Hypotension, hypoxemia, and hypercapnia may result, resulting in increased intracranial pressure and cerebral edema, resulting in cerebral hypoxia. The little one period of low blood supply and hypoxemia can cause secondary ischemic damage that is often irreversible. Airway obstruction with breathing disorders is common in TBI injured and is directly related to the severity of the injury. Intubation if GCS<8; application of mechanical breathing support; securing a venous line and administering fluids; stabilization of the cervical spine, with the placement of a cervical guard; administration of sedation and analgesia drugs; rough neurological assessment of the injured person's condition before administration of sedation; assessment of pupil size and reaction to light. Monitoring should initially include ECG, measurement of systemic blood pressure, pulse oximeter, capnograph. Performing a CT scan of the brain. Diagnostic abdominal lavage or FAST to confirm or rule out intra-abdominal bleeding.

---

### 4. Chest injuries

Chest injuries account for 25% of injury deaths. They involve the chest wall, the pleural space, the lung parenchyma and the meso-pneumonia. The casualty with a chest injury may have: rib fractures; unstable chest; pneumothorax; hemothorax; tension pneumothorax; rupture of the tracheobronchial tree; pulmonary parenchyma fractures. The most common chest injuries are rib fractures and accompanied by pulmonary contusions, hemothorax, pneumothorax and abdominal organ injuries. When the fractures are located in two parts and involve three or more sides in the same hemithorax, then we speak of an unstable chest. The hemithorax loses its stability and exhibits paradoxical mobility. An unstable chest means significant lung injury with underlying intrathoracic or intrapulmonary injury. A pneumothorax is the collection of air inside the pleural cavity. The pneumothorax can be open or closed depending on whether it communicates with the atmosphere. The continued accumulation in the pleural cavity without the possibility of exit creates a life-threatening situation for the injured person, the tension pneumothorax. Hemothorax is the collection of blood in the pleural cavity, often coexisting with pneumothorax. The initial management of chest injuries is based on clinical assessment and does not require radiological examination. Tension pneumothorax, the pressure in the pleural cavity under normal conditions is negative, the outflow of air greatly increases the pressure so that the lung cannot be expanded, the mediastinum moves to the opposite side, venous return is obstructed, and cardiac output is reduced. The diagnosis of pneumothorax is clinical and not radiological. Clinical signs for suspected pneumothorax are: shortness of breath; hypotension; absence of respiratory murmur; stretching of the cheekbones; tracheal displacement; cyanosis. Coping: direct needle decompression; placement of a chest tube. Open pneumothorax, there is communication between the thoracic cavity and the atmosphere and is usually due to penetrating trauma. Coping: conversion to a closed pneumothorax; placement of a chest tube; intubation if necessary; possible surgical treatment. Large hemothorax is usually due to transthoracic wounds with ruptured vessels, but it can also be present in blunt injuries. Diagnosis: chest injury; hemorrhagic shock; absence of respiratory murmur; flat cheekbones. Coping: absolute indication for intubation; aggressive fluid administration; placement of a chest tube; possible thoracotomy. Indications for thoracotomy: hemodynamic instability despite aggressive fluid and blood administration; blood loss >1500 ml with drainage tube placement; persistent bleeding >200 ml/h for >4 h; inability to control bleeding. Emergency thoracotomy in the

Emergency Department (ED) may be lifesaving for some titratable heart injuries. The goals of this intervention are: relief from cardiac tamponade; control of bleeding from the heart and large vessels; application of open heart massages; blockage of the aorta and redistribution of blood to the heart and brain. Unstable chest, diagnosis is clinical and not radiological: existence of fractures of two or more ribs in two or more places; existence of paradoxical mobility; possibility of hemothorax, pneumothorax or both; or display them later. Damage to intra-abdominal organs coexist in 15%. There is usually underlying lung parenchymal damage; mortality is related to lung injuries and not to chest wall injury; need for intubation for prehospital respiratory support; possible surgical treatment; need for ICU hospitalization and mechanical respiratory support. Clinical approach to the casualty with chest injury. The diagnosis of chest injury is not easy in blunt injuries and before an imaging test is performed. Evaluation parameters for the possibility of chest injury form: mechanism of injury; application of great force to the chest wall; sudden deceleration. Points: deformation of the chest cavity; fractures or trauma to the chest wall; stretching of the cheekbones; subcutaneous emphysema. Symptoms are shortness of breath, tachypnea, way of breathing, paradoxical mobility, absence of respiratory murmur, hemoptysis, hypotension, pain.

#### **4.1. Injuries of thoracic aorta**

Traumatic rupture of the thoracic aorta is immediately fatal in 75 – 90%, as the mechanism of rupture of the thoracic aorta in blunt injuries is abrupt deceleration. In the plain chest X-ray in the supine position there may be findings suggestive of a possible aortic rupture, such as: fractures of sternum and scapula; first rib and clavicle fractures; multiple rib fractures; unstable left hemithorax; findings from the mediastinum; ambiguity in the outline of the aortic arch; widening of the mediastinum > 8 cm; levin deviation; tracheal displacement; massive hemothorax; diaphragm rupture. Helical computed tomography is diagnostic for aortic injuries, while transesophageal ultrasound is easily performed in the ED without moving the injured person. Aortography is the test of choice. Coping, the surgical treatment with endovascular repair of the damage. Measures to be taken: ensure airway and breathing; administration of analgesia; blood pressure control with the goal that the systolic blood pressure does not exceed 100 mmHg.

#### **4.2. Abdominal injuries**

Intra-abdominal organ injuries can be caused by blunt and penetrating injuries. They are accompanied by significant blood loss up to shock. The presence of subdiaphragmatic air is an indication of rupture of the hollow viscera. Fractures of lower ribs are often present in splenic or liver injuries, while abnormal Levin course is present in diaphragmatic rupture. Coping: use of FAST helps us detect intra-abdominal bleeding; peritoneal lavage; computed tomography; diagnostic peritoneal lavage is considered an acceptable method for trauma victims with a sensitivity exceeding 90%, but its usefulness in trauma victims is controversial. Depending on its findings, it can be positive or negative. Indications for emergency laparotomy: hemodynamically unstable casualties with positive abdominal wash or ultrasound positive for bleeding; persistent hypotension despite fluid administration; signs of peritonitis; presence of air in the peritoneal cavity; visceral discharge from the wound; hypotension in a casualty with ditratable abdominal trauma; bleeding from stomach, rectum or genital tract in titrating wounds; shooting in abdomen; diaphragm rupture; bladder rupture.

#### **4.3. Orthopedic injuries**

Their treatment is of secondary importance in the first phase of resuscitation. The most important orthopedic injuries are pelvic fractures and long bone fractures. Plain x-ray of the pelvis should be done in every seriously injured person even if there is no evidence of pelvic injury. Pelvic fractures, depending on whether the pelvis remains stable or not, are divided into: Type A: Fixed pelvis; Type B: Partially fixed pelvis; Type C: Unstable pelvis. Unstable pelvic fractures should be stabilized with external osteosynthesis in the ED before any movement of the casualty. Casualties with pelvic fractures that continue to bleed despite fixation with external fusion should be referred for angiography and possible embolization. Angiography should also be performed in injured patients in whom contrast leakage is observed during CT. Pain should be treated promptly with intravenous opioids.

Massive hemothorax: rapid accumulation of more than 1500 ml of blood in the chest cavity. Shock with absence of respiratory murmur and bluntness on percussion. Treatment is blood volume restoration and chest cavity decompression. Immediate thoracotomy: if 1500ml of blood is immediately evacuated. Open pneumothorax, large open gaps in the chest. With each respiratory effort, air passes through the opening and leads to hypoxia and hypercapnia. Treatment: pad is fixed on three sides, while the fourth remains free. Unstable chest, asymmetrical and uncoordinated movements of the chest. Limited range of breathing. Screaming from rib or cartilage fractures. Treatment: adequate ventilation. Re-expansion of the lung. Cardiac tamponade. In shock characterized by insufficient tissue perfusion and oxygenation, we follow two steps: the first is the clinical assessment and the second is the identification of the possible cause and the mechanism of injury. Clinical signs of shock are cold and pale extremities, tachycardia and hypotension. Shock is clinically differentiated into: A. Hemorrhagic; B. Non-hemorrhagic: Cardiogenic, myocardial contusion, cardiac

tamponade, air embolism, rare myocardial infarction; Neurogenic, spinal cord injury: hypotension without tachycardia and without skin vasoconstriction, due to loss of sympathetic tone; Septic, delayed arrival – titrating abdominal injuries. Clinical differentiation from tension pneumothorax should also always be made. The most common cause of shock is bleeding that can come from the peritoneal cavity, pelvis, retroperitoneum and chest cavities. Assessment of blood loss is based on blood pressure, respiratory rate, diuresis, level of consciousness, and response to fluid administration. When during initial fluid resuscitation, the patient is unresponsive, then there is a need for immediate surgical intervention. In head trauma to avoid secondary brain damage, cerebral irrigation and cerebral blood flow should be enhanced. This is achieved by reducing elevated intracranial pressure, maintaining normal intravascular volume, mean arterial pressure, oxygenation and carbon dioxide levels. At regular intervals, the level of consciousness should be assessed with the Glasgow scale. Glasgow Scale - GCS < 8 severe CCI - craniocerebral injury, GCS = 9-12 severe CCI, GCS > 13 light CCI. Pupils should also be checked for size and reaction to light.

The grading of the Glasgow scale according to the parameters: eye opening, speech and mobility is as follows:

**Table 1** Glasgow scale

| Component tested              | Score |
|-------------------------------|-------|
| <b>Eye response</b>           |       |
| Eyes open spontaneously       | 4     |
| Eye opening to verbal command | 3     |
| Eye opening to pain           | 2     |
| No eye opening                | 1     |
| <b>Motor response</b>         |       |
| Obeys command                 | 6     |
| Localises pain                | 5     |
| Withdraws from pain           | 4     |
| Flexion response to pain      | 3     |
| Extension response to pain    | 2     |
| No motor response             | 1     |
| <b>Verbal response</b>        |       |
| Oriented                      | 5     |
| Confused                      | 4     |
| Inappropriate words           | 3     |
| Incomprehensible sounds       | 2     |
| No verbal response            | 1     |

#### 4.4. Burning disease

The severity of a burn depends largely on the extent of the burned surface. Its quantitative assessment is calculated by measuring the percentage of the burned surface on the total surface of the body. The most popular method of calculating burn surface area is based on Wallace's "rule of nines". According to him, the various areas of the body have been divided into 9% or multiples of that except for the perineum area which is estimated at 1%. Another method is based on the fact that the patient's palm constitutes about 1% of the O.E.S. The number of "palms" covering the burn surface corresponds to the percentage of damage. Both of these rules are considered inaccurate for children because the anatomical proportions are different from those of adults. The qualitative assessment of a burn wound is based on the assessment of its deep extent. Thus, the necrosis can concern only some layers of the skin (partial thickness) or all layers (full thickness). Characteristic of a partial thickness burn is the necrosis of the epidermis and perhaps the superficial layer of the dermis, so that epithelial elements from the skin appendages (hair follicles, excretory ducts of sweat and sebaceous glands) remain. From these epithelial elements will come the healing of the burn surface with the process of epithelization. On the contrary, in full-thickness burns, all layers of the epidermis and dermis are damaged, with the result that the wound heals only by the growth of granulation and then scar tissue. Depending on the depth of the damage, first to third degree burns are distinguished. The first degree burn concerns only the skin, the second degree extends to the dermis and the third degree includes all layers of the skin. Some report a fourth degree burn injury where the necrosis extends to muscle, subcutaneous fat and even bone. The clinical assessment of depth of a burn is based on four clinical signs: sensation, blistering, color and texture of the skin. Qualitative assessment. First degree (partial thickness): erythema, pain; second degree (full thickness): erythema, large blisters, swelling, pain; third degree (total

oah): parchment-like texture, variable color, bubbles small, thin or no anesthesia. Securing the airway. Complications related to the respiratory system are one of the most important causes of death in burn victims. Treatment of inhalation burn should be started before the exact diagnosis can be made. All patients must be given 100% oxygen. Ensuring an open airway is vital for fire victims, particularly those with inhalation burns, where swelling of the airways quickly develops. Delay in treating and securing an airway can lead to obstruction with catastrophic consequences. A large-sized endotracheal tube should be placed to allow drainage of secretions and bronchoscopy. Indications for immediate: patients with disorders of the level of consciousness, in a comatose state; full thickness burn localized to the face and neck; in the beginning of airway obstruction due to edema or laryngospasm; in full-thickness burns of the chest wall • in coexisting chest injuries and other serious injuries; in inhalation burn; control of the adequacy of respiratory movements in patients with circumferential burns of the chest, where an emergency escharotomy may be required. The dose used is 5 g by slow intravenous infusion. Burn damage is characterized by the development of burn shock. Burn shock is due to a combination of hypovolemia and the local and systemic secretion of a large amount of inflammatory mediators. The most popular equation for calculating administered fluids is the Parkland equation:  $R/L\ 4ml \times \text{body weight (kg)} \times EU\ \% - 50\%$  of the volume of liquids is given in the first 8 hours, the rest – in 16 hours; if estimation of EU is not immediately possible, administration of 20 ml/kg of body weight is recommended; crystalline solutions during the first hour of injury; the first 24 term it is recommended to use only crystalline solutions. Factors that increase the need for administered fluids: inhalation burn; delay in fluid administration; electrical burn; extensive extent of the burn surface; concomitant injuries; in recent years, the so-called "fluid creep" phenomenon has been described; prone to excessive fluid administration with negative systemic and local effects, such as: abdominal compartment syndrome with adverse effects on liver, kidney function and function of the gastrointestinal system; pulmonary edema; need for palliative escharotomy; prolongation of mechanical ventilation and hospital stay; increase in morbidity and mortality.

#### 4.5. Crush-syndrome and rhabdomyolysis

For the first time it has been described after the bombing of London during the Second World War. The syndrome occurs during natural disasters, wars, explosions, industrial accidents. The compression of muscle masses leads to tissue ischemia, an increase in tissue pressure, which exceeds capillary filtration pressure. After the external pressure is removed the muscle tissue is re-perfused. The ischemia-reperfusion injury mechanism is the basic pathophysiological mechanism of this syndrome. Often pressure damage is combined with vascular damage, traumatic rupture of vessels, thrombosis and embolism. Congestion syndrome is clinically manifested by hypovolemia symptoms due to large fluid accumulation in damaged tissues and symptoms resulting from the release of a large amount of toxic substances from the injured tissues. The first manifestation of the syndrome is usually hypovolemic shock, which is also the most frequent cause of death during the first 4 days after the injury. Hyperkalemia with its effect on the functioning of the heart is the second cause of early death. Myoglobin is a protein that makes up 1-3% of the dry weight of muscle tissue. Under normal conditions myoglobin binds to haptoglobin and  $\alpha$ -2 globin and is eliminated through the reticuloendothelial system. The saturation of haptoglobin (when a large amount is produced of myoglobin) leads to an increase in free myoglobin in the plasma which is excreted by the kidneys. In the renal tubules myoglobin creates complexes (casts), especially in acid environment. Urine alkalinization reduces the formation of these complexes. Myoglobin causes a direct action on the renal tubules contributing to the increase of free oxygen radicals and promoting fat peroxidation. This action of myoglobin is also limited to alkaline environment. Diagnosis. Patients usually present with: extensive soft tissue injury with ischemia-reperfusion, edematous extremities, compartment syndrome, dark urine, positive test for hemoglobin, absence of red cells in the urine, increased CPK levels in the blood. Management: aggressive fluid administration; administration of mannitol; urine alkalinization (pH > 6.5) with bicarbonate. In addition to the alkalinization of the urine, the positive effect of bicarbonate is also due to the treatment of hyperkalemia. Hypocalcemia should not be treated aggressively because the greater amount of calcium accumulates in muscle and exacerbates rhabdomyolysis. Surgical transection of the deep fascia if compartment syndrome occurs. Irreversible damage to muscle tissue and nerves is seen after 6-8 hours of ischemia. Limb amputation in cases of extensive and irreversible tissue damage.

#### 4.6. Cardiopulmonary resuscitation

Cardiopulmonary resuscitation is the procedure aimed at restoring oxygen transport to the tissues, determining the specific causes of cardiac or respiratory arrest, and maintaining whatever functions are saved. Cardiac arrest or cardiorespiratory arrest is defined as the sudden and unpredictable failure of circulation or breathing, or both, resulting in inadequate supply of oxygenated blood to vital organs and cells. Basic Life Support (BLS) or Basic Cardiopulmonary Resuscitation is maintaining the patency of the airway and supporting breathing and circulation without the use of any special equipment except possibly a protective cover. Basic CPR is attempted in CPR because it has been shown to provide a small but critical amount of blood for the needs of the heart and brain, and only CPR and defibrillation can improve survival rates in CPR. The time required to cause irreversible brain damage from the moment the heart stops functioning is about 4 minutes. In 10 minutes certain brain death occurs. The main clinical signs of cardiorespiratory

arrest are: absence of pulse in the great vessels; loss of consciousness (10-20'' after the arrest); stopping automatic ventilation (15-30'' after stopping); absence of heart sounds; dilation of the pupil (60-90'' after stopping). Ventilation and compressions. After 30 compressions: open the airway again (head extension – chin lift); administer 2 inhalations lasting 1'' each; give 30 compressions again; continue compressions and inhalations in a ratio of 30:2; stop to recheck only if the victim begins to breathe on his own. Otherwise don't stop revitalization. If during the first 2 inhalations it does not occur normal chest expansion, check mouth's victim and remove anything causing obstruction. If there is another rescuer change every 1-2 minutes for avoiding fatigue. In children we start with 5 puffs. The ratio of chest compressions to inhalations is 30:2 when the rescuer is alone. When there are two rescuers the ratio of chest compressions-inhalations is 15:2

---

## 5. Conclusion

ABC is a resuscitation sequence that was developed for assessing and treating critically injured trauma patients. It has been widely accepted and used by experts in emergency medicine and trauma to guide resuscitation and improve outcomes in trauma patient. ATLS has morphed from its original construct of describing initial care of the injured patient at a small, often rural facility, to becoming the default for trauma resuscitation in multiple settings.

---

## References

- [1] American College of Surgeons, Committee on Trauma. Advanced Trauma Life Support (ATLS) 2004.
- [2] Askitopoulou E. Emergency and Intensive care medicine. Litsas Athens 1991.
- [3] Emergency medicine II. Topics in Anesthesiology and Intensive Care Medicine. Thessaloniki 2002.
- [4] Wilson W, Grande Ch, Hoyt D. Trauma, emergency resuscitation, perioperative anesthesia, surgical management. Informa Healthcare, New York 2007.
- [5] Rabinovici R, Frankel H, Kaplan L. Trauma evaluation and resuscitation. Curr Probl surg 2003; 10.
- [6] Royal College of Surgeons of England and the British Orthopedic Association. London 2003; 40: 599-81.
- [7] McGahan J, Richards J, Gillen M. The focused abdominal sonography for trauma scan. J Ultrasound Med 2002; 21: 789 – 800.
- [8] Flint L, Meredith W, Trunkey W et al. Trauma contemporary principles and therapy. Lippincott Williams & Wilkins 2008.
- [9] World Health Organisation. Guidelines for essential trauma care 2004. Feliciano D, Mattox K, Moore E. Trauma. McGraw-Hill 2008.
- [10] Beechey A. Damage control resuscitation: A sensible approach to the exsanguinating surgical patient. Crit Care Med 2008; 36: 67-74.
- [11] Dorgers V. Airway management in emergency situations. Best Pract Res Clin Anaesthesiol 2005; 19: 699– 715.
- [12] Nolan J. Fluid resuscitation for the trauma patient. Resuscitation 2001 ; 48: 57-69.
- [13] Hunt P, Greaves I, Owens W. Emergency thoracotomy in thoracic trauma, a review. Injury 2006; 37: 1-19.
- [14] Phillips G, Runciman W, Ilesley A. Monitoring in emergency medicine. Resuscitation 1989 ; 18: 21-35.
- [15] Keel M, Trentz O. Pathophysiology of polytrauma. Injury 2005; 36: 691-709.
- [16] Lenz A, Franklin G, Cheadle W. Systemic inflammation after trauma. Injury 2007; 38: 1336 – 45.
- [17] Guidelines for the Management of Severe Traumatic Brain Injury, 3rd Edition. Journal of Neurotrauma suppl. 2007; 24: S1-106.
- [18] David Herndon, Total Burn Care, WB Saunders, 3rd ed, Harcourt 2007.
- [19] Ispactchi K, Arbabi S Advances in burn critical care. Crit Care Med 2006; 34, (9)(Suppl): S239-S244.
- [20] Liberman M, Roudsari BS. Prehospital trauma care: What do we really know? Curr Opin Crit Care 2007; 13: 691– 696.
- [21] Pierik, EG, Bruining, HA Non-steroidal anti-inflammatory drugs can disguise an acute abdomen. Eur J Surg. 1994; 160: 61-65.

- [22] Rady, MY, Kirkman, E., Cranley, J., Little, R. (1996). Nociceptive somatic nerve stimulation and skeletal muscle injury modify systemic hemodynamics and oxygen transport and utilization after resuscitation from hemorrhage. *Crit Care Med.* 1996; 24: 623-628.
- [23] Timmermann A., Sebastian G. Russo SG Hollmann MW Paramedic versus emergency physician emergency medical service: role of the anesthesiologist and the European versus the Anglo-American concept. *Current Opinion in Anesthesiology* 2008, 21:222–227.
- [24] PHTLS-Basic and Advanced Prehospital Trauma Life Support. Fifth Edition, Mosby 2003.
- [25] Hettiaratchy S, Papini R Initial management of a major burn: I-overview. *BMJ* 2004; 328(26):1555-1557.
- [26] Klein M, Hayden Douglas, Elson C et al The association between Fluid Administration and outcome Following Major Burn. A multicenter study. *Annals of Surgery* 2007; 245(4):622-628.
- [27] Pruitt BA. Protection from excessive resuscitation: “pushing the pendulum back” *J Trauma* 2000; 49: 567-8.
- [28] Palmieri TL, Jackson W, Greenhalgh DG Benefits of early tracheostomy in severely burned children. *Crit CareMed* 2002; 309(4):922-4.
- [29] Palmieri TL et al Effect of blood transfusion on the outcome after major burn injury: A multicenter study *CritCare Med* 2006; 34(6):1602-7.
- [30] Malinoski DJ, Slater MS, Mullins RJ Crush injury and rhabdomyolysis. *Crit. Care Clin* 2004; 20:171-192.
- [31] Sever MS, Vanholder R, Lameire N Management of Crush-Related Injuries after Disasters. *N. Engl J Med* 2006;354(10):1052-1063.



## Antipsychotics and anesthetic drugs

Maria I Dalamagka

### ► To cite this version:

Maria I Dalamagka. Antipsychotics and anesthetic drugs. Magna Scientia Advanced Research and Reviews, 2024, 12, pp.001 - 004. 10.30574/msarr.2024.12.1.0141 . hal-05432481

**HAL Id: hal-05432481**

**<https://hal.science/view/index/docid/5432481>**

Submitted on 26 Dec 2025

**HAL** is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



Distributed under a Creative Commons Attribution 4.0 International License

(CASE REPORT)



## Antipsychotics and anesthetic drugs

Maria I. Dalamagka \*

*Department of Anesthesiology, General Hospital of Larisa, Greece.*

Magna Scientia Advanced Research and Reviews, 2024, 12(01), 001–004

Publication history: Received on 28 July 2024; revised on 06 September 2024; accepted on 09 September 2024

Article DOI: <https://doi.org/10.30574/msarr.2024.12.1.0141>

### Abstract

Psychiatric illness is common, affecting up to 10% of the population, with around 1% diagnosed with a major psychiatric disorder. An increased mortality rate in the postoperative period for schizophrenic patients receiving chronic antipsychotic therapy has been demonstrated. According to the Diagnostic and Statistical Manual of Disease, schizophrenia-associated psychotic disorders are characterized by disturbances in emotional, behavioral and cognitive arenas, impeding on nearly every aspect of life functioning in the form of thought disorders, delusions and hallucinations. Chronic schizophrenic patients lack pain sensitivity, and have pituitary-adrenal and autonomic nerve dysfunction, abnormalities of the immune system, and water intoxication. These alterations may influence postoperative outcome. Psychotropic drugs often given in combination with each other or with other non-psychiatric drugs generally exert profound effects on the central and peripheral neurotransmitter and ionic mechanisms. The increased complications are associated with physical disorders, antipsychotic or hazardous health behaviour and interactions between antipsychotics and anaesthetic drugs. Adverse responses during anaesthesia include arrhythmias, hypotension, prolonged narcosis or coma, hyperpyrexia, post-operative ileus and post-operative confusion. Although drug interaction probably remains the most potentially serious problem current evidence suggests that psychiatric medication need not be discontinued prior to anesthesia.

**Keywords:** Schizophrenia; Psychiatric medication; Mental health; Cognitive dysfunction

### 1. Introduction

Schizophrenia is a mental disease characterized by thought disorders, delusions, and hallucinations. It is the most common psychotic disorder, accounting for approximately 20% of all persons treated for mental illness. The physical and psychological fragility of patients with psychiatric illness poses critical importance in the preoperative assessment, evaluation, and choice of premedication, which includes regular therapy, as well as concerns about polypharmacy with possible interactions of anesthetics, analgesics, and psychiatric medications. A considerable effort is to reduce risks for exacerbations or relapses of imminent illness in the postoperative period. Antipsychotic drugs are used in the treatment of psychoses such as schizophrenia and mania. They treat the acute symptoms of hallucinations and delusions and in the case of schizophrenia are used prophylactically often for long periods. Most drugs act by antagonism at the dopamine D2 receptors in the central nervous system. Neuroleptic or typical antipsychotics, chlorpromazine, haloperidol, trifluoperazine, which have a propensity to cause extrapyramidal side effects: acute dystonia, akathisia, parkinsonism and tardive dyskinesia. Atypical antipsychotics, clozapine, olanzapine, risperidone, amisulpride, quetiapine and aripiprazole, which do not have a tendency to cause extrapyramidal side effects. An increased mortality rate in the post-operative period for psychotic patients receiving chronic antipsychotic therapy has been demonstrated. Approximately 70% of schizophrenic patients taking chronic antipsychotic drugs develop postoperative psychosis emergence or confusion. The cognitive dysfunction of schizophrenia is associated with an increased frequency of postoperative confusion. Postoperative psychosis emergence or confusion has been associated with adverse clinical and economic outcomes [1-7].

\* Corresponding author: Maria I. Dalamagka

## 2. Case report

A 53-year-old, 80-kg woman underwent a total hysterectomy due to cancer, who is in chronic schizophrenia. The operation was performed with a combination of epidural and general anesthesia. Naropaine 20 ml, 0.75 % was administered epidurally and 0.1 mg Fentanyl, 200 mg Propofol and 70 mg Esmeron were given intravenously. The patient was mechanically ventilated on a Drager ventilator and with a volume control ventilation model. Sevoflurane 2% was administered to maintain anesthesia. Fluid deficit was adjusted with Ringers Lactate. Intraoperatively, she developed persistent hypotension that was treated with phenylephrine. Awakening was delayed up to one hour after the inhaled anesthetic was discontinued. Reversal of residual muscle wasting was done with Bridion. The interaction with the patient's antipsychotic medications was particularly intense.

### 2.1. Interactions between antipsychotics and anesthetic drugs

#### 2.1.1. Antipsychotics

#### 2.1.2. Tricyclic antidepressants (TCA), selective serotonin re-uptake inhibitors, atypical agents and monoamine oxidase inhibitors (MAOIs)

Atypical antidepressants include venlafaxine and mirtazapine. Both these drugs should be continued throughout the perioperative period. Abrupt cessation of antidepressants is associated with the risk of developing withdrawal symptoms. Selective serotonin reuptake inhibitors (SSRIs), TCAs: amitriptyline, imipramine, desipramine, doxepin, nortriptyline; were the most commonly used drugs for treating depression. They inhibit synaptic reuptake of norepinephrine and serotonin. Also affect neurochemical systems including histaminergic and cholinergic systems, thus they have a large range of side-effects, including postural hypotension, cardiac dysrhythmias, urinary retention, dry mouth, blurred vision and sedation. Given chronically, these drugs decrease stores of noradrenergic catecholamines, and can cause changes on the ECG, ventricular arrhythmias and refractory hypotension. Increased availability of neurotransmitters in the central nervous system can result in increased anaesthetic requirements. TCAs may result in increased response to intraoperatively administered anticholinergics. Increased availability of norepinephrine at the post-synaptic nervous system can be responsible for exaggerated blood pressure responses following administration of indirect acting vasopressors such as ephedrine. Pancuronium, ketamine, meperidine and epinephrine containing solutions should be avoided. If hypotension occurs and vasopressors are needed, direct acting drugs such as phenylephrine are recommended. SSRIs block reuptake of serotonin at the pre-synaptic membranes, with relatively little effect on adrenergic, cholinergic, histaminergic or other neurochemical systems. As a result, they are associated with few side-effects. Fluoxetine is inhibitor of certain hepatic cytochrome P-450 enzymes and their side-effects are headache, agitation and insomnia. This drug may increase the plasma concentration of drugs that depend upon hepatic metabolism for clearance, such as warfarin, theophylline, phenytoin and benzodiazepines. Cardiac antidysarrhythmic drugs are also metabolized by this enzyme system, and fluoxetine inhibition of the enzyme system may result in potentiation of their effects. SSRIs should be continued throughout the perioperative period to prevent discontinuation syndrome. The use of pethidine, tramadol, pentazocine and dextromethorphan should be avoided. Serotonin syndrome is a potentially life-threatening adverse drug reaction that results from increased serotonin levels in the brain stem and spinal cord. A large number of drugs have been associated with the serotonin syndrome (agitation and confusion, increased motor activity and autonomic instability; hyperthermia, tachycardia, labile blood pressure and diarrhoea. Seizures, rhabdomyolysis, renal failure, arrhythmias, coma and death may occur). These include SSRI, MAOI, TCAs, pethidine, tramadol and dextromethorphan. MAOIs, tranylcypromine and phenelzine, and moclobemide, act by inhibition of the metabolic breakdown of norepinephrine and serotonin by the MAO enzyme. The level of norepinephrine and serotonin is increased at the receptor site. All MAOIs are eliminated by hepatic metabolism. Opioids like morphine, fentanyl, alfentanil and remifentanyl can all be used safely. Reactions occur in patients given pethidine and dextromethorphan, both of which inhibit serotonin reuptake. Depressive reaction, which is very rare, is thought to be due to MAO inhibition of hepatic enzymes resulting in enhanced effects of all opioids (It reversed by naloxone). Indirect acting sympathomimetics may precipitate potentially fatal hypertensive crisis and are absolutely contraindicated with any MAOIs. Direct acting sympathomimetics (adrenaline, noradrenaline and phenylephrine) may have an enhanced effect due to receptor hypersensitivity. Propofol and etomidate can be used safely. Ketamine should be avoided as it causes sympathetic stimulation. Local anaesthetics containing adrenaline should be used with caution. Benzodiazepines, inhalational anaesthetic agents, anticholinergic drugs and non-steroidal anti-inflammatory drugs can be used safely in patients taking MAOIs. Therefore, dosages should be titrated. Benzodiazepine premedication can be given and sympathetic stimulation should be avoided. Adequate hydration of the patient should be ensured. Hypotension should be treated initially with intravenous fluids and then with cautious doses of phenylephrine. Pethidine and indirect acting sympathomimetics are absolutely contraindicated. Valproate is the drug of choice for treating acute episodes of Bipolar disorder. Mood stabilizers are used to treat bipolar affective disorders. Lithium and valproate remain a mainstay of treatment. In some cases, olanzapine and other antipsychotics are also used in the

beginning of treatment to control excitement and agitation. Lithium inhibits the release of thyroid hormones and results in hypothyroidism. Lithium is eliminated by the kidneys and, therefore, if renal function is compromised or there is dehydration, lithium levels rise dramatically. Toxic blood concentration produces confusion, sedation, muscle weakness tremors and slurred speech. Cardiac problems may include sinus bradycardia, sinus node dysfunction, AV block, T wave changes, hypotension and ventricular irritability. Lithium toxicity occurs when levels are  $>1.5$  mmol/L, and is exacerbated by dehydration, diuretics and renal impairment. Lithium carbonate is used to treat manic depression, but it is more effective in preventing mania than in relieving depression. Lithium prolongs neuromuscular blockade and may decrease anaesthetic requirements because it blocks brainstem release of norepinephrine, epinephrine and dopamine. Lithium discontinuation is recommended. Lithium can be stopped at once because no withdrawal symptoms occur. The association of sedation with lithium suggests that anaesthetic requirement may be decreased in these patients. Duration of both depolarizing and non-depolarizing muscle relaxants may be prolonged. Neuroleptic or typical antipsychotics (chlorpromazine, haloperidol, trifluoperazine) cause extrapyramidal side-effects like acute dystonia, akathisia, Parkinsonism and tardive dyskinesia. Clozapine causes seizures and neutropenia [8-24].

---

### 3. Discussion

Schizophrenic patients are at increased risk for perioperative complications, as their biological response to stress is impaired. Antipsychotic medication should be continued perioperatively, as abrupt withdrawal may result in recurrence of psychotic symptoms. It may also increase the incidence of postoperative confusion, which is high in those suffering with schizophrenia. Opioids like pethidine and indirectly acting sympathomimetics are contraindicated in MAOI patients. Mood stabilisers are used to treat bipolar affective disorders which have a lifetime incidence of 1% in the UK. A manic or hypomanic episode is required for the diagnosis to be made. Some patients also suffer depressive episodes. Mood stabilisers include lithium and anticonvulsant drugs such as carbamazepine and sodium valproate. Mood stabilisers should not be stopped abruptly as there is a significant risk of relapse. The management of lithium poisoning is supportive, with correction of electrolyte imbalance and convulsions. Haemodialysis is required if renal failure is present, which may need to be repeated as tissue lithium enters the circulation. Cardiac risk factors are increased in patients taking long term antipsychotic drugs due to a combination of side effects and high smoking levels in these patients.

---

### 4. Conclusion

Patients with chronic schizophrenia should continue their antipsychotics pre-operatively as abrupt withdrawal may result in recurrence of psychotic symptoms. Important effects of antipsychotic drugs include  $\alpha$ -adrenergic blockade causing postural hypotension, prolongation of QT intervals, seizures, hepatic enzyme elevation, abnormal temperature regulation, sedation and Parkinsonism-like manifestations. Drug-induced sedation may decrease anaesthetic requirement.

---

### Compliance with ethical standards

#### *Statement of informed consent*

Informed consent was obtained from all individual participants included in the study.

---

### References

- [1] Kudoh A. Perioperative management of chronic schizophrenic patients. *AnesthAnalg*. 2005, 101:1867–72.
- [2] Wik G. Effects of neuroleptic treatment on cortisol and 3-methoxy-4-hydroxyphenylethyl glycol levels in blood. *J Endocrinol*. 1995, 144:425–9.
- [3] Naudin J, Capo C, Giusano B, Mege JL, Azorin JM. A differential role for interleukin-6 and tumor necrosis factor in schizophrenia? *Schizophr Res*. 1997, 26:227–33.
- [4] Goldman MB, Robertson GL, Luchins DJ, Hedeker D, Pandey GN. Psychotic exacerbations and enhanced vasopressin secretion in schizophrenic patients with hyponatremia and polydipsia. *Arch Gen Psychiatry*. 1997, 54:443–9.
- [5] Morgan GE, Mikhail MS, Murray MJ. Morgan: Clinical anaesthesiology. 4th ed. USA: LANGE International edition, 2008. Anesthesia for patients with neurologic and psychiatric diseases, pp. 647–61.

- [6] Howland RH. Potential adverse effects of discontinuing psychotropic drugs. *J PsychosocNursMent Health Serv*. 2010, 48:9–12.
- [7] Hines RL, Marschall KE. Psychiatric disease/substance abuse/drug overdose *Stoelting's: Anaesthesia and co-existing diseases*. 5th ed. Gurgaon (India): ELSEVIER A division of Reed Elsevier India Private Ltd, 2010.
- [8] Kudoh A, Katagai H, Takazawa T. Effect of preoperative discontinuation of antipsychotics in schizophrenic patients on outcome during and after anesthesia. *Eur J Anaesth*. 2004, 21:414–6.
- [9] Sawada N, Higashi K, Yanagi F, Nishi M, Akasaka T, Kudoh J. Sudden onset of bronchospasm and persistent hypotension during spinal anesthesia in a patient on long-term psychotropic therapy. *Masui*. 1997, 46:1225–9.
- [10] Lanctot KL, Best TS, Mittman N, Liu BA, Oh PI, Einarson TR, et al. Efficacy and safety of neuroleptics in behavioral disorders associated with dementia. *J Clin Psychiatry*. 1998, 59:550–61.
- [11] Buckley NA, Sanders P. Cardiovascular adverse effects of antipsychotic drugs. *Drug Saf*. 2000, 23:215–28.
- [12] Kudoh A, Kudo T, Ishihara H, Matsuki A. Depressed pituitary-adrenal response to surgical stress in chronic schizophrenic patients. *Neuropsychobiol*. 1997, 36:112–6.
- [13] Kawachi I. Physical and psychological consequence of weight gain. *J Clin Psychiatry*. 1999, 60(Suppl 21):5–9.
- [14] Baptista T. Body weight gain induced by antipsychotic drugs: Mechanisms and management. *ActaPsychiatrScand*. 1999, 100:3–16.
- [15] Glassman AH, Bigger JT., Jr Antipsychotic drugs: Prolonged QTc interval, torsade de pointes, and sudden death. *Am J Psychiatry*. 2001, 158:1774–82.
- [16] Kudoh A, Ishihara H, Matsuki A. Pituitary-adrenal and parasympathetic function of chronic schizophrenic patients with postoperative ileus or hypotension. *Neuropsychobiol*. 1999, 39:125–30.
- [17] Patt RB, Proper G, Reddy S. The neuroleptics as adjuvant analgesics. *J Pain Sympt Manage*. 1994, 9:446–53.
- [18] Young DM. Risk factors for hypothermia in psychiatric patients. *Ann Clin Psychiatry*. 1996, 8:93–7.
- [19] Molnar G, Fava GA. Principles of medical psychiatry. Orland: Gruneand Stratton Inc, 1987. Intercurrent medical illness in the schizophrenic patients, p. 451.
- [20] O'Keeffe ST, Devlin JG. Delirium and the dexamethasone suppression test in the elderly. *Neuropsychobiol*. 1994, 30:153–6.
- [21] Van der Mast RC. Pathophysiology of delirium. *J Geriatr Psychiatry Neurol*. 1998, 11:138–45.
- [22] Allman KG, Wilson IH. *Oxford Handbook of Anaesthesia* (2nd Edition). 2006.
- [23] Peck TE, Hill SA, Williams M. *Pharmacology for Anaesthesia and Intensive Care* (3rd Edition). 2008.
- [24] Taylor D, Paton C, Kerwin R. *The Maudsley Prescribing Guidelines* (9th Edition). 2007.



## Anesthesiology complications in children

Maria I Dalamagka

### ► To cite this version:

Maria I Dalamagka. Anesthesiology complications in children. International Journal of Science and Research Archive, 2024, 12, pp.2924 - 2927. 10.30574/ijra.2024.12.1.1179 . hal-05432478

**HAL Id: hal-05432478**

**<https://hal.science/view/index/docid/5432478>**

Submitted on 26 Dec 2025

**HAL** is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



Distributed under a Creative Commons Attribution 4.0 International License

## Anesthesiology complications in children

Maria I. Dalamagka \*

*Anesthesia Department, General Hospital of Larisa, Greece.*

International Journal of Science and Research Archive, 2024, 12(01), 2924–2927

Publication history: Received on 13 May 2024; revised on 25 June 2024; accepted on 28 June 2024

Article DOI: <https://doi.org/10.30574/ijrsra.2024.12.1.1179>

### Abstract

General anesthesia is defined by the American Society of Anesthesiologists as a “drug induced loss of consciousness during which patients are not arousable, even by painful stimulation”. The perioperative period, immediately before, during, and after surgery, is a particularly critical time for pediatric patients. Even though anesthesia today is much safer than it has ever been, all anesthesia has an element of risk. General anesthesia has been safely given to children for many years. Improvements in inhalational and intravenous agents have increased the safety profile of general anesthesia and complication rates are low. General anesthesia has also allowed children to undergo painful or anxiety inducing procedures. General anesthesia is a complete loss of consciousness with amnesia, analgesia and neuromuscular blockade. It can be divided into three phases: induction, maintenance and emergence. Induction usually occurs with inhalational anesthetic in children until intravenous access occurs. Propofol, etomidate and ketamine are often used for induction. Maintenance of anesthesia occurs with inhalational or intravenous medications. Common inhalational anesthetics include nitrous oxide, sevoflurane or desflurane. Common intravenous medications used for maintenance are propofol and remifentanyl. Typical complications in pediatric anesthesia are respiratory problems, medication errors, difficulties with the intravenous puncture and pulmonary aspiration. In the postoperative setting, nausea and vomiting, pain, and emergence delirium can be mentioned as typical complications. Side effects of anesthesia may include nausea, vomiting, drowsiness, muscle soreness and sore throat. Rare but more serious complications include adverse reactions that affect breathing, allergic reactions and irregular heart rhythms. Airway and respiratory events are the most common perioperative complications in pediatric patients. Several studies have reported on the incidence of pediatric airway-related complications. Consistently reported risk factors for serious airway complications include very young age and multiple intubation attempts. Laryngospasm was the most common cause of respiratory related arrests. Other etiologies included airway obstruction, difficult intubation, esophageal intubation, and aspiration.

**Keywords:** Anaesthesia; Complications; Intubation; Laryngospasm; Vomiting; Pain.

### 1. Introduction

The purpose of general anesthesia is to suppress the conscious perception and physiologic response to noxious stimuli and to render the patient unconscious. This is done through analgesia; decreasing pain, amnesia; blunting consciousness, akinesia; preventing movement, physiologic support; maintaining cardiovascular function, fluid management, electrolyte control, and thermoregulation and vigilance. Inhalation Respiratory and airway events accounted for 60 percent of all anesthesia-related complications and occurred in 3.1 percent of all anesthetics; laryngospasm occurred in 1.2 percent, bronchospasm in 1.2 percent, postoperative stridor in 0.7 percent, and aspiration in 0.1 percent of all anesthetics. The highest rates of airway and respiratory events occurred in neonates (<1 month old) and infants (<1 year old). Other risk factors for respiratory complications included history of prematurity and reactive airway disease. The risk factors for anesthetic mortality include neonates, age less than one year, American Society of Anesthesiologists (ASA) III and IV classification, children undergoing emergency surgery, general anesthesia and cardiac surgery. Airway management problems and cardiopulmonary events account for the majority of the cases of

\* Corresponding author: Maria I. Dalamagka

anesthesia related mortality in children with comorbidities. Though there can be inter-rater variability between these scores, they have been shown to correlate with perioperative complications and mortality. Pulmonary aspiration can be a devastating event although studies have shown that the incidence of aspiration in children is relatively low - two per 10,000 cases in elective procedures and 2.2 per 10,000 in emergency operations. Laryngospasm is common in pediatric surgical patients - twice the incidence when compared to adults. Studies have found that in children from birth to nine years of age the incidence is 17 in 1000 cases. In children with obstructive lung disease or upper respiratory infections the risks can be up to 64 to 96 per 1000 cases. There is evidence to suggest that the use of supraglottic airway devices would decrease the incidence of laryngospasm. Postoperative apnea is more commonly seen in former preterm infants. All of the major societies, the American, European and Canadian Societies of Anesthesiologists agree on recommended preoperative fasting time prior to surgery. The recommendations include clear fluids until two hours before surgery, breast milk, without additives, up to four hours before surgery, infant formula: light meals; nonhuman milk up to six hours before surgery and high fat; fried foods; meat meals up to eight hours before surgery [1-14].

## 2. Anesthesiology drugs administered to children

Benzodiazepines induce hypnosis, anxiolysis, sedation, and amnesia and have anticonvulsant activity. In larger doses, they cause respiratory depression and apnea; they are synergistic with narcotics and barbiturates in their respiratory depressant effects. The most commonly used benzodiazepine in pediatric anesthesia is midazolam. Propofol has a very rapid onset and is associated with a reduced rate of postoperative nausea and vomiting. Propofol ranks as the most commonly used IV induction agent in pediatric anesthesia. Ketamine, which has a rapid onset and a short duration of action, causes central dissociation while providing analgesia and amnesia. Because ketamine increases the production of secretions, anticholinergic medications; atropine, glycopyrrolate, are usually needed. Dreams and hallucinations are the most common side effects associated with ketamine. Administration of a benzodiazepine concomitantly with ketamine decreases the incidence of dreams and hallucinations. Contraindications to the use of ketamine in children include the presence of an active upper respiratory tract infection, increased intracranial pressure, open globe injury, and the presence of psychiatric or seizure disorders. Opioids are widely used in the practice of pediatric anesthesia and pediatric perioperative medicine; they provide analgesia for painful procedures and decrease the incidence of postprocedural pain. Opioids suppress the carbon dioxide response, induce apnea, and are respiratory depressants, their use must be monitored closely, especially when they are administered with other respiratory depressing agents; inhalational anesthetics, barbiturates, benzodiazepines. Morphine and fentanyl are the most commonly used opioids in pediatric anesthesia. Morphine is a long-acting analgesic medication that provides effective analgesia. Because morphine can cause bronchospasm, secondary to histamine release, it should be used cautiously in children who suffer from asthma. Fentanyl, which is 80 to 100 times more potent than morphine, ranks as the most commonly used opioid in pediatric anesthesia. It has a shorter duration of activity and a more stable hemodynamic profile than morphine. Sufentanil, which is eight to 10 times more potent than fentanyl, is used primarily in pediatric cardiac cases. Remifentanyl, which is similar in potency to fentanyl, is considered an ultrashort-acting opioid. It is used when anesthesia needs to be rapidly induced. Because the primary narcotic effect of remifentanyl ends so quickly, additional analgesics must be used to provide postoperative pain relief. Alfentanil, which is five to 10 times less potent than fentanyl but has a faster onset of action, appears to have an increased incidence of muscle rigidity, convulsions, and prolonged respiratory depression compared with fentanyl and is therefore rarely used in pediatric anesthesia. Inhalational anesthetics have protective effects against ischemia-reperfusion injury, and recent authors suggest that these effects may also result in clinically relevant improved organ function. This protective capacity has been related to a preconditioning effect, a postconditioning effect, and also to an effect on apoptosis. With the currently used volatile agents, Sevoflurane and Desflurane, anesthetic hepatotoxicity is no longer a significant clinical problem. Sevoflurane and Desflurane are contraindicated only in the very rare cases of malignant hyperthermia susceptibility. Sevoflurane is the ideal agent for inhalation induction of anesthesia in children because it is not irritating to the airway. Sevoflurane is a very effective bronchodilator, and, despite early controversial reports about nephrotoxicity by means of the production of Compound A, a degradation product resulting from the interaction of sevoflurane with the CO<sub>2</sub> absorbent soda lime, it is actually not at all toxic to the kidney as long as a fresh gas flow is kept at no less than 2 litres per minute. Desflurane is not suitable for inhalation induction due to its irritating effect on the airways, but we use it for the maintenance of anesthesia, after tracheal intubation, even in newborns. Desflurane is a bronchodilator and has a modest negative inotropic effect; it does not have any nephrotoxic effect. Nitrous oxide can inactivate vitamin B<sub>12</sub>, provoking neurologic disorders both in patients and in operating room personnel. We never use nitrous oxide. Toxic effects are particularly possible in persons with a subclinical, preexisting deficit of vitamin B<sub>12</sub> but it is not possible to know in advance which patients are deprived of vitamin B<sub>12</sub>. Nitrous oxide also has the well-known effect of expanding any air-filled cavity. The neuromuscular blocking agents are used primarily as an adjunct to general anesthesia to facilitate endotracheal intubation and to maintain muscle relaxation during surgery. Neuromuscular blockers are the most useful drugs in our pharmacologic armamentarium but postanesthetic morbidity associated with incomplete reversal of neuromuscular blockers is still a frequent occurrence. Sugammadex, a relatively new drug, is a cyclodextrin that can

form a very tight complex with rocuronium, less with vecuronium and pancuronium, thus allowing a reversal even during a profound neuromuscular block. Anesthesia can be both induced and maintained with either boluses or continuous infusions of IV anesthetic agents. IV anesthetic agents include barbiturates, opioid narcotics, benzodiazepines, and miscellaneous products; ketamine, propofol [15-23].

### 3. Adverse Events and Discussion

Assessment of anesthetic risk in children has been estimated by the American Society of Anesthesia risk assessment score. The ASA can help identify patients preoperatively that may be at higher risk for anesthetic complications and allow adjustment to the anesthetic plan for these higher risk children. Complications associated with general anesthesia can be divided into four basic categories: postoperative nausea and vomiting (PONV), oral trauma, thermoregulation issues and cardiorespiratory complications. An estimated 40% to 50% of children experience PONV after receiving general anesthesia. The serotonin antagonists; ondansetron, dolasetron, granisetron, are considered to be first line drugs for PONV. The insertion or removal of the endotracheal tube can irritate the throat, causing a sore throat and, or hoarseness. Craniofacial anomalies include cleft lip and palate, mandibular hypoplasia, the enlarged tongue seen with Beckwith-Wiedemann syndrome and many others. It is important to anticipate difficulty in mask ventilation as well as endotracheal intubation. The most common complications are respiratory and include airway obstruction, laryngospasm or bronchospasm, desaturation and apnea. Cardiovascular complications are rare in healthy children and include arrhythmias and hypotension. Longer procedures and higher anesthetic dose increase the risk of complications. Mechanical ventilation can be a source of complications, first of all pulmonary barotrauma and volutrauma. Usually we start mechanical ventilation with a very low tidal volume; then we slowly increase it until we observe the chest moving and hear gas entering the lungs with the stethoscope. We also check the monitor, looking for an acceptable capnographic curve and a peak inspiratory pressure in the airways no higher than 15–20 cm H<sub>2</sub>O. Bronchospasm involves the constriction of the smooth muscle layers of the small airways. If left untreated, bronchospasm can progress to respiratory distress and ultimately, respiratory failure. Bronchospasm involves the constriction of the smooth muscle layers of the small airways. If left untreated, bronchospasm can progress to respiratory distress and ultimately, respiratory failure. This activity outlines the varying etiologies and clinical phenotypes, evaluation processes, and management strategies of bronchospasm. Additionally, the interdependent roles of the interprofessional team are highlighted in the care for pediatric patients with bronchospasm. Treatment of an acute asthma exacerbation targets the main components of the underlying pathophysiology: airway inflammation, smooth muscle constriction, edema, and mucus production. The standard initial therapy includes oxygen, intermittent dosing of a short acting beta 2 agonist; albuterol, an inhaled anticholinergic; ipratropium bromide, and corticosteroids. MH is an inherited disorder that is triggered by the volatile anesthetic agents and or succinylcholine. It usually occurs within the first two hours of anesthesia but can occur up to 24 hours later. MH is characterized by the development of a sudden and rapid high fever (38.5°C to 46°C, rising 1°C every five minutes), muscle rigidity, acidosis, and cardiac arrhythmias. Because MH can be fatal, aggressive therapy is warranted. The treatment of MH includes discontinuing all volatile inhalation agents, correction of the metabolic acidosis, and treatment with IV dantrolene: 2.5 mg/kg up to 10 mg/kg. Cardiorespiratory complications; respiratory depression, hypertension, arrhythmias, heart attacks, stroke, can also result from general anesthesia. Fortunately, these problems are extremely rare and are more likely related to surgical complications. The American Academy of Pediatrics Committee on Drugs (COD) notes that children tend to be particularly susceptible to oversedation with anesthesia. Incidence of perioperative anaphylaxis is estimated at 1 in 10–20,000 anesthetic procedures. It is noteworthy that in some cases bradycardia may be a life-protecting adaptive mechanism that allows the ventricle to fill despite hypovolemia. Then the treatment with atropine must be simultaneous with epinephrine and fluids; otherwise giving only atropine can result in cardiac arrest. As a matter of fact epinephrine and an expansion of intravascular volume are the key points in the perioperative management of anaphylaxis. The standard dose is 0.01–0.02 mg/Kg intravenously. Sometimes doctors are overzealous, using a high dosage due to their impatience in the treatment of an acute condition. According to recent reviews high doses can worsen a patient's post resuscitation hemodynamic condition by causing increased myocardial oxygen demand, ventricular ectopy, hypertension and myocardial necrosis. High doses do not improve survival and may be associated with a worse neurological outcome. Long term risks of general anesthesia have focused mainly on concerns for the effect of anesthesia on neurodevelopment in infants and children.

### 4. Conclusion

Minor events occur with variable frequency. The likelihood of these occurring is increased in patients under a year of age and in patients with an ASA status of 3 and above. Most events are respiratory in nature, emphasizing the importance of pulse oximetry and capnography as routine monitors in the delivery of anesthesia. Postoperative vomiting remains a common problem in older children.

## References

- [1] Barash PG, Cullen BF, and Stoelting RK. Clinical Anesthesia. Philadelphia: Lippincott Williams & Wilkins, 2006. Print
- [2] Subramanyam R, Yeramaneni S, Hossain MM, et al. Perioperative Respiratory Adverse Events in Pediatric Ambulatory Anesthesia: Development and Validation of a Risk Prediction Tool. *Anesth Analg*. 2016, 122(5):1578-85.
- [3] von Ungern-Sternberg BS, Boda K, Chambers NA, et al. Risk assessment for respiratory complications in paediatric anaesthesia: a prospective cohort study. *Lancet*. 2010, 376(9743):773-83.
- [4] Gonzalez LP, Pignaton W, Kusano PS, et al. Anesthesia-related mortality in pediatric patients: a systematic review. *Clinics (Sao Paulo)*. 2012, 67(4):381-7.
- [5] Sankar A, Johnson SR, Beattie WS, et al. Reliability of the American Society of Anesthesiologists physical status scale in clinical practice. *Br J Anaesth*. 2014.
- [6] American Society of Anesthesiologists. (2014, October). Retrieved December 17, 2016, from <http://www.asahq.org>
- [7] Lin EE, Tran KM. Anesthesia for fetal surgery. *Semin Pediatr Surg*. 2013, 22(1):50-5.
- [8] American Society of Anesthesiologists Committee. Practice guidelines for preoperative fasting and the use of pharmacologic agents to reduce the risk of pulmonary aspiration: application to healthy patients undergoing elective procedures: an updated report by the American Society of Anesthesiologists Committee on Standards and Practice Parameters. *Anesthesiology*. 2011, 114(3):495-511.
- [9] Merchant R, Chartrand D, Dain S, et al. Guidelines to the Practice of Anesthesia - Revised Edition 2016. *Can J Anaesth*. 2016, 63(1):86-112.
- [10] Smith I, Kranke P, Murat I, et al. Perioperative fasting in adults and children: guidelines from the European Society of Anaesthesiology. *Eur J Anaesthesiol*. 2011, 28(8):556-69.
- [11] Bordet F, Allaouchiche B, Lansiaux S, et al. Risk factors for airway complications during general anaesthesia in paediatric patients. *Paediatr Anaesth*. 2002, 12(9):762-9.
- [12] Anon. Guidelines for monitoring and management of pediatric patients during and after sedation for diagnostic and therapeutic procedures: Addendum. *Pediatrics*. 2002, 110:836-838.
- [13] Cote CJ, Notterman DA, et al. Adverse sedation events in pediatrics: A critical incident analysis of contributing factors. *Pediatrics*. 2000, 105:805-814.
- [14] Mondoñedo JR, McNeil JS, Amin SD, Herrmann J, Simon BA, Kaczka DW. Volatile Anesthetics and the Treatment of Severe Bronchospasm: A Concept of Targeted Delivery. *Drug Discov Today Dis Models*. 2015 Spring, 15:43-50.
- [15] Gencorelli FJ, Fields RG, Litman RS. Complications during rapid sequence induction of general anesthesia in children: A benchmark study. *Paediatr Anaesth*. 2010, 20:421-4.
- [16] Neuhaus D, Schmitz A, Gerber A, Weiss M. Controlled rapid sequence induction and intubation – An analysis of 1001 children. *Paediatr Anaesth*. 2013, 23:734-40.
- [17] Newton R, Hack H. Place of rapid sequence induction in paediatric anaesthesia. *BJA Educ*. 2016, 16:120-3.
- [18] Kulkarni K, Karnik P, Dave N, Garasia M. Ultra-modified rapid sequence induction. *Paediatr Anaesth*. 2017, 27:1278-83.
- [19] Malviya S, Voepel-Lewis T, Prochaska G, Tait AR. Prolonged recovery and delayed side effects of sedation for diagnostic imaging studies in children. *Pediatrics*. 2000, 105:E42.
- [20] Boonmak P, Boonmak S, Pattanittum P. High initial concentration versus low initial concentration sevoflurane for inhalational induction of anaesthesia. *Cochrane Database Syst Rev*. 2016 Issue 6, Art. No.:CD006837.
- [21] Lejus C, Bazin V, Fernandez M, Nguyen JM, Radošević A, Quere MF, et al. Inhalation induction using sevoflurane in children: The single-breath vital capacity technique compared to the tidal volume technique. *Anaesthesia*. 2006, 61:535-40.
- [22] Ho K, Chua WL, Lim S. NGA. A comparison between single-and double-breath vital capacity inhalation induction with 8% sevoflurane in children. *Paediatr Anaesth*. 2004, 14:457-61.
- [23] Meyers EF, Muravchick S. Anesthesia induction technics in pediatric patients: A controlled study of behavioural consequences. *Anesth Analg*. 1977, 56:538-42.



# General anesthesia in Geriatric patients in combination with Epidural anesthesia in the right colectomy

Maria I Dalamagka

## ► To cite this version:

Maria I Dalamagka. General anesthesia in Geriatric patients in combination with Epidural anesthesia in the right colectomy. GSC Biological and Pharmaceutical Sciences, 2022, 19, pp.158 - 159. 10.30574/gscbps.2022.19.2.0188 . hal-05432463

**HAL Id: hal-05432463**

**<https://hal.science/view/index/docid/5432463>**

Submitted on 26 Dec 2025

**HAL** is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



Distributed under a Creative Commons Attribution 4.0 International License

## General anesthesia in Geriatric patients in combination with Epidural anesthesia in the right colectomy

Maria I Dalamagka \*

*Department of Anesthesia, General Hospital of Larisa, Greece.*

GSC Biological and Pharmaceutical Sciences, 2022, 19(02), 158–159

Publication history: Received on 13 April 2022; revised on 14 May 2022; accepted on 16 May 2022

Article DOI: <https://doi.org/10.30574/gscbps.2022.19.2.0188>

### Abstract

Aging involves a progressive loss of functional reserve in all organ systems, to variable extend. Moreover, perioperative morbidity becomes more frequent in the elderly with steep increases after the age of 75. The elderly are more sensitive to anesthetic agents and generally require smaller doses for the same clinical effect, and drug action is usually prolonged. In the study, a 78-year-old man with Chronic obstructive pulmonary disease (COPD) and a cardiac history underwent a right colectomy with a combination of general and epidural anesthesia. The purpose of this paper is to investigate the beneficial effects of epidural anesthesia when it acts in combination with general anesthesia in elderly patients.

**Keywords:** Geriatric; Anesthesia; General; Epidural; COPD; Colectomy

### 1. Introduction

Geriatric patients are more sensitive to anesthetic agents. Less medication is usually required to achieve a desired clinical effect, and drug effect is often prolonged. The most important outcome and overall objective of perioperative care of geriatric population, is to speed recovery and avoid functional decline. The risk from anesthesia is more related with the presence of co-existing disease than with the age of the patient. Minimizing perioperative risk in geriatric patients requires thoughtful preoperative assessment of organ function and reserve, meticulous intraoperative management of coexisting disorders, and vigilant postoperative pain control [1-8].

### 2. Case Study

A 78-year-old man weighing 80 kg with a history of ulcerative colitis, with known emphysematous lung lesions, COPD and a surgical abdominal aortic aneurysm came for a right colectomy. The patient was placed in an epidural catheter at the level: lumbar 1 - lumbar 2. A test dose of 3 ml Lidocaine 2% was given and then under continuous monitoring with ECG, ART, SpO<sub>2</sub>, an additional 5 ml Naropaine 0.75% was given. Anesthesia was introduced with 0.1 mg Fentanyl, 200 mg Propofol and Esmeron 100 mg. The patient was rescued on a Drager ventilator with volume control model and PEEP 5. Bronchial aspirations were performed with 100% O<sub>2</sub> and Flexotide 250 mcg and Aeroline 100 mcg were given tracheally. An additional 2 mg of Morphine was given through the epidural catheter and a continuous infusion pump was pumped with a flow of 5 ml Naropaine 2% per hour. Maintenance of anesthesia was performed with Sevoflurane 2%, and an additional 3 mg of Morphine was given intravenously. Additionally Onda 4 mg, Dexaton 8 mg and Apotel 1g were given. The patient was hydrated with 1Lt Plasmalite slow intravenous administration and 1Lt Lactated Ringer's injection intravenous infusion. The awakening was reversed with Bridium, as repeated doses of Esmeron were given during the four-hour surgery.

\* Corresponding author: Maria I Dalamagka  
Department of Anesthesia, General Hospital of Larisa, Greece.

### 3. Management and outcome

The combination of general and epidural anesthesia offers the advantage of hemodynamic stability with satisfactory analgesia and reduced intravenous opioid administration intraoperatively, resulting in the patient waking up at the end of the surgery, without being transferred to an intensive care unit.

### 4. Discussion

Functional capacities of the respiratory system are all reduced in the geriatric patients. Decrease in chest wall compliance and the strength of respiratory muscles, making the lungs more difficult to ventilate and declining in maximum inspiratory and expiratory force. Increased alveolar compliance with collapse of small airways and subsequent alveolar hypoventilation, air trapping leading to ventilation perfusion mismatch. The hemodynamic effects of regional anesthesia may be associated with reduced blood loss in pelvic and lower extremity operations. More important, the patient maintains his airway and pulmonary function. The continuous injection of local anesthetic through the epidural catheter enables good postoperative analgesia and the avoidance of complications in respiratory patients with or without cardiac substrate.

### 5. Conclusion

The combination of general and epidural anesthesia is a good choice in the elderly, especially when they undergo major abdominal surgeries as the dose of intravenous opioids is reduced and satisfactory intraoperative and postoperative analgesia is achieved.

### Compliance with ethical standards

#### *Acknowledgments*

General Hospital of Larisa.

#### *Statement of informed consent*

Written informed consent was obtained from the patient for publication of this case report.. A copy of the written consent is available.

### References

- [1] Knudson RJ, Lebowitz MD, Holberg CJ, Burrows B. Changes in the normal maximal expiratory flow-volume curve with growth and aging. *Am Rev Respir Dis.* 1983; 27: 725–734?
- [2] Sorbini CA, Grassi V, Solinas E, Muiesan G. Arterial oxygen tension in relation to age in healthy subjects. *Respiration.* 1968; 25: 3–13.
- [3] Ergina PL, Gold SL, Meakins JL. Perioperative care of the elderly patient. *World J Surg.* 1993; 17: 192–198.
- [4] Kim KI, Park KH, Koo KH, Han HS, Kim CH. Comprehensive geriatric assessment can predict postoperative morbidity and mortality in elderly patients undergoing elective surgery. *Arch Gerontol Geriatr.* 2013; 56 (3): 507–512.
- [5] Edelstein SB, Metry JE. Anesthesia considerations for the geriatric patient. *Curr Geri Rep.* 2017; 6(3): 115–121.
- [6] Incalzi RA, Scarlata S, Pennazza G, Santonico M, Pedone C. Chronic obstructive pulmonary disease in the elderly. *Eur J Intern Med.* 2014; 25(4): 320–328.
- [7] Novis BK, Roizen MF, Aronson S, Thisted RA. Association of preoperative risk factors with postoperative acute renal failure. *Anesth Analg.* 1994; 78(1): 143–149.
- [8] Lim BG, Lee IO. Anesthetic management of geriatric patients. *Korean J Anesthesiol.* 2020; 73(1): 8–29.

## Autoimmune diseases in children and anesthesia

Maria I Dalamagka \*

*Department of Anesthesiology, General Hospital of Larisa, Greece.*

Magna Scientia Advanced Biology and Pharmacy, 2025, 14(01), 045-052

Publication history: Received on 26 December 2024; revised on 13 February 2025; accepted on 16 February 2025

Article DOI: <https://doi.org/10.30574/msabp.2025.14.1.0022>

### Abstract

Autoimmune diseases in children may be linked to genetic, environmental, and hormonal factors. It's estimated that 5% of children worldwide have at least one autoimmune condition, with many experiencing multiple autoimmune challenges. Around 80 autoimmune diseases are present. However, autoimmune diseases in children are rare. When they occur they can be challenging to diagnose and difficult to treat. Advances in our knowledge of the immune system are uncovering connections between inflammation and many different diseases. All autoimmune diseases have relapsing and remitting tendencies. Autoimmune diseases can affect almost any part of the body, though they often target connective tissues (skin, muscle and joints). Symptoms can range from fatigue and mild rashes to rare, serious side effects, like seizures. Diagnosis can be difficult, because many symptoms tend to come and go and are frequently nonspecific. They occur in different kinds of autoimmune diseases as well as other types of illnesses, like infection and cancer. Autoimmune diseases occur most often in females by a 3-to-1 margin over males. Organ-specific disorders (also called localized) focus on one organ or a specific type of tissue: Addison's disease affects the adrenal glands; Autoimmune hepatitis affects the liver; Crohn's disease affects the gastrointestinal tract; Multiple sclerosis (MS) affects the central nervous system; Type 1 diabetes affects the pancreas; Ulcerative colitis affects the gastrointestinal tract. Non-organ-specific disorders (also called systemic) cause problems throughout the body: Juvenile dermatomyositis affects the skin and muscles; Juvenile idiopathic arthritis affects the joints and sometimes the skin and lungs; Lupus affects the joints, skin, liver, kidneys, heart, brain, and other organs; Scleroderma affects the skin, joints, intestine, and sometimes the lungs. Pediatric autoimmune neuropsychiatric disorders associated with streptococcal infections (PANDAS) are rare neurobehavioral disorders in children. Affected children may have a diverse array of perioperative manifestations including compulsive behavior, agitation, and abnormal movements. Intravenous anesthetics have anti-inflammatory properties, which in most septic cases are useful for patients. The anti-inflammatory effects of ketamine may be related to the suppression of TNF production by macrophage in the presence of bacteria. Propofol inhibits the phagocytosis and chemotaxis of human monocytes through GABAA receptors. Inhalational anesthetics, in a dose-dependent manner, suppress cytokine release, reduces lymphocyte proliferation, induce apoptosis of the lymphocytes, and inhibits the function of neutrophils. Inhalational anesthetics influence the endocrine response from the hypothalamus-pituitary-adrenal axis and indirectly through the secretions of hormones, such as glucocorticoids and catecholamines. Long-term administration of general anesthesia drugs, due to their effects on cytokines, can lead to disease progression in patients with immune deficiency.

**Keywords:** Autoimmune diseases; General anesthesia; Children; Immune system; PANDAS; OCD

### 1. Introduction

The production of antibodies to streptococcal antigens associated with the M-protein of Group A beta hemolytic streptococcus (GABHS) has been shown to cross-react with epitopes on neuronal tissue. It has been postulated that the pathogenesis of Sydenham's chorea results from immune complex disease produced by nondestructive anti-streptococcal antibodies that localize to the basal ganglia and striatal areas of the brain. Structural and functional

\* Corresponding author: Maria I Dalamagka

neuroimaging studies have demonstrated abnormalities of the basal ganglia structures and their related corticostriato-thalamocortical circuitry in the pathobiology of obsessive compulsive disorder (OCD). The PANDAS subgroup is identified by five clinical criteria: presence of OCD and/or tic disorder; prepubertal symptom onset; episodic course characterized by acute, severe onset and dramatic symptom exacerbations; neurological abnormalities (e.g., choreiform movements) present during symptom exacerbations; temporal relation between GABHS infections and symptom exacerbations. The major distinguishing feature of the PANDAS subgroup is the temporal association between neuropsychiatric symptom exacerbations and GABHS infections. Anesthetic Implications: PANDAS are rare and affected children have significant anesthetic implications. The abnormal behavior and movement of children with PANDAS are a real challenge to anesthesiologists. We have described the first case in anesthesia literature of a patient with PANDAS who required anesthesia for a series of therapeutic plasmapheresis. The children usually have motor or vocal tics, obsessions, and/or compulsions. Often communication with them is a major problem. In addition to these symptoms, children may also become moody, irritable or show concerns about separating from parents. In our patient, premedication with oral ketamine and midazolam worked well to separate the patient from his parents and gain an intravenous access. Propofol infusion at usual doses provided optimum sedation during plasmapheresis and the recovery from propofol was favorable. With frequent plasmapheresis, there is a potential for coagulopathy from washout/dilution of plasma clotting factors. Evaluating coagulation status, especially after the first few plasmaphereses may identify patients at risk of bleeding. In conclusion, we described our anesthetic management of a case of PANDAS in a young boy who underwent a series of therapeutic plasmapheresis. Anesthesiologists should be aware of the abnormal behavior and movements with this disorder and potential anesthetic implications. Atopic dermatitis is one of the most common skin disorders in childhood, and general anaesthesia has a risk of developing potential complications in various organs and anaphylaxis and allergies. However, the relationship between exposure to general anaesthesia and atopic dermatitis remains unknown. General anaesthesia (GA) has a risk of developing potential complications in various organs and anaphylaxis and allergies. General anaesthesia could affect various immune responses, including Th1 and Th2 immunity, which might also affect cells that play an important role in the pathogenesis of atopic dermatitis. The number of patients undergoing GA in the paediatric population is increasing and thus interest in whether GA could be a risk factor for various diseases has increased. In particular, previous studies revealed that exposure to GA in early childhood and multiple exposures increased the risk of developmental delay and attention deficit hyperactivity disorder. Conversely, autistic disorders or learning disabilities were not associated with a single exposure to GA. Also, it has been reported that exposure to GA in early life reduced the risk of asthma, allergic rhinitis, and AD. Congenital myasthenic syndromes are genetically, mainly recessive, transmitted disorders. This group is clinically very heterogeneous, children present shortly after birth with a feeding problem and muscular hypotonia. As a severe symptom is respiratory insufficiency is well described and children often need artificial ventilation. A few children respond to therapy with acetylcholinesterase-blockers. The defect could be located presynaptic (eg. CholinacetylTransferase Defect CHAT), synaptic (eg. mutations in the gene encoding for Collagen Q COLQ) and postsynaptic (eg. disturbances in fast or slow channels in acetylcolinreceptors or mutations in the gene encoding for the Rapsyn gene). A special feature in CHAT defects are recurrent apnea in children suffering from simple infectious diseases. These children require fast intubation and artificial ventilation. Transient Neonatal Myasthenia gravis occurs in approximately 12% of the newborns of mothers with MG and improves within the first weeks of life as the antibody concentration decreases. The first signs are often muscular hypotonia, feeding problems and respiratory [1-24].

---

## 2. Autoimmune diseases

The exact causes of autoimmune conditions in children are not fully understood. Research suggests there may be a complex interplay between environmental factors, genetics, and increased intestinal permeability. A disease is defined as autoimmune if the tissue injury is caused by an adaptive immune response to self-antigen. Development of the disease by transferring the active component of the immune response to an appropriate recipient (usually an animal model, or in some instances materno-fetal transfer of autoantibody) is the best proof that a disease is caused by autoimmunity. Autoimmune diseases can be considered on a continuum between those which are organ specific, and those which are non-organ specific. There are over 100 identified autoimmune conditions. Some commonly known in children include: Type 1 diabetes, the immune system affects the beta cells of the pancreas, impacting insulin production; Celiac disease, the immune system reacts to gliadin, a protein in gluten-containing foods, affecting the small intestine lining; Juvenile idiopathic arthritis, the immune system produces markers that primarily affect the joints; Pediatric lupus, the immune system targets various tissues, including the lungs, skin, and kidneys. Autoantibodies are commonly regarded as being intrinsically involved in the tissue injury associated with autoimmune diseases. Whilst this may be true in a number of important examples (e.g. systemic lupus erythematosus (SLE), Goodpasture's disease, and myasthenia gravis), this is not true of all autoimmune diseases. For example, autoantibodies may form secondary to the tissue damage associated with the underlying disease (e.g. cardiac antibodies produced after myocardial infarction). Corticosteroids, cytotoxic drugs and other immunosuppressant agents are an important part of the therapeutic approach to many non-malignant autoimmune disorders. Thus, treatment protocols for these diseases in pediatrics are often based on adult practice, but

despite the similarities in disease processes, the most widely used treatments have different effects in children. Some of the side effects of chronic steroid use, including linear growth deceleration, bone demineralization, and chronic weight issues, are more consequential in children than in adults. Although steroids remain a cornerstone of therapy in JDM and are useful in many cases of CIDP and JMG, other immunomodulatory therapies with similar efficacy may be used more frequently. Early treatment with intravenous immunoglobulin (IVIG) or plasma exchange (PE) is recommended for any child with Guillain-Barré syndrome who has difficulty ambulating. Azathioprine is the most widely used steroid-sparing therapy in childhood CIDP (Chronic inflammatory demyelinating polyradiculoneuropathy). Patients with juvenile myasthenia gravis (JMG) can be offered three main categories of disease-modifying therapy: acetylcholinesterase inhibitors (AChE-I), medical immunomodulation or immunosuppression, and surgical intervention (thymectomy). Corticosteroids (prednisone, prednisolone, methylprednisolone) are effective immunosuppressants with good evidence for their efficacy in MG. Methotrexate is often used as an alternative immunosuppressant agent in adults with MG who are unable to tolerate azathioprine. Cyclophosphamide therapy has been used for severe, treatment-resistant adult MG. Rituximab has been used with promising results in adults with severe, refractory MG. Plasma exchange (PE) is an alternative to IVIG for treating acute exacerbations. The immune system relies on a large number of genes for its normal development. In children affected by combined immunodeficiency (CID), a defect in one or more of these genes results in the absence or malfunction of a protein necessary for normal functioning of the immune system. Different types of CID are named based on the particular protein or gene that is affected. Some of the better-known types include Wiskott-Aldrich syndrome, DOCK8 deficiency and recombination activating gene (RAG) deficiency. Some live vaccines might need to be avoided, particularly BCG, rotavirus, measles/mumps/rubella (MMR) and chickenpox (VZV). Haematopoietic stem cell transplantation In some cases, haematopoietic stem cell transplant (HSCT) (including bone marrow transplant, BMT) offers the potential for long-term cure of CID. This treatment is most well-established for defined types of CID that are known to be life-limiting because of predictably severe complications. Gene therapy aims to correct the underlying genetic abnormality by replacing the faulty gene in immune cells with a normal copy. The commonest condition affecting thymus development is called 22q11 deletion syndrome (also known as DiGeorge syndrome). In the small fraction (<1%) of DiGeorge patients whose thymus is completely missing, thymic transplant may need to be considered [25-63].

### 3. Anesthesia - Anesthetic Drugs and immune system

General anesthesia, which is a drug-induced deliberated state, is the most commonly used strategy in anesthesia care. There are various drugs to ensure unconsciousness, analgesia, amnesia, and loss of reflexes of the autonomic nervous system. General anesthesia usually induces using intravenous anesthetics, inhalational (volatile) anesthetics or a combination of both. Also, opioids and benzodiazepines are often used as adjuvants in general anesthesia. In some cases, ventilation of the patients should be controlled. Anesthesia and drugs that administer to induce the anesthesia, affect the immune system, mainly immune cells, during the perioperative period. The effects of anesthesia on B-lymphocytes, T-lymphocytes, NK cells, macrophages, erythrocytes, and leukocytes is well documented. Anesthesia affects the pro-inflammatory and anti-inflammatory cytokines' secretion. The main Pro-inflammatory cytokines include IL-6, IL-8, IL-1, and tumor necrosis factor- $\alpha$  (TNF- $\alpha$ ). The main anti-inflammatory cytokine includes IL-10. The levels of interferon Gamma (IFN- $\gamma$ ) and TNF are changed by general anesthesia. Evaluation of the effects of general anesthesia agents on the immune system helps to improve the management of anesthesia. The most common method to produce general anesthesia is the intravenous anesthesia (IV anesthesia). Sodium thiopental, propofol, and ketamine are examples of drugs which commonly use in IV anesthesia. Ketamine and sodium thiopental reduce the number of T-helper cells and NK cell activities and increase the T-inhibiting cells. Sodium thiopental and ketamine inhibit the release of IL-1, IL-6, TNF- $\alpha$ , and IL-8. It is well-documented that low dose ketamine, as an N methyl-D-aspartate (NMDA) receptors antagonist, reduces the lifetime of IL-6. In addition, these drugs increase the level of IL-10. However, according to the results reported by Song et al., ketamine inhibits the production of anti-inflammatory cytokines. Sodium thiopental is a rapid induction intravenous anesthetic agent which affects the GABA- A receptors that their presence in immune cells is confirmed. In a study, the effect of propofol and sodium thiopental on Th1/Th2 balance is compared by measuring the levels of IFN- $\gamma$ , IL-4, and IL-2. Compared to propofol, sodium thiopental reduce the concentration of IFN- $\gamma$  and IL-4 without affecting the concentration of IL-2. Propofol (2,6-diisopropylphenol) is a commonly used intravenous anesthetic agent. Besides, it uses for maintenance of general anesthesia and sedation in the intensive care unit. Propofol causes rapid induction of and recovery from anesthesia. This drug has anti-oxidant and anti-inflammatory characteristics. Information about whether anesthesia with propofol increases or decreases the IL-6 levels is controversial. Propofol in vitro inhibits IL-6 production by stimulated lipoproteins. The study conducted by Gonzalez-Correa et al. reported that propofol decreased the concentration of IL-1, TNF- $\alpha$ , and IL-6 cytokines. Propofol increases the concentration of IL-10. A higher concentration of anti-inflammatory cytokines in patients who received propofol is the reason for its anti-inflammatory properties. Jin et al. reported that Injection of high dose propofol increased the ratio of IFN- $\gamma$ /IL-4, while low dose propofol did not change the concentration of the cytokine. Volatile anesthetics can inhibit the release of IFN in animals, reduce the number of NK cells, and the production of cytokines in the human body.

Desflurane increases the number of neutrophil cell and pro-inflammatory cytokines expression in alveolar macrophages. Desflurane produces more pro-inflammatory responses compared to sevoflurane. Evidence suggests that opioids affect the immune system, although the results are contradictory. Both the suppressive and stimulant effects of the opioids on the immune system are reported by various studies. Opioids cause a significant reduction in the concentration of TNF- $\alpha$ , IL-1, and IL-6. Morphine produces TGF- $\beta$  by stimulating lymphocytes, monocytes, and macrophages. These effects are mainly the result of long-term treatment, which leads to an increase in the concentration of IL-4 and IL-5 as well as the reduction of IL-2 and IFN- $\gamma$ . Fentanyl exhibited cytotoxicity against NK cells. It is found that morphine, remifentanyl, fentanyl, methadone, and codeine alter the immune system more than oxycodone, hydrocodone, and tramadol. Remifentanyl decreases IFN- $\gamma$  concentrations. IL-6, TNF, IL-10, and IL-2 concentrations do not change during treatments with remifentanyl or fentanyl. Midazolam is more common than other benzodiazepines, because if administered intravenously causes less pain, and it's a short-acting drug compared to others. Midazolam can inhibit the production of IL-2 and IL-8 and produces an immunosuppression effect. Midazolam can inhibit the activity of TNF- $\alpha$  [64-81].

---

#### 4. Discussion

One of the ventilation methods during general anesthesia is controlled ventilation. Controlled ventilation, through damaging alveolar cells and mechanical stress, suppresses the stability of the ventilation/perfusion ratio, which results in the release of inflammatory mediators and worsening of the gas exchange during the postoperative period. The other factor that causes lung tissue damage and increases the release of cytokines is hypoxia during surgeries. Sevoflurane reduces the inflammatory response during one-lung ventilation in chest surgery and may be preferable in patients who the level of their pro-inflammatory cytokines is more than expected. Treatment of childhood autoimmune disorders is based upon published prospective and retrospective cohort studies, expert opinion, pediatric randomized controlled trials (particularly for Guillain-Barré syndrome and dermatomyositis), and extrapolation of results from adult studies. Early diagnosis and initiation of treatment can significantly reduce long-term morbidity for these diseases. In addition to affecting quality of life, pediatric autoimmune conditions may increase the risk of various health issues, including cardiometabolic concerns, cancer, and mental health challenges. Conventional treatments can provide some relief but may not address all underlying factors of autoimmunity.

---

#### 5. Conclusion

Autoimmune neuromuscular disorders in childhood include Guillain-Barré syndrome and its variants, chronic inflammatory demyelinating polyradiculoneuropathy (CIDP), juvenile myasthenia gravis (JMG), and juvenile dermatomyositis (JDM), along with other disorders rarely seen in childhood. Intravenous anesthetics have anti-inflammatory properties, which in most septic cases are useful for patients. The anti-inflammatory effects of ketamine may be related to the suppression of TNF production by macrophage in the presence of bacteria. Propofol inhibits the phagocytosis and chemotaxis of human monocytes through GABAA receptors. Inhalational anesthetics, in a dose-dependent manner, suppress cytokine release, reduces lymphocyte proliferation, induce apoptosis of the lymphocytes, and inhibits the function of neutrophils. Inhalational anesthetics influence the endocrine response from the hypothalamus-pituitary-adrenal axis and indirectly through the secretions of hormones, such as glucocorticoids and catecholamines. Long-term administration of general anesthesia drugs, due to their effects on cytokines, can lead to disease progression in patients with immune deficiency. Many autoimmune diseases share common pathogenic mechanisms. However, there are limited studies quantifying the coexistence of autoimmune diseases.

---

#### References

- [1] Yang EJ, Beck KM, Sekhon S, Bhutani T, Koo J. The impact of pediatric atopic dermatitis on families: a review. *Pediatr Dermatol* 2019; 36: 66–71.
- [2] Kantor R, Thyssen J, Paller A, Silverberg J. Atopic dermatitis, atopic eczema, or eczema? A systematic review, meta-analysis, and recommendation for uniform use of 'atopic dermatitis'. *Allergy* 2016; 71: 1480–1485.
- [3] Çetinkaya PG, Şahiner ÜM. Childhood atopic dermatitis: current developments, treatment approaches, and future expectations. *Turk J Med Sci* 2019; 49: 963–984.
- [4] Ha J, Lee SW, Yon DK. Ten-Year trends and prevalence of asthma, allergic rhinitis, and atopic dermatitis among the Korean population, 2008–2017. *Clin Exp Pediatr* 2020; 63: 278.
- [5] Pols DH, Bohnen AM, Nielen MM, Korevaar JC, Bindels PJ. Risks for comorbidity in children with atopic disorders: an observational study in Dutch general practices. *BMJ open* 2017; 7: e018091.

- [6] Gilaberte Y, Pérez-Gilaberte JB, Poblador-Plou B, Bliiek-Bueno K, Gimeno-Miguel A, Prados-Torres A. Prevalence and comorbidity of atopic dermatitis in children: a large-scale population study based on real-world data. *J Clin Med* 2020; 9: 1632.
- [7] Huang AH, Roh YS, Sutaria N, Choi J, Williams KA, Canner JK, et al. Real-world comorbidities of atopic dermatitis in the pediatric ambulatory population in the United States. *J Am Acad Dermatol* 2021; 85: 893–900.
- [8] Dharmage SC, Lowe A, Matheson MC, Burgess J, Allen K, Abramson MJ. Atopic dermatitis and the atopic march revisited. *Allergy* 2014; 69: 17–27.
- [9] Harris M, Chung F. Complications of general anesthesia. *Clin Plast Surg* 2013; 40: 503–513.
- [10] Stepanovic B, Sommerfield D, Lucas M, von Ungern-Sternberg BS. An update on allergy and anaphylaxis in pediatric anesthesia. *Paediatr Anaesth* 2019; 29: 892–900.
- [11] Kim JC, Kim DC, Choi YW, Lee ES, Choi JW. Association of chronic spontaneous urticaria with the first exposure to general anaesthesia. *Clin Exp Allergy* 2022; 52: 990–993.
- [12] Kuo H-C, Yang Y-L, Ho S-C, Guo MM-H, Jiang J-H, Huang Y-H. General anesthesia exposure in early life reduces the risk of allergic diseases: a nationwide population-based cohort study. *Medicine* 2016; 95: e4269.
- [13] Palacios AS, Ponce MO, Pérez AR, Medina FS, Marrero JG. Modification of mediators of immune reaction after general anaesthesia. *Allergol Immunopathol (Madr)* 2004; 32: 352–360.
- [14] Lee J, Lee JS, Park SH, Shin SA, Kim K. Cohort profile: The National Health Insurance Service-National Sample Cohort (NHIS-NSC), South Korea. *Int J Epidemiol* 2017; 46: e15.
- [15] Yoon J, Choi JW. Primary cicatricial alopecia in a single-race Asian population: a 10-year nationwide population-based study in South Korea. *J Dermatol* 2018; 45: 1306–1311.
- [16] Soh BW, Kim SM, Kim YC, Choi GS, Choi JW. Increasing prevalence of alopecia areata in South Korea. *J Dermatol* 2019; 46: e331–e332.
- [17] Kim JC, Lee ES, Choi JW. Impact of alopecia areata on psychiatric disorders: a retrospective cohort study. *J Am Acad Dermatol* 2020; 82: 484–486.
- [18] 18. Kim JC, Choi JW. Impact of alopecia areata on subsequent pregnancy rate: a retrospective cohort study. *Australas J Dermatol* 2021; 62: e121–e123.
- [19] Lee H, Kim YC, Choi JW. Alopecia areata is not a risk factor for heart diseases: a 10-year retrospective cohort study. *PLoS One* 2021; 16: e0250216.
- [20] Sundararajan V, Quan H, Halfon P, Fushimi K, Luthi JC, Burnand B, et al. Cross-national comparative performance of three versions of the ICD-10 Charlson index. *Med Care* 2007; 45: 1210–1215.
- [21] Tzong KY, Han S, Roh A, Ing C. Epidemiology of pediatric surgical admissions in US children: data from the HCUP kids inpatient database. *J Neurosurg Anesthesiol* 2012; 24: 391–395.
- [22] von Ungern-Sternberg BS, Boda K, Chambers NA, Rebmann C, Johnson C, Sly PD, et al. Risk assessment for respiratory complications in paediatric anaesthesia: a prospective cohort study. *The Lancet* 2010; 376: 773–783.
- [23] Ko W-R, Huang J-Y, Chiang Y-C, Nfor ON, Ko P-C, Jan S-R, et al. Risk of autistic disorder after exposure to general anaesthesia and surgery: a nationwide, retrospective matched cohort study. *Eur J Anaesthesiol* 2015; 32: 303–310.
- [24] Tsai C-J, Lee CT-C, Liang SH-Y, Tsai P-J, Chen VC-H, Gossop M. Risk of ADHD after multiple exposures to general anesthesia: a nationwide retrospective cohort study. *J Atten Disord* 2018; 22: 229–239.
- [25] Feng Y-P, Yang T-S, Chung C-H, Chien W-C, Wong C-S. Early childhood general anesthesia exposure associated with later developmental delay: a national population-based cohort study. *PLoS One* 2020; 15: e0238289.
- [26] Ng YT, Chew FT. A systematic review and meta-analysis of risk factors associated with atopic dermatitis in Asia. *World Allergy Organ J* 2020; 13: 100477.
- [27] Mortz C, Andersen K, Dellgren C, Barington T, Bindslev-Jensen C. Atopic dermatitis from adolescence to adulthood in the TOACS cohort: prevalence, persistence and comorbidities. *Allergy* 2015; 70: 836–845.
- [28] Czarnowicki T, He H, Krueger JG, Guttman-Yassky E. Atopic dermatitis endotypes and implications for targeted therapeutics. *J Allergy Clin Immunol* 2019; 143: 1–11.

- [29] Prescott SL, Macaubas C, Smallacombe T, Holt BJ, Sly PD, Holt PG. Development of allergen-specific T-cell memory in atopic and normal children. *The Lancet* 1999; 353: 196–200.
- [30] Kawamoto N, Kaneko H, Takemura M, Seishima M, Sakurai S, Fukao T, et al. Age-related changes in intracellular cytokine profiles and Th2 dominance in allergic children. *Pediatr Allergy Immunol* 2006; 17: 125–133.
- [31] Jafarzadeh A, Hadavi M, Hassanshahi G, Rezaeian M, Vazirinejad R. General anesthetics on immune system cytokines: a narrative review article. *Anesth Pain Med* 2020; 10: e103033.
- [32] Lee J, Lozano-Ruiz B, Yang FM, Fan DD, Shen L, González-Navajas JM. The multifaceted role of Th1, Th9, and Th17 cells in immune checkpoint inhibition therapy. *Front Immunol* 2021; 12: 625667.
- [33] Jin Z, Mendu SK, Birnir B. GABA is an effective immunomodulatory molecule. *Amino Acids* 2013; 45: 87–94.
- [34] Snider LA, Seligman LD, Ketchen BR et al. Tics and problem behaviors in schoolchildren: prevalence, characterization, and associations. *Pediatrics* 2002; 110: 331–336.
- [35] National Institute of Mental Health. Pediatric and Developmental Neuro-psychiatry Branch. <http://intramural.nimh.nih.gov/pdn/web.htm>, Posted January 26, 2005.
- [36] Swedo SE. Sydenham's chorea. A model for childhood autoimmune neuropsychiatric disorders. *JAMA* 1994; 272: 1788–1791.
- [37] Zai G, Bezchlibnyk YB, Richter MA et al. Myelin oligodendrocyte glycoprotein (MOG) gene is associated with obsessive-compulsive disorder. *Am J Med Genet B Neuropsychiatr Genet* 2004; 129B: 64–68.
- [38] Snider LA, Swedo SE. PANDAS: current status and directions for research. *Mol Psychiatry* 2004; 9: 900–907.
- [39] Lougee L, Perlmutter SJ, Nicolson R et al. Psychiatric disorders in first-degree relatives of children with pediatric autoimmune neuropsychiatric disorders associated with streptococcal infections (PANDAS). *J Am Acad Child Adolesc Psychiatry* 2000; 39: 1120–1126.
- [40] Giedd JN, Rapoport JL, Leonard HL et al. Case study: acute basal ganglia enlargement and obsessive-compulsive symptoms in an adolescent boy. *J Am Acad Child Adolesc Psychiatry* 1996; 35: 913–915.
- [41] Citak EC, Gucuyener K, Karabacak NI et al. Functional brain imaging in Sydenham's chorea and streptococcal tic disorders. *J Child Neurol* 2004; 19: 387–390.
- [42] Swedo SE, Leonard HL, Garvey M et al. Pediatric autoimmune neuropsychiatric disorders associated with streptococcal infections: clinical description of the first 50 Cases. *Am J Psychiatry* 1998; 155: 265–271.
- [43] van Toorn R, Weyers HH, Schoeman JF. Distinguishing PANDAS from Sydenham's chorea: case report and review of the literature. *Eur J Paediatr Neurol* 2004; 8: 211–216.
- [44] Snider LA, Sachdev V, MaCkaronis JE et al. Echocardiographic findings in the PANDAS subgroup. *Pediatrics* 2004; 114: 748–751.
- [45] Swedo SE, Grant PJ. Annotation: PANDAS: a model for human autoimmune disease. *J Child Psychol Psychiatry* 2005; 46: 227–234.
- [46] Storch EA, Gerdes AC, Adkins JW et al. Behavioral treatment of a child with PANDAS. *J Am Acad Child Adolesc Psychiatry* 2004; 43: 510–511.
- [47] Naguib M, Koorn R. Interactions between psychotropics, anaesthetics and electroconvulsive therapy: implications for drug choice and patient management. *CNS Drugs* 2002; 16: 229–247.
- [48] Kudoh A, Katagai H, Takase H et al. Effect of preoperative discontinuation of antipsychotics in schizophrenic patients on outcome during and after anaesthesia. *Eur J Anaesthesiol* 2004; 21: 414–416.
- [49] Coulter DM, Bate A, Meyboom RH et al. Antipsychotic drugs and heart muscle disorder in international pharmacovigilance: data mining study. *BMJ* 2001; 322: 1207–1209.
- [50] Bookheimer S, Schrader LM, Rausch R et al. Reduced anesthetization during the intracarotid amobarbital (Wada) test in patients taking carbonic anhydrase-inhibiting medications. *Epilepsia* 2005; 46: 236–243.
- [51] Perlmutter SJ, Leitman SF, Garvey MA et al. Therapeutic plasma exchange and intravenous immunoglobulin for obsessive-compulsive disorder and tic disorders in childhood. *The Lancet* 1999; 354: 1153–1159.
- [52] Orvidas LJ, Slattey MJ. Pediatric autoimmune neuropsychiatric disorders and streptococcal infections: role of otolaryngologist. *Laryngoscope* 2001; 111: 1515–1519.

- [53] Murphy ML, Pichichero ME. Prospective identification and treatment of children with pediatric autoimmune neuropsychiatric disorder associated with group A streptococcal infection (PANDAS). *Arch Pediatr Adolesc Med* 2002; 156: 356–361.
- [54] Heubi C, Shott SR. PANDAS: pediatric autoimmune neuropsychiatric disorders associated with streptococcal infections – an uncommon, but important indication for tonsillectomy. *Int J Pediatr Otorhinolaryngol* 2003; 67: 837–840.
- [55] Wood L, Jacobs P. The effect of serial therapeutic plasmapheresis on platelet count, coagulation factors, plasma immunoglobulin, and complement levels. *J Clin Apheresis* 1986; 3: 124–128.
- [56] Qudah AA, Shahar E, Logan WJ, Murphy EG. Neonatal Guillain-Barre syndrome. *Pediatr Neurol.* 1988;4(4):255–256. doi: 10.1016/0887-8994(88)90042-2.
- [57] Vajsar J, Fehlings D, Stephens D. Long-term outcome in children with Guillain-Barre syndrome. *J Pediatr.* 2003;142(3):305–309. doi: 10.1067/mpd.2003.115.
- [58] Bradshaw DY, Jones HR., Jr Guillain-Barre syndrome in children: clinical course, electrodiagnosis, and prognosis. *Muscle Nerve.* 1992;15(4):500–506. doi: 10.1002/mus.880150415.
- [59] Nguyen DK, Agenarioti-Belanger S, Vanasse M. Pain and the Guillain-Barre syndrome in children under 6 years old. *J Pediatr.* 1999;134(6):773–776. doi: 10.1016/S0022-3476(99)70297-0.
- [60] Roodbol J, de Wit MC, Walgaard C, de Hoog M, Catsman-Berrevoets CE, Jacobs BC. Recognizing Guillain-Barre syndrome in preschool children. *Neurology.* 2011;76(9):807–810. doi: 10.1212/WNL.0b013e31820e7b62.
- [61] Asbury AK, Cornblath DR. Assessment of current diagnostic criteria for Guillain-Barre syndrome. *Ann Neurol.* 1990;27(Suppl):S21–24. doi: 10.1002/ana.410270707.
- [62] Yikilmaz A, Doganay S, Gumus H, Per H, Kumandas S, Coskun A. Magnetic resonance imaging of childhood Guillain-Barre syndrome. *Childs Nerv Syst.* 2010;26(8):1103–1108. doi: 10.1007/s00381-010-1197-8.
- [63] Nagasawa K, Kuwabara S, Misawa S, Fujii K, Tanabe Y, Yuki N, Hattori T, Kohno Y. Electrophysiological subtypes and prognosis of childhood Guillain-Barre syndrome in Japan. *Muscle Nerve.* 2006;33(6):766–770. doi: 10.1002/mus.20520.
- [64] Chen L, Deng H, Cui H, Fang J, Zuo Z, Deng J, et al. Inflammatory responses and inflammation-associated diseases in organs. *Oncotarget.* 2018;9(6):7204–18. doi: 10.18632/oncotarget.23208.
- [65] Akdis CA. Therapies for allergic inflammation: Refining strategies to induce tolerance. *Nat Med.* 2012;18(5):736–49. doi: 10.1038/nm.2754.
- [66] Abbas AK, Lichtman AH, Pillai S. Cellular and molecular immunology. 8th ed. US: Elsevier Sanders; 2014.
- [67] Mahmoud K, Ammar A. Immunomodulatory effects of anesthetics during thoracic surgery. *Anesthesiol Res Pract.* 2011;2011:317410. doi: 10.1155/2011/317410.
- [68] Jafarzadeh A, Larussa T, Nemati M, Jalapour S. T cell subsets play an important role in the determination of the clinical outcome of *Helicobacter pylori* infection. *Microb Pathog.* 2018;116:227–36. doi: 10.1016/j.micpath.2018.01.040.
- [69] Nemati M, Malla N, Yadav M, Khorramdelazad H, Jafarzadeh A. Humoral and T cell-mediated immune response against trichomoniasis. *Parasite Immunol.* 2018;40(3) doi: 10.1111/pim.12510.
- [70] Kalayci D, Dikmen B, Kacmaz M, Taspinar V, Ornek D, Turan O. Plasma levels of interleukin-10 and nitric oxide in response to two different desflurane anesthesia flow rates. *Braz J Anesthesiol.* 2014;64(4):292–8. doi: 10.1016/j.bjane.2013.06.008.
- [71] Jafarzadeh A, Shokri F. The antibody response to HBs antigen is regulated by coordinated Th1 and Th2 cytokine production in healthy neonates. *Clin Exp Immunol.* 2003;131(3):451–6. doi: 10.1046/j.1365-2249.2003.02093.
- [72] Lenz A, Franklin GA, Cheadle WG. Systemic inflammation after trauma. *Injury.* 2007;38(12):1336–45. doi: 10.1016/j.injury.2007.10.003.
- [73] Pirbudak Cocelli L, Ugur MG, Karadasli H. Comparison of effects of low-flow sevoflurane and desflurane anesthesia on neutrophil and T-cell populations. *Curr Ther Res Clin Exp.* 2012;73(1-2):41–51. doi: 10.1016/j.curtheres.2012.02.005.

- [74] Gong L, Qin Q, Zhou L, Ouyang W, Li Y, Wu Y, et al. Effects of fentanyl anesthesia and sufentanil anesthesia on regulatory T cells frequencies. *Int J Clin Exp Pathol*. 2014;7(11):7708–16.
- [75] Alsina E, Matute E, Ruiz-Huerta AD, Gilsanz F. The effects of sevoflurane or remifentanil on the stress response to surgical stimulus. *Curr Pharm Des*. 2014;20(34):5449–68. doi: 10.2174/1381612820666140325105723.
- [76] Brown EN, Lydic R, Schiff ND. General anesthesia, sleep, and coma. *N Engl J Med*. 2010;363(27):2638–50. doi: 10.1056/NEJMr0808281.
- [77] Miller RD. *Miller's anesthesia*. Philadelphia: Elsevier; 2015.
- [78] Melamed R, Bar-Yosef S, Shakhar G, Shakhar K, Ben-Eliyahu S. Suppression of natural killer cell activity and promotion of tumor metastasis by ketamine, thiopental, and halothane, but not by propofol: mediating mechanisms and prophylactic measures. *Anesth Analg*. 2003;97(5):1331–9. doi: 10.1213/01.ane.0000082995.44040.07.
- [79] Schneemilch CE, Hachenberg T, Ansorge S, Ittenson A, Bank U. Effects of different anaesthetic agents on immune cell function in vitro. *Eur J Anaesthesiol*. 2005;22(8):616–23. doi: 10.1017/s0265021505001031.
- [80] Snyder GL, Greenberg S. Effect of anaesthetic technique and other perioperative factors on cancer recurrence. *Br J Anaesth*. 2010;105(2):106–15. doi: 10.1093/bja/aeq164.
- [81] Welters ID, Hafer G, Menzebach A, Muhling J, Neuhauser C, Browning P, et al. Ketamine inhibits transcription factors activator protein 1 and nuclear factor-kappaB, interleukin-8 production, as well as CD11b and CD16 expression: studies in human leukocytes and leukocytic cell lines. *Anesth Analg*. 2010;110(3):934–41. doi: 10.1213/ANE.0b013e3181c95cfa.



# Combined Spinal-Epidural (CSE) technique plus ketamine in research laparotomies with vertical incision: Case report

Dalamamagka Maria, Maria I Dalamagka

## ► To cite this version:

Dalamamagka Maria, Maria I Dalamagka. Combined Spinal-Epidural (CSE) technique plus ketamine in research laparotomies with vertical incision: Case report. GSC Advanced Research and Reviews , 2022, 10, pp.176 - 178. 10.30574/gscarr.2022.10.3.0073 . hal-05432464

**HAL Id: hal-05432464**

**<https://hal.science/view/index/docid/5432464>**

Submitted on 26 Dec 2025

**HAL** is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



Distributed under a Creative Commons Attribution 4.0 International License

## Combined Spinal-Epidural (CSE) technique plus ketamine in research laparotomies with vertical incision: Case report

Maria I Dalamagka \*

*Anesthesia Department, General Hospital of Larisa, Greece.*

GSC Advanced Research and Reviews, 2022, 10(03), 176–178

Publication history: Received on 05 February 2022; revised on 17 March 2022; accepted on 19 March 2022

Article DOI: <https://doi.org/10.30574/gscarr.2022.10.3.0073>

### Abstract

The combined spinal-epidural (CSE) technique, a comparatively new anesthetic choice, includes an initial subarachnoid injection followed by epidural catheter placement and subsequent administration of epidural medications. Clinical studies have demonstrated that the CSE technique provides excellent surgical conditions as quickly as with single-shot subarachnoid block conditions that are better than with epidural block alone. A 72-year-old man with pelvic tumor undergoing laparotomy surgery underwent combined regional anesthesia (Th12 - O1) plus small doses of Ketamine via intravenous infusion. The combined spinal-epidural technique along with intravenous Ketamine offers a better analgesic and hemodynamic result.

**Keywords:** CSE; Anesthesia; Epidural; Spinal; Ketamine; Laparotomies

### 1. Introduction

The combination (CSE) allows for rapid relief of pain or induction of regional anesthesia by the rapid onset of spinal drugs and subsequent administration of medications for prolonged anesthesia [1-4]. In addition, postoperative analgesia via the epidural catheter can be delivered for extended periods [5-6]. Clinical studies have demonstrated that the CSE technique provides excellent surgical conditions as quickly as the single-shot subarachnoid block and with advantages in comparison to the conventional epidural block. The CSE technique has been described in the medical literature for use in general surgery, orthopedics, trauma surgery of a lower limb, and urological and gynecological surgery [7-11].

### 2. Case study

A 72-year-old man with a known history of obstructive pulmonary disease and pelvic tumor, undergoing laparotomy surgery, underwent combined regional anesthesia (Th12 - O1) with subarachnoid infusion of 2.6 ml chirocaine 0.5%, 200 µg morphine and 0.1 ml Lidocaine 2%. Epidural infusion by placement of an epidural catheter with a test dose of 3 ml Lidocaine 2% and 5 ml Naropaine 0.75% and 15 minutes later an additional 5 ml Naropaine 0.75%. The patient was also given: Onda 4 mg, Dormicum 1 mg and Ketamine 20 mg plus 20 mg, 15 minutes later, by intravenous infusion. The patient was constantly monitored with ECG, NBP, and SpO2 monitoring. Hydrated with 1 lt Plasmalite and then titrated with 1 lt Ringers Lactate.

\* Corresponding author: Dalamagka Maria  
Anesthesia Department, General Hospital of Larisa, Greece.

### 3. Management and Outcome

Combined subarachnoid - epidural anesthesia was given because the goal was to rapidly begin the blockade with subarachnoid anesthesia and the incision was extensive and the laparotomy particularly laborious due to the location of the tumor. It was enhanced with epidural infusion of the drug. Blockade rise to a satisfactory level so that intense surgical manipulations do not cause discomfort to the patient. In subarachnoid blockade, chirocaine 0.5% was preferred because it is distinguished for its hemodynamic stability and was combined with a small dose of 2% lidocaine for faster onset of blockade, as well as morphine 200 mcg for better blockade and postoperative analgesia. After the installation of the subarachnoid block, the height of the block (its dermatome installation) as well as the hemodynamic stability of the patient were checked and after it was documented, then 5 ml Naropain 0.75% was given. After 15 minutes and after maintaining a satisfactory blood pressure, an additional 5 ml of 0.75% Naropain was given. Ketamine was also given as an intravenous infusion, which induces dissociative anesthesia and provides analgesia without suppressing the respiratory system.

### 4. Discussion

Combined subarachnoid - epidural anesthesia is ideal anesthesia for patients who are unable to undergo general anesthesia, such as patients with chronic respiratory disease, who would be referred to an Intensive Care Unit as their respiratory tract does not allow their good postoperative course. The patient in the study was a respiratory patient with chronic obstructive pulmonary disease, who could not be released from the ventilator if he was under general anesthesia. In addition, epidural catheter dilation adds a complete analgesic effect postoperatively, as epidural doses can be given postoperatively or a continuous infusion drug pump can be connected epidurally and a satisfactory level of analgesia maintained.

### 5. Conclusion

It has been found that combined local anesthesia excels and offers better exclusion and the combination with ketamine leads to a better hemodynamic and analgesic result.

### Compliance with ethical standards

#### *Acknowledgments*

General Hospital of Larisa

#### *Statement of informed consent*

Written informed consent was obtained from the patient for publication of this case report.. A copy of the written consent is available.

### References

- [1] Rodgers A, Walker N, Schug S, et al. Reduction of postoperative mortality and morbidity with epidural or spinal anaesthesia. *BMJ* 2000; 321: 1493–1504.
- [2] Buhre W, Rossaint R: Perioperative management and monitoring in anaesthesia. *Lancet* 2003; 362: 1839–1846.
- [3] Kehlet H, Wilmore DW: Multimodal strategies to improve surgical outcome. *Am J Surg* 2002; 183: 630–641.
- [4] Holmström B, Laugaland K, Rawal N, et al: Combined spinal epidural block versus spinal and epidural block for orthopaedic surgery. *Can J Anesth* 1993; 40: 601–606.
- [5] Stienstra R, Dahan A, Alhadi ZRB, et al: Mechanism of action of an epidural top-up in combined spinal epidural anaesthesia. *Anesth Analg* 1996; 83: 382–386.
- [6] Stienstra R, Dilrosun-Alhadi BZR, Dahan A, et al: The epidural top-up in combined spinal-epidural anaesthesia: The effect of volume versus dose. *Anesth Analg* 1999; 88: 810–814.
- [7] Blanshard HJ, Cook TM: Use of combined spinal-epidural by obstetric anaesthetists. *Anaesthesia* 2004; 59(9):922–923.

- [8] Kopacz DJ: Spinal 2-chloroprocaine: Minimum effective dose. *Reg Anesth Pain Med* 2005; 30(1): 36–42.
- [9] Hamza J, Smida M, Benhamou D, Cohen SE: Parturient's posture during epidural puncture affects the distance from skin to epidural space. *J Clin Anesth* 1995; 7: 1–4.
- [10] Yun EM, Marx GF, Santos AC: The effects of maternal position during induction of combined spinal-epidural anesthesia for caesarean delivery. *Anesth Analg* 1998; 87: 614–618.
- [11] Lewis NL, Ritchie EL, Downer JP, Nel MR: Left lateral vs. supine, wedged position for development of block after combined spinal-epidural anaesthesia for caesarean section. *Anaesthesia* 2004; 59: 894–898.



# Combination of ketamine, corticosteroids and sevoflurane inhibits the risk of bronchospasm in intubated children under general anesthesia

Maria I Dalamagka

## ► To cite this version:

Maria I Dalamagka. Combination of ketamine, corticosteroids and sevoflurane inhibits the risk of bronchospasm in intubated children under general anesthesia. GSC Advanced Research and Reviews , 2022, 11, pp.165 - 166. 10.30574/gscarr.2022.11.1.0108 . hal-05432459

**HAL Id: hal-05432459**

**<https://hal.science/view/index/docid/5432459>**

Submitted on 26 Dec 2025

**HAL** is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



Distributed under a Creative Commons Attribution 4.0 International License

## Combination of ketamine, corticosteroids and sevoflurane inhibits the risk of bronchospasm in intubated children under general anesthesia

Maria I Dalamagka \*

*Department of Anesthesia, General Hospital of Larisa, Greece.*

GSC Advanced Research and Reviews, 2022, 11(01), 165–166

Publication history: Received on 09 March 2022; revised on 17 April 2022; accepted on 17 April 2022

Article DOI: <https://doi.org/10.30574/gscarr.2022.11.1.0108>

### Abstract

Asthma in children is associated with significant morbidity. Children with severe asthma are at increased risk for adverse outcomes including medication-related side effects, life-threatening exacerbations, and impaired quality of life. In the study, an asthmatic child with a recent cold, received general anesthesia for emergency surgery centered on sevoflurane, corticosteroids and ketamine. The purpose of this study is to demonstrate the beneficial effects of the combination of sevoflurane, ketamine and corticosteroids in asthmatic children and to prevent complications when they are given general anesthesia for emergency surgery.

**Keywords:** Corticosteroids; Sevoflurane; Bronchospasm; Ketamine; Anesthesia; Astma

### 1. Introduction

Asthma is a chronic respiratory disease that affects people of all ages and is characterized by episodic and reversible attacks of wheezing, chest tightness, shortness of breath, and coughing. According to the ATS/ERS guideline, severe asthma is defined as asthma which requires treatment with high dose inhaled corticosteroids (ICS) plus a second controller (and/or systemic corticosteroid) to prevent it from becoming “uncontrolled” or remains “uncontrolled” despite this therapy [1-7].

### 2. Case Study

A 6-year-old child weighing 30 kg with a recent cold, nasal congestion and hearing of both amphibians and with a known history of frequent asthma attacks underwent emergency appendectomy. It was given intravenously before the introduction of anesthesia solu medrol 60 mg, Dexaton 3 mg, onda 3 mg. Introduction to anesthesia was by intravenous administration of Fentanyl 60 mcg, Propofol 120 mg, Esmeron 30 mg. After intubation, intrabronchial aspiration was performed, intrabronchial Flixotide 50 mcg και Aerolin 100 mcg was given and mechanically ventilated with 50% N2O and 2% sevoflurane. Ketamine 6 mg was given intravenously, plus 30 mcg Fentanyl, Apotel 350 mg, and morphine 1.5 mg. The monitoring included ECG, NBP, SpO2 and the ventilation model in Drager machine was Volume Control. Towards the end of the operation and with pure inhaled oxygen he was put in a Pressure Control model and the awakening was done smoothly after intravenous Bridium 0.6 mg.

### 3. Management and Outcome

The common denominator underlined in all forms of asthma is bronchial hyperresponsiveness to various stimuli. Inhaled glucocorticoids have long been used as a first-line treatment for persistent pediatric asthma, as they are the

\* Corresponding author: Maria I Dalamagka  
Anesthesia Department, General Hospital of Larisa, Greece.

most effective intervention for the treatment of asthma. Thus, Solu medrol (kg x 2), Dexatone (up to 0.1x kg) and Flixotide 50 mcg and Aerolin 100 mcg were administered intravenously. Ketamine also causes bronchodilation and was administered at its appropriate titrated dose (0.2 x kg), as anesthesia was maintained with sevoflurane which does not irritate the respiratory system.

---

#### 4. Discussion

Corticosteroids have inhibitory properties in many effects on many stem cells and inflammatory cells, which are activated in asthma. Inhaled steroids reduce the number and activation of inflammatory cells in the epithelium and submucosa by clogging the damaged epithelium and, potentially, inhibiting the production of proinflammatory cytokines, and reducing the survival time of the epithelium. This action of corticosteroids in combination with the ketamine which has a bronchodilator effect and the property of sevoflurane as it does not irritate the airway, they eliminate bronchospasm and prevent possible laryngospasm.

---

#### 5. Conclusion

In conclusion, the combination of Ketamine, corticosteroids and sevoflurane has been shown to inhibit possible complications such as bronchospasm when asthmatic children implied in general anesthesia.

---

#### Compliance with ethical standards

##### *Acknowledgments*

General Hospital of Larisa.

##### *Statement of informed consent*

Written informed consent was obtained from the patient for publication of this case report. A copy of the written consent is available.

---

#### References

- [1] KF Chung, SE Wenzel, JL Brozek, A Bush, M Castro, PJ Sterk, et al. International ERS/ATS guidelines on definition, evaluation and treatment of severe asthma Eur Respir J. 2014; 43; 343-373.
- [2] HS Zahran, CM Bailey, SA Damon, PL Garbe, PN Breyse. Vital signs: asthma in children – United States, 2001-2016 MMWR Morb Mortal Wkly Rep. 2018; 67: 149-155.
- [3] WG Teague, BR Phillips, JV Fahy, SE Wenzel, AM Fitzpatrick, WC Moore. et al. Baseline features of the severe asthma Research program (SARP III) cohort: differences with age J Allergy Clin Immunol Pract. 2018; 6: 545-554.
- [4] F Rabe, P Nair, G Brusselle, JF Maspero, M Castro, L. Sher et al. Efficacy and safety of dupilumab in glucocorticoid-dependent severe asthma N Engl J Med. 2018; 378: 2475-2485.
- [5] S Gonem, R Berair, A Singapuri, R Hartley, MFM Laurencin, G Bacher, et al. Fevipiprant, a prostaglandin D2 receptor 2 antagonist, in patients with persistent eosinophilic asthma: a single-centre, randomised, double-blind, parallel-group, placebo-controlled trial Lancet Respir Med. 2016; 4: 699-707.
- [6] H Farne, DJ Jackson, SL Johnston Are emerging PGD2 antagonists a promising therapy class for treating asthma? Expert Opin Emerg Drugs. 2016; 21: 359-364.
- [7] J Corren, JR Parnes, L Wang, M Mo, SL Roseti, JM Griffiths, et al. Tezepelumab in adults with uncontrolled asthma N Engl J Med. 2017; 377: 936-946.



## Acupuncture for low back pain and migraine

Maria I Dalamagka

### ► To cite this version:

Maria I Dalamagka. Acupuncture for low back pain and migraine. World Journal of Advanced Research and Reviews, 2024, 22, pp.1269 - 1274. 10.30574/wjarr.2024.22.2.1540 . hal-05432453

**HAL Id: hal-05432453**

**<https://hal.science/view/index/docid/5432453>**

Submitted on 26 Dec 2025

**HAL** is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



Distributed under a Creative Commons Attribution 4.0 International License

## Acupuncture for low back pain and migraine

Maria I. Dalamagka \*

*Department of Anaesthesia, General Hospital of Larisa, Greece.*

World Journal of Advanced Research and Reviews, 2024, 22(02), 1269–1274

Publication history: Received on 08 April 2024; revised on 16 May 2024; accepted on 18 May 2024

Article DOI: <https://doi.org/10.30574/wjarr.2024.22.2.1540>

### Abstract

Acupuncture has been increasingly used as an integrative or complementary therapy for pain. It is well-tolerated with little risk of serious adverse effects. Traditional acupuncture and nontraditional techniques, such as electroacupuncture and dry needling, often result in reported pain improvement. Acupuncture is an ancient Chinese medical practice that involves inserting thin needles into specific points on the body. Acupuncture can be used to treat a wide range of health conditions, including migraines and low back pain. Acupuncture works by stimulating the nervous system, releasing endorphins and increasing blood flow to the affected area. Acupuncture may also help to reduce inflammation and improve circulation in the body. Controlled trials have been published on pain syndromes, such as acupuncture for acute and chronic low back pain, knee osteoarthritis, headache, myofascial pain, neck pain, and fibromyalgia. For some conditions, enough data are available for systematic evaluations or meta-analyses. Acupuncture may provide modest benefits in the treatment of chronic low back pain, tension headache and chronic headache, migraine headache prophylaxis, and myofascial pain. The benefits of acupuncture for people with back pain has been well researched compared to many other conditions, with more higher quality clinical trials. This systematic review will focus on the use of acupuncture and its role in the treatment of low back pain to help better guide physicians in their practice. To determine the effects of a policy of 'use acupuncture' on headache, health status and chronic pain. The results of this study will determine if acupuncture, alone or as an adjunct to pharmacotherapy provides effective, safe and acceptable pain relief for patients.

**Keywords:** Acupuncture; Pain; Migraine; Therapy; Medicine; Complementary.

### 1. Introduction

Acupuncture is commonly used to treat pain. In traditional Chinese medicine concepts Meridian and vital energy Qi are part of the theoretical basis for the application of acupuncture in strictly specific acupoints. These studies demonstrate that the penetration of a needle through the skin, either acupuncturists point or not having physiological activities. The gate control theory and the release of endogenous opioids have been proposed as the causative mechanisms leading to obvious analgesic effect of acupuncture [1]. Back pain is a common condition that leads to a weakening and disability with an estimated prevalence length at a rate of 70-85% [2]. The non-specific back pain represents the majority of cases. Although 90% of patients show improvement at 1 month, the majority continues to be symptomatic for 1 year, with a rate of 21 to 25% on terms of pain and disability. In summary, the back pain is among the most costly diseases in the UK, which according to the findings in other countries, leading to a total cost of £ 10,668 million. (Including direct healthcare costs and indirect costs, such informal care and lost productivity) [3]. The Royal College of General Physicians suggests that back pain should be transferred from secondary to primary care and the goal should be a rapid return to normal functionality. There is intense debate how to achieve a return to normal activities. Meanwhile, the complementary and alternative medicine (CAM) has proven that acupuncture is a powerful treatment, which is associated with a clinically significant improvement in back pain. Two randomized controlled trials have evaluated the economic costs, one in the UK and one in Germany, show that acupuncture is relatively cost-effective in terms of quality

\* Corresponding author: Maria I. Dalamagka

of life for low back pain. These considerations seem to have translated into action, as a growing number of physicians in England offer their patients access to acupuncture. In addition, public health has a strong interest in the use of acupuncture, for example, a survey in the US showed that the most patients with back pain were most likely to experience acupuncture if they did not have to pay [3]. Acupuncture is of increasing importance for the treatment of headache in Western medicine [4]. In 2001 a major review of the Cochrane concluded that acupuncture is important in the treatment of idiopathic headache, but the quantity and quality of the evidence was not sufficient [4]. Little is known, however, about the relationship between the diagnostic criteria of the International Headache Society and the diagnosis of traditional Chinese medicine in primary headache disorders [4]. The authors reported that they needed well-designed, large-scale studies to evaluate the effectiveness and cost-effectiveness of acupuncture [4]. In 2005 two large, high quality studies in patients with headache noticed a slight difference between the effect of acupuncture and placebo, but there was a significant difference between acupuncture and placebo group with no intervention. This result differed from that of a large systematic review comparing all interventions of acupuncture [2].

## 2. Results

In research review of all types of placebo by Matia Madsen et al. [1], thirteen studies were selected (3025 patients), which included various pain conditions such as knee osteoarthritis, tension headache, migraine, low back pain, fibromyalgia, abdominal pain due to scars, postoperative pain, and pain during colonoscopy. They included all studies of acupuncture, as traditional acupuncture and electroacupuncture and excluded studies that used transcutaneous electrical nerve stimulation and applying pressure to acupuncture points (acupressure). The studies were not blinded (blinded) concerning the application of acupuncture and «placebo» acupuncture by clinicians. The duration of treatment ranged from one day to 12 weeks. A small but statistically significant difference was found between the acupuncture and «placebo» acupuncture. A moderate difference was found between the «placebo» acupuncture and the control group, as large studies make reference both to small, and in the important actions of placebo. In a blinded randomized controlled study conducted in Germany (340 outpatients) by Michael Haake [2] and his collaborators, included 1162 patients aged 18 to 86 years with a history of chronic low back pain for an average of 8 years. Patients underwent 10 sessions of 30 minutes, either real acupuncture (according to the principles of traditional Chinese medicine), or sham acupuncture (where applied superficial acupuncture at non acupuncture points) or conventional treatment (a combination of medications, physical therapy and exercise). Primary endpoint was the response after six months, defined as a 33% improvement. Patients who have not followed the process blind or have recourse to other non-permitted concomitant treatments during the monitoring were classified as non-responders, regardless of symptom improvement. At six months, response rate was 47.6% in the real acupuncture group, 44.2% in the sham acupuncture group and 27.4% in the conventional therapy group. Almost half of the patients in the acupuncture group and only one quarter of patients in the conventional therapy benefited. This study also found clear superiority of acupuncture compared with conventional treatment guideline, but showed no superiority over «sham» acupuncture for at least six months. The authors of both studies concluded that acupuncture is an affordable and effective treatment for back pain. To evaluate the effect of acupuncture in nonspecific back pain conducted systematic review of randomized controlled trials (RCTs) from Jing Yuan et al. [3] twenty three studies (n=6359) were included and classified into 5 categories comparator (1. Acupuncture versus control group (no intervention), 2. acupuncture versus sham acupuncture, 3. acupuncture versus conventional treatment 4. acupuncture and conventional therapy versus conventional therapy, 5. acupuncture and conventional treatment versus sham acupuncture and conventional treatment). There is a moderate number of elements proving that acupuncture is more effective compared to the control group and strong evidence that there is no significant difference between the acupuncture and the sham acupuncture in terms of short-term pain relief. There is strong evidence that acupuncture may be a useful adjunct to other forms of conventional therapy for nonspecific low back pain, but the effectiveness of acupuncture compared with other forms of conventional therapies requires further exploration. A systematic review by Sidney Rubinstein et al. [5] evaluated the effect of chiropractic (SMT), acupuncture and herbal medicine in chronic nonspecific low back pain. Thirty-five randomized controlled trials (8 SMT, 20 with acupuncture, herbology 7) examined 8,298 patients. The major findings are based on low quality data suggest that chiropractic did not provide a most beneficial clinical effect compared to the sham, or any other intervention in the treatment of non-specific lower back pain. There are indications however, that acupuncture provides a short-term clinically significant effect when compared to the control group or when added to another intervention. Although there are some encouraging results for the use of herbs in single studies, the lack of homogeneity among studies did not allow a cumulative assessment of the effect. These results are also in agreement with recent reviews on acupuncture and herbology. It is necessary randomized trials with low risk of bias and adequate sample size. Thirty-five randomized trials (2861 patients) that include acupuncture in adults with nonspecific subacute or chronic low back pain, or dry needling for myofascial pain syndromes in the lumbar, included in a systematic review by Andrea Furlan [6]. Of the thirty-five RCT that included, 20 published in English, seven in Japanese, five Chinese, one Norwegian, one in Polish and one German. There were only three trials of acupuncture for treating low back pain. They did not make valid conclusions, because of small sample size and low methodological quality of the studies. There are signs of pain and

functional improvement regarding chronic low back pain after acupuncture therapy compared with no therapeutic intervention or sham-treatment. These effects were only observed immediately after the end of sessions and short-term monitoring. However, there are clear recommendations about the most effective acupuncture technique. Although, a growing number of clinical studies verify the effect of Chinese acupuncture on headaches, the relationship between the diagnostic criteria of the International Headache Society and the diagnosis of traditional Chinese medicine in primary headache disorders is not clarified. In a prospective, randomized, multicenter, double-blind, controlled trial by Hans-Christoph Diener [7] et al, 960 patients had 2-6 migraine episodes per month randomized to real-acupuncture group (n=313); sham acupuncture (n=339), or conventional treatment group (n=308). Patients received 10 acupuncture treatment sessions in 6 consecutive weeks or prophylactic medicine. The proportion of responders, defined as a decrease of 50% in days with migraine, 26 weeks after randomization. The results showed 47% decrease in the real acupuncture group; 39% in the sham acupuncture group and 40% in the standard therapy group. The results of treatment for migraine did not differ between patients treated with sham acupuncture, real acupuncture or standard treatment. In a Cochrane review by Linde [8] and his associates included randomized trials with a follow-up period of at least eight weeks after randomization, compared the clinical results of acupuncture with a control group; a group of sham acupuncture or other intervention in migraine patients. It was evaluated in 22 studies (4419 participants), investigating whether acupuncture is effective in the prophylaxis of migraine. Six studies evaluated the addition of acupuncture to basic care (which usually includes treating only the acute headache), and found that patients who received acupuncture had fewer headaches. Fourteen studies compared real acupuncture with inadequate or false acupuncture, in which the insertion of the needles were in false places or needles did not penetrate the skin. In these studies, the patients in both groups had fewer headaches, but there was no difference between the two treatments. In four studies acupuncture compared with prophylactic treatment, the patients in the acupuncture group reported a greater improvement and fewer side effects. In summary, the studies show that migraine patients benefit from acupuncture, although the correct placement of the needle appears to be less important than normally considered by acupuncturists. An update of the Cochrane review by Klaus Linde [9] and coworkers included eleven (2317 participants) randomized studies in order to compare the clinical effects of acupuncture on the tension headache, compared to control group (treatment of acute single headache or standard therapy); group of sham acupuncture or other treatment. Two large studies regarding acute headache compared acupuncture with the standard treatment. Both were statistically significant and with short-term (up to 3 months) benefits of acupuncture in relation to the control group in terms of the response; the number of days with headache and pain intensity. Longterm benefits (over three months) are not investigated. Six studies compared acupuncture with sham acupuncture and five of the six provided data for meta-analysis. They found small but statistically significant benefits of acupuncture in relation to the sham needling in patient outcomes. Three of the four studies that compared acupuncture with physiotherapy, massage or relaxation methods had methodological or other reported shortages. The authors conclude that acupuncture may be a useful non-pharmacological tool in treating patients with frequent episodic or chronic tension headache. To evaluate acupuncture in osteoarthritis of the knee was performed randomized, controlled study by Hanns-Peter Scharf [10] and his associates and included 1007 patients who had chronic pain for at least 6 months (criteria of the American College of Rheumatology and score 2 or 3 with KellgrenLawrence). The treatment consisted up to 6 physiotherapy sessions and as needed anti-inflammatory drugs plus 10 sessions traditional Chinese acupuncture, or 10 sessions sham acupuncture or 10 medical visits within 6 weeks. The success rate was defined with at least 36% improvement in score index osteoarthritis of the University Western Ontario and McMaster (WOMAC) at 26 weeks. The success rates were 53.1% for the traditional Chinese acupuncture, 51% for the sham acupuncture and 29.1% for conservative treatment. Groups of acupuncture had higher success rates than conservative therapy groups. No difference between Chinese traditional acupuncture and sham acupuncture. There was no statistically significant difference between the traditional Chinese acupuncture and sham acupuncture, suggesting that the observed differences could be due to effects of placebo, a different doctor-patient relationship in various groups, or a physiological effect of acupuncture, whether made according with the principles of traditional Chinese acupuncture. Eight randomized clinical trials (RCT) of acupuncture were included in the systematic review by Lee [11] and his associates in order to evaluate acupuncture in the treatment of rheumatoid arthritis. Four RCT compared the effects of acupuncture or electro acupuncture with sham acupuncture or no penetration of needles and failed to demonstrate specific effects of acupuncture in pain or in other outcome measures. An RCT comparing acupuncture with indomethacin showed beneficial effects of acupuncture on the overall response rate. In conclusion, the RCT sham-acupuncture group (with or without penetration of the needles) failed to show specific effects of acupuncture compared to control group, in patients with rheumatoid arthritis. In summary, the findings provide no convincing evidence that acupuncture with or without moxa are beneficial in the treatment of rheumatoid arthritis. We assessed the methodological quality of primary studies, using a modified scale Jadad. The duration of the activities were low in most studies (<3 months), in addition to a study. The blindness weakness can lead to overestimation of the treatment effect. This systematic review did not found superiority of real acupuncture compared to sham acupuncture. The sham acupuncture without penetration of the needles was found to be superior to the placebo tablets. The major problem with clinical trials of acupuncture is to find the appropriate placebo. The placebo acupuncture include minimal or superficial introduction of needles or sham acupuncture or without penetrating

needles [12]. However there is no universally acceptable placebo. This data showed no evidence that the presence or absence of De Qi exerts a significant effect on clinical outcomes. The observed effects of sham acupuncture needle penetration may be due to a physiological activity of the insertion of the needle or therapeutic relationship. The Manheimer et al. [13] conducted a meta-analysis of 22 studies of acupuncture for back pain published before September 2004. This analysis found that acupuncture in a short time was statistically and clinically superior to sham needling (4 trials with 343 patients) and without any additional treatment (eight trials with 586 patients). They concluded that the evidence was insufficient to demonstrate the short-term efficacy of acupuncture compared with other treatment. The Ammendolia [14] carried out a systematic review on acupuncture for back pain, including studies published until July 2006. It included 19 studies (4998 patients), most of which included acupuncture by inserting needles. This review compared acupuncture with a control group - no intervention (three studies); with sham acupuncture (7 studies); with other therapies (4 studies) and as adjunctive therapy to other treatments (7 trials). Evaluate the improvement in the pain and the functional status at various points in time: immediately after treatment (<1 week), short term (up to 3 months), in the interval (3-12 months), and long term (>1 time). Immediately after treatment, three studies found that acupuncture is more effective compared to the control group in improving pain and functional situation. Acupuncture was rarely greater than sham acupuncture for pain relief (eg in one of the six studies immediately after treatment, in 2 of the 3 studies on short-term results). The functional status between the acupuncture and the sham acupuncture was similar in short-term observation. Acupuncture has been evaluated as an adjunct therapy in 7 studies. The primary treatment was exercise (two studies), standard treatment (3 studies), physical therapy (one study) and orthopedic treatment (one study). Briefly in all studies and in all time monitoring periods, the complementary application of acupuncture led to better results. In a systematic review with acupuncture and electroacupuncture on the common peripheral osteoarthritis, the Kwon [15] found that in 10 of 18 trials, acupuncture showed a greater reduction of pain compared with various other control groups. The studies showed a wide variation in terms of the number of sessions, ranging from 5 to 45 total sessions, within 2-26 weeks and 1-5 sessions per week. They found that acupuncture is more effective compared with no intervention (2 studies), or relative to the sham control group (3 out of 4 trials). The White [16] conducted a systematic review of acupuncture for chronic knee pain. They included 13 studies, of which 8 were considered sufficient and thus were included in a meta-analysis. They found that acupuncture is superior compared to the sham control group (five trials with 1334 patients) and compared with the control group without any additional treatment (4 trials with 927 patients), in improving the functional status and reduction of pain in the short and long term period. The Manheimer [17] published the most recent systematic review of acupuncture for osteoarthritis, including studies published prior to February 2007. They included 11 studies, of which 9 used for a metaanalysis. They found that acupuncture has relatively short clinical improvement in pain and the functional status compared to the standard treatment. Compared to the sham control group, acupuncture associated with clinically irrelevant (but statistically significant) and long-term improvement in pain and functional situation. They interpreted the difference of the results regarding the beneficial effect of acupuncture, as a consequential action of placebo or the expected expectations. Following the publication of Manheimer, two large studies published by Foster [18] and Williamson [19]. The study used counseling and exercise, also applied acupuncture, up to six sessions over three weeks or the same number of sham sessions acupuncture, without penetrating needles. There seemed extra pain reduction after treatment and at 6 or 12 months [18]. In patients who were standby for knee replacement, the application of acupuncture and physiotherapy for six weeks resulted in a brief decrease of osteoarthritis. However the beneficial effect of treatment was not maintained in monitoring (follow up) at 12 weeks [19]. According to Lizhou Liou and coll. (seven systemic reviews) acupuncture in treatment of chronic low back pain provides shortterm clinically relevant benefits for pain relief and functional improvement compared with no treatment or acupuncture plus another conventional intervention [20].

The search by Li ZQ, Chen H et al, encompassed 7,190 relevant studies, including 1,263 randomized controlled trials (RCTs) and 1,293 systematic reviews and meta-analyses. The results indicated a gradual increase in the number of annual publications on acupuncture analgesia in clinical practice. Among countries and institutions, China (2,139) and Chengdu University of Traditional Chinese Medicine (258) ranked first. Liang FR (89 articles) was the most prolific author, while MacPherson H (604) was the most cited author. MEDICINE (455) was the most productive journal, and Pain (2,473/0.20) ranked first in both the frequency and centrality of cited journals. Notably, the most frequently cited reference was a systematic review of individual patient data on acupuncture carried out for chronic pain that was published by Vickers Andrew J in 2012 (156). Burst analysis identified frontier research areas for 2010-2020, encompassing network meta-analysis, case reports, dry needling, lumbar disc herniation, cancer, post-herpetic neuralgia, insomnia, and bibliometric analysis. These studies outlines current trends and potential future research hotspots in clinical acupuncture analgesia over the past decade. Findings emphasize the necessity for enhanced international collaboration to improve research output and translation [21-22].

### 3. Conclusion

In summary, studies show promising results for acupuncture as an alternative treatment for low back pain and migraine. Acupuncture appears to be superior compared with "no action" or usual care in patients with chronic low back pain, osteoarthritis, or headache. Acupuncture has a beneficial safety profile, with relatively few side effects. Data suggest that acupuncture is a cost-effective treatment. The results support the use of acupuncture as an alternative or adjunctive treatment with or without analgesic drugs in the context of a multifactorial treatment.

### References

- [1] MV Madsen, PC Gøtzsche, A Hróbjartsson (2009) Acupuncture treatment for pain: systematic review of randomised clinical trials with acupuncture, placebo acupuncture, and no acupuncture groups. *BMJ* 2009;338:a3115doi:10.1136/bmj.a3115.
- [2] Haake M, Müller HH, Schade-Brittinger C, Basler HD, Schäfer H, et al. (2007) German Acupuncture Trials (GERAC) for chronic low back pain: randomized, multicenter, blinded, parallel-group trial with 3 groups. *Arch Intern Med* 167: 1892-1898.
- [3] Yuan J, Purepong N, Kerr DP, Park J, Bradbury I, et al. (2008) Effectiveness of acupuncture for low back pain: a systematic review. *Spine (Phila Pa 1976)* 33: E887-900.
- [4] Böwing G, Zhou J, Endres HG, Coeytaux RR, Diener HC, et al. (2010) Differences in Chinese diagnoses for migraine and tension-type headache: an analysis of the German acupuncture trials (GERAC) for headache. *Cephalalgia* 30: 224-232.
- [5] Rubinstein SM, van Middelkoop M, Kuijpers T, Ostelo R, Verhagen AP, et al. (2010) A systematic review on the effectiveness of complementary and alternative medicine for chronic non-specific low-back pain. *Eur Spine J* 19: 1213-1228.
- [6] Furlan AD, van Tulder M, Cherkin D, Tsukayama H, Lao L, et al. (2005) Acupuncture and dry-needling for low back pain: an updated systematic review within the framework of the cochrane collaboration. *Spine (Phila Pa 1976)* 30:944-963.
- [7] Diener HC, Kronfeld K, Boewing G, Lungenhausen M, Maier C, et al. (2006) Efficacy of acupuncture for the prophylaxis of migraine: a multicentre randomised controlled clinical trial. *Lancet Neurol* 5: 310-316.
- [8] Linde K, Allais G, Brinkhaus B, Manheimer E, Vickers A, et al. (2009) Acupuncture for migraine prophylaxis. *Cochrane Database Syst Rev* 1: CD001218.
- [9] Linde K, Allais G, Brinkhaus B, Manheimer E, Vickers A, et al. (2009) Acupuncture for tension-type headache. *Cochrane Database Syst Rev* 1: CD007587.
- [10] Scharf HP, Mansmann U, Streitberger K, Witte S, Krämer J, et al. (2006) Acupuncture and knee osteoarthritis: a three-armed randomized trial. *Ann Intern Med* 145: 12-20.
- [11] Lee MS, Shin BC, Ernst E (2008) Acupuncture for rheumatoid arthritis: a systematic review. *Rheumatology (Oxford)* 47: 1747-1753.
- [12] Sherman KJ, Coeytaux RR (2009) Acupuncture for Improving Chronic Back Pain, Osteoarthritis and Headache. *J Clin Outcomes Manag* 16: 224-230.
- [13] Manheimer E, White A, Berman B, Forys K, Ernst E (2005) Meta-analysis: acupuncture for low back pain. *Ann Intern Med* 142: 651-663.
- [14] Ammendolia C, Furlan AD, Imamura M, Irvin E, van Tulder M (2008) Evidence informed management of chronic low back pain with needle acupuncture. *Spine J* 8: 160-172.
- [15] Kwon YD, Pittler MH, Ernst E (2006) Acupuncture for peripheral joint osteoarthritis: a systematic review and meta-analysis. *Rheumatology (Oxford)* 45: 1331-1337.
- [16] White A, Foster NE, Cummings M, Barlas P (2007) Acupuncture treatment for chronic knee pain: a systematic review. *Rheumatology (Oxford)* 46: 384-390.
- [17] Manheimer E, Linde K, Lao L, Bouter LM, Berman BM (2007) Meta-analysis: acupuncture for osteoarthritis of the knee. *Ann Intern Med* 146: 868-877.

- [18] Foster NE, Thomas E, Barlas P, Hill JC, Young J, et al. (2007) Acupuncture as an adjunct to exercise based physiotherapy for osteoarthritis of the knee: randomised controlled trial. *BMJ* 335: 436.
- [19] Williamson L, Wyatt MR, Yein K, Melton JT (2007) Severe knee osteoarthritis: a randomized controlled trial of acupuncture, physiotherapy (supervised exercise) and standard management for patients awaiting knee replacement. *Rheumatology (Oxford)* 2007 46(9): 1445-1449.
- [20] Liu L, Skinner M, McDonough S, Mabire L, Baxter GD (2015) Acupuncture for Low Back Pain: An Overview of Systematic Reviews. *Evidence-Based Complementary and Alternative Medicine* 2015.
- [21] Global trends of acupuncture clinical research on analgesia from 2010 to 2023: a bibliometric and visualization analysis. Li ZQ, et al. *Front Neurol*. 2024. PMID: 38665996
- [22] Durable Effects of Acupuncture for Knee Osteoarthritis: A Systematic Review and Meta-analysis. Chen H, et al. *Curr Pain Headache Rep*. 2024. PMID: 38635021 Review.



# Epilepsy in children and anesthesia

Maria I Dalamagka

## ► To cite this version:

Maria I Dalamagka. Epilepsy in children and anesthesia. Magna Scientia Advanced Research and Reviews, 2024, 12, pp.033 - 038. 10.30574/msarr.2024.12.2.0180 . hal-05432454

**HAL Id: hal-05432454**

**<https://hal.science/view/index/docid/5432454>**

Submitted on 26 Dec 2025

**HAL** is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



Distributed under a Creative Commons Attribution 4.0 International License

## Epilepsy in children and anesthesia

Maria I Dalamagka \*

*Anesthesiology Department, General Hospital of Larisa, Greece.*

Magna Scientia Advanced Research and Reviews, 2024, 12(02), 033–038

Publication history: Received on 15 September 2024; revised on 04 November 2024; accepted on 06 November 2024

Article DOI: <https://doi.org/10.30574/msarr.2024.12.2.0180>

### Abstract

Epilepsy, also called seizure disorder, is the most common childhood brain disorder in the United States. The aetiology of epilepsy in children is multifactorial with congenital, metabolic, infective, and problems associated with prematurity being common causes. Nearly 3 million Americans have epilepsy. About 450,000 of them are under 17 years old. About 1 in 200 children (0.5%) have epilepsy, a neurological condition where children have a predisposition to recurrent, unprovoked seizures. There are many different types of epilepsy, especially in infancy, childhood and adolescence. Epilepsy can be thought of in terms of either: the site of seizure origin in the brain (generalised or focal seizures), or the underlying cause. Genetic epilepsies (formerly called idiopathic or primary epilepsies) occur in an otherwise normal person and are due to a genetic predisposition to seizures. Some epilepsies are due to an underlying abnormality of the brain structure or chemistry (formerly called symptomatic or secondary epilepsies). Other epilepsies have no known cause. Epilepsy is commonly diagnosed in children and can be confused with other conditions. An accurate diagnosis is essential. A seizure is an excessive surge of electrical activity in the brain that can cause a variety of symptoms, depending on which parts of the brain are involved. Seizures can be provoked or unprovoked. Provoked seizures, caused by fever in a young child or severe hypoglycemia, are not considered to be forms of epilepsy. Unprovoked seizures have no clear cause but can be related to genetics or brain injury. When a child has two or more unprovoked seizures, epilepsy is often the diagnosis. Despite advances in antiepileptic medication therapy, a significant number of pediatric patients with epilepsy have seizures that are not well controlled. Antiepileptic medications interact with anesthetic agents, and common anesthetics can precipitate or suppress seizure activity. There are important pharmacokinetic and pharmacodynamic interactions between AEDs and drugs commonly used in anaesthesia. These affect both drug efficacy and the risk of seizure activity intraoperatively.

**Keywords:** Epilepsy; Seizures; Anesthesia; Children; Status epilepticus

### 1. Introduction

Seizures occur when there is a momentary 'imbalance' within electrical and chemical circuits in the brain, such that groups of brain cells act in an excessive fashion. This may create a temporary disturbance in the way the brain controls awareness and responsiveness and may cause unusual sensations or abnormal movements and postures. The International League against Epilepsy (ILAE) has classified seizures into focal (or partial) seizures which arise from one hemisphere and generalized seizures which show electrographic seizure onset over both hemispheres. Lamotrigine and carbamazepine are considered drugs of choice in focal epilepsies, while valproate is probably the most effective drug for primary generalized seizures. If the initial antiepileptic drug (AED) results in adverse effects, an alternative AED is tried as monotherapy. If, on the other hand, seizures continue in spite of adequate doses, combination therapy is often necessary. Seizures may be primary (idiopathic) or more commonly secondary to other conditions. Primary epilepsy has a genetic predisposition with a 1.5–3% risk of paternal inheritance and a 3–9% risk of maternal inheritance. Secondary seizures may be caused by prenatal, perinatal, or post-natal events. Absence epilepsy causes short moments. Rolandic epilepsy (self-limited epilepsy with centrotemporal spikes) affecting an estimated 15% of children diagnosed

\* Corresponding author: Maria I Dalamagka

with epilepsy. Juvenile myoclonic epilepsy: common during puberty, myoclonic epilepsy seizures cause uncontrolled muscle movements. Infantile spasms (West syndrome) are a severe type of epilepsy that affects infants. Lennox-Gastaut syndrome (LGS) can cause seizures. Risk factors for epilepsy in children are: history of epilepsy in their biological family history (genetic predisposition); traumatic brain injury; brain infection; other neurological issues. Epileptic seizures (“fits”) can present in very different ways: Some only last a few seconds and are hardly noticeable, while others cause severe convulsions (jerking and shaking), sometimes in all of the body. In children, the symptoms are often not diagnosed properly at first. Improvements in epilepsy treatment in recent years have made the condition more manageable. Many new anti-seizure medications are available and more are being tested. In addition to newer medications, alternative treatments such as surgical procedures, medical devices, and dietary therapies, are also available for children and teens who continue to have seizures while on medication. A number of studies have suggested that a ketogenic diet can reduce the number of seizures in children who don’t respond well to medication [1-7].

## 2. Mechanism of epilepsy

The mechanism of epilepsy is related to (i) loss of post-synaptic inhibition with loss of inhibitory  $\gamma$ -aminobutyric acid (GABA) activity, (ii) new excitatory synaptic connections with increased release of the excitatory amino acid glutamate, and (iii) appearance of pacemaker neurones with abnormal voltage-mediated calcium currents, all leading to abnormal firing of the neurones. Seizures may be partial (focal), or generalized, or they may start as partial and progress to generalized. Partial seizures can be simple, not associated with impaired conscious level or complex (associated with impaired consciousness. Generalized seizures are associated with bilateral symmetrical electric brain activity. They present with transient impaired consciousness (absence or ‘petit mal’ seizures) or generalized muscle contractions (tonic, clonic, or tonic-clonic ‘grand mal’ seizures), which are usually associated with impaired consciousness. Several forms of epilepsy: petit mal epilepsy, infantile spasms, myoclonic epilepsy, benign partial epilepsy, and benign childhood epilepsy which carries an excellent prognosis [8-9].

### 2.1. Types of seizures

Some are very short, lasting only a few seconds, while others can last a few minutes. Some can cause uncontrollable jerking movements, while others cause them to be confused or stare blankly. Seizures are categorized into two main types: focal, which are also called partial seizures, and generalized. Another, rarer, type is progressive myoclonic epilepsy. Infantile-onset epilepsy syndromes and childhood-onset epilepsy syndromes and seizure disorders are both categorized by the age at which symptoms began, among other factors. Finally, genetic and neurologic disorders can also lead to seizures in children. Focal seizures begin with an abnormal electrical discharge in one region of the brain. They are further categorized by their effect on a child’s consciousness, responsiveness, and memory. Temporal lobe seizures are the most common type of epilepsy in both children and adults. The temporal lobe is located beneath the temples, on either side of the head. It is responsible for memory, emotions, interpreting sounds, and understanding language. Frontal lobe epilepsy is the second most common form of epilepsy. The frontal lobe is located beneath the forehead and is the part of the brain responsible for decision-making, problem-solving, and emotions. The occipital lobe is located at the back of the brain, behind the parietal and temporal lobes. This is the site of the brain’s visual system. Parietal lobe seizures are also relatively rare in children. The parietal lobe, located near the center of the brain, is responsible for processing information about the senses of touch, pain, and space. Symptoms of gelastic epilepsy may include brief, repeated seizures characterized by uncontrollable laughter that has no known cause. Generalized seizures begin with a widespread, excessive electrical discharge that involves both hemispheres, or sides, of the brain at the same time. Symptoms include blinking and staring, loss of muscle tone, stiffening of limbs, and, when the entire brain is involved, rhythmic, full-body jerking. Generalized epilepsies can be divided into two categories: idiopathic generalized epilepsy, in which the child is otherwise behaviorally and neurologically normal between seizures, and developmental and epileptic encephalopathies, in which intellectual and developmental problems occur between seizures. Progressive myoclonic epilepsy is rare and frequently results from hereditary metabolic disorders or neurodegenerative conditions, such as neuronal ceroid lipofuscinosis, Lafora body disease, and mitochondrial encephalopathy. When a disorder has a constellation of features that tend to occur together, it is termed a syndrome. Some children with seizure disorders have certain characteristics in common and have an epilepsy syndrome—defined by the age at which seizures start, seizure types, presence or absence of developmental delay, and findings on the electroencephalogram (EEG). Benign familial neonatal seizures cause recurrent seizures in newborns. The seizures usually begin when the infant is about 3 days old and are brief, lasting 1 to 2 minutes. Ohtahara syndrome is a rare type of epilepsy that develops in newborns, often within the first two weeks of life. The seizures are primarily tonic seizures but may also include partial seizures and myoclonic seizures. Ohtahara syndrome is often caused by metabolic disorders or brain damage, although in many babies the cause cannot be determined. Infantile spasms, also known as West syndrome, are rare. This is a severe type of epilepsy syndrome that begins in children who are 3 to 12 months old. Dravet syndrome is a severe type of epilepsy syndrome often resulting from a gene mutation that causes abnormalities in sodium channels in the brain, which play

a role in nerve cell communication. Seizures generally begin before a child is 1 year old and can be difficult to control. These types of seizures often affect a baby's cognitive development. Certain other types of seizures and epilepsy syndromes more commonly start in childhood. They include febrile seizures, Landau-Kleffner syndrome, Lennox-Gastaut syndrome, Rasmussen syndrome, benign Rolandic epilepsy, benign occipital epilepsy, childhood absence epilepsy, and juvenile myoclonic epilepsy. Febrile seizure occurs when a child between 6 months and 6 years old has a tonic-clonic seizure plus a high fever, which may occur as a result of a viral illness. Landau-Kleffner syndrome, also called acquired epileptic aphasia, is a rare disorder in which a child loses their ability to speak and understand others' speech. Lennox-Gastaut syndrome is an uncommon form of epilepsy that causes difficult-to-control seizures, including tonic, atonic, prolonged absences, and generalized convulsions. Almost all children who have Lennox-Gastaut syndrome have cognitive and developmental delays. Rasmussen syndrome is rare and usually begins in children who are 14 months to 14 years old. The condition is associated with progressive neurologic deterioration and seizures. Benign Rolandic epilepsy, also called benign epilepsy of childhood with centrotemporal spikes, is one of the most common childhood seizure disorders. Seizures typically begin when children are 2 to 13 years old. There are two subtypes of benign occipital epilepsy—Panayiotopoulos syndrome and Gastaut-type syndrome. The type depends on your child's age when symptoms begin. Panayiotopoulos syndrome begins when a child is 3 to 5 years old, while Gastaut-type syndrome can start at any age throughout childhood but tends to peak around the age of 8 or 9. Absence seizures are generalized seizures that usually occur in children who are 5 to 9 years old. Many disorders that affect the structure and function of the brain in early life can lead to epilepsy, causing seizures in children. Some genetic and neurologic disorders may lead to seizures, along with other neurologic, development, and behavioral symptoms. Rett syndrome is a neurodevelopmental disorder that causes seizures. This genetic condition typically affects girls and begins when a child is 6 to 18 months old. Angelman syndrome is a type of epilepsy syndrome that causes learning difficulties, speech delays, and certain behavioral characteristics, such as a cheerful mood and bursts of sudden and unexplained laughter. The condition is a genetic disorder that usually develops when babies are 6 to 12 months old. Seizures occur in children with Sturge-Weber syndrome, a congenital neurological disorder, due to abnormalities in the blood vessels lining the brain. Children with Sturge-Weber syndrome often have a port wine stain birthmark on the forehead and upper eyelid of one side of the face. There is a greater likelihood of intellectual impairment when seizures start before a child is 2 years old and are resistant to treatment. Symptoms of FOXG1 syndrome typically develop in the second month of life and include irritability and seizures. Delayed development, difficulty walking and sitting, intellectual disability, and problems with speech and vision are also symptoms of the disorder. Most children with this syndrome have microcephaly, or a smaller head size. Children who have Dup15q syndrome, a developmental disorder, have weak muscle tone, delays in sitting and walking, and problems with speech, language, and social interaction. Seizures, which may be difficult to control, typically develop between the ages of 6 months and 9 years. Children who have KBG syndrome are short for their age and may have underdeveloped bones of the spine, arms, leg, head, and face. They may also experience developmental and intellectual delays. Symptoms may include seizures, starting as early as infancy or as late as the teen years. Seizures may respond to medication. Some children with KBG syndrome outgrow seizures [10–15].

## 2.2. Anesthesia

Seizures can occur under anaesthesia and are managed by immediate deepening of anaesthesia, administration of an anticonvulsant agent, and subsequent correction of any reversible precipitating factor. Seizures are episodes of abnormal synchronized electrical brain activity. It is estimated that 2% of all people will experience a seizure during their lifetime. Epilepsy is the most common serious neurological disorder, with a prevalence of 0.5–1% of the population. While the traditional antiepileptic drugs (AEDs) still play a significant role in treatment of seizures, there has been an influx of newer agents over the last 20 yr, which are now in common usage. Patients with epilepsy often require anaesthesia for elective and emergency surgery. Appropriate perioperative management of AED therapy is vital in maintaining seizure control in these patients. Anaesthesia is usually maintained with a volatile agent, with or without nitrous oxide and an opioid. Less frequently, total i.v. anaesthesia (TIVA) is used. Phenytoin, phenobarbital, and carbamazepine cause hepatic enzyme induction. This results in increased metabolism of halogenated anaesthetics, which increases the risk of halothane hepatitis and urinary fluoride excretion. Enzyme induction may cause increased opioid requirements, whereas gabapentin has a morphine-sparing effect. Anaesthetic agents with epileptogenic potential, e.g. ketamine and alfentanil, and those with epileptogenic metabolites, e.g. meperidine, should be avoided. Seizures under anaesthesia, should be managed by immediate deepening of anaesthesia, administration of an anticonvulsant agent such as propofol, thiopental, or a benzodiazepine, administration of 100% oxygen, and correction of any reversible precipitating factors, including hypoxia, hypercarbia, hypocapnoea, hyponatraemia, and hypoglycaemia. Normal saline solution should be used in these patients, with close monitoring of the acid–base status. Plasma levels of AEDs must be checked if there is any delay in restarting them or if perioperative seizures occur. Seizures are more common in the postoperative period. They may be precipitated by the use of proconvulsant anaesthetic agents, hypoxia, hypercapnoea, electrolyte disturbances (hyponatraemia, hypocalcaemia and hypomagnesaemia),

hypoglycaemia, uraemia, subtherapeutic levels of AEDs, or local anaesthetic toxicity. The occurrence of epileptic seizures after an intracranial neurosurgical operation is a well-known phenomenon, occurring in 5% to 20% of patients undergoing brain surgery. Several risk factors have been propounded to explain such an event the type of lesion, low grade gliomas, meningioma, abscess, aneurysm, or chronic subdural hemorrhage; its location; an age less than 2 years; some comorbidities, e.g., postoperative electrolytic imbalance in children, presence of cognitive impairment in adults; cortical damage related to the surgical approach. Many anaesthetic agents affect the propensity to seizures, both in patients with epilepsy and in those with no prior history of seizures. In patients taking AEDs, drug interactions and maintenance dosing of AEDs during periods of starvation are important considerations in the perioperative period [16–22].

### 2.3. Status epilepticus

Status epilepticus is defined as seizure lasting more than 30 min, or two or more seizures without complete recovery of consciousness between them. Status epilepticus is a neurological emergency associated with 25% overall mortality; 8% in children, 30% in adults, and 40–50% in elderly. Status epilepticus is a neurological emergency associated with significant mortality in children. In the early phase, the increased metabolic demands of the brain are met by an increase in arterial pressure, increased cerebral blood flow, increased minute ventilation, and increased blood glucose level. After 30 min, cerebral auto-regulation fails, with decreased cerebral blood flow, increased intracranial pressure, and a decrease in arterial pressure. Treatment guidelines are as follow: Supportive (ABC); Ensure patent airway; Ensure adequate ventilation and oxygenation; Secure large-bore i.v. access to facilitate blood tests (full blood count, clotting, glucose, AED levels, renal and liver function tests, calcium, and magnesium) and administration of i.v. fluids; Perform arterial blood gas analyses; Use i.v. vasopressors if required; Specific: consider aetiology and exclude alternative cause of seizures; First line: benzodiazepines; i.v. lorazepam 0.1 mg kg<sup>-1</sup> over 30–60 s. This can be repeated once if seizures continue for 10 min. If i.v. access is not established; rectal diazepam 0.5 mg kg<sup>-1</sup>. If seizures continue for 10 min, lorazepam is used if i.v. access established, otherwise rectal paraldehyde 0.4 mg kg<sup>-1</sup> in an equal volume of olive oil; Second line: phenytoin 15–20 mg kg<sup>-1</sup> i.v. over 20 min with ECG and arterial pressure monitoring. If i.v. access is not yet established, use an intra-osseous needle. If the patient already on phenytoin, give phenobarbital 20 mg kg<sup>-1</sup> over 20 min and send blood for phenytoin level. Use rectal paraldehyde if not already given; Third line (refractory seizures): induction of general anaesthesia (GA) with thiopental 4–5 mg kg<sup>-1</sup>, followed by 3–5 mg kg<sup>-1</sup> h<sup>-1</sup> infusion or midazolam 0.1 mg kg<sup>-1</sup>, followed by 2–20 µg kg<sup>-1</sup> min<sup>-1</sup> infusion and transfer to an intensive care unit.

### 2.4. Local anaesthetic toxicity

Local anaesthetic toxicity can present with tonic–clonic seizures, and if this is suspected in a child, it should be managed as follows: Stop injecting local anaesthetic and call for help; ABC: maintain the airway and, if necessary, secure with a tracheal tube. Give 100% oxygen and ensure adequate ventilation. Confirm or establish i.v. access; Control seizures: give benzodiazepine, thiopental, or propofol in small incremental doses; Management of cardiac arrest associated with LA injection: Start cardiopulmonary resuscitation bearing in mind that prolonged resuscitation may be necessary and cardiac arrhythmias may be refractory; Consider treatment with lipid emulsion: Give an i.v. bolus injection of Intralipid® 20% 1.5 ml kg<sup>-1</sup> over 1 min followed by an infusion of 0.25 ml kg<sup>-1</sup> min<sup>-1</sup>. Repeat the bolus injection twice at 5 min intervals if adequate circulation has not been restored. If adequate circulation still not restored after another 5 min, increase rate to 0.5 ml kg<sup>-1</sup> min<sup>-1</sup>. Continue infusion until a stable and adequate circulation restored; Consider the use of cardiopulmonary bypass if available.

## 3. Discussion

Traditional AEDs exert antiseizure activity by the following mechanisms: reduce the inward voltage-gated positive currents (Na<sup>+</sup>, Ca<sup>2+</sup>); increase inhibitory neurotransmitter activity (GABA); decrease excitatory neurotransmitter activity (glutamate, aspartate). Pharmacological treatment with antiepileptic drugs (AEDs) depends on the type and frequency of seizures, age of the patient, and drug side-effects. Therapy is usually started with one anticonvulsant drug at a dose likely to result in therapeutic plasma level, after which the dose can be optimized as required. AEDs should be continued up to the time of the surgery and should be restarted as soon as possible after operation. Traditional anticonvulsant drugs including phenobarbital, phenytoin, carbamazepine, and sodium valproate are potent hepatic enzyme inducers, whereas felbamate, not available in the UK, is a hepatic enzyme inhibitor. Sodium valproate, carbamazepine, and ethosuximide can be associated with hepatotoxic effects. The latter three drugs, and primidone, are associated with thrombocytopenia and platelet abnormalities. Topiramate is associated with asymptomatic high anion gap metabolic acidosis. In a child with epileptic seizures, a recording of brainwave activity (EEG) and a picture of the brain (MRI) may be obtained, where necessary. Common generalized onset seizures: Generalized tonic-clonic seizures (previously called "grand mal seizures") involve abnormal electrical activity in the whole brain. They are the most dramatic type of seizure; they cause rhythmic and sometimes violent jerking movements in both sides of the body with

loss of consciousness. These seizures usually last for 2 to 3 minutes and will almost always end on their own. Absence seizures (previously called "petit mal seizures") are very short episodes with a vacant stare or a brief (few seconds) lapse of attention. They may also include other subtle symptoms like eyelid fluttering, rapid eye blinking or lip smacking. Focal onset seizures. Focal onset seizures, previously called "complex partial seizures", involve abnormal electrical activity in one part of the brain. During these seizures, a person may be aware of what is going on (called a "focal aware seizure"), or they may not be aware of what is happening, called a "focal seizure with impaired awareness". Epilepsy surgery encompasses a wide variety of treatments performed by neurosurgeons to eliminate the source of seizures, prevent the spread of seizures, stimulate the brain to stop seizures, or otherwise change seizure circuits. The goal of surgery is to eliminate seizures or reduce the number and severity of seizures.

---

#### 4. Conclusion

Seizure occurs when one or more parts of the brain has a burst of abnormal electrical signals that interrupt normal signals. There are many types of seizures. Each can cause different kinds of symptoms. These range from slight body movements to loss of consciousness and convulsions. The most common treatment of recurrent seizures is pharmacologic with antiepileptic medication (AED). Approximately 70% of patients respond to AED management. Those who continue to have seizures are considered to have medically refractory epilepsy, and are at risk for adverse effects on quality of life, brain development, learning, language, and injury. For children with uncontrolled epilepsy, that is epilepsy in which seizures are not adequately controlled by medication, other treatments are available, including: rectal diazepam or intranasal/ buccal midazolam, for treatment of prolonged seizures and seizure clusters; new antiepileptic medications, available in clinical drug trials or through special access schemes; epilepsy surgery; ketogenic diet; vagus nerve stimulation; alternative therapies

---

#### Compliance with ethical standards

##### *Disclosure of conflict of interest*

No conflict of interest to be disclosed.

---

#### References

- [1] Stefen H Halasz P Gil-Nagel Aet al. Recent advances in the diagnosis and treatment of epilepsy, Eur J Neurol, 2001, vol. 8 (pg. 519-39).
- [2] Brorson SO Wranne L Long-term prognosis in childhood epilepsy: survival and seizure prognosis, Epilepsia, 1987, vol. 28 (pg. 324-30).
- [3] Vining EP Long-term health consequences of epilepsy diet treatments, Epilepsia, 2008, vol. 49 Suppl. 8 (pg. 27-9).
- [4] Stokes T Shaw EJ Juarez-Garcia A Camosso-Stefinovic J Baker R, Clinical Guidelines and Evidence Review for the Epilepsies: Diagnosis and Management in Adults and Children in Primary and Secondary Care, 2004 London Royal College of General Practitioners (NICE guidelines).
- [5] Shenkman Z Rockoff MA Eldredge EA Korf BR Black PM Soriano SG Anaesthetic management of children with tuberous sclerosis, Paediatr Anaesth, 2002, vol. 12 (pg. 700-4).
- [6] Cuignet O Pirsona J Soudon O Zizi ME Effects of gabapentin on morphine consumption and pain in severely burned patients, Burns, 2007, vol. 33 (pg. 81-6).
- [7] Association of Anaesthetists of Great Britain and Ireland (AAGBI) Management of severe local anaesthetic toxicity guideline [www.aagbi.org/publications/guidelines/docs/latoxicity07.pdf](http://www.aagbi.org/publications/guidelines/docs/latoxicity07.pdf).
- [8] Engel J. The current place of epilepsy surgery. Curr Opin Neurol. 2018;31(2):192–7.
- [9] Wiebe S, Jette N. Pharmacoresistance and the role of surgery in difficult to treat epilepsy. Nat Rev Neurol. 2012;8(12):669–77.
- [10] Jayalakshmi S, Vooturi S, Gupta S, Panigrahi M. Epilepsy surgery in children. Neurol India. 2017;65(3):9.
- [11] Soriano SG, Bozza P. Anesthesia for epilepsy surgery in children. Childs Nerv Syst. 2006;22(8):834–43.

- [12] Valencia I, Pfeifer H, Thiele EA. General anesthesia and the ketogenic diet: clinical experience in nine patients. *Epilepsia*. 2002;43(5):525–9.
- [13] Foy P.M., Copeland G.P., Shaw M.D. The Incidence of Postoperative Seizures. *Acta Neurochir*. 1981;55:253–264. doi: 10.1007/BF01808441.
- [14] Grobelny B.T., Ducruet A.F., Zacharia B.E., Hickman Z.L., Andersen K.N., Sussman E., Carpenter A., Connolly E.S. Preoperative Antiepileptic Drug Administration and the Incidence of Postoperative Seizures Following Bur Hole-Treated Chronic Subdural Hematoma. *J. Neurosurg*. 2009;111:1257–1262. doi: 10.3171/2009.6.JNS0928.
- [15] Kern K., Schebesch K.M., Schlaier J., Hansen E., Feigl G.C., Brawanski A.T., Lange M. Levetiracetam Compared to Phenytoin for the Prevention of Postoperative Seizures after Craniotomy for Intracranial Tumours in Patients without Epilepsy. *J. Clin. Neurosci*. 2012;19:99–100. doi: 10.1016/j.jocn.2011.07.021.
- [16] Kombogiorgas D., Jatavallabhula N.S., Sgouros S., Josan V., Walsh A.R., Hockley A.D. Risk Factors for Developing Epilepsy after Craniotomy in Children. *Child's Nerv. Syst*. 2006;22:1441–1445. doi: 10.1007/s00381-006-0117-4.
- [17] Sen R.D., Nistal D., McGrath M., Barros G., Shenoy V.S., Sekhar L.N., Levitt M.R., Kim L.J. De Novo Epilepsy after Microsurgical Resection of Brain Arteriovenous Malformations. *Neurosurg. Focus*. 2022;53:E6. doi: 10.3171/2022.4.FOCUS2288.
- [18] Sughrue M.E., Rutkowski M.J., Chang E.F., Shangari G., Kane A.J., McDermott M.W., Berger M.S., Parsa A.T. Postoperative Seizures Following the Resection of Convexity Meningiomas: Are Prophylactic Anticonvulsants Indicated? Clinical Article. *J. Neurosurg*. 2011;114:705–709. doi: 10.3171/2010.5.JNS091972.
- [19] Zhang B., Zhao G., Yang H.-F., Wang D., Yu J.-L., Huang H.-Y. Assessment of Risk Factors for Early Seizures Following Surgery for Meningiomas Using Logistic Regression Analysis. *J. Int. Med. Res*. 2011;39:1728–1735. doi: 10.1177/147323001103900515.
- [20] Ben-Ari et al. Effects of seizures on developmental processes in the immature brain *Lancet Neurol* (2006).
- [21] S. Noachtar et al. Epilepsy surgery; a critical review *Epilepsy Behav* (2009).
- [22] E. Ornstein et al. Accelerated recovery from doxacurium-induced neuromuscular blockade in patients receiving chronic anticonvulsant therapy *J Clin Anesth* (1991).



# Etomidate as anesthetic agent in asthmatic child with food allergy

Maria I Dalamagka

## ► To cite this version:

Maria I Dalamagka. Etomidate as anesthetic agent in asthmatic child with food allergy. GSC Advanced Research and Reviews , 2024, 19, pp.001 - 003. 10.30574/gscarr.2024.19.1.0125 . hal-05432449

**HAL Id: hal-05432449**

**<https://hal.science/view/index/docid/5432449>**

Submitted on 26 Dec 2025

**HAL** is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



Distributed under a Creative Commons Attribution 4.0 International License

## Etomidate as anesthetic agent in asthmatic child with food allergy

Maria I. Dalamagka \*

*Department of Anaesthesia, General Hospital of Larisa, Greece.*

GSC Advanced Research and Reviews, 2024, 19(01), 001–003

Publication history: Received on 23 February 2024; revised on 30 March 2024; accepted on 02 April 2024

Article DOI: <https://doi.org/10.30574/gscarr.2024.19.1.0125>

### Abstract

Children with bronchial asthma and respiratory infections have significant peri-operative implications for the anaesthesiologists. With improvements in medical knowledge and anaesthesia techniques, morbidity and mortality associated with an asthmatic child undergoing surgery have come down. Etomidate is an intravenous anesthetic agent whose clinical effects are the result of potentiation of the gamma-amino butyric acid inhibitory neurotransmitter system with the alteration of transmembrane chloride conductance. Physicians commonly use etomidate for adult rapid-sequence intubation, but the manufacturer does not recommend its use for children under 10 years of age due to a lack of data. Ketamine has been suggested as an alternative agent given its limited effects on hemodynamic function related to the release of endogenous catecholamines. A child aged 11 years and weighing 40 kg, with egg allergy and a more general allergic predisposition, with a history of asthma treated with inhalants and a recent respiratory infection with antibiotics, came with a diagnosis of acute abdomen. Etomidate was used as an anesthetic agent in combination with ketamine, using rocuronium and sevoflurane. Corticosteroids and aminophylline were given to prevent bronchospasm. Ketamine and etomidate, both of which provide effective sedation with limited effects on hemodynamic function, have become increasingly popular as induction agents.

**Keywords:** Etomidate; Ketamine; Allergic; Astma; Child; Intubation

### 1. Introduction

Asthma represents one of the most common chronic diseases in children with an increasing incidence reported worldwide. The key to successful anaesthetic outcome involves thorough pre-operative assessment and optimisation of the child's pulmonary status, leading to airway obstruction. Even though the association between airway inflammation and hyper-responsiveness is consistent, the exact mechanism involved is not completely elucidated [1].

First introduced into clinical practice in 1972, etomidate has a long history of use as an intravenous anesthetic and sedative. Like propofol, etomidate has a hypnotic effect but does not provide any analgesia. Its popularity in clinical practice is the result of its beneficial effects on intracerebral dynamics with limited effects on hemodynamic function. When compared to other commonly used anesthetic induction agents such as propofol or the barbiturates, etomidate is generally devoid of adverse effects on cardiovascular. These properties have led to its use as an anesthetic induction in both adult and pediatric patients with altered myocardial performance, congenital heart disease, or hypovolemia. In contrary to the large amount of clinical experience with its use in the adult population, the clinical experience in the pediatric-aged patient is somewhat limited. Resuscitation guidelines from the American Academy of Pediatrics state "etomidate should not be routinely used when intubating an infant or child with septic shock." In the case, that it is used, recognition of adrenal suppression as a consequence is advocated [2-12].

\* Corresponding author: Maria I. Dalamagka

---

## 2. Case Study

A child aged 11 years and weighing 40 kg, with egg allergy and a more general allergic predisposition, with a history of asthma treated with inhalants and a recent respiratory infection with antibiotics, came with a diagnosis of acute abdomen. The child was given preoperatively 4 mg of ondansetron, 4 mg of Dexatone, 80 mg of salbutamol and 1 mg of Dormicum. After 3 minutes of preoxygenation, rapid induction was performed with Fentanyl 0.15 mg, ketamine 10 mg, Propofol 12 mg, and Esmeron 50 mg and with open-label sevoflurane 3%. After intubation with tube 6, he was placed on mechanical ventilation on a Dräger ventilator and volume control ventilation model, as anesthesia was maintained with 3% sevoflurane. Morphine 2 mg and Apotel 600 mg were given intraoperatively. Aminophylline 180 mg was additionally given intraoperatively. The duration of the surgery was two hours. Awakening with pure oxygen was done with Bridion 150 mg.

### 2.1. Management and outcome

Etomidate, a safe and effective anesthetic induction agent in both adult and pediatric patients with altered myocardial performance, congenital heart disease, or hypovolemia. Recent concern has been expressed regarding its effects on the endogenous production of corticosteroids and the impact of that effect on patient outcomes. Rapid sequence intubation is generally used in emergency airway management to protect the airway from passive regurgitation of gastric contents, along with a rapid acting neuromuscular blocking agent. Ketamine and etomidate, both of which provide effective sedation with limited effects on hemodynamic function, have become increasingly popular as induction agents. In this case study, etomidate was used as an anesthetic agent in combination with ketamine, using rocuronium and sevoflurane. Corticosteroids and aminophylline were given to prevent bronchospasm.

---

## 3. Discussion

Pentobarbital is associated with more frequent side effects and parental concerns compared to etomidate. Propofol is the most commonly administered intravenous agent for anaesthesia in children. However, there are concerns that the emulsified preparation may not be safe in children with an allergy to egg, peanut, soybean or other legumes. The propofol is mixed in a liquid containing soybean oil and a substance called egg lecithin. Lecithin is a fatty substance found in some plant and animal tissues. Genuine serious allergic reaction to propofol is rare and is not reliably predicted by a history of food allergy. However, there are concerns that the emulsified preparation may not be safe in children with an allergy to egg, peanut, soybean or other legumes. The rate of reaction to propofol may be higher in food-allergic children than the 1 in 60,000 seen in the general population. In children less than 10 years old, etomidate seems to produce minimal hemodynamic changes, and appears to have a low risk of clinically important adrenal insufficiency, myoclonus, and status epilepticus.

---

## 4. Conclusion

Etomidate remains an effective agent for anesthetic induction in the operating room as well as for sedation during endotracheal intubation in the ICU and emergency room setting. In addition to beneficial effects on intracerebral dynamics, it has limited effects on hemodynamic function even in the setting of hypovolemia, congenital heart disease, and depressed myocardial function. Although it results in the depression of adrenal function for up to 24 h following a single dose, there is controversy regarding the clinical impact of this effect. These concerns have led to the recommendation that it not be administered to patients with sepsis or at risk of sepsis. Etomidate shows a very good anesthetic agent profile when combined with ketamine and sevoflurane in asthma and food allergies in children over 10 years of age.

---

## Compliance with ethical standards

### *Statement of ethical approval*

The present research work does not contain any studies performed on animals/humans subjects by any of the authors.

### *Statement of informed consent*

Informed consent was obtained from all individual participants included in the study.

---

## References

- [1] Tesse R, Borrelli G, Mongelli G, Mastroianni V, Cardinale F. Treating Pediatric Asthma According Guidelines. *Front Pediatr*. 2018, 6:234.
- [2] Bano S, Akhtar S, Zia N, Khan UR, Haq AU. Pediatric endotracheal intubations for airway management in the emergency department. *Pediatr Emerg Care*. 2012, 28:1129–31.
- [3] Ching KY, Baum CR. Newer agents for rapid sequence intubation: Etomidate and rocuronium. *Pediatr Emerg Care*. 2009, 25:200–7.
- [4] Zuckerbraun NS, Pitetti RD, Herr SM, Roth KR, Gaines BA, King C. Use of etomidate as an induction agent for rapid sequence intubation in a pediatric emergency department. *Acad Emerg Med*. 2006, 13:602–9.
- [5] Sokolove PE, Price DD, Okada P. The safety of etomidate for emergency rapid sequence intubation of pediatric patients. *Pediatr Emerg Care*. 2000, 16:18–21.
- [6] Guldner G, Schultz J, Sexton P, Fortner C, Richmond M. Etomidate for rapid-sequence intubation in young children: Hemodynamic effects and adverse events. *Acad Emerg Med*. 2003, 10:134–9.
- [7] Kay B. Total intravenous anesthesia with etomidate. I. A trial in children. *Acta Anesthesiol Belg*. 1977, 28:107–13.
- [8] Kay B. Total intravenous anesthesia with etomidate. II. Evaluation of a practical technique for children. *Acta Anesthesiol Belg*. 1977, 28:115–21.
- [9] Ma YH, Li YW, Ma L, Cao CH, Liu XD. Anesthesia for stem cell transplantation in autistic children: A prospective, randomized, double-blind comparison of propofol and etomidate following sevoflurane inhalation. *Exp Ther Med*. 2015, 9:1035–9.
- [10] Mandt MJ, Roback MG, Bajaj L, Galinkin JL, Gao D, Wathen JE. Etomidate for short pediatric procedures in the emergency department. *Pediatr Emerg Care*. 2012, 28:898–904.
- [11] Lee-Jayaram JJ, Green A, Siembieda J, Gracely EJ, Mull CC, Quintana E, et al. Ketamine/midazolam versus etomidate/fentanyl: Procedural sedation for pediatric orthopedic reductions. *Pediatr Emerg Care*. 2010, 26:408–12.
- [12] Vinson DR, Bradbury DR. Etomidate for procedural sedation in emergency medicine. *Ann Emerg Med*. 2002, 39:592–8.



## Acute Pain and Analgesia in children

Maria I Dalamagka

### ► To cite this version:

Maria I Dalamagka. Acute Pain and Analgesia in children. GSC Advanced Research and Reviews , 2024, 19, pp.161 - 164. 10.30574/gscarr.2024.19.1.0153 . hal-05432479

**HAL Id: hal-05432479**

**<https://hal.science/view/index/docid/5432479>**

Submitted on 26 Dec 2025

**HAL** is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



Distributed under a Creative Commons Attribution 4.0 International License



## Acute Pain and Analgesia in children

Maria I. Dalamagka \*

*Department of Anesthesia, General Hospital of Larisa, Greece.*

GSC Advanced Research and Reviews, 2024, 19(01), 161–164

Publication history: Received on 13 March 2024; revised on 21 April 2024; accepted on 24 April 2024

Article DOI: <https://doi.org/10.30574/gscarr.2024.19.1.0153>

### Abstract

Acute pain is a complex process involving activation of nociceptors, chemical mediators and inflammation. All the major children's hospitals now have dedicated pain services to provide evaluation and immediate treatment of pain in any child. Children experience pain in a similar way to adults. Pain intensity in children depends on the surgical procedure itself but also on numerous other factors such as age, emotional state or the level of anxiety associated with the hospital stay. The perception and communication of the child's pain depends on his or her intellectual and social development. Expression of pain therefore relies on the child's ability to understand, quantitate and communicate it. Previous pain experiences or chronic diseases that required many medical procedures may significantly change the pain threshold in paediatric patients. Moreover, genetic predispositions and environmental effects are significant. The American Academy of Pediatrics and the American Pain Society have reiterated the importance of a multidisciplinary approach in order to eliminate pain in children. In all pediatric settings, an adequate assessment is the initial stage in a proper clinical approach to pain, especially in the emergency departments; therefore, an increasing number of age-related tools have been validated. Systemic opioids, nonsteroidal anti-inflammatory agents and regional analgesics alone or combined with additives are currently used to provide effective postoperative analgesia. These modalities are best utilized when combined as a multimodal approach to treat acute pain in the perioperative setting. The total number of behaviors that are present in the child are used to estimate his/her pain at the time of assessment. Pain assessment is important to facilitate effective postoperative pain management in these vulnerable children. The purpose of this study is to investigate acute pain as well as analgesia and availability of the used analgesic drugs in children. A multimodal approach to preventing and treating pain is usually used.

**Keywords:** Analgesia; Pain; Children; Drugs; Anesthesia; Surgery

### 1. Introduction

Pain is initiated when specialized nerves, called nociceptors, are activated in response to adverse chemical, thermal or mechanical stimulus. Activation can be direct due to trauma or indirect via biochemical mediators released from damaged tissues and circulation. These mediators can further augment the pain process by up-regulating pain receptors and recruiting additional surrounding nociceptors into activity. Mediators include, but are not limited to, prostaglandins, bradykinins, histamine, serotonin and arachidonic acid. The severity of the pain sensed is dependent on the number of receptors stimulated, the duration of the stimulus and the amount of mediators released locally. Once the nociceptor is depolarized, a signal is sent from the periphery into the dorsal horn of the spinal cord, where pain signals are integrated to elicit spinal reflexes such as withdrawal of the affected area, muscle spasms, and to release additional mediators within adjacent spinal segments and relay information to higher cortical areas. Nociceptors are divided into two major nerve groups based on presence or absence of myelination. Myelinated A-delta fibers transmit the signal rapidly and are responsible for the initial sharp pain changing later to burning or soreness. Unmyelinated C fibers are relatively slower in speed and are associated with deep aching or throbbing types of pain that follows the initial sharp pain. Both types of pain fibers then cross the midline and stimulate the ascending pain fibers in the spinothalamic tract.

\* Corresponding author: Maria I. Dalamagka

Substance P is one of the key neurotransmitters relaying the pain signal from the periphery and the spinothalamic tract. Fibers in the spinothalamic tract terminate in the thalamus, limbic system and brain stem. Further information is transmitted to multiple cortical areas of the brain responsible for localization and pain perception. Descending pain fibers are in turn activated from the cerebral cortex via efferent pathway to the spinal cord and periphery and act to decrease the intensity of the pain signal via enkephalin, serotonin and gamma aminobutyric acid (GABA) neurotransmitters. The autonomic nervous system can also be activated by pain. This occurs in large measure in the dorsal horn, and is responsible for the associated symptoms seen with pain such as nausea, sweating, alteration of heart rate and blood pressure. Nociception refers to the process in which specialized fibers in the peripheral nervous system transmit impulses from the periphery to the brain's higher centers through the spinothalamic tracts due to the connection of synapses within the spinal cord's dorsal. In the past, several studies have revealed the existence of distinct classes of nociceptors that, when activated by different stimuli, transmit signals through two basic types of afferent fibers: C fibers and A-delta fibers [1-3].

Pain is an unpleasant physical and emotional sensation that may be experienced in a unique manner by each child, due to differences in pain perception. Exposure to severe pain without adequate pain management has negative long-term consequences, including increased morbidity. Successful pain management in children involves identifying and assessing the pain, followed by prompt control of the pain through pharmacological management and resolution of the underlying cause. If unmanaged, pain can lead to anxiety and stress, and in the long-term this can impact on the psychosocial health and development of a child. Children are at increased risk for adverse drug effects from analgesic therapy if proper vigilance and monitoring are neglected. Analgesics and anesthetics are responsible for the majority of adverse drug effects in hospitalized children. application of the World Health Organization (WHO) pain ladder [4-9].

---

## 2. Clinical base of analgesia drugs

In children, acetaminophen is frequently the initial treatment for mild to moderate pain. A central effect of acetaminophen is the activation of descending serotonergic pathways. Its exact mode of action has not yet been determined. Ibuprofen is the preferred non-steroidal anti-inflammatory drug (NSAID) in children. The practice of combining paracetamol and ibuprofen or alternating doses has gained popularity. Although acceptable, this is not routinely recommended in children as there is currently a lack of evidence to support the safety or efficacy of this practice. Due to their mechanisms of action, using paracetamol and ibuprofen together theoretically increases the risk of renal and hepatic toxicity. Codeine and tramadol are not recommended in a general practice setting for use in children, as other analgesic options with better safety data are available. Codeine metabolism is even less predictable in children. Tramadol metabolism is also individually variable, resulting in different levels of the active component and uncertainty in dosage. Morphine is the prototype opioid that was first isolated from opium in 1804. Intravenous morphine is often the first drug used in most Emergency Departments to treat severe pain. Fentanyl is a purely synthetic, short-acting opioid, utilized for acute pain management and considered to be 50–100 times more potent than morphine. Intranasal fentanyl has a greater safety profile than intravenous morphine, without any difference in the effectiveness of pain control. It can be administered by intravenous or intranasal routes, and by transdermal patches. Transdermal fentanyl has been studied in children with chronic pain secondary to malignant and non-malignant diseases and was proved to be a valid alternative to more invasive administrations. Ketamine is an N-methyl-D-aspartate antagonist that prevents the central sensitization of nociceptors and inhibits peripheral nociception. It may be used to provide anaesthesia, sedation or analgesia. The effects of ketamine are dose-dependent, with analgesia without sedation occurring at around 10–30% of the dissociative dose. Nitrous oxide (N<sub>2</sub>O) is a useful agent for pediatric procedural sedation, because it provides a rapid onset and offset of sedation. A nasal or face mask or muzzle is used to self-administer N<sub>2</sub>O via a demand valve system or continuous flow. It can be administered as 50% or 70% with oxygen for both analgesia and sedation; the higher dose provides a similar sedation depth with no increase in adverse events. Despite its efficacy, some studies indicate that N<sub>2</sub>O alone provides limited pain relief during extremely excruciating procedures, such as fracture reduction. The most frequent adverse events reported are nausea, vomiting, agitation and dizziness. A non-pharmacological approach to pain includes psychological, behavioral and physical interventions used as adjuncts to pharmacological treatment and consist of physical comfort measures and distracting activities [10-15].

---

## 3. Management and Outcomes

There are still no universal tools for pain assessment that could be used in all children; therefore, the available tools should be adjusted to individual age groups. The scales most commonly used in young children rely on pain assessment conducted by the medical personnel and mainly involve the assessment of their behaviour, FLACC (Face, legs, Activity, Cry and Consolability) and/or physiological parameters, CRIES (Crying, Requirement for O<sub>2</sub>, increased heart rate,

expression, Sleeplessness). in older children, it is possible to use self-report scales, the VAS (Visual Analogue Scale), the NRS (Numeric Rating Scale), or the FAcES pain Rating Scale. Some of the scales enable not only recognition of pain but also determination of its intensity the simplest and most common classification of pain intensity divides it into three levels: mild – 1–3 VAS/NRS/ FIACC points; moderate – 4–6 VAS/NRS/FIACC points; and strong/severe – 7 or more VAS/NRS/FIACC points. It should be remembered that pain intensity has to be assessed both at rest and in dynamic conditions. Surgical procedures and painful diagnostic procedures are the main causes of this unpleasant and dangerous phenomenon for hospitalized children. It should be remembered that maintaining homeostasis in a child undergoing surgery is also affected by provision of an adequate level of analgesia and sedation as well as nerve conduction block within the surgical site. Inadequate treatment of acute pain may have both short- and long-term repercussions. Neuroimaging studies have discovered long-lasting modifications in brain structure and connectivity correlating with the amount of acute pain exposure during the perinatal period and with subsequent cognitive and behavioral effects in adult age. The American Academy of Pediatrics and the American Pain Society (2001) reiterated the need to eliminate pain in pediatric patients, using a multidisciplinary method. An ideal way to provide better postoperative analgesia in children is the use of multiple analgesics with additive or synergistic effects, yet with different side effect profiles so that adequate analgesia can be provided with the least amount of side effects.

---

#### 4. Discussion

Children experience pain in much the same way as adults do, but may manifest or display that pain in a different way. Pain assessment tools can be considered, but these tools are subjective and may under or over estimate pain. Pain for children is often emotionally complex, and the involvement of parents and caregivers can add to the difficulty of management. These factors, along with a cautious approach to giving analgesia to children, can lead to pain being under-treated in some situations.

---

#### 5. Conclusion

In order to facilitate the choice of the appropriate drug, a treatment algorithm based on a ladder approach can be used. Moreover, non-pharmacological techniques should be considered to alleviate anxiety and distress in pediatric age. Various considerations should be made to select the most appropriate analgesic medication. The initial assessment of pain represents the key stage in its management and relies on appropriate scales tailored to the patient's age and cognitive level. Multimodal analgesia may include pharmacology (basic analgesics, opioids, and adjuvant analgesia), regional anesthesia, rehabilitation, psychological approaches, spirituality, and integrative modalities, which act synergistically for more effective acute pediatric pain control with fewer side effects than any single analgesic or modality.

---

#### References

- [1] Armstrong, D. Bradykinin, Kallidin and Kallikrein. Vol. 25 of Handbook of Experimental Pharmacology (Erdos, E.G., editor, ed.). Berlin: Springer-Verlag, 1970.
- [2] Barber, T.X. Toward a theory of pain: relief of chronic pain by prefrontal leucotomy, opiates, placebos, and hypnosis. *Psychological Bulletin* 56:430-460, 1959.
- [3] Basbaum, A.I., and Fields, H.L. Endogenous pain control systems: brainstem spinal pathways and endorphin circuitry. *Annual Review of Neuroscience* 7:309-338, 1984.
- [4] Crellin D, Sullivan TP, Babl FE, O'Sullivan R, Hutchinson A. Analysis of the validation of existing behavioral pain and distress scales for use in the procedural setting. *Paediatr Anaesth* 2007; 17: 720-733. doi: 10.1111/j.1460-9592.2007.02218.x.
- [5] Vargas-Schaffer G. Is the WHO analgesic ladder still valid? Twenty-four years of experience. *Can Fam Physician*. 2010; 56:514–517.
- [6] Berde CB, Sethna NF. Analgesics for the treatment of pain in children. *New Engl J Med*. 2002;347(14):1094–1103.
- [7] Odell S, Logan D. Pediatric pain management: the multidisciplinary approach. *Journal of Pain Research*. 2013;6: 785–790.
- [8] Stevens BJ, et al. Pain assessment and intensity in hospitalized children in Canada. *J Pain*. 2012;13(9):857–65.
- [9] Berde CB, Sethna NF. Analgesics for the treatment of pain in children. *N Engl J Med*. 2002; 347:1094–1103.

- [10] Malviya S, Polaner DM, Berde C. Acute pain. In: Cote CJ, Lerman J, Todres ID, editors. A practice of anesthesia for infants and children. Philadelphia: Saunders Elsevier; 2009. pp. 939–978.
- [11] Verghese ST, Hannallah RS. Postoperative Pain Management in Children. *Anesthesiology Clinics N Am.* 2005; 23:163–184.
- [12] Anderson BJ, Holford NH, Woollard GA, et al. Perioperative pharmacodynamics of acetaminophen analgesia in children. *Anesthesiology.* 1999; 90:411–421.
- [13] Romsing J, Hertel S, Harder A, et al. Examination of acetaminophen for outpatient management of postoperative pain in children. *Paediatr Anaesth.* 1998; 8:235–239.
- [14] Hannallah RS. “Outpatient Anesthesia” In: Coté C, Todres ID, editors. A practice of anesthesia for infants and children. Philadelphia: WB Saunders; 2001. pp. 55–67.
- [15] van der Marel CD, Anderson BJ, Romsing J, et al. Diclofenac and metabolite pharmacokinetics in children. *Paediatr Anaesth.* 2004; 14:443–451.



## Pain control & acupuncture

Maria I Dalamagka

### ► To cite this version:

Maria I Dalamagka. Pain control & acupuncture. World Journal of Advanced Research and Reviews, 2022, 14, pp.602 - 607. 10.30574/wjarr.2022.14.3.0625 . hal-05432477

**HAL Id: hal-05432477**

**<https://hal.science/view/index/docid/5432477>**

Submitted on 26 Dec 2025

**HAL** is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



Distributed under a Creative Commons Attribution 4.0 International License

## Pain control & acupuncture

Maria I. Dalamagka \*

*Department of Anesthesia, General Hospital of Larisa, Greece.*

World Journal of Advanced Research and Reviews, 2022, 14(03), 602–607

Publication history: Received on 20 May 2022; revised on 23 June 2022; accepted on 25 June 2022

Article DOI: <https://doi.org/10.30574/wjarr.2022.14.3.0625>

### Abstract

Acupuncture is used mainly to relieve discomfort associated with a variety of diseases and pain conditions. Over the years there has been substantial debate about whether acupuncture really works for chronic pain. Research from an international team of experts adds to the evidence that it does provide real relief from common forms of pain. Despite wide use in clinical practice, acupuncture remains a controversial treatment for chronic pain. Referral for a course of acupuncture treatment is a reasonable option for a patient with chronic pain. The aim of the study was to assess the effectiveness of acupuncture as a treatment of chronic pain, within the context of the methodological quality of the studies. Complementary medicine databases, bibliographies and articles were searched. Our objective was to determine the effect size of acupuncture for chronic pain conditions. We searched MEDLINE and the Cochrane Central Registry of Controlled Trials randomized trial. We included randomized trials of acupuncture needling versus either sham acupuncture or no acupuncture control. We conclude that acupuncture is effective for the treatment of chronic pain, with treatment effects persisting over time.

**Keywords:** Acupuncture; Chronic pain; Chinese medicine; Back pain

### 1. Introduction

Acupuncture is commonly used to treat pain. In traditional Chinese medicine concepts Meridian and vital energy Qi are part of the theoretical basis for the application of acupuncture in strictly specific acupoints. These studies demonstrate that the penetration of a needle through the skin, either acupuncturists point or not having physiological activities. The gate control theory, and the release of endogenous opioids have been proposed as the causative mechanisms leading to obvious analgesic effect of acupuncture [1].

Back pain is a common condition that leads to a weakening and disability with an estimated prevalence length at a rate of 70-85% [2]. The non-specific back pain represents the majority of cases. Although 90% of patients show improvement at 1 month, the majority continues to be symptomatic for 1 year, with a rate of 21 to 25% on terms of pain and disability. In summary, the back pain is among the costliest diseases in the UK, which according to the findings in other countries, leading to a total cost of £ 10,668 million. (Including direct healthcare costs and indirect costs, such informal care and lost productivity) [3]. The Royal College of General Physicians suggests that back pain should be transferred from secondary to primary care and the goal should be a rapid return to normal functionality. There is intense debate how to achieve a return to normal activities. Meanwhile, the complementary and alternative medicine (CAM) has proven that acupuncture is a powerful treatment, which is associated with a clinically significant improvement in back pain. Two recent randomized controlled trials have evaluated the economic costs, one in the UK and one in Germany, show that acupuncture is relatively cost-effective in terms of quality of life for low back pain. These considerations seem to have translated into action, as a growing number of physicians in England offer their patients access to acupuncture. In addition, public health has a strong interest in the use of acupuncture, for example, a recent

\* Corresponding author: Maria I. Dalamagka  
Department of Anesthesia, General Hospital of Larisa, Greece.

survey in the US showed that the most patients with back pain were most likely to experience acupuncture if they did not have to pay [3]. Acupuncture is of increasing importance for the treatment of headache in Western medicine [4]. In 2001 a major review of the Cochrane concluded that acupuncture is important in the treatment of idiopathic headache, but the quantity and quality of the evidence was not sufficient [4]. Little is known, however, about the relationship between the diagnostic criteria of the International Headache Society and the diagnosis of traditional Chinese medicine in primary headache disorders [4]. The authors reported that they needed well-designed, large-scale studies to evaluate the effectiveness and cost-effectiveness of acupuncture [4]. In 2005 two large, high quality studies in patients with headache noticed a slight difference between the effect of acupuncture and placebo, but a significant difference between the acupuncture and placebo group with no intervention. This result differed from that of a large systematic review comparing all interventions of acupuncture compared to placebo groups where there was no interference in which only a small to moderate analgesic effect of placebo, as he could not make a clear distinction, both for obvious submission of patients and because of weakness blindness of the study groups, which did not apply to any interference [1]. For migraine by Linde et al. found that the real and sham acupuncture was superior to a waiting list control and Diener et al. demonstrated that for migraine acupuncture 11 sessions, either real or sham for a period of six weeks It was as effective as the standard prophylactic medical treatment on a daily basis for more than six months. For tension headache by Endres et al. showed that acupuncture therapy for more than six weeks reduced the number of headache days by 15.6 to 6.0 per month. The Coeytaux et al. showed that medical management of headache had better results when there was additional use acupuncture and Witt et al. found that acupuncture is an affordable and effective treatment in patients with primary headache. For tension headache demonstrated limited effectiveness of real acupuncture compared with sham acupuncture in a meta-analysis of all randomized controlled trials until August 2007 [4]. National and international guidelines do not suggest concomitant therapy. Although several treatments have been tried, including behavioral changes, NSAIDs, and physical therapy, randomized controlled trials have failed to show that any of these treatments is significantly more effective for the relief of pain or to improve functionality. So acupuncture is increasingly used as an alternative therapy [2].

## 2. Results

In research done to date review of all types of placebo by Matia Madsen et al. [1] thirteen studies were selected (3025 patients), which included various pain conditions such as knee osteoarthritis, tension headache, migraine, low back pain, fibromyalgia, abdominal pain due to scars, postoperative pain, and pain during colonoscopy. They included all studies of acupuncture, as traditional acupuncture and electroacupuncture and excluded studies that used transcutaneous electrical nerve stimulation and applying pressure to acupuncture points (acupressure). The studies were not blinded (blinded) concerning the application of acupuncture and «placebo» acupuncture by clinicians. The duration of treatment ranged from one day to 12 weeks. A small but statistically significant difference was found between the acupuncture and «placebo» acupuncture. A moderate difference was found between the «placebo» acupuncture and the control group, as large studies make reference both to small, and in the important actions of placebo. In a blinded randomized controlled study conducted in Germany (340 outpatients) by Michael Haake [2] and his collaborators, included 1162 patients aged 18 to 86 years with a history of chronic low back pain for an average of 8 years. Patients underwent 10 sessions of 30 minutes, either real acupuncture (according to the principles of traditional Chinese medicine), or sham acupuncture (where applied superficial acupuncture at non acupuncture points) or conventional treatment (a combination of medications, physical therapy and exercise). Primary endpoint was the response after six months, defined as a 33% improvement. Patients who have not followed the process blind or have recourse to other non-permitted concomitant treatments during the monitoring were classified as non-responders, regardless of symptom improvement. At six months, response rate was 47.6% in the real acupuncture group, 44.2% in the sham acupuncture group and 27.4% in the conventional therapy group. Almost half of the patients in the acupuncture group and only one quarter of patients in the conventional therapy benefited. This study also found clear superiority of acupuncture compared with conventional treatment guideline, but showed no superiority over «sham» acupuncture for at least six months. The authors of both studies concluded that acupuncture is an affordable and effective treatment for back pain. To evaluate the effect of acupuncture in nonspecific back pain conducted systematic review of randomized controlled trials (RCTs) from Jing Yuan et al. [3] twenty-three studies (n=6359) were included and classified into 5 categories comparator (1. Acupuncture versus control group (no intervention), 2. acupuncture versus sham acupuncture, 3. acupuncture versus conventional treatment 4. acupuncture and conventional therapy versus conventional therapy, 5. acupuncture and conventional treatment versus sham acupuncture and conventional treatment). There is a moderate number of elements proving that acupuncture is more effective compared to the control group and strong evidence that there is no significant difference between the acupuncture and the sham acupuncture in terms of short-term pain relief. There is strong evidence that acupuncture may be a useful adjunct to other forms of conventional therapy for nonspecific low back pain, but the effectiveness of acupuncture compared with other forms of conventional therapies requires further exploration. A systematic review by Sidney Rubinstein et al. [5] evaluated the effect of chiropractic (SMT), acupuncture and herbal medicine in chronic nonspecific low back pain. Thirty-five

randomized controlled trials (8 SMT, 20 with acupuncture, herbology 7) examined 8,298 patients. The major findings are based on low quality data suggest that chiropractic did not provide a most beneficial clinical effect compared to the sham, or any other intervention in the treatment of non-specific lower back pain. There are indications however, that acupuncture provides a short-term clinically significant effect when compared to the control group or when added to another intervention. Although there are some good results for the herb into short single studies, the lack of homogeneity among studies did not allow a cumulative assessment of the effect. These results are also in agreement with recent reviews on acupuncture and herbology. It is necessary randomized trials with low risk of bias and adequate sample size. Thirty-five randomized trials (2861 patients) that include acupuncture in adults with nonspecific subacute or chronic low back pain, or dry needling for myofascial pain syndromes in the lumbar, included in a systematic review by Andrea Furlan [6]. Of the thirty-five RCT that included, 20 published in English, seven in Japanese, five Chinese, one Norwegian, one in Polish and one German. There were only three trials of acupuncture for treating low back pain. They did not make valid conclusions, because of small sample size and low methodological quality of the studies. There are signs of pain and functional improvement regarding chronic low back pain after acupuncture therapy compared with no therapeutic intervention or sham-treatment. These effects were only observed immediately after the end of sessions and short-term monitoring. However, there are clear recommendations about the most effective acupuncture technique. Although, a growing number of clinical studies verify the effect of Chinese acupuncture on headaches, the relationship between the diagnostic criteria of the International Headache Society and the diagnosis of traditional Chinese medicine in primary headache disorders is not clarified. In a prospective, randomized, multicenter, double-blind, controlled trial by Hans-Christoph Diener [7] et al, 960 patients had 2-6 migraine episodes per month randomized to real-acupuncture group (n=313); sham acupuncture (n=339), or conventional treatment group (n=308). Patients received 10 acupuncture treatment sessions in 6 consecutive weeks or prophylactic medicine. The proportion of responders, defined as a decrease of 50% in days with migraine, 26 weeks after randomization. The results showed 47% decrease in the real acupuncture group; 39% in the sham acupuncture group and 40% in the standard therapy group. The results of treatment for migraine did not differ between patients treated with sham acupuncture, real acupuncture or standard treatment. In a Cochrane review by Linde [8] and his associates included randomized trials with a follow-up period of at least eight weeks after randomization, compared the clinical results of acupuncture with a control group; a group of sham acupuncture or other intervention in migraine patients. It was evaluated in 22 studies (4419 participants), investigating whether acupuncture is effective in the prophylaxis of migraine. Six studies evaluated the addition of acupuncture to basic care (which usually includes treating only the acute headache), and found that patients who received acupuncture had fewer headaches. Fourteen studies compared real acupuncture with inadequate or false acupuncture, in which the insertion of the needles were in false places or needles did not penetrate the skin. In these studies, the patients in both groups had fewer headaches, but there was no difference between the two treatments. In four studies acupuncture compared with prophylactic treatment, the patients in the acupuncture group reported a greater improvement and fewer side effects. In summary, the studies show that migraine patients benefit from acupuncture, although the correct placement of the needle appears to be less important than normally considered by acupuncturists. An update of the Cochrane review by Klaus Linde [9] and coworkers included eleven (2317 participants) randomized studies in order to compare the clinical effects of acupuncture on the tension headache, compared to control group (treatment of acute single headache or standard therapy); group of sham acupuncture or other treatment. Two large studies regarding acute headache compared acupuncture with the standard treatment. Both were statistically significant and with short-term (up to 3 months) benefits of acupuncture in relation to the control group in terms of the response; the number of days with headache and pain intensity. Longterm benefits (over three months) are not investigated. Six studies compared acupuncture with sham acupuncture and five of the six provided data for meta-analysis. They found small but statistically significant benefits of acupuncture in relation to the sham needling in patient outcomes. Three of the four studies that compared acupuncture with physiotherapy, massage or relaxation methods had methodological or other reported shortages. The authors conclude that acupuncture may be a useful non-pharmacological tool in treating patients with frequent episodic or chronic tension headache. To evaluate acupuncture in osteoarthritis of the knee was performed randomized, controlled study by Hanns-Peter Scharf [10] and his associates and included 1007 patients who had chronic pain for at least 6 months (criteria of the American College of Rheumatology and score 2 or 3 with Kellgren- Lawrence). The treatment consisted up to 6 physiotherapy sessions and as needed anti-inflammatory drugs plus 10 sessions traditional Chinese acupuncture, or 10 sessions sham acupuncture or 10 medical visits within 6 weeks. The success rate was defined with at least 36% improvement in score index osteoarthritis of the University Western Ontario and McMaster (WOMAC) at 26 weeks. The success rates were 53.1% for the traditional Chinese acupuncture, 51% for the sham acupuncture and 29.1% for conservative treatment. Groups of acupuncture had higher success rates than conservative therapy groups. No difference between Chinese traditional acupuncture and sham acupuncture. There was no statistically significant difference between the traditional Chinese acupuncture and sham acupuncture, suggesting that the observed differences could be due to effects of placebo, a different doctor-patient relationship in various groups, or a physiological effect of acupuncture, whether made according with the principles of traditional Chinese acupuncture. Eight randomized clinical trials (RCT) of acupuncture were included in the systematic review by Lee [11] and his associates in order to evaluate acupuncture in the treatment

of rheumatoid arthritis. Four RCT compared the effects of acupuncture or electro acupuncture with sham acupuncture or no penetration of needles and failed to demonstrate specific effects of acupuncture in pain or in other outcome measures. An RCT comparing acupuncture with indomethacin showed beneficial effects of acupuncture on the overall response rate. In conclusion, the RCT sham-acupuncture group (with or without penetration of the needles) failed to show specific effects of acupuncture compared to control group, in patients with rheumatoid arthritis. In summary, the findings provide no convincing evidence that acupuncture with or without moxa are beneficial in the treatment of rheumatoid arthritis. We assessed the methodological quality of primary studies, using a modified scale Jadad. The duration of the activities was low in most studies (<3 months), in addition to a study. The blindness weakness can lead to overestimation of the treatment effect. This systematic review did not found superiority of real acupuncture compared to sham acupuncture. The sham acupuncture without penetration of the needles was found to be superior to the placebo tablets. The major problem with clinical trials of acupuncture is to find the appropriate placebo. The placebo acupuncture includes minimal or superficial introduction of needles or sham acupuncture or without penetrating needles [12]. However, there is no universally acceptable placebo. This data showed no evidence that the presence or absence of De Qi exerts a significant effect on clinical outcomes. The observed effects of sham acupuncture needle penetration may be due to a physiological activity of the insertion of the needle or therapeutic relationship. The Manheimer et al. [13] conducted a meta-analysis of 22 studies of acupuncture for back pain published before September 2004. This analysis found that acupuncture in a short time was statistically and clinically superior to sham needling (4 trials with 343 patients) and without any additional treatment (eight trials with 586 patients). They concluded that the evidence was insufficient to demonstrate the short-term efficacy of acupuncture compared with other treatment. The Ammendolia [14] carried out a systematic review on acupuncture for back pain, including studies published until July 2006. It included 19 studies (4998 patients), most of which included acupuncture by inserting needles. This review compared acupuncture with a control group - no intervention (three studies); with sham acupuncture (7 studies); with other therapies (4 studies) and as adjunctive therapy to other treatments (7 trials). Evaluate the improvement in the pain and the functional status at various points in time: immediately after treatment (<1 week), short term (up to 3 months), in the interval (3-12 months), and long term (>1 time). Immediately after treatment, three studies found that acupuncture is more effective compared to the control group in improving pain and functional situation. Acupuncture was rarely greater than sham acupuncture for pain relief (eg in one of the six studies immediately after treatment, in 2 of the 3 studies on short-term results). The functional status between the acupuncture and the sham acupuncture was similar in short-term observation. Acupuncture has been evaluated as an adjunct therapy in 7 studies. The primary treatment was exercise (two studies), standard treatment (3 studies), physical therapy (one study) and orthopedic treatment (one study). Briefly in all studies and in all time monitoring periods, the complementary application of acupuncture led to better results. In a systematic review with acupuncture and electroacupuncture on the common peripheral osteoarthritis, the Kwon [15] found that in 10 of 18 trials, acupuncture showed a greater reduction of pain compared with various other control groups. The studies showed a wide variation in terms of the number of sessions, ranging from 5 to 45 total sessions, within 2-26 weeks and 1-5 sessions per week. They found that acupuncture is more effective compared with no intervention (2 studies), or relative to the sham control group (3 out of 4 trials).

The White [16] conducted a systematic review of acupuncture for chronic knee pain. They included 13 studies, of which 8 were considered sufficient and thus were included in a meta-analysis. They found that acupuncture is superior compared to the sham control group (five trials with 1334 patients) and compared with the control group without any additional treatment (4 trials with 927 patients), in improving the functional status and reduction of pain in the short and long term period. The Manheimer [17] published the most recent systematic review of acupuncture for osteoarthritis, including studies published prior to February 2007. They included 11 studies, of which 9 used for a metaanalysis. They found that acupuncture has relatively short clinical improvement in pain and the functional status compared to the standard treatment. Compared to the sham control group, acupuncture associated with clinically irrelevant (but statistically significant) and long-term improvement in pain and functional situation. They interpreted the difference of the results regarding the beneficial effect of acupuncture, as a consequential action of placebo or the expected expectations. Following the publication of Manheimer, two large studies published by Foster [18] and Williamson [19]. The study used counseling and exercise, also applied acupuncture, up to six sessions over three weeks or the same number of sham sessions acupuncture, without penetrating needles. There seemed extra pain reduction after treatment and at 6 or 12 months [18]. In patients who were standby for knee replacement, the application of acupuncture and physiotherapy for six weeks resulted in a brief decrease of osteoarthritis. However the beneficial effect of treatment was not maintained in monitoring (follow up) at 12 weeks [19].

According to Lizhou Liou and coll. (seven systemic reviews) acupuncture in treatment of chronic low back pain provides shortterm clinically relevant benefits for pain relief and functional improvement compared with no treatment or acupuncture plus another conventional intervention [20].

### 3. Conclusion

In summary, acupuncture appears to be superior compared with "no action" or usual care in patients with chronic low back pain, osteoarthritis, or headache. Acupuncture has a beneficial safety profile, with relatively few side effects. Data suggest that acupuncture is a cost-effective treatment. Our results provide the most robust evidence to date that acupuncture may be valuable in patients who prefer over other treatment options or concern the use of analgesic drugs.

### Compliance with ethical standards

#### *Acknowledgments*

General Hospital of Larisa.

### References

- [1] Madsen MV, Gøtzsche PC, Hróbjartsson A. Acupuncture treatment for pain: systematic review of randomised clinical trials with acupuncture, placebo acupuncture, and no acupuncture groups. *BMJ* 2009;338:a3115.
- [2] Haake M, Müller HH, Schade-Brittinger C, Basler HD, Schäfer H, et al. German Acupuncture Trials (GERAC) for chronic low back pain: randomized, multicenter, blinded, parallel-group trial with 3 groups. *Arch Intern Med* 2007; 167: 1892-1898.
- [3] Yuan J, Purepong N, Kerr DP, Park J, Bradbury I, McDonough S. Effectiveness of acupuncture for low back pain: a systematic review. *Spine (Phila Pa 1976)*. 2008 Nov 1;33(23):E887-900.
- [4] Böwing G, Zhou J, Endres HG, Coeytaux RR, Diener HC, et al. Differences in Chinese diagnoses for migraine and tension-type headache: an analysis of the German acupuncture trials (GERAC) for headache. *Cephalalgia* 2010; 30: 224-232.
- [5] Rubinstein SM, van Middelkoop M, Kuijpers T, Ostelo R, Verhagen AP, et al. A systematic review on the effectiveness of complementary and alternative medicine for chronic non-specific low-back pain. *Eur Spine J* 2010; 19: 1213-1228.
- [6] Furlan AD, van Tulder M, Cherkin D, Tsukayama H, Lao L, et al. Acupuncture and dry-needling for low back pain: an updated systematic review within the framework of the cochrane collaboration. *Spine (Phila Pa 1976)* 2005; 30: 944-963.
- [7] Diener HC, Kronfeld K, Boewing G, Lungenhausen M, Maier C, et al. Efficacy of acupuncture for the prophylaxis of migraine: a multicentre randomised controlled clinical trial. *Lancet Neurol* 2006; 5: 310-316.
- [8] Linde K, Allais G, Brinkhaus B et al.. Acupuncture for migraine prophylaxis. *Cochrane Database Syst Rev* 2009; (1):CD001218.
- [9] Linde K, Allais G, Brinkhaus B, Manheimer E, Vickers A, White AR. Acupuncture for tension-type headache. *Cochrane Database Syst Rev*. 2009 Jan 21;(1):CD007587.
- [10] Scharf HP, Mansmann U, Streitberger K, Witte S, Krämer J, et al. Acupuncture and knee osteoarthritis: a three-armed randomized trial. *Ann Intern Med* 2006; 145: 12-20.
- [11] Lee MS, Shin BC, Ernst E Acupuncture for rheumatoid arthritis: a systematic review. *Rheumatology (Oxford)* 2008; 47: 1747-1753.
- [12] Sherman KJ, Coeytaux RR Acupuncture for Improving Chronic Back Pain, Osteoarthritis and Headache. *J Clin Outcomes Manag* 2009; 16: 224-230.
- [13] Manheimer E, White A, Berman B, Forsys K, Ernst E Meta-analysis: acupuncture for low back pain. *Ann Intern Med* 2005; 142: 651-663.
- [14] Ammendolia C, Furlan AD, Imamura M, Irvin E, van Tulder M Evidence-informed management of chronic low back pain with needle acupuncture. *Spine J* 2008; 8: 160-172.
- [15] Kwon YD, Pittler MH, Ernst E Acupuncture for peripheral joint osteoarthritis: a systematic review and meta-analysis. *Rheumatology (Oxford)* 2006; 45: 1331-1337.

- [16] White A, Foster NE, Cummings M, Barlas P Acupuncture treatment for chronic knee pain: a systematic review. *Rheumatology (Oxford)* 2007; 46: 384-390.
- [17] Manheimer E, Linde K, Lao L, Bouter LM, Berman BM Meta-analysis: acupuncture for osteoarthritis of the knee. *Ann Intern Med* 2007; 146: 868-877.
- [18] Foster NE, Thomas E, Barlas P, Hill JC, Young J, et al. Acupuncture as an adjunct to exercise based physiotherapy for osteoarthritis of the knee: randomised controlled trial. *BMJ* 2007; 335: 436.
- [19] Williamson L, Wyatt MR, Yein K, Melton JT. Severe knee osteoarthritis: a randomized controlled trial of acupuncture, physiotherapy (supervised exercise) and standard management for patients awaiting knee replacement. *Rheumatology (Oxford)*. 2007 Sep;46(9):1445-9.
- [20] Liu L, Skinner M, McDonough S, Mabire L, Baxter GD. Acupuncture for low back pain: an overview of systematic reviews. *Evid Based Complement Alternat Med*. 2015;2015:328196.



# Multiple sclerosis an chronic demyelinating disease and anesthesia management

Maria I Dalamagka

## ► To cite this version:

Maria I Dalamagka. Multiple sclerosis an chronic demyelinating disease and anesthesia management. Magna Scientia Advanced Research and Reviews, 2025, 15, pp.180 - 184. 10.30574/msarr.2025.15.1.0124 . hal-05432465

**HAL Id: hal-05432465**

**<https://hal.science/view/index/docid/5432465>**

Submitted on 26 Dec 2025

**HAL** is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



Distributed under a Creative Commons Attribution 4.0 International License

# Multiple sclerosis an chronic demyelinating disease and anesthesia management

Maria I. Dalamagka \*

*Department of Anesthesiology, General Hospital of Larissa, Greece.*

Magna Scientia Advanced Research and Reviews, 2025, 15(01), 180-184

Publication history: Received on 16 September 2025; revised on 25 October 2025; accepted on 27 October 2025

Article DOI: <https://doi.org/10.30574/msarr.2025.15.1.0124>

## Abstract

Multiple sclerosis (MS) is an immune-mediated disease of the central nervous system (CNS) and the most common chronic demyelinating disease affecting approximately 2.8 million people worldwide. Its high prevalence worldwide makes it probable for an anesthesia professional to take care of a patient diagnosed with MS at some point during their career. MS is an immune-mediated disease characterized by inflammation, demyelination, and axonal damage in the central nervous system. The course of MS is characterized by exacerbations and remissions at unpredictable intervals over a period of several years. Progression of disability due to MS is highly variable among patients. General, epidural, and spinal anesthesia are generally deemed safe when used in patients with the disease. General anesthesia is the most common anesthetic technique used. There is more controversy surrounding spinal anesthesia, but it seems spinal anesthesia can be administered safely without any additional increased risk of disease relapse. The purpose of the preoperative assessment is to ensure the patient undergoes a safe and adequate anaesthetic experience by selecting the most suitable type of anaesthesia. This assessment must include the full documentation of the neurological deficits detailed by the neurologist in charge of the patient. The use of an established score such as the "Expanded Disability Status Scale" score or "EDSS" score should also be used in order to evaluate the degree of disability. Besides neurological data, the preoperative assessment conducted by the anaesthesiologist should also document any respiratory, cardiac, renal, hepatic, autonomic or any other medical comorbidities. There is a lack of published guidelines on the management of patients with multiple sclerosis (MS) undergoing procedures that require anaesthesia. The anaesthetic management covers preoperative assessment, choice of anaesthetic techniques and agents, side-effects of drugs used during anaesthesia and their potential impact on the disease evolution and drug interactions. The choice of the anaesthetic technique used should be made after careful assessment of the risks and benefits for each specific patient and his preference must also be considered. The association of general anaesthesia and epidural with low concentrations of local anaesthetic are considered safe. Post-operative MS relapses are not affected by the choice of the anaesthetic technique, and that all techniques, including general anaesthesia, may be safely used.

**Keywords:** Multiple Sclerosis; MS Relapses; Disability; Neurological Condition; Anesthesia

## 1. Introduction

Inflammation, demyelination, and axonal damage are the major pathologic mechanisms that cause clinical manifestations. The characteristic neuropathology of MS is the presence of focal demyelinated plaques within the central nervous system with variable inflammation and scarring (gliosis) that can be seen on magnetic resonance imaging (MRI). The varying location of plaque formation dictates the clinical manifestations that patients may have. Brainstem and cranial nerve involvement can lead to optic neuritis, diplopia, and trigeminal neuralgia. Cerebellar involvement can cause vertigo and gait disturbances. Cerebrum and spinal cord involvement causes cognitive dysfunction, limb paresthesia, motor weakness, and bladder dysfunction. Symptoms of MS can eventually persist and lead to severe disability from visual loss, ataxia, skeletal muscle weakness, incontinence, and paralysis. Multiple sclerosis is a demyelinating disease of the CNS that has implications for anaesthesia. Triggers of MS relapses include

\* Corresponding author: Maria I. Dalamagka

infections and fever of other causes, while surgical procedures and emotional stress are still controversial. The interdisciplinary management of MS should optimise medical follow-up and allow him/her to be fully informed. Treatment goals include decreasing relapses and magnetic resonance imaging activity while minimizing permanent disability [1-6].

---

## 2. Relapsing multiple sclerosis

MS is an acquired demyelinating neurological condition characterized by neurological remission of the symptoms, followed by neurological deficit and progressive disability as time passes. Relapsing Multiple Sclerosis (RMS) is characterised by acute episodes of neurological dysfunction, that are not triggered by infections or raised temperature or other metabolic change, with partial or complete remission. Secondary Progressive Multiple Sclerosis, which is characterised by chronic deterioration of neurological function without remission. Primary Progressive Multiple Sclerosis, evolution from onset. In addition, the Clinically Isolated Syndrome while radiologically isolated syndrome can be considered as a MS pre-stage syndrome. The symptoms people with MS describe are numerous and varied, depending on the regions of the CNS affected. There is no unique clinical phenotype. Frequently observed problems include impaired eyesight, impaired balance and coordination, motor and sensory abnormalities, fatigue, urinary and faecal incontinence and cognitive impairment. Genetic and environmental factors is thought to be involved in people developing MS. Multiple sclerosis is a type of demyelinating encephalomyelitis of the central nervous system with symptoms affecting sensory and/or motor nerves that typically undergo a relapsing/remitting course. Neurological dysfunction characterizes the common symptoms of MS, which include paralysis, sensory disturbances, autonomic disturbances, lack of coordination, and visual impairment. Patients with multiple sclerosis (MS) are at an increased risk for exacerbation of their signs and symptoms in the postoperative period. A thorough preoperative neurological assessment is critical to assess for worsening of signs and symptoms in the postoperative period. The neurological assessment must cover the clinical presentation of the disease, the most recent MRI results, the duration of the disease, the presence/absence and timing of the last acute exacerbation or MS relapse, as well as a complete clinical examination evaluating the degree of neurologic impairment. Risk factors that trigger multiple sclerosis relapses include infection, fever and trauma, while surgery itself and emotional stress are more controversial. Certain medications used by patients with multiple sclerosis, such as baclofen or clonazepam may trigger OSA by causing excessive relaxation of the pharyngeal muscles. Prolonged steroid therapy, eventually used for other comorbidities, needs to be avoided since it may lead to muscle wasting and osteoporosis and hence increase the risk of injury. The main treatment goal for an acute exacerbation of MS is to reduce the recovery time. Corticosteroids are used for acute relapses for their immunomodulatory and anti-inflammatory properties to improve axonal nerve conduction, decrease edema, and restore the blood-brain barrier. Adrenocorticotrophic hormone (ACTH) is an alternative for patients who do not respond to or tolerate corticosteroids. ACTH stimulates endogenous corticotropin production to regulate anti-inflammatory and immunomodulatory functions. Plasmapheresis is considered second-line therapy for MS and aims to remove pathogenic antibodies and proinflammatory mediators from the circulatory system. Patients with relapsing-remitting MS should begin disease-modifying therapy (DMT) to decrease relapse rate and slow the accumulation of CNS lesions. High efficacy DMTs include monoclonal antibodies such as natalizumab, ocrelizumab, and ofatumumab. The main disadvantage of these therapies is the risk of severe adverse events including serious infections and breakthrough disease with cessation of therapy. Low risk DMTs include interferon beta and glatiramer. The advantage of initiating therapy with these DMTs is the long-term evidence of safety and rarity of serious adverse events [7-24].

---

## 3. Anesthesia considerations

Preoperative Assessment: A thorough history and physical examination is necessary to determine an appropriate anesthetic plan and safe postoperative management. A comprehensive preoperative evaluation should assess the patient's baseline neurological signs and symptoms, triggers, progression of disease, and medications. Regarding baseline impairment, MS can cause respiratory dysfunction, autonomic nervous system dysfunction, and conduction abnormalities. The broad spectrum of medications that patients with MS can be treated with which can create many potential anesthetic drug interactions perioperatively. If patients were treated with oral or intravenous corticosteroids recently for an exacerbation, the anesthesiologist should consider stress-dose steroids. Muscle relaxants, specifically baclofen, can cause muscle weakness and increased sensitivity to nondepolarizing muscle relaxants. Preoperative testing is usually dictated by patient comorbidities. General anaesthesia has been considered as controversial considering its potential to induce a relapse. Neuraxial blockade are also controversial in MS, as local anaesthetics used in the presence of a damaged blood-brain barrier and demyelinated axons are theoretically more likely to generate neurotoxicity. However, neuraxial blockade has never been associated with any complications. Multiple sclerosis relapses after epidural anaesthesia were related to higher concentrations of local anesthetic administered for longer periods of time. Epidural and subarachnoid anaesthesia are not contraindicated, provided the doses of local anaesthetics used are kept

to a minimum. Historically, the use of regional anesthesia techniques for this population of patients has been contraindicated for fear of further deteriorating the neurological evolution (patients with a history of neurological involvement may be more susceptible to injury when exposed to a secondary insult such as mechanical trauma, toxicity of the local anesthetic agent, neuronal ischemia). Preoxygenation is important during induction of anaesthesia, as multiple sclerosis lesions may affect the respiratory centres in the medulla oblongata or the cervical or thoracic spinal cord causing diaphragmatic weakness or paralysis and therefore altering respiratory function. There is no preference as regards to the induction agents used (inhalational versus IV). Sevoflurane and desflurane are reported to be safe. Nevertheless, we recommend avoiding the use of nitrous oxide due to its potential to inhibit vitamin B12 metabolites, which play a role in myelin formation and maintenance. There is no relationship between inhaled anaesthetic agents and the development or the aggravation of MS symptoms. Moreover, inhalational anaesthetics do not appear to have adverse effects on nerve conduction. The use of depolarizing neuromuscular blocking agents should be avoided. With nondepolarizing neuromuscular blockers, patients with MS may exhibit a prolonged response if they have skeletal muscle weakness. Pain management with opioids doesn't seem to provoke any modification in the course of the disease. Peri and post-operative spasticity can be easily managed with an association of preoperative oral medications such as baclofen or tizanidine, regional or neuraxial anaesthetic techniques, and intravenous muscle relaxants. A dexmedetomidine infusion, which is a selective  $\alpha$ -2 agonist, for postoperative anxiolysis and possible antispasticity effect could also be beneficial. Finally, IV dantrolene, a ryanodine receptor-1 antagonist, was used to control postoperative exacerbated spasticity if needed. Close perioperative monitoring of hemodynamic, respiratory function, neuromuscular and temperature is mandatory. MS may present with respiratory impairment secondary to brainstem or cerebellar dysfunction, resulting in the lack of coordination of the respiratory muscle function. MS may present with respiratory impairment secondary to brainstem or cerebellar dysfunction, resulting in the lack of coordination of the respiratory muscle function. Breathing disorders that need to be identified in MS are obstructive sleep apnea syndrome (OSA), and more rarely central sleep apnea, insomnia or narcolepsy. Baseline autonomic assessment in patients with more advanced disease and those on a large number of medications known to affect the autonomic nervous system. Anesthetic goals include maintaining normothermia, fluid homeostasis, and normal hemodynamics during the perioperative period. In patients with motor weakness or paralysis, succinylcholine should be avoided as it can result in profound hyperkalemia. Spinal anesthesia has been associated with an exacerbation of the disease. General anesthesia, regional nerve blocks, and epidural anesthesia have not been directly associated with exacerbations. Specific attention is required for post-operative complications, infection, delayed wound healing and venous thrombosis [25-37].

---

#### 4. Discussion

Irrespective of the type of anesthetic performed and medications used, patients are at an increased risk of release in the postoperative period. Complications such as fever and infection have been shown to exacerbate the signs and symptoms of MS. Patients with respiratory involvement at baseline are more susceptible to pulmonary issues such as atelectasis, hypoventilation, and airway obstruction. Blood pressure fluctuations can occur in the postoperative period in patients with autonomic nervous system involvement. Residual motor blockade may be more common in patients with MS. Vigilance is key for safe postoperative management of a patient with MS. There are two main patterns of disease with MS: relapsing-remitting and progressive disease. The relapsing-remitting phenotype of MS accounts for approximately 85% of cases and is defined by MS attacks (flares, relapses, exacerbations) followed by full or incomplete recovery of symptoms. The progressive type is much less common and is characterized by continued deterioration over time. It can be further delineated into the primary progressive and secondary progressive type of MS. Primary progressive MS is the accumulation of disease with occasional plateaus or temporary minor improvements. Secondary progressive MS describes patients with relapsing-remitting disease that eventually develop worsening disease with or without additional flares. There are several case reports associating spinal anesthesia to MS symptom exacerbation, but there have also been studies that show no link between spinal anesthesia and relapses. MS is not a contraindication to spinal anesthesia, but the anesthesiologist should consider the possibility of inciting an MS flare. Epidural anesthesia is considered a safe technique in patients with MS, and several studies in the obstetric population have shown no difference in relapse rate using epidural anesthesia. Peripheral nerve blocks are regarded to be safe in MS, a disease of the CNS, though studies are limited. Regardless of the type of anesthetic and medications used, patients may experience exacerbation of their disease postoperatively. There is no association between the post pregnancy relapse rate and anaesthesia procedures or drugs used.

---

#### 5. Conclusion

Multiple sclerosis (MS) is a demyelinating disease of the central nervous system characterized by varying initial symptoms such as visual impairment, bowel/bladder dysfunction, trigeminal neuralgia, cognitive problems, gait

disorder, and motor paralysis. The stress of surgery and anesthesia may worsen symptoms and incite an MS flare, but this correlation remains unpredictable. An elevated body temperature has been shown to be a trigger for an exacerbation by causing conduction block in demyelinated nerves, and an increase in body temperature as little as 1 degree Celsius can cause an MS flare. The anesthesiologist should meticulously monitor the patient's temperature and maintain normothermia. Regarding general anesthesia, there is no evidence to suggest superiority between inhaled anesthetics and intravenous anesthetics. No complications have been reported with the use of sevoflurane, desflurane, nitrous oxide, propofol, midazolam, dexmedetomidine, ketamine, and opioids. The use of premedication may be advantageous in patients with MS as stress is a reported trigger for disease exacerbation. Since patients with MS have an upregulation of nicotinic acetylcholine receptors characteristic of upper motor neuron lesions, succinylcholine should be avoided as it can cause a profound hyperkalemic response. With nondepolarizing neuromuscular blockers, patients with MS may exhibit a prolonged response if they have skeletal muscle weakness. The concern with neuraxial anesthesia is the possibility of exposing demyelinated areas of the spinal cord with local anesthetics and their potential neurotoxic effects.

---

## Compliance with ethical standards

### *Disclosure of conflict of interest*

The author has no conflicts of interest to declare.

---

## References

- [1] A.F. Dalmás, C. Texier, A.S. Ducloy-Bouthors, R. Krivosic-Horber. Obstetrical analgesia and anaesthesia in multiple sclerosis. *Ann Fr Anesth Reanim*, 22 (2003), pp. 861-864.
- [2] T.T. Horlocker. Complications of spinal and epidural anesthesia. *Anesthesiol Clin N Am*, 18 (2000), pp. 461-485.
- [3] J. Noseworthy, C. Lucchetti, M. Rodríguez, B. Weinshenker. Multiple sclerosis. *N Engl J Med*, 343 (2000), pp. 938-952 <http://dx.doi.org/10.1056/NEJM200009283431307>.
- [4] A. Comptson, A. Coles. Multiple sclerosis. *Lancet*, 359 (2002), pp. 1221-1231 [http://dx.doi.org/10.1016/S0140-6736\(02\)08220-X](http://dx.doi.org/10.1016/S0140-6736(02)08220-X).
- [5] D. Vizcarra, J. Kawano, C. Castañeda, A. Chereque, M. Tipismana, A. Bernabé, et al. Prevalencia de Esclerosis Múltiple en Lima – Perú. *Rev Med Hered*, 20 (2009), pp. 146-150.
- [6] D. Vizcarra-Escobar, L. Cava-Prado, M. Tipismana-Barbaran. Descripción clínico epidemiológica de una serie de pacientes. *Rev Neurol*, 41 (2005), pp. 591-595.
- [7] M. Vercauteren, L. Heytens. Anaesthetic considerations for patients with a preexisting neurological deficit: are neuraxial techniques safe?. *Acta Anaesthesiol Scand*, 51 (2007), pp. 831 -838.
- [8] E. Drake, M. Drake, J. Bird, R. Russell. Obstetric regional blocks for women with multiple sclerosis: a survey of UK experience. *Int J Obstet Anesth*, 15 (2006), pp. 115-123 <http://dx.doi.org/10.1016/j.ijoa.2005.10.010>.
- [9] P. Levesque, T. Marsepolt, P. Ho, P.H. Venutolo, J.M. Lesouefjm. Multiple sclerosis revealed by spinal anesthesia. *Ann Fr Anesth Reanim*, 7 (1988), pp. 68-70.
- [10] Pasternak JJ, Lanier WL. Diseases affecting the brain. In: Hines RL, Marschall KE. *Stoelting's Anesthesia and Co-Existing Disease*. 7th Edition. Philadelphia: Elsevier; 2022. 296-298.
- [11] Butterworth JF, Mackey DC, Wasnick JD. Anesthesia for patients with neurological and psychiatric diseases. In: Morgan and Mikhail's *Clinical Anesthesiology*. 6th Edition. McGraw Hill; 2018. 627-8.
- [12] Dorotta IR, Schubert A. Multiple sclerosis and anesthetic implications. *Curr Opin Anaesthesiol*. 2002;15(3):365-70.
- [13] De Lott, L.B. · Zerafa, S. · Shedden, K. ... Multiple sclerosis relapse risk in the postoperative period: effects of invasive surgery and anesthesia *Mult. Scler*. 2020; 26:1437-1440.
- [14] Dobson, R. · Giovannoni, G. Multiple sclerosis - a review *Eur. J. Neurol*. 2019; 26:27-40.
- [15] Dorotta, I.R. · Schubert, A. Multiple sclerosis and anesthetic implications *Curr. Opin. Anaesthesiol*. 2002; 15:365-370.

- [16] Drust, A. · Bartels, C. · Hachenberg, T. [Anesthesia and multiple sclerosis: what needs to be considered?] *Anesthesiol. Intensivmed. Notfallmed. Schmerzther.* 2016; 51:458-467.
- [17] Elan Pharmaceutical/Athena Neurosciences Inc, South San Francisco, CA, 2023 Product Information. Tysabri (natalizumab).
- [18] Fambrough, D.M. Control of acetylcholine receptors in skeletal muscle *Physiol. Rev.* 1979; 59:165-227.
- [19] Finucane, B.T. · Terblanche, O.C. Prolonged duration of anesthesia in a patient with multiple sclerosis following paravertebral block *Can. J. Anaesth.* 2005; 52:493-497.
- [20] Flachenecker, P. · Reiners, K. · Krauser, M. ... Autonomic dysfunction in multiple sclerosis is related to disease activity and progression of disability *Mult. Scler.* 2001; 7:327-334.
- [21] Freedman, R.L. · Lucas, D.N. MBRRACE-UK: saving lives, improving mothers' care - implications for anaesthetists *Int. J. Obstet. Anesth.* 2015; 24:161-173.
- [22] Gardner, J.S. · Blough, D. · Drinkard, C.R. ... Tramadol and seizures: a surveillance study in a managed care population *Pharmacotherapy.* 2000; 20:1423-1431.
- [23] Gerçek, A. · Baykan, N. · Dağcinar, A. ... Does general anesthesia potentiate the GABA-ergic action of intrathecal baclofen? *J. Neurosurg. Anesthesiol.* 2004; 16:323-324.
- [24] Gold, S.M. · Mohr, D.C. · Huitinga, I. ... The role of stress-response systems for the pathogenesis and progression of MS *Trends Immunol.* 2005; 26:644-652.
- [25] Grasso, M.G. · Lubich, S. · Guidi, L. ... Cerebellar deficit and respiratory impairment: a strong association in multiple sclerosis? *Acta Neurol. Scand.* 2000; 101:98-103.
- [26] Guthrie, T.C. · Nelson, D.A. Influence of temperature changes on multiple sclerosis: critical review of mechanisms and research potential *J. Neurol. Sci.* 1995; 129:1-8.
- [27] Gutman, J.M. · Kim, K. · Schwarzkopf, R. ... Total hip and knee arthroplasty in patients with multiple sclerosis *Int. J. MS Care.* 2018; 20:244-250.
- [28] Havrdova, E. · Galetta, S. · Hutchinson, M. ... Effect of natalizumab on clinical and radiological disease activity in multiple sclerosis: a retrospective analysis of the Natalizumab safety and efficacy in relapsing-remitting multiple sclerosis (AFFIRM) study *Lancet Neurol.* 2009; 8:254-260.
- [29] Hebl, J.R. Ultrasound-guided regional anesthesia and the prevention of neurologic injury *Anesthesiology.* 2008; 108:186-188.
- [30] Hebl, J.R. · Horlocker, T.T. · Schroeder, D.R. Neuraxial anesthesia and analgesia in patients with preexisting central nervous system disorders *Anesth. Analg.* 2006; 103:223-228.
- [31] Hedström, A.K. · Hillert, J. · Olsson, T. ... Exposure to anaesthetic agents does not affect multiple sclerosis risk *Eur. J. Neurol.* 2013; 20:735-739.
- [32] Hellwig, K. · Verdun di Cantogno, E. · Sabidó, M. A systematic review of relapse rates during pregnancy and postpartum in patients with relapsing multiple sclerosis *Ther. Adv. Neurol. Disord.* 2021; 14, 17562864211051012.
- [33] Honarmand, A. · Safavi, M.R. Comparison of prophylactic use of midazolam, ketamine, and ketamine plus midazolam for prevention of shivering during regional anaesthesia: a randomized double-blind placebo controlled trial *Br. J. Anaesth.* 2008; 101:557-562.
- [34] Hughes, K.E. · Nickel, D. · Gurney-Dunlop, T. ... Total knee arthroplasty in multiple sclerosis *Arthroplast. Today.* 2016; 2:117-122.
- [35] Ingrosso, M. · Cirillo, V. · Papasso, A. ... Femoral and sciatic nerves block (BiBlock) in orthopedic traumatologic lower limbs surgery in patients with multiple sclerosis *Minerva Anesthesiol.* 2005; 71:223-226.
- [36] Lublin, F.D. · Reingold, S.C. · Cohen, J.A. ... Defining the clinical course of multiple sclerosis: the 2013 revisions *Neurology.* 2014; 83:278-286.
- [37] Makris, A. · Piperopoulos, A. · Karmaniolou, I. Multiple sclerosis: basic knowledge and new insights in perioperative management *J. Anesth.* 2014; 28:267-278.



# COVID-19 and mechanical ventilation

Maria I Dalamagka

## ► To cite this version:

Maria I Dalamagka. COVID-19 and mechanical ventilation. GSC Advanced Research and Reviews , 2024, 19, pp.204 - 207. 10.30574/gscarr.2024.19.2.0186 . hal-05432471

**HAL Id: hal-05432471**

**<https://hal.science/view/index/docid/5432471>**

Submitted on 26 Dec 2025

**HAL** is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



Distributed under a Creative Commons Attribution 4.0 International License

## COVID-19 and mechanical ventilation

Maria I. Dalamagka \*

*Department of Anaesthesia, General Hospital of Larisa, Greece.*

GSC Advanced Research and Reviews, 2024, 19(02), 204–207

Publication history: Received on 12 April 2024; revised on 18 May 2024; accepted on 20 May 2024

Article DOI: <https://doi.org/10.30574/gscarr.2024.19.2.0186>

### Abstract

Mechanical ventilation (MV) is used to treat patients with severe coronavirus disease 2019 (COVID-19). This severe respiratory illness, typically develops 8 days after symptom onset and when it does not respond to non-invasive respiratory support, it requires advanced respiratory support, including high concentrations of inspired oxygen and mechanical ventilation. Such therapies are also required for the acute respiratory distress syndrome (ARDS), which has been widely studied over several decades. Obesity as a disease causes a restrictive lung disease and is a sufficient predisposing factor for difficult ventilation. Higher BMI patients are more likely to be young, with single organ failure, less chronic comorbidity but with increased severity of hypoxemia at presentation. Severe respiratory failure from coronavirus disease 2019 pneumonia not responding to non-invasive respiratory support requires mechanical ventilation. Although ventilation can be a life-saving therapy, it can cause further lung injury if airway pressure and flow and their timing are not tailored to the respiratory system mechanics of the individual patient. The phenomenon of “hard lung” is observed as the ventilation of the intubated patient is very arduous and recruitment requires a lot of effort. Coronavirus disease 2019-induced acute respiratory distress syndrome is more severe in morbidly obese patients. This relationship between BMI and mortality was investigated by several observational studies, but the relationship was not universally observed. Some studies found increased BMI was associated with an increased risk of requiring intubation and ventilation, but with no clear relationship with mortality. The combination is quite difficult as these patients oppose the ventilator. Our objective was to determine the association between MV for treatment of COVID-19.

**Keywords:** COVID-19; Ventilation; Obesity; Pneumonia; ARDS; MV

### 1. Introduction

The prevalence of adult obesity and severe obesity in 2017 to 2018 has increased since 2009 to 2010 and is now 42% and 9%, respectively [1]. Obesity is a global disease with at least 2.8 million people dying each year as a result of being overweight or obese according to the world health organization figures. Obesity is affecting most of the physiological processes and modifying the functions of the system including the immune system [2]. It is crucial to understand the effect of obesity on the course of infection to prevent or mitigate the morbidities and mortality [3,4]. In the current COVID-19 era, bariatric teams are aware of the potential risks and thus stressing the extra caution and appropriate management of these patients [5]. Knowing the scale of the obesity problem in the world, we anticipate difficult times for this group of patients in Europe, America, Middle East and rest of the world with a high rate of obesity [6]. In 2009, a significant percentage of admissions to the hospitals and mortality because of H1N1 Influenza virus infection was due to obesity, an estimated 151,700–575,400 total deaths was reported [7,8]. Patients with coronavirus disease 2019 often suffer severe symptoms, from viral pneumonia to respiratory distress. This respiratory distress can lead to alveolar damage and fibrosis in the lungs, which reduces oxygen saturation in the blood. COVID-19-related respiratory distress has been associated with higher rates of mortality and intensive care unit (ICU) admission than respiratory distress associated with other illnesses. The recommended treatment for patients suffering from severe respiratory

\* Corresponding author: Maria I. Dalamagka.

distress due to COVID-19 is mechanical ventilation (MV), which provides oxygen to these critically ill patients and removes carbon dioxide from the blood. The use of MV treatment for severe COVID-19 is common [9-15].

During inspiration, the lung expands owing to a positive transpulmonary pressure. In spontaneous ventilation, this gradient is produced by a negative pleural pressure created by the inspiratory muscles, mainly the diaphragm. In contrast, controlled mechanical ventilation relies upon a positive airway pressure driving gas into the lungs, with the positive transpulmonary gradient dependent upon an increased alveolar pressure, and passive movement of the chest wall. The fundamental ventilator parameters that can be set are airway pressure and flow and their timing, which need to match the patient's respiratory system resistance and elastance (inverse of compliance). At each time point during inspiration, airway pressure is determined by the equation of motion and equals the sum of end-expiratory alveolar pressure, the product of flow and resistance to flow, and the product of tidal volume and elastance of the respiratory system. Mechanical ventilation can be delivered in mandatory mode or in assisted mode to support spontaneous breathing. In the latter modality, the patient's inspiratory effort triggers the delivery of breaths, the work of breathing is shared in various proportions between the respiratory muscles and the mechanical ventilator, and the transpulmonary pressure is generated by a combination of a negative pleural pressure and a positive alveolar pressure. This ventilatory support can be necessary to sustain life in the acute phase of the disease while the immune system fights the viral infection, yet can cause harm to the patient if the levels of positive pressure are not tailored to the associated lung mechanics (ventilator-induced lung injury). Ventilator-induced lung injury is well studied in classical ARDS, a syndrome associated with a distinct histopathological entity termed diffuse alveolar damage (DAD). Diffuse alveolar damage is a widespread, heterogeneous inflammatory reaction comprising alveolar infiltrates with leucocytes and proteinaceous deposits, damage to alveolar pneumocytes, the basement membrane and endothelium, and patchy areas of haemorrhage [16-17].

---

## 2. Case Report

A 55-year-old man (was admitted to the intensive care unit due to respiratory failure), with a body weight of 130 kg and coronavirus disease 2019 pneumonia. The patient was intubated and put into mechanical ventilation with model ACV, FiO<sub>2</sub> 70%, which gradually dropped to 50%, PEEP 18 which gradually dropped to 14, Tidal Volume 500 ml which gradually dropped to 480 ml and respiratory rate 30/min which gradually dropped to 25/min, and I: E ½ to 1/3 and finally to 1/4. Suppression of the patient was achieved by drip co-administration of Ultiva, Diprivan, Esmeron, Dormicum, and Levophed in titrated doses for the best possible ventilation of the patient. Obesity in combination with COVID-19 leads to increased peak, up to 60 and high airway resistance. Satisfactory dosing of Esmeron 60 mg/h and placement of low tidal volumes <500 ml, as well as respiratory rate < 30/min, PEEP < 16, appears to improve ventilator ventilation when co-administered: Ultiva, Diprivan, Dormicum at satisfactory levels in combination with Levophed to maintain hemodynamic stability.

### 2.1. Management and Outcomes

The pathophysiology of severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) infection can lead to a pattern of lung injury in patients with severe COVID-19 pneumonia typically associated with two distinct phenotypes, along a temporal and pathophysiological continuum, characterized by different levels of elastance, ventilation-to-perfusion ratio, right-to-left shunt, lung weight and recruitability. Understanding the underlying pathophysiology, duration of symptoms, radiological characteristics and lung mechanics at the individual patient level is crucial for the appropriate choice of mechanical ventilation settings to optimize gas exchange and prevent further lung injury. The choice of tidal volume should be based on obtaining a driving pressure < 14 cmH<sub>2</sub>O, ensuring the avoidance of hypoventilation in patients with preserved compliance and of excessive strain in patients with smaller lung volumes and lower lung compliance. The level of positive end-expiratory pressure (PEEP) should be informed by the measurement of the potential for lung recruitability, where patients with greater recruitability potential may benefit from higher PEEP levels. Prone positioning is often beneficial and should be considered early.

---

## 3. Discussion

Severe respiratory failure from coronavirus disease 2019 pneumonia not responding to non-invasive respiratory support requires mechanical ventilation. Although ventilation can be a life-saving therapy, it can cause further lung injury if airway pressure and flow and their timing are not tailored to the respiratory system mechanics of the individual patient. Studies suggest that MV patients have a greater hazard of inpatient readmission and all-cause mortality compared to non-MV patients. Whether this is due to a difference in severity of illness, for which MV may be a proxy or a consequence of MV itself, patients who are ventilated appear to be at greater risk for adverse outcomes following discharge.

#### 4. Conclusion

This severe respiratory illness (COVID-19), when it does not respond to non-invasive respiratory support, it requires advanced respiratory support, including high concentrations of inspired oxygen and mechanical ventilation. COVID-19 pneumonia is a pathophysiological entity distinct from classical ARDS and requires different ventilatory management. The hypoxaemia is predominantly underpinned by an increased dead space and  $V/Q$  mismatch and is not correlated with the healthy lung volume. Tidal volumes should not be limited stringently, but should be more liberal as long as the driving pressure is limited. PEEP levels should not be titrated in relationship to hypoxaemia but on the potential for recruitability of the lung, which is not necessarily high if there is minimal air-space disease. Thus it appeared that low respiratory rate < 30/min and low tidal volume < 500 ml with a desired PEEP less than or equal to 14, has a beneficial effect on obese patients.

#### Compliance with ethical standards

##### *Statement of informed consent*

Informed consent was obtained from all individual participants included in the study.

#### References

- [1] Hales K, Carroll MD, Fryar CD, Ogden CL (2020) Prevalence of obesity and severe obesity among adults: United States, 2017–2018. NCHS Data Brief, no. 360: 1-8.
- [2] Fischer-Posovszky P, Moller P (2020) The immune system of adipose tissue: obesity-associated inflammation. *Pathologie* 41: 224-229.
- [3] Houdek MT, Griffin AM, Ferguson PC, Wunder JS (2019) Morbid obesity increases the risk of postoperative wound complications, infection, and repeat surgical procedures following upper extremity limb salvage surgery for Soft tissue sarcoma. *Hand (N Y)* 14: 114-120.
- [4] Zhou X, Ye Y, Tang G, Zhou X (2018) Obesity and infection, accompanying phenomenon or causal association? *Clin Microbiol Infect* 24: 668.
- [5] Dietz W, Santos-Burgoa C (2020) Obesity and its implications for COVID-19 mortality. *Obesity (Silver Spring)* 28: 1005.
- [6] Bluher M (2019) Obesity: global epidemiology and pathogenesis. *Nat Rev Endocrinol* 15: 288-298.
- [7] Louie JK, Acosta M, Samuel MC (2011) A novel risk factor for a novel virus: obesity and 2009 pandemic influenza A (H1N1). *Clin Infect Dis* 52: 301-312.
- [8] Dawood FS, Iuliano AD, Reed C (2012) Estimated global mortality associated with the first 12 months of 2009 pandemic influenza A H1N1 virus circulation: a modelling study. *Lancet Infect Dis* 12: 687-695.
- [9] Gibson PG, Qin L, Puah SH. COVID-19 acute respiratory distress syndrome (ARDS): clinical features and differences from typical pre-COVID-19 ARDS. *Medical Journal of Australia*. 2020;213(2):54–6. e1. doi: 10.5694/mja2.50674.
- [10] Xu Z, Shi L, Wang Y, Zhang J, Huang L, Zhang C, et al. Pathological findings of COVID-19 associated with acute respiratory distress syndrome. *The Lancet respiratory medicine*. 2020;8(4):420–2. doi: 10.1016/S2213-2600(20)30076-X.
- [11] Fan E, Beitler JR, Brochard L, Calfee CS, Ferguson ND, Slutsky AS, et al. COVID-19-associated acute respiratory distress syndrome: is a different approach to management warranted? *The Lancet Respiratory Medicine*. 2020. doi: 10.1016/S2213-2600(20)30304-0.
- [12] Lepper PM, Muellenbach RM. Mechanical ventilation in early COVID-19 ARDS. *EClinicalMedicine*. 2020;28. doi: 10.1016/j.eclinm.2020.100616.
- [13] Kluge S, Janssens U, Spinner CD, Pfeifer M, Marx G, Karagiannidis C. Recommendations on inpatient treatment of patients with COVID-19. *Deutsches Ärzteblatt International*. 2021;118(1–2):1.

- [14] Elsayed HH, Hassaballa AS, Ahmed TA, Gumaa M, Sharkawy HY, Moharram AA. Variation in outcome of invasive mechanical ventilation between different countries for patients with severe COVID-19: A systematic review and meta-analysis. PloS one. 2021;16(6):e0252760. doi: 10.1371/journal.pone.0252760.
- [15] Poor H. Basics of mechanical ventilation: Springer; 2018.
- [16] Amato, M. B. P. , Meade, M. O. , Slutsky, A. S. , Brochard, L. , Costa, E. L. V. , Schoenfeld, D. A. , Stewart, T. E. , Briel, M. , Talmor, D. , Mercat, A. , Richard, J.-C. M. , Carvalho, C. R. R. , & Brower, R. G. (2015). Driving pressure and survival in the acute respiratory distress syndrome. New England Journal of Medicine, 372, 747–755. 10.1056/NEJMsa1410639.
- [17] ARDSnet (2000). Ventilation with lower tidal volumes as compared with traditional tidal volumes for acute lung injury and the acute respiratory distress syndrome. New England Journal of Medicine, 342, 1301–1308. 10.1056/NEJM200005043421801.



# Acupuncture for Depression and Psychological Disorders

Maria Dalamagka

## ► To cite this version:

Maria Dalamagka. Acupuncture for Depression and Psychological Disorders. Advances in Complementary and Alternative Medicine, 2019, 4, 10.31031/acam.2019.04.000577 . hal-05432466

**HAL Id: hal-05432466**

**<https://hal.science/view/index/docid/5432466>**

Submitted on 26 Dec 2025

**HAL** is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



# Acupuncture for Depression and Psychological Disorders



**Dalamagka M\***

*Department of Anesthesiology, Greece*

**\*Corresponding author:** Maria Dalamagka, Department of Anesthesiology, General Hospital of Larissa, Greece

**Submission:** : May 03, 2018; **Published:** January 30, 2019

## Abstract

Acupuncture therapy has been known as a practice related to oriental medicine, and recently has been detected as a potential therapeutic tool for which there is good scientific evidence. Depression and anxiety are usually classified as mental illnesses and it is more useful to think of them as disturbances in brain health, which is directly related to the physical makeup and brain mechanisms and emotional and relational issues. The ancient Chinese practice of acupuncture could be used for the treatment of depression and anxiety, instead of drugs. The purpose of this review was to summarize the existing evidence on acupuncture as a treatment for anxiety and depression. Search of the literature on acupuncture treatment was limited to specific mental health conditions: depression and anxiety disorder. The literature review focused on systematic reviews. Search focused on the latest versions. The available resources provide some evidence that acupuncture is an effective treatment for these conditions. The findings of the studies show that acupuncture can play an important role in the treatment of depression and anxiety.

## Introduction

Acupuncture is one of the most popular complementary and alternative therapies. Acupuncture is a treatment method that originated more than 3,000 years in China and practiced in most of the world. This method is often used as a routine therapy in China, Japan, Korea and Taiwan, and since the late 1970s has gained popularity in the United States and elsewhere in the western world [1]. Acupuncture is increasingly used in psychiatric disorders. The effect of acupuncture in depression (including depressive neurosis and depression following stroke) has been repeatedly demonstrated in controlled studies [2-7]. Traumatic stress disorder develops after a stressful event or situation which has an exceptionally threatening or catastrophic nature, which is likely to cause significant discomfort. Stress disorder classified as an anxiety disorder and is usually defined as the clustering of three clusters of symptoms, ie, revival, marked avoidance, and hyperarousal [8,9].

A high-quality RCT evaluated the effect of acupuncture in cognitive behavioral therapy and a waiting list control [10]. No statistical difference was observed between acupuncture and behavioral therapy. But acupuncture therapy was statistically superior to waiting list control in four outcome measures. The high quality RCT showed that acupuncture had a statistically significant effect compared to a control queue, although no statistical difference was found between acupuncture and behavioral therapy. Also, the therapeutic effect of acupuncture was similar to treatment with behavioral therapy-based test.

Kim et al. [11] conducted a systematic review and meta-analysis of the effectiveness of acupuncture for stress disorder with 4 RCT (n=543) and 2 uncontrolled clinical studies (n=103). The review included patients with stress disorder regardless of gender, age, ethnicity or external therapy clinics or inpatient treatment. Interventions evaluated were classical acupuncture, electro acupuncture and auricular acupuncture and acupuncture combined with moxibustion. Comparison was made for acupuncture and its variants to controls without treatment, acupuncture and virtual conventional therapies for patients with stress disorder. The results were evaluated using the respective scales of stress disorder, as clinician administered stress disorder scale, the scale and scale depression anxiety.

The final results points are not clearly defined. Duration of treatment varies in tests 1-12 weeks. The main findings of this review were: no statistically significant difference was found between acupuncture and cognitive behavioral therapy, but acupuncture was statistically superior to waiting list control. No difference was observed between the group of acupuncture and oral selective serotonin reuptake-inhibitors. Clinical and experimental data suggest that at least some of acupuncture clinical effects mediated by the central nervous system. Functional MRI studies on acupuncture on specific acupoints have not shown significant regulatory effects in the limbic system and subcortical structures. Functional MRI reducing neuronal signal to the limbic system is

not surprising in view of the therapeutic effects of acupuncture for disorders including stress disorder, anxiety, depression, and substance abuse. Hui et al noted that the stimulation of acupuncture at a specific acupuncture point (ST36) produced a reduction in neuronal activity, especially the limbic and paralimbic structures and limbic brain regions. [9-13]. Smith, Hay & MacPherson [14] systematically reviewed 30 published and unpublished randomized trials with 2,812 participants. The systematic review comparing acupuncture with sham acupuncture, no treatment, medication and other psychotherapies or usual care. Patients groups differed in each reviewed studies, however most of the clinical studies with male and female and focused on hospitalized patients with a clinical diagnosis of depression or depression with co-morbidity. The meta-analysis included the following conditions: depression, depressive disorder, dysthymic disorder. The details include acupuncture needle acupuncture, laser and electro acupuncture. The comparator arm in the studies varied and included a number of methods, such as invasive and non-invasive acupuncture control, mock electro acupuncture, laser acupuncture model, without treatment, medication and structured psychotherapy. Majority of the trials compared manual and electro acupuncture with medication. The primary endpoints were the reduction in severity of depression, measured by self-rating scales or a clinician rating scales and an improvement in depression defined as remission versus any recession. Secondary outcomes evaluated variable in the different trials were quality of life indicators such as sleep, emotional outcomes, adverse events and change of use of drugs. Acupuncture compared to waiting list control (2 RCT), and acupuncture versus sham acupuncture (1 RCT) have shown promising results in reducing the severity of depression, but the number of trials was very low. The bias for the two trials in acupuncture and waiting list control was low. The tests in this review, which compared manual and electro acupuncture with drugs found no difference between groups. Two studies found that acupuncture combined with drugs are more beneficial alone medication. The findings of this study was overall similar to findings from previous systematic reviews done by Mukaino et al. [15] and Leo et al. [16].

Pilkington et al. [17] conducted a systematic review of the effectiveness of acupuncture for anxiety and anxiety disorders in 12 controlled trials (n=1.134). Studies included 10 RCT (n=1.010) and 2 non-randomized clinical studies (n=124). Four randomized controlled studies were performed in patients with generalized anxiety disorder and anxiety neurosis and six randomized trials were conducted in patients with status anxiety. Majority of these trials were of short duration (4-6 weeks). Studies include TCM, western acupuncture and acupressure acupuncture and evaluated in conjunction with other treatments such as medication, behavioral desensitization, biofeedback and relaxation etc. Patients in adults and in children with anxiety or anxiety disorders both inpatient hospital outpatients' facilities. The results in patients with generalized anxiety indicate that efficacy of acupuncture is

comparable to drug therapy.

Brattberg [18] explores Emotional Freedom Technique results with fibromyalgia, and Karatzias et al. [19], to investigate Tapping results with stress disorder, both used the Hospital Anxiety and Depression Scale and found very significant improvement before/after treatment both anxiety and depression. An RCT investigating the use of Tapping in depression therapy students showed that after four group sessions 90 minutes on foot, the average score for the Inventory Depression Beck went well at the range moderate depression to well below the cutoff depression. The strong statistical significance and the large size of the effect in this study confirms a number of uncontrolled reductions up studies on depression and other measures following group Tapping treatments. Surprising finding was reported by Rowe [20] was that a watch Tapping workshop produced significant, permanent reductions in the severity of the psychological symptoms of participants, as measured on a standardized self-report. Rowe [20] used the Symptom Checklist to measure global changes in psychological functioning after participating in 18-hour Tapping weekend workshop where participants self-implemented method as a way of learning is. The Symptom Checklist was administered to 102 participants one month before the workshop at the beginning of the workshop, at the end of the workshop, a month after the meeting, and six months after the workshop. A highly significant reduction ( $p < 0.0005$ ), found from the pre-workshop to post-workshop for all measures of psychological distress (depression, anxiety, compulsive, somatization, hostility, paranoia, interpersonal sensitivity, phobic anxiety and psychotism). Scores decreased anxiety continued during the six-month follow-up. Subsequent studies by the Church & Brooks [21] and Hoffman & Brooks [22] which was built in this project support and multiply the findings of Rowe. Church & Brooks [21] administered Symptom Checklist at healthcare workers in five separate conferences. Each participated as part of the conference, a Tapping workshop four hours that included two hours training and two hours of self-application. As with Rowe, differences before and after test was highly significant ( $p < 0.001$ ) for symptom scales, and global severity index. Also, as with Rowe, most of the improvements carried out in the monitoring. Hoffman & Brooks [22] granted participants in four additional Tapping workshops (n=207), each driven by a different professional, and also found strong pre and post differences immediately after the workshop ( $p < .001$ ) and for monitoring.

## Conclusion

The findings of these studies suggest that acupuncture could play an important role in the treatment of depression and anxiety.

## References

1. Ernst E (2001) The desktop guide to complementary and alternative medicine: An evidence base approach. Mosby, New York, USA.
2. Hou DF (1996) Clinical observation of therapeutic effect of baihui (GV20)-yintang (EX-HN3) electro-acupuncture in 30 cases of post-

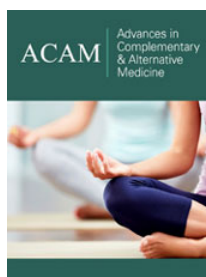
- apoplectic depression. *Chinese Acupuncture and Moxibustion* 16(8): 432-433.
3. Li CD (1994) Treating post-stroke depression with antidepressive acupuncture therapy: A clinical study of 21 cases. *International Journal of Clinical Acupuncture* 5(4): 389-393.
  4. Luo HC, Jia YK, Li Z (1985) Electro-acupuncture vs amitriptyline in the treatment of depressive states. *J Tradit Chin Med* 5(1): 3-8.
  5. Luo HC (1988) Clinical observation of electro-acupuncture on 133 patients with depression in comparison with tricyclic amitriptyline. *Chinese Journal of Integrated Traditional and Western Medicine* 8(2): 77-80.
  6. Yang X (1994) Clinical observation of needling extrameridian points in treating mental depression. *J Tradit Chin Med* 14(1):14-18.
  7. Zhang B (1996) A control study of clinical therapeutic effects of laser-acupuncture on depressive neurosis. *World Journal of Acupuncture-Moxibustion* 6(2):12-17.
  8. Reed RV, Fazel M, Goldring L (2012) Post-traumatic stress disorder. *BMJ* 344: e3790.
  9. Shalev AY (2001) What is posttraumatic stress disorder? *J Clin Psychiatry* 62(suppl 17): 4-10.
  10. Hollifield M, Sinclair-Lian N, Warner TD, Hammerschlag R (2007) Acupuncture for posttraumatic stress disorder: A randomized controlled pilot trial. *J Nerv Ment Dis* 195(6): 504-513.
  11. Kim YD1, Heo I, Shin BC, Crawford C, Kang HW, et al. (2013) Acupuncture for post-traumatic stress disorder: a systematic review of randomized controlled trials and prospective clinical trials. *Evid Based Complement Alternat Med* 2013: 615857.
  12. Hui K, Liu J, Makris N, Gollub RL, Chen AJ, et al. (2000) Acupuncture modulates the limbic system and subcortical gray structures of the human brain: evidence from fMRI studies in normal subjects. *Hum Brain Mapp* 9(1): 13-25.
  13. Hui K, Liu J, Marina O, Napadow V, Haselgrove C, et al. (2005) The integrated response of the human cerebro-cerebellar and limbic systems to acupuncture stimulation at ST 36 as evidenced by fMRI. *Neuroimage* 27(3): 479-496.
  14. Smith CA, Hay PP, MacPherson H (2010) Acupuncture for depression. *Cochrane Database of Syst Rev* 2010(1): CD004046.
  15. Mukaino Y, Park J, White A, Ernst E (2005) The effectiveness of acupuncture for depression-a systematic review of randomised controlled trials. *Acupunct Med* 23(2): 70-76.
  16. Leo JR, Ligot (2007) A systematic review of randomized controlled trials of acupuncture in the treatment of depression. *J Affect Disord* 97(1-3): 13-22.
  17. Pilkington K, Kirkwood G, Rampes H, Cummings M, Richardson J (2007) Acupuncture for anxiety and anxiety disorders: a systematic literature review. *Acupunct Med* 25(1-2): 1-10.
  18. Brattberg G (2008) Self-administered EFT (ss) in individuals with fibromyalgia: A randomized trial. *Integrative Medicine: A Clinician's Journal* 7(4): 30-35.
  19. Karatzias, T, Power K, Brown K, McGoldrick T, Begum M, et al. (2011) A controlled comparison of the effectiveness and efficiency of two psychological therapies for posttraumatic stress disorder: Eye movement desensitization and reprocessing vs. emotional freedom techniques. *J Nerv Ment Dis* 199(6): 372-378.
  20. Rowe JE (2005) The effects of EFT on long-term psychological symptoms. *Counseling and Clinical Psychology* 2(3): 104-111.
  21. Church D, Brooks AJ (2010) The effect of a brief EFT (Emotional Freedom Techniques) self-intervention on anxiety, depression, pain and cravings in healthcare workers. *Integrative Medicine: A Clinician's Journal* 9(5): 40-44.
  22. Hoffman PJ, Brooks AJ (2011) Psychological symptom change after group application of Emotional Freedom Techniques (EFT). *Energy Psychology: Theory, Research, & Treatment* 3(1): 33-38.



Creative Commons Attribution 4.0  
International License

For possible submissions Click Here

[Submit Article](#)



## Advances in Complementary & Alternative Medicine

### Benefits of Publishing with us

- High-level peer review and editorial services
- Freely accessible online immediately upon publication
- Authors retain the copyright to their work
- Licensing it under a Creative Commons license
- Visibility through different online platforms



## Quality of life in kidney transplantation

Maria Dalamagka

### ► To cite this version:

Maria Dalamagka. Quality of life in kidney transplantation. International Journal of Anesthesia and Relaxation, 2018, 10.13140/RG.2.2.36805.15846 . hal-05432543

**HAL Id: hal-05432543**

**<https://hal.science/view/index/docid/5432543>**

Submitted on 26 Dec 2025

**HAL** is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



Distributed under a Creative Commons Attribution 4.0 International License



## The Quality of Life in Kidney Transplantation

Dalamagka Maria

Anaesthesiology Department, General Hospital of Larissa

### Abstract

The health-related quality of life (HRQL) contains many aspects under patients' health as physical, psychological, social function and a general prosperity. Numerous clinical studies have established the importance of quality of life in various diseases. It is extremely popular to assess the quality of life in clinical trials as a measure of the subjective state of health of patients. The HRQL, also gradually recognized as an important measure of outcome after organ transplantation. Along with other indicators related to the improvement of patient and graft survival, quality of life has been assessed as a valid outcome measure. Investigations on the quality of life are aimed to lead to a broader view of subjective health, consider that health is a puzzle of the general welfare. Kidney transplantation is the treatment of choice in end stage renal failure. Progress in renal transplantation and immunosuppressive therapies have increased significantly in recent decades, resulting in allograft survival rates at one year is now over 90%. The main goal of transplantation is to achieve maximum quality and longevity while minimizing the impact of disease and health care costs. In general, the quality of life improved after successful renal transplantation compared with patients on dialysis, and the result was more pronounced in men than in women. In these studies it seems clear that renal transplantation is not only cheaper replacement therapy over time, but also linked to lower mortality and better quality of life for patients.

### Corresponding author: Dalamagka Maria

Anaesthesiology Department, General Hospital of Larissa, 41221, Greece

E-mail: [mary.dalamagka@gmail.com](mailto:mary.dalamagka@gmail.com)

**Citation:** Dalamagka Maria, (2018), The Quality of Life in Kidney Transplantation. Int J Ane & Rel. 2-2, 12-18

**Copyright:** ©(2018) Dalamagka Maria. This is an open-access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.

**Received:** June 13, 2018

**Accepted:** June 23, 2018

**Published:** July 30, 2018

### Introduction

As already described in the 80s by Mulley [106], there are many pitfalls in assessing the quality of life. Mulley argues that defining "health" is difficult and can be affected by a large number of variables [4]. The health-related quality of life (HRQL) contains many aspects under patients' health as physical, psychological, social function and a general well-being [11-13]. Numerous clinical studies have established the importance of quality of life in various diseases, and is extremely popular to assess the quality of life in clinical trials as a measure of the subjective state of health of patients. The HRQL also gradually recognized as an

important outcome measure after organ transplantation. Along with other indicators related to the improvement of patient and graft survival, quality of life has been assessed as a valid outcome measure. Investigations on the quality of life are aimed to lead to a broader view of subjective health, consider that health is a puzzle of the general welfare. The pieces of this puzzle consisting of psychological and social aspects of well-being, in addition to physical and mental health. Some of these pieces are evaluated either subjective or objective basis to some of them in two dimensions [1].

Kidney transplantation is the treatment of choice in end stage renal failure. The transplantation may be the focus for increasing survival and to maximize quality of life (Muehrer and Becker 2005). However, there are certain factors that may affect the quality of life after transplantation, such as side effects from highly immunosuppressive drugs, the presence of common disease states and the possibility of rejection. Patients with end stage renal failure have compromised symptomatology (Davison 2006, Murphy 2009) with the most common symptoms are fatigue, sleep disturbances, loss of appetite, itching, pain and impaired well-being. Other symptoms that have been reported are cramping, abdominal pain, palpitations, edema (Murtagh 2007). Symptoms defined by the gravity of the intensity, their duration time and frequency, and the degree of discomfort (Jablonski 2007) [5].

Progress in renal transplantation and immunosuppressive therapies have increased significantly in recent decades, resulting in allograft survival rates at one year is now over 90%. The main goal of transplantation is to achieve maximum quality and longevity while minimizing the impact of disease and health care costs. Examples of such measurements are weighted on the quality of life years gained, years of life without disease, or the equivalent healthy years per unit cost of care. In renal transplantation the cost of care is not limited to

Citation: Dalamagka Maria, (2018), The Quality of Life in Kidney Transplantation. Int J Ane & Rel. 2-2, 12-18

the transplant procedure, and in the treatment of side effects, some of which are caused by the immunosuppressive therapy. After the first successful kidney transplant in the early 1950s, the immunosuppressive therapies significantly improved, as the most revolutionary development was the introduction of cyclosporine in early 1980. Along with better patient care and new immunosuppressive regimens, mean renal survival allografts are constantly increasing. Step by step with these advances, greater attention has been given to long-term QOL. However, so far the HRQL was evaluated only in a limited number of clinical studies, as subjective health status [1].

However, it is generally accepted that patients with a functional renal allograft have improved quality of life, compared to patients on dialysis [14], [15].

To assess the effects of the disease on quality of life, specific evaluation tools. These tools are sensitive enough to identify the disease changes, but are not suitable for comparing different diseases. Special tools to evaluate the quality of life in kidney transplant is the kidney transplant questionnaire (KTQ) [16], the quality of life in renal disease (KDQOL) [17] and the final renal failure in the transplantation center stage symptoms checklist (ESRDSC-TM) [18].

The KTQ contains 26 questions in five areas (somatic symptoms, depression, fatigue, interpersonal relations, frustration) each of which can be rated on a scale from 1 to 7, wherein the lower score represents the lowest quality of life (QOL). For ultimately all points summarize therefore the maximum score is 182 points, and the lower 26 degrees. Also, the questionnaire should be in the mother tongue of the patient [19].

The KDQOL originally developed for patients with chronic renal failure and those undergoing dialysis. However recent documents use this tool to assess patients after transplantation, as well as to compare these patients with patients on dialysis [20]. The original KDQOL covering 11 areas with different number of elements. The symptoms / problems area includes 34 items, the effect of renal disease on daily life contains 20 items, the burden of kidney disease four elements, cognitive function 6 elements, the employment situation four elements, the quality of social interaction 4 data, sleep quality 9 items, social support 4 data and patient satisfaction two elements. The choices of answers is a Likert scale, where higher score represents better quality of life.

ESRDSC-TM [1] specifically developed to assess the effects of immunosuppressive drugs on quality of life. The questions are rated on a five-point Likert scale, where again higher scores representing better quality of life (QOL).

The authors [1] tested over 400 transplant patients and reviewed the retest correlation in a subset of 88 patients in one year and found sufficient validity. So far no method has proven to be ideal for measuring the quality of life in all circumstances. It has been shown that different quality of life results can be obtained in the same population, though different tools used. General tools used for comparisons between groups and studies to evaluate the effects of various diseases in the quality of life (QOL). These tools are used in research and are as follows: the profile of the effects of the disease -Sickness Impact Profile (SIP), the modified data 36 Overview of medical outcomes (36-item short-form of Medical Outcomes Survey, SF-36), and the Nottingham health profile (NHP)

With more than 200 publications, the SF-36 [1] is one of the most widely used tools for evaluating the quality of life. The SF-36 questionnaire is an independent research, contains 36 elements and requires a few minutes to complete. Includes a multidimensional scale that assesses eight health areas: 1) limitations in physical activities because of health

problems, 2) limitations in social activities because of physical or emotional problems, 3) limitations in usual activities due to disruption of physical health, 4) physical pain, 5) general mental health, 6) limitations in usual activities due to emotional problems, 7) vitality, 8) general perceptions of health. It has proven the validity of [28], the sensitivity to changes of therapy [29], and the suitability of patients with end stage renal failure [30].

The health-related quality of life is increasingly important outcome measures, especially in chronic diseases. The quality of life expressed as a quantitative measure of the ability of preference of the person on an outcome, which is also defined as associated with the outcome of the individual utility [4].

When evaluating studies using methods based on preference (preference-based), kidney transplantation is associated with a higher quality of life than dialysis and peritoneal dialysis [81-84].

TTO index (time balancing, time trade-off) [4] originally developed by Torrance and colleagues [85] and tested for reliability and validity in patients with end stage renal failure by Churchill and his associates [81]. The question posed to the patient is thought to renal disease in the last 2-3 weeks and then decide between two hypothetical options: either to remain in his current state of health, with a specific life expectancy, or exchange with a number of years to live in full health.

The SG method (standard risk, standard gamble) [4] from the expected utility and require the patient to take a decision involving either stay in the current mode, or can be subjected to a hypothetical treatment [86]. This therapy has two possible outcomes: 1) there is the possibility of immediate death, and 2) whether the patient will survive cured and live in full health.

The EQ-5D (EuroQol-5D) [4] consists of two parts. The first part, the EQ-5D profile consists of five elements: mobility, self-care, usual activities, pain / discomfort, and anxiety / depression, each with three levels of functionality: no problem, some problems and extreme problems. The second part of the questionnaire is the EQ-VAS, which ranks health from 0 (worst health state imaginable) to 100 (best health state you can imagine).

The HUI (index usefulness of health) [4], the most recent version of which is HUI3, which identifies health with eight features: ambulation, dexterity, cognition, emotion, pain and discomfort, vision, hearing and speech.

The general quality of life measures [5] is as follows: SF-36, SF-20, SF-12, QWB-SA, SIP. The utilitarian health measurements include the EQ-5D, the SF-6D, and the HUI. Specific measurements of quality of life in renal disease using multidimensional specific questionnaires, such as for life quality clearance Index (QLI-D), the quality of life in kidney hospital Long form (KDQOL-LF), the quality of life in kidney hospital Short form (KDQOL-SF), the questionnaire renal disease (KDQ), Questionnaire renal transplantation (KTQ), the profile of quality of life in renal disease (RQLP), Questionnaire CHOICE health experience (CHEQ) and Survey of quality of life, personalized for renal disease. Evidence response elements the SF-36 was found in a study with patients after renal transplant patients receiving other immunosuppressive drugs (Russ 2007). In measurements the SF-36 as part of KDQOL, patients with renal transplantation showed higher scores in the areas of general health and vitality compared with patients on the waiting list (Sureshkumar 2005).

The SF-20 [5] assesses health in six areas, namely, body pain, general health perception, physical functioning, mental health, social functioning and functionality of the individual's role. Meers and his colleagues (1992) using the SF-20 in patients on hemodialysis,

peritoneal dialysis and renal transplantation were significantly higher rating in all areas for patients with renal transplantation.

The profile of the disease impact<sup>[5]</sup> has 12 sectors, reflecting the focus of disability on quality of life with a total score ranging from 0 to 100. The physical, psychosocial and total score can be calculated.

Statistically significant difference of the scores was observed before and after transplantation in 293 patients as expected (Cetingok and colleagues, 2004)

Indicator D for quality of life (QLI-D)<sup>[5]</sup> developed in the USA during the 1980s as a measure of morbidity (Ferrans and Powers, 1985). A statistically significant difference was observed in patients before and after transplantation, as expected (Cetingok 2004).

The renal transplant Questionnaire (KTQ)<sup>[5]</sup> was developed by Laupacis and coworkers (1993). It includes five areas: physical symptoms, fatigue, uncertainty / fear, appearance and emotions. There were significant differences in the patients' rating before and after transplantation. Scores of KTQ showed great sensitivity in a group of patients with renal transplant receiving other immunosuppressive therapy (Russ et al, 2007).

## Results

In 1998 Matas<sup>[21]</sup> and his colleagues described the QOL and to evaluate used the SF-36. The authors were able to have at their disposal 446 patients evaluated once, 632 twice evaluated and 53 patients evaluated three times. The patients were referred to the time frame between the first and tenth year after transplantation. The rating of the SF-36 was not significantly changed in recent years after transplantation and were consistently lower compared to the normal population of the US. It is interesting to note that diabetic and non-diabetic patients had the same score for mental health scales, while non-diabetics had better score in physical function and general health.

Matas<sup>[22]</sup> and his colleagues recently published a temporal relationship between adverse effects of immunosuppressive drugs in kidney transplant recipients and QOL. In this huge study 4247 patients were enrolled and evaluated by a QOL questionnaire. The authors conducted a multivariate analysis, which showed that emotional problems, reduced sexual interest and headache were the main factors that negatively affected the QOL in these patients. Preliminary data from the self-reported health information program entitled "Transplant Training Centre" published by Hricik<sup>[23]</sup> and his associates. The first results obtained from 3676 patients were similar to those published in the final report a year later than 4247 patients.

Franke<sup>[24]</sup> and his colleagues evaluated the HRQL in patients with end stage renal failure. The study investigated the differences in quality of life in patients on the waiting list for a kidney transplant, while maintained on dialysis, and kidney transplant recipients. The result was evaluated by the SF-36 questionnaire and special tools for disease (End Stage Renal Disease Symptom Checklist-Transplantation Module). In this study, the group of 80 hemodialysis patients at the waiting list for a kidney transplant showed decreased satisfaction with social support, while 222 patients after successful renal transplantation, increased social support. Similarly, psychological distress was higher among patients maintained on dialysis, compared with transplant patients.

In a similar study by Jofre<sup>[14]</sup> and his colleagues also observed improvement in 88 of 93 patients after successful renal transplantation. The authors<sup>[1]</sup> used the Karnovsky scale and the Sickness Impact Profile (in the sickness impact profile) as assessment tools. Note that although each patient after transplantation showed improved rating, particularly the male population showed significant improvement in the global rating. As expected, older patients with more comorbidities

showed less improvement than younger ones.

In a randomized, open-label study in Europe, Australia and Canada, Oberbauer<sup>[25]</sup> and his colleagues investigated the outcomes of quality of life in patients after renal transplantation. In this study, 430 patients with renal transplantation, randomized three months after transplant, or continue the treatment with cyclosporin and sirolimus, or be withdrawn from the cyclosporine in a four week period. Quality of life was measured after randomisation in one and two years after the transplant, using specially tools for disease, such as KTQ and the SF-36 questionnaire. The vitality scores were higher in accordance with the SF-36 questionnaire in free ciclosporin group at two years compared to baseline, but was reduced in the combination group.

In a meta-analysis conducted by Ylian Liem<sup>[2]</sup> and his colleagues included 52 studies (Level 2b evidence according to based on the evidence the medical classification of the Oxford Centre<sup>[18]</sup>) and the quality of life was examined in 92 patient groups with renal replacement therapy, as assessed by the SF-36<sup>[31,32,33,34-80]</sup>, also the quality of life was examined in 44 groups of hemodialysis patients (30 372 patients), 20 groups of patients with peritoneal dialysis (3262 patients) and 28 groups of patients with renal transplantation (2948 patients). The majority of patients were men and so there was no statistically significant difference in the gender distribution among the three treated groups. The prevalence of diabetes was 24% in dialysis group, 17% in the PD group and 7% in the group with renal transplant patients, with a significant difference in the comparison between patients on dialysis and patients with renal transplantation. The average treatment time was 44.1 months for patients on dialysis, 24.3 months for patients on peritoneal dialysis and 63.8 months for patients with renal transplantation. Comparing all treatment groups, dialysis patients and patients with renal transplantation were significantly different average time for treatment. In addition to mental health, quality of life as assessed by the SF-36 questionnaire was higher among patients with renal transplantation compared to dialysis patients.

In a meta-analysis of the emotional distress and psychological well-being in hemodialysis, peritoneal dialysis and after renal transplant patients, the Cameron<sup>[6]</sup> and his colleagues reported comparable differences in quality of life.

Raiesifar<sup>[3]</sup> and his colleagues conducted a descriptive, analytical study, conducted in 220 renal transplant patients in clinical transplantation and nephrology two selected hospitals of Tehran city in the year 2009. A kidney transplant questionnaire (KTQ- 25) was used for data collection. The reliability KTQ-25 was set at 0.93 by Cronbach Alpha method and the questionnaire was completed by patients. The result of the study showed that the level of QOL of renal transplant patients is moderate.

The 27 studies included in the meta-analysis by Ylian Liem<sup>[4]</sup> was at least level 2b evidence in accordance with Based on the evidence medical classification Oxford center<sup>[24]</sup>. Quality of life was assessed by a single measure in most studies: three studies used the VAS (visual analogue scale)<sup>[87,91,92]</sup>, eight studies the TTO index (time balancing, time trade-off)<sup>[82,89,93 -98]</sup>, a study of the SG method (standard risk, standard gamble)<sup>[104]</sup>, and nine studies the EQ-5D (EuroQol-5D)<sup>[83,84,88,90,99-103]</sup>

The average utility rate of patients on hemodialysis and peritoneal dialysis was significantly lower than in patients with renal transplantation, when comparing the EQ-5D index. Mean values of the index TTO patients on hemodialysis, peritoneal dialysis and renal transplantation were not statistically significantly different, although the usefulness of in hemodialysis patients tended to be lower compared to peritoneal dialysis patients and renal transplant patients.

The conclusion of a systematic review and meta-analysis of alternative forms of renal replacement therapy is that there are no statistically significant differences, although the quality of life tended to be higher for patients with renal transplantation and lower in hemodialysis patients.

Cameron <sup>[105]</sup> and his colleagues reported fewer emotional disorders and greater psychological well-being for patients with renal transplantation compared with patients on hemodialysis and peritoneal dialysis.

The Dylan Smith <sup>[6]</sup> and his colleagues evaluated 307 patients from a waiting list for a cadaveric transplant or kidney-pancreas transplant and 195 patients one year after a successful transplant. A subsample interviewed patients before and after transplantation. The survey included measurements of quality of life, ie a comprehensive assessment and even individual sectors, including health, work and activities.

The results showed that quality of life improved after transplantation, but projections of patients before transplantation overestimated the magnitude of improvement. In addition, patients had predicted great improvements in specific areas of their lives, who have not changed. These results were confirmed by analysis perspective. Further analyzes showed that patients after renal transplantation, when asked to do recall their memory and to rate their quality of life before the transplant received lower scores compared to that were mentioned before transplantation. According to the bias effects, patients greatly overestimated the benefits of a successful kidney transplant, both in terms of the forecasts were made for their life after treatment, and memories preserved for their lives before transplantation.

Shah <sup>[7]</sup> used a previously validated scale score (QLS), in which patients directly asked about their quality of life, as follows: "Considering all aspects of your life, physical, emotional, social, spiritual and economic in the last two days, the quality of your life is ... ". The QLS ranging from 0 ( "very bad") to 10 ( "excellent"). Patients came when he was free from serious medical problems. Fifty renal transplant recipients involved. Psychosocial and medical variables included the inventory index of depressed mood (Beck Depression Inventory), the effects questionnaire sickness, the Multidimensional social support scale (Multidimensional Scale of Perceived Social Support), the time of transplantation, age, creatinine, hemoglobin and albumin levels. Of the patients 64% were African-American and 48% were women. The perception of a better quality of life is associated with lower rates of depression and minor effects of the disease and a higher perception of social support and life satisfaction. The life quality perception is not associated with age, the time of transplantation, creatinine, hemoglobin or albumin levels. The conclusion was that the QLS is a quick measurement tool for the subjective quality of life in renal transplant recipients and the correlation with psychosocial factors, interest in this patient group.

The average number of patients after renal transplantation displays an improved quality of life, but this is not the case in all recipients. The purpose of the study, conducted by Bohlke <sup>[8]</sup> and his colleagues was to identify factors associated with quality of life after renal transplantation. The study population was assessed by the SF-36 were analyzed 272 renal transplant recipients operating. Hypertension, diabetes, high serum creatinine levels and low hematocrit were independently and significantly associated with a lower rating for the SF-36. The rating for the physical component was worse for women, for patients with low income, the unemployed and patients with higher serum creatinine. Among the variables studied, comorbidity and graft function were the main factors associated with the physical component, and socio-demographic variables, and graft function were

the main factors that determine the mental component. Additional factors such as the personality and the environment can affect the quality of life.

In the general population and in patients on dialysis, sleep disorders affect health-related quality of life. Eryilmaz <sup>[9]</sup> examined the prevalence of these sleep disorders in patients with renal transplantation, and the relationship between sleep quality and related quality of life health. The Pittsburgh Sleep Quality Index (PSQI) for measuring the quality of sleep, the WHOQOL-BREF index for quality of life and the inventory ratio of depressed mood (Beck Depression Inventory) were applied to 100 patients with renal transplantation. Thirty (30%) subjects had severe sleep disorders ( "poor sleepers"). Sleep disorders are not so common in renal transplantation compared to dialysis patients, but definitely have a greater rate than the general population. Sleep disorders appear to be part of the symptomatology of depression. The severity of depression and low educational level play a major role in the quality of life of these patients, despite the quality of sleep.

A controlled program live unrelated donors (LURDs) kidney transplant began in 1988 in Iran. Nejatisafa <sup>[10]</sup> examined the LURDs in order to investigate the size of the stressful situations in which they were patients before donation and their quality of life after donation. Five hundred donors participated. The study included donors, in which the donation was made at least three months before the study. The scale Paykel Life Events and the short version of the global organization quality of life (WHOQOL-Bref) used in the study. Complete data were available for 424 donors. 95% of respondents reported at least a stressful situation during the six months before the kidney donation. The three most common events in their life that trigger the stress experienced in their daily life were rising costs of living, low income and household tasks. The most stressful events were the loss of their job, financial problems and the death of a member of their family. Participants reported more stressful events with an average total score of stress double compared to the findings of a study preceding the normal population. The rating of donors in all four areas of the WHOQOL-BREF was low. The quality of life of the donors may be low and may be at risk of experiencing more stressful events in their lives. For this reason, medical supervision should be continued following the donation, with emphasis on mental health and psychosocial problems.

## Conclusions

There is evidence for the use of the SF-36 <sup>[5]</sup> in patients with chronic renal failure. Yet more evidence is required before widespread use to evaluate the outcome and quality. The EQ-5D <sup>[5]</sup> is favored among the preferred measurements, as there is more evidence, but this ratio applicable to the above. Since multidimensional specific renal disease measurements in KDQOL includes most evidence. Given this overlap between the SF-36 and KDQOL, there is some benefit when used in the same survey, unlike the combination of EQ-5D and KDQOL provides additional information about the perception of patients for kidney disease. While the idea of using a short questionnaire based on severity would be very good, the main benefit would only control or recognition of symptoms. The SF-36 <sup>[5]</sup> is the only general measurement with good properties and functional characteristics. Furthermore psychometric criteria can be reproduced when administered as autonomous as possible, and when used in conjunction with KDQOL questionnaire. The EQ-5D appears to have a favorable use, since three of the four studies that used the EQ-5D were conducted in the UK. The evidence show high response rates.

Although, clinical studies relating to quality of life after kidney transplantation is relatively rare, the few published studies yielded rather similar results. In general, the quality of life improved after

successful renal transplantation compared with patients on dialysis, and the result was more pronounced in men than in women. In these studies it seems clear that renal transplantation is not only cheaper replacement therapy over time, associated with a lower mortality and quality of life of patients. <sup>[1,26,27]</sup>

## References

1. Wolfgang Fiebiger, Christa Mitterbauer and Rainer Oberbauer: Health-related quality of life outcomes after kidney transplantation. *Health and Quality of Life Outcomes* 2004, 2:2, <http://www.hqlo.com/content/2/1/2>
2. Ylian S. Liem, MD, MSc, Johanna L. Bosch, PhD, Lidia R. Arends, MSc, MA, Majanka H. Heijlenbroek-Kal, PhD, M. G. Myriam Hunink, MD, PhD : *Quality of Life Assessed with the Medical Outcomes Study Short Form 36-Item Health Survey of Patients on Renal Replacement Therapy: A Systematic Review and Meta-Analysis*. *Value in Health*, Volume 10, Issue 5, Pages 390-397
3. Tayebi A. , Raiesifar A. , Ebadi A. , Eynollahi B. , Rafiyan Z. , Keyvanloo F. , *Review the renal transplantation patients' quality of life by using kidney transplantation questionnaire (KTQ-25)* *Iranian Journal of Critical Care Nursing*, 2010; 3 (3) :13-14
4. Ylian S. Liem, MD, MSc, Johanna L. Bosch, PhD, M. G. Myriam Hunink, MD, PhD: *Preference-Based Quality of Life of Patients on Renal Replacement Therapy: A Systematic Review and Meta-Analysis*. *Value in Health*, Volume 11 , Number 4 , Pages 733-741, 2008
5. Elizabeth Gibbons , Ray Fitzpatrick: *A STRUCTURED REVIEW OF PATIENT-REPORTED OUTCOME MEASURES FOR PEOPLE WITH CHRONIC KIDNEY DISEASE*, 2010, Patient-reported Outcome Measurement Group , Department of Public Health , University of Oxford (<http://phi.uhce.ox.ac.uk/>)
6. Dylan Smith, George Loewenstein, Christopher Jepson, Aleksandra Jankovich, Harold Feldman, Peter Ubel : *Mispredicting and misremembering: patients with renal failure overestimate improvements in quality of life after a kidney transplant*. *Health psychology : official journal of the Division of Health Psychology*, *American Psychological Association* 2008;27(5):653-8.
7. V S Shah, A Ananth, G K Sohal, W Bertges-Yost, A Eshelman, R K Parasuraman, K K Venkat : *Quality of life and psychosocial factors in renal transplant recipients*. *Transplant Proc.* 2006 Jun;38(5):1283-5.
8. Bohlke M, Marini SS, Rocha M, Terhorst L, Gomes RH, Barcellos FC, Irigoyen MC, Sesso R : *Factors associated with health-related quality of life after successful kidney transplantation: a population-based study*. *Qual Life Res.* 2009 Nov;18(9):1185-93. Epub 2009 Sep 10.
9. Eryilmaz MM, Ozdemir C, Yurtman F, Cilli A, Karaman T : *Quality of sleep and quality of life in renal transplantation patients*. *Transplant Proc.* 2005 Jun;37(5):2072-6.
10. Nejatisafa AA, Mortaz-Hedjri S, Malakoutian T, Arbabi M, Hakemi MS, Haghighi AN, Mohammadi MR, Fazel I : *Quality of life and life events of living unrelated kidney donors in Iran: a multicenter study* . *Transplantation*. 2008 Oct 15;86(7):937-40.
11. Patrick DL, Chiang YP: *Measurement of health outcomes in treatment effectiveness evaluations: conceptual and methodological challenges*. *Med Care* 2000, 38:1114-25.
12. Revicki DA, Osoba D, Fairclough D, Barofsky I, Berzon R, Leidy NK, Rothman M: *Recommendations on health-related quality of life research to support labeling and promotional claims in the United States*. *Qual Life Res* 2000, 9:887-900.
13. Testa MA, Simonson DC: *Assesment of quality-of-life outcomes*. *N Engl J Med* 1996, 334:835-840.
14. Jofre R, Lopez-Gomez JM, Moreno F, Sanz-Guajardo D, Valderrabano F: *Changes in quality of life after renal transplantation*. *Am J Kidney Dis* 1998, 32:93-100.
15. Keown P: *Improving quality of life—the new target for transplantation*. *Transplantation* 2001, 72:S67-74).
16. Laupacis A, Muirhead N, Keown P, Wong C: *A disease-specific questionnaire for assessing quality of life in patients on hemodialysis*. *Nephron* 1992, 60:302-306.
17. Hays RD, Kallich JD, Mapes DL, Coons SJ, Carter WB: *Development of the kidney disease quality of life (KDQOL) instrument*. *Qual Life Res* 1994, 3:329-338.
18. Franke GH, Reimer J, Kohnle M, Luetkes P, Maehner N, Heemann U: *Quality of life in end-stage renal disease patients after successful kidney transplantation: development of the ESRD symptom checklist-transplantation module*. *Nephron* 1999, 83:31-39
19. Rebollo P, Ortega F, Ortega T, Valdes C, Garcia-Mendoza M, Gomez E: *Spanish validation of the "Kidney Transplant Questionnaire": a useful instrument for assessing health related quality of life in kidney transplant patients*. *Health Qual Life Outcomes* 2003, 1:56.
20. Bakewell AB, Higgins RM, Edmunds ME: *Does ethnicity influence perceived quality of life of patients on dialysis and following renal transplant?* *Nephrol Dial Transplant* 2001, 16:1395-1401.
21. Matas AJ, McHugh L, Payne WD, Wrenshall LE, Dunn DL, Gruessner RW, Sutherland DE, Najarian JS: *Long-term quality of life after kidney and simultaneous pancreas-kidney transplantation*. *Clin Transplant* 1998, 12:233-242
22. Matas AJ, Halbert RJ, Barr ML, Helderma JH, Hricik DE, Pirsch JD, Schenkel FA, Siegal BR, Liu H, Ferguson RM: *Life satisfaction and adverse effects in renal transplant recipients: a longitudinal analysis*. *Clin Transplant* 2002, 16:113-121)
23. Hricik DE, Halbert RJ, Barr ML, Helderma JH, Matas AJ, Pirsch JD, Schenkel FA, Siegal B, Ferguson RM: *Life satisfaction in renal transplant recipients: preliminary results from the transplant learning center*. *Am J Kidney Dis* 2001, 38:580-587
24. Franke GH, Reimer J, Philipp T, Heemann U: *Aspects of quality of life through end-stage renal disease*. *Qual Life Res* 2003, 12:103-115
25. Oberbauer R, Hutchison B, Eris J, Arias M, Claesson K, Mota A, Kreis H, Kleinman L, Wang F, Chen J, Revicki DA: *Health-related quality-of-life outcomes of sirolimus-treated kidney transplant patients after elimination of cyclosporine A: results of a 2- year randomized clinical trial*. *Transplantation* 2003, 75:1277-1285)
26. Evans RW, Manninen DL, Garrison LP Jr, et al. *The quality of life of patients with end-stage renal disease*. *N Engl J Med* 1985;312:553-9.
27. Rebollo P, Ortega F. *New trends on health related quality of life assessment in end-stage renal disease patients*. *Int Urol Nephrol* 2002;33:195-202.
28. Gomez-Besteiro MI, Santiago-Perez MI, Alonso- Hernandez A, et al. *Validity and reliability of the SF-36 questionnaire in patients on the waiting list for a kidney transplant and transplant patients*. *Am J Nephrol* 2004;24:346-51. *Renal Replacement Therapy Patients' Quality of Life* 395
29. Levin NW, Lazarus JM, Nissenson AR. *National Cooperative rHu Erythropoietin Study in patients with chronic renal failure—an interim report*. *The National Cooperative rHu Erythropoietin Study Group*. *Am J Kidney Dis* 1993;22:3-12.
30. Kurtin PS, Davies AR, Meyer KB, et al. *Patient-based health status measures in outpatient dialysis. Early experiences in developing an outcomes assessment program*. *Med Care* 1992;30:MS136-49.
31. DeOreo PB. *Hemodialysis patient-assessed functional health status predicts continued survival, hospitalization, and dialysis-attendance compliance*. *Am J Kidney Dis* 1997;30:204-12
32. Diaz-Buxo JA, Lowrie EG, Lew NL, et al. *Quality-of-life evaluation*

using Short Form 36: comparison in hemodialysis and peritoneal dialysis patients. *Am J Kidney Dis* 2000;35:293–300.

33. Kurtin PS, Davies AR, Meyer KB, et al. Patient-based health status measures in outpatient dialysis. Early experiences in developing an outcomes assessment program. *Med Care* 1992;30:MS136–49.

34. Shield CF III, McGrath MM, Goss TF. Assessment of health-related quality of life in kidney transplant patients receiving tacrolimus (FK506)-based versus cyclosporine-based immunosuppression. FK506 Kidney Transplant Study Group. *Transplantation* 1997;64:1738–43.

35. Arogundade FA, Zayed B, Daba M, Barsoum RS. Correlation between Karnofsky Performance Status Scale and Short-Form Health Survey in patients on maintenance hemodialysis. *J Natl Med Assoc* 2004;96:1661–7.

36. Baiardi F, Degli Esposti E, Cocchi R, et al. Effects of clinical and individual variables on quality of life in chronic renal failure patients. *J Nephrol* 2002;15: 61–7.

37. Bro S, Bjorner JB, Tofte-Jensen P, et al. A prospective, randomized multicenter study comparing APD and CAPD treatment. *Perit Dial Int* 1999;19:526–33.

38. Buemi M, Caccamo C, Floccari F, et al. Correlation between quality of life assessment and a personality neurobiologic model in dialyzed patients. *J Nephrol* 2003;16:895–902.

39. Carmichael P, Popoola J, John I, et al. Assessment of quality of life in a single centre dialysis population using the KDQOL-SF questionnaire. *Qual Life Res* 2000;9:195–205.

40. Chang ST, Chen CL, Chen CC, et al. Enhancement of quality of life with adjustment of dry weight by echocardiographic measurement of inferior vena cava diameter in patients undergoing chronic hemodialysis. *Nephron Clin Pract* 2004;97:c90–7.

41. Chen YC, Hung KY, Kao TW, et al. Relationship between dialysis adequacy and quality of life in longterm peritoneal dialysis patients. *Perit Dial Int* 2000;20:534–40.

42. Chiang CK, Peng YS, Chiang SS, et al. Health-related quality of life of hemodialysis patients in Taiwan: a multicenter study. *Blood Purif* 2004;22:490–8.

43. Cleemput I, Kesteloot K, Moons P, et al. The construct and concurrent validity of the EQ-5D in a renal transplant population. *Value Health* 2004;7:499–509.29 deWit GA, Merkus MP, Krediet RT, de Charro FT. A comparison of quality of life of patients on automated and continuous ambulatory peritoneal dialysis. *Perit Dial Int* 2001;21:306–12.

44. Franke GH, Reimer J, Kohnle M, et al. Quality of life in end-stage renal disease patients after successful kidney transplantation: development of the ESRD symptom checklist—transplantation module. *Nephron* 1999;83:31–9.

45. Fujisawa M, Ichikawa Y, Yoshiya K, et al. Assessment of health-related quality of life in renal transplant and hemodialysis patients using the SF-36 health survey. *Urology* 2000;56:201–6.

46. Fukuhara S, Lopes AA, Bragg-Gresham JL, et al. Health-related quality of life among dialysis patients on three continents: the Dialysis Outcomes and Practice Patterns Study. *Kidney Int* 2003;64:1903–10.

47. Goller JL, McMahon JM, Rutledge C, et al. Dialysis adequacy and self-reported health status in a group of CAPD patients. *Adv Perit Dial* 1997;13:128–33.

48. Groothoff JW, Grootenhuis MA, Offringa M, et al. Quality of life in adults with end-stage renal disease since childhood is only partially impaired. *Nephrol Dial Transplant* 2003;18:310–17.

49. Hamilton G, Locking-Cusolito H. Hemodialysis adequacy and quality of life: how do they relate? *Cannt J* 2003;13:24–9.

50. Hays RD, Kallich JD, Mapes DL, et al. Development of the kidney disease quality of life (KDQOL) instrument. *Qual Life Res* 1994;3:329–38.

51. Iliescu EA, Coe H, McMurray MH, et al. Quality of sleep and health-related quality of life in haemodialysis patients. *Nephrol Dial Transplant* 2003;18:126–32.

52. Juergensen PH, Zemchenkov A, Watnick S, et al. Comparison of quality-of-life assessment in Russia and the United States in chronic peritoneal dialysis patients. *Adv Perit Dial* 2002;18:55–7.

53. Khan IH, Garratt AM, Kumar A, et al. Patients' perception of health on renal replacement therapy: evaluation using a new instrument. *Nephrol Dial Transplant* 1995;10:684–9.

54. Lee SY, Lee HJ, Kim YK, et al. Neurocognitive function and quality of life in relation to hematocrit levels in chronic hemodialysis patients. *J Psychosom Res* 2004;57:5–10.

55. Manns B, Johnson JA, Taub K, et al. Quality of life in patients treated with hemodialysis or peritoneal dialysis: what are the important determinants? *Clin Nephrol* 2003;60:341–51.

56. Manu MA, Radulescu S, Harza M, et al. Quality of life assessed by SF-36 health survey in renal transplant patients. *Transplant Proc* 2001;33:1927–8. 396 Liem et al.

57. Matas AJ, McHugh L, Payne WD, et al. Long-term quality of life after kidney and simultaneous pancreaskidney transplantation. *Clin Transplant* 1998;12: 233–42.

58. Merkus MP, Jager KJ, Dekker FW, et al. Quality of life in patients on chronic dialysis: self-assessment 3 months after the start of treatment. The Necosad Study Group. *Am J Kidney Dis* 1997;29:584–92.

59. Molsted S, Aadahl M, Schou L, Eidemak I. Self-rated health and employment status in chronic haemodialysis patients. *Scand J Urol Nephrol* 2004;38:174–8.

60. Morton AR, Meers C, Singer MA, et al. Quantity of dialysis: quality of life—what is the relationship? *Asaio J* 1996;42:M713–17.

61. Oberbauer R, Hutchison B, Eris J, et al. Healthrelated quality-of-life outcomes of sirolimus-treated kidney transplant patients after elimination of cyclosporine A: results of a 2-year randomized clinical trial. *Transplantation* 2003;75:1277–85.

62. Ozminkowski RJ, White AJ, Hassol A, Murphy M. General health of end stage renal disease program beneficiaries. *Health Care Financ Rev* 1997;19:121–44.

63. Painter P, Carlson L, Carey S, et al. Low-functioning hemodialysis patients improve with exercise training. *Am J Kidney Dis* 2000;36:600–8.

64. Painter PL, Topp KS, Krasnoff JB, et al. Healthrelated fitness and quality of life following steroid withdrawal in renal transplant recipients. *Kidney Int* 2003;63:2309 16.

65. Paniagua R, Amato D, Vonesh E, et al. Health-related quality of life predicts outcomes but is not affected by peritoneal clearance: The ADEMEX Trial. *Kidney Int* 2005;67:1093–104.

66. Perneger TV, Leski M, Chopard-Stoermann C, Martin PY. Assessment of health status in chronic hemodialysis patients. *J Nephrol* 2003;16:252–9.

67. Pucheu S, Consoli SM, D'Auzac C, et al. Do health causal attributions and coping strategies act as moderators of quality of life in peritoneal dialysis patients? *J Psychosom Res* 2004;56:317–22.

68. Rebollo P, Ortega F, Baltar JM, et al. Health-related quality of life (HRQOL) in end stage renal disease (ESRD) patients over 65 years. *Geriatr Nephrol Urol* 1998;8:85–94.

69. Sesso R, Rodrigues-Neto JF, Ferraz MB. Impact of socioeconomic status on the quality of life of ESRD patients. *Am J Kidney Dis* 2003;41:186–95.

70. Sloan RS, Kastan B, Rice SI, et al. Quality of life during and between hemodialysis treatments: role of l-carnitine supplementation. *Am J Kidney Dis* 1998;32:265–72.
71. Sureshkumar KK, Mubin T, Mikhael N, et al. Assessment of quality of life after simultaneous pancreaskidney transplantation. *Am J Kidney Dis* 2002;39: 1300–6.
72. Taji Y, Morimoto T, Okada K, et al. Effects of intravenous ascorbic acid on erythropoiesis and quality of life in unselected hemodialysis patients. *J Nephrol* 2004;17:537–43.
73. Tanriverdi N, Ozcurumez G, Colak T, et al. Quality of life and mood in renal transplantation recipients, donors, and controls: preliminary report. *Transplant Proc* 2004;36:117–19.
74. Taskapan H, Ates F, Kaya B, et al. Psychiatric disorders and large interdialytic weight gain in patients on chronic haemodialysis. *Nephrology (Carlton)* 2005; 10:15–20.
75. Tawney KW, Tawney PJ, Hladik G, et al. The life readiness program: a physical rehabilitation program for patients on hemodialysis. *Am J Kidney Dis* 2000;36:581–91.
76. Turk S, Guney I, Altintepe L, et al. Quality of life in male hemodialysis patients. Role of erectile dysfunction. *Nephron Clin Pract* 2004;96:c21–7.
63. Unruh M, Benz R, Greene T, et al. Effects of hemodialysis dose and membrane flux on health-related quality of life in the HEMO Study. *Kidney Int* 2004;66:355–66.
77. Vazquez I, Valderrabano F, Fort J, et al. Psychosocial factors and health-related quality of life in hemodialysis patients. *Qual Life Res* 2005;14:179–90.
78. Walters BA, Hays RD, Spritzer KL, et al. Healthrelated quality of life, depressive symptoms, anemia, and malnutrition at hemodialysis initiation. *Am J Kidney Dis* 2002;40:1185–94.
79. Wight JP, Edwards L, Brazier J, et al. The SF36 as an outcome measure of services for end stage renal failure. *Qual Health Care* 1998;7:209–21.
80. Wu AW, Fink NE, Marsh-Manzi JV, et al. Changes in quality of life during hemodialysis and peritoneal dialysis treatment: generic and disease specific measures. *J Am Soc Nephrol* 2004;15:743–53
81. Churchill DN, Torrance GW, Taylor DW, et al. Measurement of quality of life in end-stage renal disease: the time trade-off approach. *Clin Invest Med* 1987;10:14–20.
82. Molzahn AE, Northcott HC, Dossetor JB. Quality of life of individuals with end stage renal disease. Perceptions of patients, nurses, and physicians. *Anna J* 1997;24:325–33; discussion 334–5.
83. Sennfalt K, Magnusson M, Carlsson P. Comparison of hemodialysis and peritoneal dialysis—a cost-utility analysis. *Perit Dial Int* 2002;22:39–47.
84. Lee AJ, Morgan CL, Conway P, Currie CJ. Characterisation and comparison of health-related quality of life for patients with renal failure. *Curr Med Res Opin* 2005;21:1777–83.
85. Torrance GW, Thomas WH, Sackett DL. A utility maximization model for evaluation of health care programs. *Health Serv Res* 1972;7:118–33.
86. Von Neumann J, Morgenstern O. *Theory of Games and Economic Behaviour* (3rd ed.). New York: Wiley, 1953.
87. Wolcott DL, Nissenson AR. Quality of life in chronic dialysis patients: a critical comparison of continuous ambulatory peritoneal dialysis (CAPD) and hemodialysis. *Am J Kidney Dis* 1988;11:402–12.
88. Manns B, Johnson JA, Taub K, et al. Quality of life in patients treated with hemodialysis or peritoneal dialysis: what are the important determinants? *Clin Nephrol* 2003;60:341–51
89. Bass EB, Wills S, Fink NE, et al. How strong are patients' preferences in choices between dialysis modalities and doses? *Am J Kidney Dis* 2004;44:695–705.
90. Wasserfallen JB, Halabi G, Saudan P, et al. Quality of life on chronic dialysis: comparison between haemodialysis and peritoneal dialysis. *Nephrol Dial Transplant* 2004;19:1594–9
91. Forsberg A, Lorenzon U, Nilsson F, Backmana L. Pain and health related quality of life after heart, kidney, and liver transplantation. *Clin Transplant* 1999;13: 453–60.
92. Hays RD, Kallich JD, Mapes DL, et al. Development of the kidney disease quality of life (KDQOL) instrument. *Qual Life Res* 1994;3:329–38.
93. Canadian Erythropoietin Study Group. Association between recombinant human erythropoietin and quality of life and exercise capacity of patients receiving haemodialysis. *BMJ* 1990;300:573–8
94. Churchill DN, Wallace JE, Ludwin D, et al. A comparison of evaluative indices of quality of life and cognitive function in hemodialysis patients. *Control Clin Trials* 1991;12(Suppl.):S159–67.
95. Harris DC, Chapman JR, Stewart JH, et al. Low dose erythropoietin in maintenance haemodialysis: improvement in quality of life and reduction in true cost of haemodialysis. *Aust N Z J Med* 1991;21:693–700.
96. Laupacis A, Keown P, Pus N, et al. A study of the quality of life and cost-utility of renal transplantation. *Kidney Int* 1996;50:235–42.
97. Sesso R, Yoshihiro MM, Ajzen H. Late diagnosis of chronic renal failure and the quality of life during dialysis treatment. *Braz J Med Biol Res* 1996; 29:1283–9.
98. Sesso R, Yoshihiro MM. Time of diagnosis of chronic renal failure and assessment of quality of life in haemodialysis patients. *Nephrol Dial Transplant* 1997; 12:2111–6.
99. Cleemput I, Kesteloot K, De Geest S, et al. Health professionals' perceptions of health status after renal transplantation: a comparison with transplantation candidates' expectations. *Transplantation* 2003;76: 176–82.
100. Greiner W, Obermann K, Graf v.d. Schulenburg J-M. Socio-economic evaluation of kidney-transplantation in Germany. *Arch Hell Med* 2001;18:147–55.
101. Moons P, Vanrenterghem Y, Van Hooff JP, et al. Health-related quality of life and symptom experience in tacrolimus-based regimens after renal transplantation: a multicentre study. *Transpl Int* 2003;16:653–64.
102. Polsky D, Weinfurt KP, Kaplan B, et al. An economic and quality-of-life assessment of basiliximab vs antithymocyte globulin immunoprophylaxis in renal transplantation. *Nephrol Dial Transplant* 2001;16: 1028–33.
103. Roderick P, Nicholson T, Armitage A, et al. An evaluation of the costs, effectiveness and quality of renal replacement therapy provision in renal satellite units in England and Wales. *Health Technol Assess* 2005; 9:1–178.
104. McFarlane PA, Bayoumi AM, Pierratos A, Redelmeier DA. The quality of life and cost utility of home nocturnal and conventional in-center hemodialysis. *Kidney Int* 2003;64:1004–11.
105. Cameron JI, Whiteside C, Katz J, Devins GM. Differences in quality of life across renal replacement therapies: a meta-analytic comparison. *Am J Kidney Dis* 2000;35:629–37.
106. Mulley AG, Jr. Assessing patients' utilities. Can the ends justify the means? *Med Care* 1989; 27(Suppl.):S269–81.



## Neuropathic Pain

Maria Dalamagka

### ► To cite this version:

| Maria Dalamagka. Neuropathic Pain. Archives of Anesthesiology, 2018. hal-05432534

**HAL Id: hal-05432534**

**<https://hal.science/view/index/docid/5432534>**

Submitted on 26 Dec 2025

**HAL** is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



Distributed under a Creative Commons Attribution 4.0 International License

## Neuropathic Pain

Dalamagka Maria\*

Anesthesiology department, General hospital of Larissa, Larissa, Greece.  
mary.dalamaga@gmail.com

**\*Corresponding Author:** Dalamagka Maria, Anesthesiology department, General hospital of Larissa, Larissa, Greece.

### Abstract

**Background and purpose:** Neuropathic pain (NP) is often refractory to pharmacologic and non-interventional treatment. Neuropathic pain can be distinguished from non-neuropathic pain by two factors. In neuropathic pain there is no transduction (conversion of a nociceptive stimulus into an electrical impulse). Finally, injury to major nerves is more likely than injury to non-nervous tissue to result in chronic pain. This review focuses on the clinical treatment of neuropathic pain as well aims to improve the care of adults with neuropathic pain.

**Methods:** Clinical treatment of neuropathic pain depends on clinical evaluation, as treatment can be pharmacological or not; invasive treatment or other therapies.

**Results:** Neuropathic pain because of its heterogeneity, there is often uncertainty regarding the nature and exact location of a lesion or health condition associated with neuropathic pain.

**Conclusions:** Both peripheral and central nervous system mechanisms contribute to the persistence of most types of neuropathic pain. Patients with conditions as diverse as diabetic polyneuropathy, human immunodeficiency virus (HIV) sensory neuropathy, poststroke syndromes, and multiple sclerosis frequently experience daily pain that greatly impairs their quality of life.

**Implications:** Progress in basic science will lead to a greater understanding of the pathophysiologic mechanisms of neuropathic pain.

### INTRODUCTION

Neuropathic pain (NP) has been defined by the International Association for the Study of Pain as pain arising as a direct consequence of a lesion or disease affecting the somatosensory nervous system.<sup>1</sup> According to an Institute of Medicine report, one in three Americans experiences chronic pain more than the total number affected by heart disease, cancer, and diabetes combined.<sup>2</sup> The prevalence of NP is 6-8% in the general population, but associated factors are advancing age, female gender and lower socioeconomic groups.<sup>3</sup> Around 15-25% of people with chronic pain are currently thought to have neuropathic pain.<sup>3,4</sup>

### PHYSIOLOGY AND MECHANISM

The generation of pain in response to tissue injury

involves four basic elements. Transduction, a function of nociceptors that converts noxious stimulation to nociceptive signals; Transmission, a process that sends nociceptive signals along nerve fibers from the site of injury to the central nervous system (CNS); Transformation or plasticity, a mechanism that modulates nociceptive signals at synaptic sites and at the level of the CNS through ascending, descending, or regional facilitation and inhibition; Perception, a key component of the clinical pain experience that integrates cognitive and affective (emotional) responses.

Neuropathic pain can be classified according to the underlying disease (diabetic neuropathy, multiple sclerosis), the location of the lesion (peripheral nerve damage, spinal cord) and the underlying

## Neuropathic Pain

mechanism. Characteristics of neuropathic pain are automatic start, independent stimulus, continuous burning, breakthrough pain, electric shock-like pain, dysaesthesia (abnormal and unpleasant sensation), paraesthesia, hyperalgesia (increased response to normally painful stimuli), allodynia (pain from non-painful stimuli). In neuropathic pain, tissue damage directly affects the nervous system, resulting in the generation of ectopic discharges that bypass transduction.<sup>5</sup> The pain is initiated or caused by a primary lesion or dysfunction in the nervous system, and yet its cause is heterogeneous and commonly may be inflammatory (for example, post herpetic neuralgia), metabolic, or ischaemic.<sup>6</sup> The main approach is to divide peripheral neuropathic pain into stimulus evoked pain or stimulus independent pain (spontaneous pain).<sup>7</sup>

## EVALUATION AND TREATMENT OF NEUROPATHIC PAIN

Clinical evaluation of pain includes: historical and onset of symptoms; localization, quality, intensity and duration of pain; evaluation of the impact of pain on ordinary activities and any sleep disorders; effect of pain in the psychological realm of patient; answer or not previous treatments.

Non pharmacological treatments for neuropathic pain includes: maintain activity and labor employment; mobilisation, exercise, TENS, Acupuncture. Pharmacological treatments includes: analgesics such as Paracetamol, mild opioid (tramadol, codeine); non-steroidal anti-inflammatory drugs; opioids, antidepressants, antiepileptics, local factors such as Lidocaine 5%. Invasive techniques such as Neuromodulation in which a series of treatments aimed at changing the perception of pain after stimulation or inhibition of neural pathways. Another treatment is induction of bone marrow; in resistant neuropathic pain are inserted one or two epidural electrodes, which are then connected to an electrical source, such as that of the pacemaker. Finally another option is pulsed radiofrequency in which a single needle is inserted in the area of neural damage. The resulting heat creates a temporary impairment of the nervous tissue.<sup>8</sup>

## CONCLUSIONS

Injury to the peripheral or central nervous system results in maladaptive changes in neurons along the nociceptive pathway that can cause neuropathic pain. Neuropathic pain is very challenging to manage because of its heterogeneity. There is often uncertainty regarding the nature and exact location of a lesion or health condition associated with neuropathic pain. Chronic neuropathic pain is common in clinical practice. Patients with conditions as diverse as diabetic polyneuropathy, human immunodeficiency virus (HIV) sensory neuropathy, poststroke syndromes, and multiple sclerosis frequently experience daily pain that greatly impairs their quality of life.

Both peripheral and central nervous system mechanisms contribute to the persistence of most types of neuropathic pain. Interest in the mechanisms and treatment of chronic neuropathic pain has increased during the past several years, and this is likely to result in significant treatment advances in the future. Progress in basic science will lead to a greater understanding of the pathophysiologic mechanisms of neuropathic pain.

## REFERENCES

- [1] Treede RD, Jensen TS, Campbell JN et al. *Neurology* 2008; 70: 1630-5
- [2] Research, Care, and Education. *Relieving pain in America. A blueprint for transforming prevention, care, education and research.* National Academies Press, 2011. [http://books.nap.edu/openbook.php?record\\_id=13172&page=1](http://books.nap.edu/openbook.php?record_id=13172&page=1).
- [3] Torrance N, Smith BH, Bennett MI et al.. The epidemiology of chronic pain of predominantly neuropathic origin. Results from a general population survey. *J Pain* 2006;7:281-9.
- [4] Bouhassira D, Lantéri-Minet M, Attal N et al.. Prevalence of chronic pain with neuropathic characteristics in the general population. *Pain* 2008;136:380-7.
- [5] Devor M. Neuropathic pain and injured nerve: peripheral mechanisms, *Br Med Bull* 1991;47:619-30.
- [6] Merskey H, Bogduk N. Classification of chronic pain. Descriptions of chronic pain syndromes and definitions of pain

## Neuropathic Pain

---

- [7] Woolf CJ, Mannion RJ. Pain: neuropathic pain: aetiology, symptoms, mechanisms and management. *Lancet* 1999; 353: 1959–1964
- [8] Neuropathic pain—pharmacological management: NICE clinical guideline 173 (November 2013)

**Citation:** Dalamagka Maria. *Neuropathic Pain. Archives of Anesthesiology. 2018; 1(2): 28-30.*

**Copyright:** © 2018 Dalamagka Maria. *This is an open access article distributed under the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited.*



# Obesity in children and general anesthesia

Maria I Dalamagka

## ► To cite this version:

Maria I Dalamagka. Obesity in children and general anesthesia. GSC Biological and Pharmaceutical Sciences, 2022, 19, pp.308 - 309. 10.30574/gscbps.2022.19.2.0206 . hal-05432493

**HAL Id: hal-05432493**

**<https://hal.science/view/index/docid/5432493>**

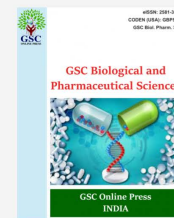
Submitted on 26 Dec 2025

**HAL** is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



Distributed under a Creative Commons Attribution 4.0 International License



(CASE REPORT)



## Obesity in children and general anesthesia

Maria I Dalamagka \*

*Department of Anaesthesia, General Hospital of Larisa, Larisa, Thessaly, Greece.*

GSC Biological and Pharmaceutical Sciences, 2022, 19(02), 308–309

Publication history: Received on 25 April 2022; revised on 27 May 2022; accepted on 29 May 2022

Article DOI: <https://doi.org/10.30574/gscbps.2022.19.2.0206>

### Abstract

Childhood obesity has reached epidemic levels in developed as well as in developing countries. Overweight and obesity in childhood are known to have significant impact on both physical and psychological health. Obese 10-year-old boy undergoes general anesthesia for emergency peritonitis surgery. The purpose of this study was to study the intraoperative behavior of obese children as they are accompanied by respiratory and metabolic disorders.

**Keywords:** Obesity; Anesthesia; Allergia; Children; Bridion; Astma

### 1. Introduction

The World Health Organization (WHO) has identified childhood obesity as “one of the most serious public health challenges of the 21st century.” In 2016, WHO found that globally, the number of overweight children under age 5 topped 41 million, and the number between ages 5 and 19 was 340 million. Socio-cultural factors have also been found to influence the development of obesity. Our society tends to use food as a reward, as a means to control others, and as part of socializing. Nearly 30% of obese children suffer from asthma. Obese children are more likely to suffer from respiratory infections which can necessitate a change in timing of elective surgery. Patients with bronchial asthma have an increased risk of developing intra-operative bronchospasm, especially if they are receiving general anesthesia with an endotracheal tube. Bronchospasm occurs because as bronchoconstriction becomes more severe it takes a progressively longer time for the alveoli to empty their carbon dioxide during exhalation [1-5].

### 2. Case Study

A child aged 10 years and 60 kg came with an acute abdomen image for appendectomy. The child had allergic bronchial asthma, with a known allergic predisposition, known egg allergy and frequent asthma attacks two years ago. Preoperative administration of Solu medrol 120 mg, Dexaton 4 mg and Onda 4 mg was performed. Introduction to anesthesia was performed with Dormicum 1 mg, Ketamine 10 mg, Fentanyl 0.1 mg, Hypnomidate 18 mg, Esmeron 50 mg. After intubation, entered to mechanic ventilation, with a volume control ventilation model and a Primus Drager ventilator. Maintenance of anesthesia was achieved with Sevoflurane 2.5%, additional Fentanyl 0.1 mg and morphine 3 mg were given. The duration of the surgery was one hour. At the end of the surgery, Bridion of 200 mg was given with pure inhaled oxygen and the awakening was delayed for half an hour.

### 3. Management and Outcome

The concept of cross-reactivity between drugs used in the perioperative setting and food is often quoted, but usually not supported by evidence. Pre-existing allergic diseases and other confounding non-allergic but related clinical entities need to be carefully considered during the preoperative evaluation, as they may result in perioperative life-threatening

\* Corresponding author: Maria I Dalamagka

Department of Anaesthesia, General Hospital of Larisa, Larisa, Thessaly, Greece.

conditions. For this reason it was given as a hypnotic drug Dormicum, Ketamine and Hypnomidate, instead of Propofol, although studies have shown that it can be used. The awakening was delayed due to the recirculation of medicines and drugs are not calculated based on ideal body weight because the child had acute pain and anxiety as it would lead to complications in the induction of anesthesia. Also given intravenous Solu Medrol and Dexaton removing sensitization.

---

#### 4. Discussion

Expiratory reserve volume is decreased in obesity. Total lung capacity is usually preserved except with morbid obesity, excessive central adiposity or with obesity- hypoventilation syndrome. Obese patient have decreased FRC which may lead to atelectasis. Atelectasis means incomplete expansion. Obesity leads to an increased metabolic rate, increased oxygen demand, increased CO<sub>2</sub> production and increased alveolar ventilation. These is decreased chest wall compliance due to adipose tissue that lies over the thorax. Lung volumes are decreased as the increased abdominal mass forces the diaphragm cephalad. For this reason Bridion was given to the awakening, as the remaining muscle relaxation would lead to life-threatening situations for the patient.

---

#### 5. Conclusion

Obese children have special needs for anesthesia in emergency surgeries and should take into account the metabolic and respiratory disorders that may occur as a result of obesity.

---

#### Compliance with ethical standards

##### *Acknowledgments*

General Hospital of Larisa.

##### *Statement of ethical approval*

The present research work does not contain any studies performed on animals/humans subjects by any of the authors'.

##### *Statement of informed consent*

Written informed consent was obtained from the patient for publication of this case report. A copy of the written consent is available.

---

#### References

- [1] Popkin BM, Doak CM. The obesity epidemic is a worldwide phenomenon. *Nutr Rev.* 1998; 56: 106–14.
- [2] Bhave S, Bavdekar A, Otiv M. IAP National Task Force for Childhood, Prevention of Adult Diseases: Childhood Obesity. IAP National Task Force for Childhood Prevention of Adult Diseases: Childhood Obesity. *Indian Pediatr.* 2004; 41: 559–75.
- [3] Raj M, Sundaram KR, Paul M, Deepa AS, Kumar RK. Obesity in Indian children: Time trends and relationship with hypertension. *Natl Med J India.* 2007; 20: 288–93.
- [4] Brenn BR. Anesthesia for Pediatric Obesity. *Anesthesiol Clin North America.* 1 DEC 2005; 23(4): 745-64
- [5] Blaho K, Winbery S, Merigian K. Pharmacological considerations for the pediatric patient. *Optom Clin.* 1996; 5: 61-90.



## Antipsychotics and anesthetic drugs

Maria I Dalamagka

### ► To cite this version:

Maria I Dalamagka. Antipsychotics and anesthetic drugs. Magna Scientia Advanced Research and Reviews, 2024, 12, pp.001 - 004. 10.30574/msarr.2024.12.1.0141 . hal-05432481

**HAL Id: hal-05432481**

**<https://hal.science/view/index/docid/5432481>**

Submitted on 26 Dec 2025

**HAL** is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



Distributed under a Creative Commons Attribution 4.0 International License

(CASE REPORT)



## Antipsychotics and anesthetic drugs

Maria I. Dalamagka \*

*Department of Anesthesiology, General Hospital of Larisa, Greece.*

Magna Scientia Advanced Research and Reviews, 2024, 12(01), 001–004

Publication history: Received on 28 July 2024; revised on 06 September 2024; accepted on 09 September 2024

Article DOI: <https://doi.org/10.30574/msarr.2024.12.1.0141>

### Abstract

Psychiatric illness is common, affecting up to 10% of the population, with around 1% diagnosed with a major psychiatric disorder. An increased mortality rate in the postoperative period for schizophrenic patients receiving chronic antipsychotic therapy has been demonstrated. According to the Diagnostic and Statistical Manual of Disease, schizophrenia-associated psychotic disorders are characterized by disturbances in emotional, behavioral and cognitive arenas, impeding on nearly every aspect of life functioning in the form of thought disorders, delusions and hallucinations. Chronic schizophrenic patients lack pain sensitivity, and have pituitary-adrenal and autonomic nerve dysfunction, abnormalities of the immune system, and water intoxication. These alterations may influence postoperative outcome. Psychotropic drugs often given in combination with each other or with other non-psychiatric drugs generally exert profound effects on the central and peripheral neurotransmitter and ionic mechanisms. The increased complications are associated with physical disorders, antipsychotic or hazardous health behaviour and interactions between antipsychotics and anaesthetic drugs. Adverse responses during anaesthesia include arrhythmias, hypotension, prolonged narcosis or coma, hyperpyrexia, post-operative ileus and post-operative confusion. Although drug interaction probably remains the most potentially serious problem current evidence suggests that psychiatric medication need not be discontinued prior to anesthesia.

**Keywords:** Schizophrenia; Psychiatric medication; Mental health; Cognitive dysfunction

### 1. Introduction

Schizophrenia is a mental disease characterized by thought disorders, delusions, and hallucinations. It is the most common psychotic disorder, accounting for approximately 20% of all persons treated for mental illness. The physical and psychological fragility of patients with psychiatric illness poses critical importance in the preoperative assessment, evaluation, and choice of premedication, which includes regular therapy, as well as concerns about polypharmacy with possible interactions of anesthetics, analgesics, and psychiatric medications. A considerable effort is to reduce risks for exacerbations or relapses of imminent illness in the postoperative period. Antipsychotic drugs are used in the treatment of psychoses such as schizophrenia and mania. They treat the acute symptoms of hallucinations and delusions and in the case of schizophrenia are used prophylactically often for long periods. Most drugs act by antagonism at the dopamine D2 receptors in the central nervous system. Neuroleptic or typical antipsychotics, chlorpromazine, haloperidol, trifluoperazine, which have a propensity to cause extrapyramidal side effects: acute dystonia, akathisia, parkinsonism and tardive dyskinesia. Atypical antipsychotics, clozapine, olanzapine, risperidone, amisulpride, quetiapine and aripiprazole, which do not have a tendency to cause extrapyramidal side effects. An increased mortality rate in the post-operative period for psychotic patients receiving chronic antipsychotic therapy has been demonstrated. Approximately 70% of schizophrenic patients taking chronic antipsychotic drugs develop postoperative psychosis emergence or confusion. The cognitive dysfunction of schizophrenia is associated with an increased frequency of postoperative confusion. Postoperative psychosis emergence or confusion has been associated with adverse clinical and economic outcomes [1-7].

\* Corresponding author: Maria I. Dalamagka

## 2. Case report

A 53-year-old, 80-kg woman underwent a total hysterectomy due to cancer, who is in chronic schizophrenia. The operation was performed with a combination of epidural and general anesthesia. Naropaine 20 ml, 0.75 % was administered epidurally and 0.1 mg Fentanyl, 200 mg Propofol and 70 mg Esmeron were given intravenously. The patient was mechanically ventilated on a Drager ventilator and with a volume control ventilation model. Sevoflurane 2% was administered to maintain anesthesia. Fluid deficit was adjusted with Ringers Lactate. Intraoperatively, she developed persistent hypotension that was treated with phenylephrine. Awakening was delayed up to one hour after the inhaled anesthetic was discontinued. Reversal of residual muscle wasting was done with Bridion. The interaction with the patient's antipsychotic medications was particularly intense.

### 2.1. Interactions between antipsychotics and anesthetic drugs

#### 2.1.1. Antipsychotics

#### 2.1.2. Tricyclic antidepressants (TCA), selective serotonin re-uptake inhibitors, atypical agents and monoamine oxidase inhibitors (MAOIs)

Atypical antidepressants include venlafaxine and mirtazapine. Both these drugs should be continued throughout the perioperative period. Abrupt cessation of antidepressants is associated with the risk of developing withdrawal symptoms. Selective serotonin reuptake inhibitors (SSRIs), TCAs: amitriptyline, imipramine, desipramine, doxepin, nortriptyline; were the most commonly used drugs for treating depression. They inhibit synaptic reuptake of norepinephrine and serotonin. Also affect neurochemical systems including histaminergic and cholinergic systems, thus they have a large range of side-effects, including postural hypotension, cardiac dysrhythmias, urinary retention, dry mouth, blurred vision and sedation. Given chronically, these drugs decrease stores of noradrenergic catecholamines, and can cause changes on the ECG, ventricular arrhythmias and refractory hypotension. Increased availability of neurotransmitters in the central nervous system can result in increased anaesthetic requirements. TCAs may result in increased response to intraoperatively administered anticholinergics. Increased availability of norepinephrine at the post-synaptic nervous system can be responsible for exaggerated blood pressure responses following administration of indirect acting vasopressors such as ephedrine. Pancuronium, ketamine, meperidine and epinephrine containing solutions should be avoided. If hypotension occurs and vasopressors are needed, direct acting drugs such as phenylephrine are recommended. SSRIs block reuptake of serotonin at the pre-synaptic membranes, with relatively little effect on adrenergic, cholinergic, histaminergic or other neurochemical systems. As a result, they are associated with few side-effects. Fluoxetine is inhibitor of certain hepatic cytochrome P-450 enzymes and their side-effects are headache, agitation and insomnia. This drug may increase the plasma concentration of drugs that depend upon hepatic metabolism for clearance, such as warfarin, theophylline, phenytoin and benzodiazepines. Cardiac antidysarrhythmic drugs are also metabolized by this enzyme system, and fluoxetine inhibition of the enzyme system may result in potentiation of their effects. SSRIs should be continued throughout the perioperative period to prevent discontinuation syndrome. The use of pethidine, tramadol, pentazocine and dextromethorphan should be avoided. Serotonin syndrome is a potentially life-threatening adverse drug reaction that results from increased serotonin levels in the brain stem and spinal cord. A large number of drugs have been associated with the serotonin syndrome (agitation and confusion, increased motor activity and autonomic instability; hyperthermia, tachycardia, labile blood pressure and diarrhoea. Seizures, rhabdomyolysis, renal failure, arrhythmias, coma and death may occur). These include SSRI, MAOI, TCAs, pethidine, tramadol and dextromethorphan. MAOIs, tranylcypromine and phenelzine, and moclobemide, act by inhibition of the metabolic breakdown of norepinephrine and serotonin by the MAO enzyme. The level of norepinephrine and serotonin is increased at the receptor site. All MAOIs are eliminated by hepatic metabolism. Opioids like morphine, fentanyl, alfentanil and remifentanyl can all be used safely. Reactions occur in patients given pethidine and dextromethorphan, both of which inhibit serotonin reuptake. Depressive reaction, which is very rare, is thought to be due to MAO inhibition of hepatic enzymes resulting in enhanced effects of all opioids (It reversed by naloxone). Indirect acting sympathomimetics may precipitate potentially fatal hypertensive crisis and are absolutely contraindicated with any MAOIs. Direct acting sympathomimetics (adrenaline, noradrenaline and phenylephrine) may have an enhanced effect due to receptor hypersensitivity. Propofol and etomidate can be used safely. Ketamine should be avoided as it causes sympathetic stimulation. Local anaesthetics containing adrenaline should be used with caution. Benzodiazepines, inhalational anaesthetic agents, anticholinergic drugs and non-steroidal anti-inflammatory drugs can be used safely in patients taking MAOIs. Therefore, dosages should be titrated. Benzodiazepine premedication can be given and sympathetic stimulation should be avoided. Adequate hydration of the patient should be ensured. Hypotension should be treated initially with intravenous fluids and then with cautious doses of phenylephrine. Pethidine and indirect acting sympathomimetics are absolutely contraindicated. Valproate is the drug of choice for treating acute episodes of Bipolar disorder. Mood stabilizers are used to treat bipolar affective disorders. Lithium and valproate remain a mainstay of treatment. In some cases, olanzapine and other antipsychotics are also used in the

beginning of treatment to control excitement and agitation. Lithium inhibits the release of thyroid hormones and results in hypothyroidism. Lithium is eliminated by the kidneys and, therefore, if renal function is compromised or there is dehydration, lithium levels rise dramatically. Toxic blood concentration produces confusion, sedation, muscle weakness tremors and slurred speech. Cardiac problems may include sinus bradycardia, sinus node dysfunction, AV block, T wave changes, hypotension and ventricular irritability. Lithium toxicity occurs when levels are  $>1.5$  mmol/L, and is exacerbated by dehydration, diuretics and renal impairment. Lithium carbonate is used to treat manic depression, but it is more effective in preventing mania than in relieving depression. Lithium prolongs neuromuscular blockade and may decrease anaesthetic requirements because it blocks brainstem release of norepinephrine, epinephrine and dopamine. Lithium discontinuation is recommended. Lithium can be stopped at once because no withdrawal symptoms occur. The association of sedation with lithium suggests that anaesthetic requirement may be decreased in these patients. Duration of both depolarizing and non-depolarizing muscle relaxants may be prolonged. Neuroleptic or typical antipsychotics (chlorpromazine, haloperidol, trifluoperazine) cause extrapyramidal side-effects like acute dystonia, akathisia, Parkinsonism and tardive dyskinesia. Clozapine causes seizures and neutropenia [8-24].

---

### 3. Discussion

Schizophrenic patients are at increased risk for perioperative complications, as their biological response to stress is impaired. Antipsychotic medication should be continued perioperatively, as abrupt withdrawal may result in recurrence of psychotic symptoms. It may also increase the incidence of postoperative confusion, which is high in those suffering with schizophrenia. Opioids like pethidine and indirectly acting sympathomimetics are contraindicated in MAOI patients. Mood stabilisers are used to treat bipolar affective disorders which have a lifetime incidence of 1% in the UK. A manic or hypomanic episode is required for the diagnosis to be made. Some patients also suffer depressive episodes. Mood stabilisers include lithium and anticonvulsant drugs such as carbamazepine and sodium valproate. Mood stabilisers should not be stopped abruptly as there is a significant risk of relapse. The management of lithium poisoning is supportive, with correction of electrolyte imbalance and convulsions. Haemodialysis is required if renal failure is present, which may need to be repeated as tissue lithium enters the circulation. Cardiac risk factors are increased in patients taking long term antipsychotic drugs due to a combination of side effects and high smoking levels in these patients.

---

### 4. Conclusion

Patients with chronic schizophrenia should continue their antipsychotics pre-operatively as abrupt withdrawal may result in recurrence of psychotic symptoms. Important effects of antipsychotic drugs include  $\alpha$ -adrenergic blockade causing postural hypotension, prolongation of QT intervals, seizures, hepatic enzyme elevation, abnormal temperature regulation, sedation and Parkinsonism-like manifestations. Drug-induced sedation may decrease anaesthetic requirement.

---

### Compliance with ethical standards

#### *Statement of informed consent*

Informed consent was obtained from all individual participants included in the study.

---

### References

- [1] Kudoh A. Perioperative management of chronic schizophrenic patients. *AnesthAnalg*. 2005, 101:1867–72.
- [2] Wik G. Effects of neuroleptic treatment on cortisol and 3-methoxy-4-hydroxyphenylethyl glycol levels in blood. *J Endocrinol*. 1995, 144:425–9.
- [3] Naudin J, Capo C, Giusano B, Mege JL, Azorin JM. A differential role for interleukin-6 and tumor necrosis factor in schizophrenia? *Schizophr Res*. 1997, 26:227–33.
- [4] Goldman MB, Robertson GL, Luchins DJ, Hedeker D, Pandey GN. Psychotic exacerbations and enhanced vasopressin secretion in schizophrenic patients with hyponatremia and polydipsia. *Arch Gen Psychiatry*. 1997, 54:443–9.
- [5] Morgan GE, Mikhail MS, Murray MJ. Morgan: Clinical anaesthesiology. 4th ed. USA: LANGE International edition, 2008. Anesthesia for patients with neurologic and psychiatric diseases, pp. 647–61.

- [6] Howland RH. Potential adverse effects of discontinuing psychotropic drugs. *J PsychosocNursMent Health Serv.* 2010, 48:9–12.
- [7] Hines RL, Marschall KE. Psychiatric disease/substance abuse/drug overdose *Stoelting's: Anaesthesia and co-existing diseases.* 5th ed. Gurgaon (India): ELSEVIER A division of Reed Elsevier India Private Ltd, 2010.
- [8] Kudoh A, Katagai H, Takazawa T. Effect of preoperative discontinuation of antipsychotics in schizophrenic patients on outcome during and after anesthesia. *Eur J Anaesth.* 2004, 21:414–6.
- [9] Sawada N, Higashi K, Yanagi F, Nishi M, Akasaka T, Kudoh J. Sudden onset of bronchospasm and persistent hypotension during spinal anesthesia in a patient on long-term psychotropic therapy. *Masui.* 1997, 46:1225–9.
- [10] Lanctot KL, Best TS, Mittman N, Liu BA, Oh PI, Einarson TR, et al. Efficacy and safety of neuroleptics in behavioral disorders associated with dementia. *J Clin Psychiatry.* 1998, 59:550–61.
- [11] Buckley NA, Sanders P. Cardiovascular adverse effects of antipsychotic drugs. *Drug Saf.* 2000, 23:215–28.
- [12] Kudoh A, Kudo T, Ishihara H, Matsuki A. Depressed pituitary-adrenal response to surgical stress in chronic schizophrenic patients. *Neuropsychobiol.* 1997, 36:112–6.
- [13] Kawachi I. Physical and psychological consequence of weight gain. *J Clin Psychiatry.* 1999, 60(Suppl 21):5–9.
- [14] Baptista T. Body weight gain induced by antipsychotic drugs: Mechanisms and management. *ActaPsychiatrScand.* 1999, 100:3–16.
- [15] Glassman AH, Bigger JT., Jr Antipsychotic drugs: Prolonged QTc interval, torsade de pointes, and sudden death. *Am J Psychiatry.* 2001, 158:1774–82.
- [16] Kudoh A, Ishihara H, Matsuki A. Pituitary-adrenal and parasympathetic function of chronic schizophrenic patients with postoperative ileus or hypotension. *Neuropsychobiol.* 1999, 39:125–30.
- [17] Patt RB, Proper G, Reddy S. The neuroleptics as adjuvant analgesics. *J Pain Sympt Manage.* 1994, 9:446–53.
- [18] Young DM. Risk factors for hypothermia in psychiatric patients. *Ann Clin Psychiatry.* 1996, 8:93–7.
- [19] Molnar G, Fava GA. Principles of medical psychiatry. Orland: Gruneand Stratton Inc, 1987. Intercurrent medical illness in the schizophrenic patients, p. 451.
- [20] O'Keeffe ST, Devlin JG. Delirium and the dexamethasone suppression test in the elderly. *Neuropsychobiol.* 1994, 30:153–6.
- [21] Van der Mast RC. Pathophysiology of delirium. *J Geriatr Psychiatry Neurol.* 1998, 11:138–45.
- [22] Allman KG, Wilson IH. *Oxford Handbook of Anaesthesia* (2nd Edition). 2006.
- [23] Peck TE, Hill SA, Williams M. *Pharmacology for Anaesthesia and Intensive Care* (3rd Edition). 2008.
- [24] Taylor D, Paton C, Kerwin R. *The Maudsley Prescribing Guidelines* (9th Edition). 2007.



## Anesthesiology complications in children

Maria I Dalamagka

### ► To cite this version:

Maria I Dalamagka. Anesthesiology complications in children. International Journal of Science and Research Archive, 2024, 12, pp.2924 - 2927. 10.30574/ijra.2024.12.1.1179 . hal-05432478

**HAL Id: hal-05432478**

**<https://hal.science/view/index/docid/5432478>**

Submitted on 26 Dec 2025

**HAL** is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



Distributed under a Creative Commons Attribution 4.0 International License

## Anesthesiology complications in children

Maria I. Dalamagka \*

*Anesthesia Department, General Hospital of Larisa, Greece.*

International Journal of Science and Research Archive, 2024, 12(01), 2924–2927

Publication history: Received on 13 May 2024; revised on 25 June 2024; accepted on 28 June 2024

Article DOI: <https://doi.org/10.30574/ijrsra.2024.12.1.1179>

### Abstract

General anesthesia is defined by the American Society of Anesthesiologists as a “drug induced loss of consciousness during which patients are not arousable, even by painful stimulation”. The perioperative period, immediately before, during, and after surgery, is a particularly critical time for pediatric patients. Even though anesthesia today is much safer than it has ever been, all anesthesia has an element of risk. General anesthesia has been safely given to children for many years. Improvements in inhalational and intravenous agents have increased the safety profile of general anesthesia and complication rates are low. General anesthesia has also allowed children to undergo painful or anxiety inducing procedures. General anesthesia is a complete loss of consciousness with amnesia, analgesia and neuromuscular blockade. It can be divided into three phases: induction, maintenance and emergence. Induction usually occurs with inhalational anesthetic in children until intravenous access occurs. Propofol, etomidate and ketamine are often used for induction. Maintenance of anesthesia occurs with inhalational or intravenous medications. Common inhalational anesthetics include nitrous oxide, sevoflurane or desflurane. Common intravenous medications used for maintenance are propofol and remifentanyl. Typical complications in pediatric anesthesia are respiratory problems, medication errors, difficulties with the intravenous puncture and pulmonary aspiration. In the postoperative setting, nausea and vomiting, pain, and emergence delirium can be mentioned as typical complications. Side effects of anesthesia may include nausea, vomiting, drowsiness, muscle soreness and sore throat. Rare but more serious complications include adverse reactions that affect breathing, allergic reactions and irregular heart rhythms. Airway and respiratory events are the most common perioperative complications in pediatric patients. Several studies have reported on the incidence of pediatric airway-related complications. Consistently reported risk factors for serious airway complications include very young age and multiple intubation attempts. Laryngospasm was the most common cause of respiratory related arrests. Other etiologies included airway obstruction, difficult intubation, esophageal intubation, and aspiration.

**Keywords:** Anaesthesia; Complications; Intubation; Laryngospasm; Vomiting; Pain.

### 1. Introduction

The purpose of general anesthesia is to suppress the conscious perception and physiologic response to noxious stimuli and to render the patient unconscious. This is done through analgesia; decreasing pain, amnesia; blunting consciousness, akinesia; preventing movement, physiologic support; maintaining cardiovascular function, fluid management, electrolyte control, and thermoregulation and vigilance. Inhalation Respiratory and airway events accounted for 60 percent of all anesthesia-related complications and occurred in 3.1 percent of all anesthetics; laryngospasm occurred in 1.2 percent, bronchospasm in 1.2 percent, postoperative stridor in 0.7 percent, and aspiration in 0.1 percent of all anesthetics. The highest rates of airway and respiratory events occurred in neonates (<1 month old) and infants (<1 year old). Other risk factors for respiratory complications included history of prematurity and reactive airway disease. The risk factors for anesthetic mortality include neonates, age less than one year, American Society of Anesthesiologists (ASA) III and IV classification, children undergoing emergency surgery, general anesthesia and cardiac surgery. Airway management problems and cardiopulmonary events account for the majority of the cases of

\* Corresponding author: Maria I. Dalamagka

anesthesia related mortality in children with comorbidities. Though there can be inter-rater variability between these scores, they have been shown to correlate with perioperative complications and mortality. Pulmonary aspiration can be a devastating event although studies have shown that the incidence of aspiration in children is relatively low - two per 10,000 cases in elective procedures and 2.2 per 10,000 in emergency operations. Laryngospasm is common in pediatric surgical patients - twice the incidence when compared to adults. Studies have found that in children from birth to nine years of age the incidence is 17 in 1000 cases. In children with obstructive lung disease or upper respiratory infections the risks can be up to 64 to 96 per 1000 cases. There is evidence to suggest that the use of supraglottic airway devices would decrease the incidence of laryngospasm. Postoperative apnea is more commonly seen in former preterm infants. All of the major societies, the American, European and Canadian Societies of Anesthesiologists agree on recommended preoperative fasting time prior to surgery. The recommendations include clear fluids until two hours before surgery, breast milk, without additives, up to four hours before surgery, infant formula: light meals; nonhuman milk up to six hours before surgery and high fat; fried foods; meat meals up to eight hours before surgery [1-14].

---

## 2. Anesthesiology drugs administered to children

Benzodiazepines induce hypnosis, anxiolysis, sedation, and amnesia and have anticonvulsant activity. In larger doses, they cause respiratory depression and apnea; they are synergistic with narcotics and barbiturates in their respiratory depressant effects. The most commonly used benzodiazepine in pediatric anesthesia is midazolam. Propofol has a very rapid onset and is associated with a reduced rate of postoperative nausea and vomiting. Propofol ranks as the most commonly used IV induction agent in pediatric anesthesia. Ketamine, which has a rapid onset and a short duration of action, causes central dissociation while providing analgesia and amnesia. Because ketamine increases the production of secretions, anticholinergic medications; atropine, glycopyrrolate, are usually needed. Dreams and hallucinations are the most common side effects associated with ketamine. Administration of a benzodiazepine concomitantly with ketamine decreases the incidence of dreams and hallucinations. Contraindications to the use of ketamine in children include the presence of an active upper respiratory tract infection, increased intracranial pressure, open globe injury, and the presence of psychiatric or seizure disorders. Opioids are widely used in the practice of pediatric anesthesia and pediatric perioperative medicine; they provide analgesia for painful procedures and decrease the incidence of postprocedural pain. Opioids suppress the carbon dioxide response, induce apnea, and are respiratory depressants, their use must be monitored closely, especially when they are administered with other respiratory depressing agents; inhalational anesthetics, barbiturates, benzodiazepines. Morphine and fentanyl are the most commonly used opioids in pediatric anesthesia. Morphine is a long-acting analgesic medication that provides effective analgesia. Because morphine can cause bronchospasm, secondary to histamine release, it should be used cautiously in children who suffer from asthma. Fentanyl, which is 80 to 100 times more potent than morphine, ranks as the most commonly used opioid in pediatric anesthesia. It has a shorter duration of activity and a more stable hemodynamic profile than morphine. Sufentanil, which is eight to 10 times more potent than fentanyl, is used primarily in pediatric cardiac cases. Remifentanyl, which is similar in potency to fentanyl, is considered an ultrashort-acting opioid. It is used when anesthesia needs to be rapidly induced. Because the primary narcotic effect of remifentanyl ends so quickly, additional analgesics must be used to provide postoperative pain relief. Alfentanil, which is five to 10 times less potent than fentanyl but has a faster onset of action, appears to have an increased incidence of muscle rigidity, convulsions, and prolonged respiratory depression compared with fentanyl and is therefore rarely used in pediatric anesthesia. Inhalational anesthetics have protective effects against ischemia-reperfusion injury, and recent authors suggest that these effects may also result in clinically relevant improved organ function. This protective capacity has been related to a preconditioning effect, a postconditioning effect, and also to an effect on apoptosis. With the currently used volatile agents, Sevoflurane and Desflurane, anesthetic hepatotoxicity is no longer a significant clinical problem. Sevoflurane and Desflurane are contraindicated only in the very rare cases of malignant hyperthermia susceptibility. Sevoflurane is the ideal agent for inhalation induction of anesthesia in children because it is not irritating to the airway. Sevoflurane is a very effective bronchodilator, and, despite early controversial reports about nephrotoxicity by means of the production of Compound A, a degradation product resulting from the interaction of sevoflurane with the CO<sub>2</sub> absorbent soda lime, it is actually not at all toxic to the kidney as long as a fresh gas flow is kept at no less than 2 litres per minute. Desflurane is not suitable for inhalation induction due to its irritating effect on the airways, but we use it for the maintenance of anesthesia, after tracheal intubation, even in newborns. Desflurane is a bronchodilator and has a modest negative inotropic effect; it does not have any nephrotoxic effect. Nitrous oxide can inactivate vitamin B<sub>12</sub>, provoking neurologic disorders both in patients and in operating room personnel. We never use nitrous oxide. Toxic effects are particularly possible in persons with a subclinical, preexisting deficit of vitamin B<sub>12</sub> but it is not possible to know in advance which patients are deprived of vitamin B<sub>12</sub>. Nitrous oxide also has the well-known effect of expanding any air-filled cavity. The neuromuscular blocking agents are used primarily as an adjunct to general anesthesia to facilitate endotracheal intubation and to maintain muscle relaxation during surgery. Neuromuscular blockers are the most useful drugs in our pharmacologic armamentarium but postanesthetic morbidity associated with incomplete reversal of neuromuscular blockers is still a frequent occurrence. Sugammadex, a relatively new drug, is a cyclodextrin that can

form a very tight complex with rocuronium, less with vecuronium and pancuronium, thus allowing a reversal even during a profound neuromuscular block. Anesthesia can be both induced and maintained with either boluses or continuous infusions of IV anesthetic agents. IV anesthetic agents include barbiturates, opioid narcotics, benzodiazepines, and miscellaneous products; ketamine, propofol [15-23].

### 3. Adverse Events and Discussion

Assessment of anesthetic risk in children has been estimated by the American Society of Anesthesia risk assessment score. The ASA can help identify patients preoperatively that may be at higher risk for anesthetic complications and allow adjustment to the anesthetic plan for these higher risk children. Complications associated with general anesthesia can be divided into four basic categories: postoperative nausea and vomiting (PONV), oral trauma, thermoregulation issues and cardiorespiratory complications. An estimated 40% to 50% of children experience PONV after receiving general anesthesia. The serotonin antagonists; ondansetron, dolasetron, granisetron, are considered to be first line drugs for PONV. The insertion or removal of the endotracheal tube can irritate the throat, causing a sore throat and, or hoarseness. Craniofacial anomalies include cleft lip and palate, mandibular hypoplasia, the enlarged tongue seen with Beckwith-Wiedemann syndrome and many others. It is important to anticipate difficulty in mask ventilation as well as endotracheal intubation. The most common complications are respiratory and include airway obstruction, laryngospasm or bronchospasm, desaturation and apnea. Cardiovascular complications are rare in healthy children and include arrhythmias and hypotension. Longer procedures and higher anesthetic dose increase the risk of complications. Mechanical ventilation can be a source of complications, first of all pulmonary barotrauma and volutrauma. Usually we start mechanical ventilation with a very low tidal volume; then we slowly increase it until we observe the chest moving and hear gas entering the lungs with the stethoscope. We also check the monitor, looking for an acceptable capnographic curve and a peak inspiratory pressure in the airways no higher than 15–20 cm H<sub>2</sub>O. Bronchospasm involves the constriction of the smooth muscle layers of the small airways. If left untreated, bronchospasm can progress to respiratory distress and ultimately, respiratory failure. Bronchospasm involves the constriction of the smooth muscle layers of the small airways. If left untreated, bronchospasm can progress to respiratory distress and ultimately, respiratory failure. This activity outlines the varying etiologies and clinical phenotypes, evaluation processes, and management strategies of bronchospasm. Additionally, the interdependent roles of the interprofessional team are highlighted in the care for pediatric patients with bronchospasm. Treatment of an acute asthma exacerbation targets the main components of the underlying pathophysiology: airway inflammation, smooth muscle constriction, edema, and mucus production. The standard initial therapy includes oxygen, intermittent dosing of a short acting beta 2 agonist; albuterol, an inhaled anticholinergic; ipratropium bromide, and corticosteroids. MH is an inherited disorder that is triggered by the volatile anesthetic agents and or succinylcholine. It usually occurs within the first two hours of anesthesia but can occur up to 24 hours later. MH is characterized by the development of a sudden and rapid high fever (38.5°C to 46°C, rising 1°C every five minutes), muscle rigidity, acidosis, and cardiac arrhythmias. Because MH can be fatal, aggressive therapy is warranted. The treatment of MH includes discontinuing all volatile inhalation agents, correction of the metabolic acidosis, and treatment with IV dantrolene: 2.5 mg/kg up to 10 mg/kg. Cardiorespiratory complications; respiratory depression, hypertension, arrhythmias, heart attacks, stroke, can also result from general anesthesia. Fortunately, these problems are extremely rare and are more likely related to surgical complications. The American Academy of Pediatrics Committee on Drugs (COD) notes that children tend to be particularly susceptible to oversedation with anesthesia. Incidence of perioperative anaphylaxis is estimated at 1 in 10–20,000 anesthetic procedures. It is noteworthy that in some cases bradycardia may be a life-protecting adaptive mechanism that allows the ventricle to fill despite hypovolemia. Then the treatment with atropine must be simultaneous with epinephrine and fluids; otherwise giving only atropine can result in cardiac arrest. As a matter of fact epinephrine and an expansion of intravascular volume are the key points in the perioperative management of anaphylaxis. The standard dose is 0.01–0.02 mg/Kg intravenously. Sometimes doctors are overzealous, using a high dosage due to their impatience in the treatment of an acute condition. According to recent reviews high doses can worsen a patient's post resuscitation hemodynamic condition by causing increased myocardial oxygen demand, ventricular ectopy, hypertension and myocardial necrosis. High doses do not improve survival and may be associated with a worse neurological outcome. Long term risks of general anesthesia have focused mainly on concerns for the effect of anesthesia on neurodevelopment in infants and children.

### 4. Conclusion

Minor events occur with variable frequency. The likelihood of these occurring is increased in patients under a year of age and in patients with an ASA status of 3 and above. Most events are respiratory in nature, emphasizing the importance of pulse oximetry and capnography as routine monitors in the delivery of anesthesia. Postoperative vomiting remains a common problem in older children.

## References

- [1] Barash PG, Cullen BF, and Stoelting RK. Clinical Anesthesia. Philadelphia: Lippincott Williams & Wilkins, 2006. Print
- [2] Subramanyam R, Yeramaneni S, Hossain MM, et al. Perioperative Respiratory Adverse Events in Pediatric Ambulatory Anesthesia: Development and Validation of a Risk Prediction Tool. *Anesth Analg*. 2016, 122(5):1578-85.
- [3] von Ungern-Sternberg BS, Boda K, Chambers NA, et al. Risk assessment for respiratory complications in paediatric anaesthesia: a prospective cohort study. *Lancet*. 2010, 376(9743):773-83.
- [4] Gonzalez LP, Pignaton W, Kusano PS, et al. Anesthesia-related mortality in pediatric patients: a systematic review. *Clinics (Sao Paulo)*. 2012, 67(4):381-7.
- [5] Sankar A, Johnson SR, Beattie WS, et al. Reliability of the American Society of Anesthesiologists physical status scale in clinical practice. *Br J Anaesth*. 2014.
- [6] American Society of Anesthesiologists. (2014, October). Retrieved December 17, 2016, from <http://www.asahq.org>
- [7] Lin EE, Tran KM. Anesthesia for fetal surgery. *Semin Pediatr Surg*. 2013, 22(1):50-5.
- [8] American Society of Anesthesiologists Committee. Practice guidelines for preoperative fasting and the use of pharmacologic agents to reduce the risk of pulmonary aspiration: application to healthy patients undergoing elective procedures: an updated report by the American Society of Anesthesiologists Committee on Standards and Practice Parameters. *Anesthesiology*. 2011, 114(3):495-511.
- [9] Merchant R, Chartrand D, Dain S, et al. Guidelines to the Practice of Anesthesia - Revised Edition 2016. *Can J Anaesth*. 2016, 63(1):86-112.
- [10] Smith I, Kranke P, Murat I, et al. Perioperative fasting in adults and children: guidelines from the European Society of Anaesthesiology. *Eur J Anaesthesiol*. 2011, 28(8):556-69.
- [11] Bordet F, Allaouchiche B, Lansiaux S, et al. Risk factors for airway complications during general anaesthesia in paediatric patients. *Paediatr Anaesth*. 2002, 12(9):762-9.
- [12] Anon. Guidelines for monitoring and management of pediatric patients during and after sedation for diagnostic and therapeutic procedures: Addendum. *Pediatrics*. 2002, 110:836-838.
- [13] Cote CJ, Notterman DA, et al. Adverse sedation events in pediatrics: A critical incident analysis of contributing factors. *Pediatrics*. 2000, 105:805-814.
- [14] Mondoñedo JR, McNeil JS, Amin SD, Herrmann J, Simon BA, Kaczka DW. Volatile Anesthetics and the Treatment of Severe Bronchospasm: A Concept of Targeted Delivery. *Drug Discov Today Dis Models*. 2015 Spring, 15:43-50.
- [15] Gencorelli FJ, Fields RG, Litman RS. Complications during rapid sequence induction of general anesthesia in children: A benchmark study. *Paediatr Anaesth*. 2010, 20:421-4.
- [16] Neuhaus D, Schmitz A, Gerber A, Weiss M. Controlled rapid sequence induction and intubation – An analysis of 1001 children. *Paediatr Anaesth*. 2013, 23:734-40.
- [17] Newton R, Hack H. Place of rapid sequence induction in paediatric anaesthesia. *BJA Educ*. 2016, 16:120-3.
- [18] Kulkarni K, Karnik P, Dave N, Garasia M. Ultra-modified rapid sequence induction. *Paediatr Anaesth*. 2017, 27:1278-83.
- [19] Malviya S, Voepel-Lewis T, Prochaska G, Tait AR. Prolonged recovery and delayed side effects of sedation for diagnostic imaging studies in children. *Pediatrics*. 2000, 105:E42.
- [20] Boonmak P, Boonmak S, Pattanittum P. High initial concentration versus low initial concentration sevoflurane for inhalational induction of anaesthesia. *Cochrane Database Syst Rev*. 2016 Issue 6, Art. No.:CD006837.
- [21] Lejus C, Bazin V, Fernandez M, Nguyen JM, Radosevic A, Quere MF, et al. Inhalation induction using sevoflurane in children: The single-breath vital capacity technique compared to the tidal volume technique. *Anaesthesia*. 2006, 61:535-40.
- [22] Ho K, Chua WL, Lim S. NGA. A comparison between single-and double-breath vital capacity inhalation induction with 8% sevoflurane in children. *Paediatr Anaesth*. 2004, 14:457-61.
- [23] Meyers EF, Muravchick S. Anesthesia induction technics in pediatric patients: A controlled study of behavioural consequences. *Anesth Analg*. 1977, 56:538-42.



# Pathology associated with kidney failure and anesthesia

Maria I Dalamagka

## ► To cite this version:

Maria I Dalamagka. Pathology associated with kidney failure and anesthesia. Magna Scientia Advanced Research and Reviews, 2024, 12, pp.106 - 109. 10.30574/msarr.2024.12.1.0149 . hal-05432456

**HAL Id: hal-05432456**

**<https://hal.science/view/index/docid/5432456>**

Submitted on 26 Dec 2025

**HAL** is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



Distributed under a Creative Commons Attribution 4.0 International License

## Pathology associated with kidney failure and anesthesia

Maria I. Dalamagka \*

*Department of Anesthesia, General Hospital of Larisa, Greece.*

Magna Scientia Advanced Research and Reviews, 2024, 12(01), 106–109

Publication history: Received on 10 August 2024; revised on 26 September 2024; accepted on 28 September 2024

Article DOI: <https://doi.org/10.30574/msarr.2024.12.1.0149>

### Abstract

Chronic kidney disease (CKD) is defined as either a glomerular filtration rate (GFR) of  $<60 \text{ ml min}^{-1} 1.73 \text{ m}^{-2}$  for 3 months or more, irrespective of cause, or kidney damage leading to a decrease in GFR, present for 3 months or more. The damage may manifest as abnormalities in the composition of blood or urine, on radiological imaging, or in histology. CKD is classified into five stages depending on GFR, ranging from Stage 1 - normal GFR to Stage 5 - established renal failure. Patients with kidney disease undergoing surgery and anesthesia are at high risk for increased adverse events that include, cardiovascular complications and mortality, and further deterioration of renal function and development of acute kidney injury (AKI) due to perioperative injuries that are caused by hemodynamic instability, hypovolemia, or drug toxicity. Special consideration should be placed on preventing further deterioration of renal function as well as protection of existing renal function in patients with moderate to severe impairment from the effects of anesthetics and pain medications. Analgesics such as nonsteroidal anti-inflammatory drugs (NSAIDs) can contribute to a reduction of the residual renal function in CKD and should be avoided. Renal IR injury is a risk factor for acute renal failure and delayed graft function. Pathogenic factors for renal IR include, but are not limited to, the following: oxidative stress, inflammation, cellular necrosis, and apoptosis. Further clinical studies are required to address the optimal medication regimen that can be used for postoperative pain management in the more severe stages of CKD, including hemodialysis.

**Keywords:** Chronic kidney disease; GFR; Acute kidney injury; Renal function.

### 1. Introduction

More than 1.8 million people in the UK have a diagnosis of chronic kidney disease- CKD with around a further million thought to be undiagnosed. The estimated disease prevalence in the UK of CKD stages 3–5 is thought to be between 4.3% and 8.5% and this appears to be increasing. The global prevalence of CKD is 9.1%. In 2017, CKD was recognised as the 12th leading cause of death, responsible for 1.2 million deaths each year worldwide. Deaths attributable to CKD are set to increase further, with predictive modelling suggesting that by 2040 2.2 million people will die annually because of CKD. Chronic kidney disease is defined by the Renal Association as an abnormality of kidney structure or function that lasts more than 3 months. It only becomes evident when fewer than 40% of nephrons are functioning. Tests which enable diagnosis include electrolyte abnormalities due to tubular disorders, proteinuria (albumin to creatinine ratio (ACR)  $> 3 \text{ mg/mmol}$ ), haematuria of renal origin, histological or radiological abnormalities in structure and abnormal function with resultant raised creatinine and/or cystatin C (eGFR  $<60 \text{ ml/min/1.73 m}^2$ ) on more than two occasions 90 days apart. Glomerular filtration rate (GFR) is the internationally accepted measure to express renal function. However, GFR is not routinely measured because it is a complex procedure that involves measurement of plasma or urinary clearance of an exogenous marker such as inulin. More commonly, eGFR is used: this uses serum creatinine, cystatin, or both, age, and sex to mathematically derive the eGFR. According to the NIDDK, more than 30 million American adults may have CKD. Safe anesthetic management requires an understanding of CKD pathophysiology to prevent aggravation of pre-existing disease. Patients who are dialysis-dependent are at particularly high risk of requiring emergency medical and surgical interventions due to frequent presence of comorbidities such as acute

\* Corresponding author: Maria I. Dalamagka

cardiovascular events, heart failure, vascular-access-related infections, endocarditis, bowel ischemia/bleeding, limb ischemia/necrosis, bone fractures, and cancer. Chronic kidney disease is associated with an increased risk of ischaemic heart disease. Most patients do not progress to end-stage renal failure but die as a result of fatal cardiovascular complications. Left ventricular hypertrophy can occur because of chronic volume overload, pressure overload, or both. Patients undergoing renal replacement therapy (RRT) may exhibit calciphylaxis, which describes accumulation of calcium in small blood vessels. Patients with end-stage renal failure were considered hypocoagulable. Acquired uraemic platelet dysfunction and thromboasthenia decrease platelet adhesion and increase vessel wall fragility, which contribute to bleeding diatheses [1-9].

### 1.1. Anesthetic management of patients in CKD

Depending on the patient's status and the surgical procedure, the preoperative evaluation may require close communication between the primary care physician, nephrologist, surgeon, and anesthesiologist to determine if a patient is optimized for surgery. The following assessments are recommended for patients with CKD: comorbid conditions, severity of CKD assessed by level of kidney function, complications related to level of kidney function, risk for loss of kidney function, and risk for cardiovascular disease. The risk for cardiovascular complications should be promptly evaluated. The patient should undergo a routine electrocardiogram. All present preoperative abnormalities, such as anemia, hyperkalemia, and metabolic acidosis, should be preoperatively corrected. A hemoglobin value of 10 g/dl is strongly recommended. Calcium chloride, insulin and dextrose, sodium bicarbonate, and resins can be used to correct hyperkalemia. If the patient is under dialysis treatment, the final dialysis prior to surgery should be scheduled 12-24 hours before surgery. The anesthetic management of patients suffering from CKD is complex. Due to delayed gastric emptying and neuropathy, there is risk of gastric acid aspiration. Gastric aspiration prophylaxis can be managed using sodium citrate, metoclopramide, anti-H<sub>2</sub> drugs, and rapid induction. Short-acting anesthetic drugs are recommended (propofol, remifentanyl, cisatracurium, vecuronium). Sevoflurane can deteriorate renal function by fluoride ion and compound A production, so isoflurane remains the preferred anesthetic inhalator agent. When selecting a neuromuscular blocking agent (NMBA) for use in patients with CKD, the anesthesiologist should consider the impact of renal impairment on the elimination of the drug, the potential for drug accumulation with incremental doses, and the production of active metabolites. To prevent postoperative residual curarization (PORC), long-acting NMBAs should be avoided. Opioids may be used, as they have no direct toxic effects on the kidney. They do, however, have an antidiuretic effect, and they may cause urinary retention. Patients with Stage 5 CKD who have undergone renal transplantation require immunosuppression. These drugs are usually given by the oral route. If enteral administration is inappropriate, then IV administration with dose adjustment will be required. Desflurane and isoflurane are metabolised to a minimal extent and there are no concerns regarding their use in CKD. Atracurium is the neuromuscular blocking agent (NMBA) of choice in patients with CKD. Its metabolism is unique. Atracurium undergoes ester hydrolysis and Hofmann degradation, both of which are independent of renal function. Cisatracurium can also be used in patients with CKD. It is predominantly eliminated by Hofmann degradation. Sugammadex has been used successfully in clinical research to reverse blockade from aminosteroid NMBAs in patients with severe renal impairment. Opioids have no direct nephrotoxic effects. They do have an antidiuretic effect, which may manifest clinically as urinary retention. Remifentanyl is not dependent on renal function for its elimination. An appropriate dose reduction of morphine and Fentanyl should be considered. Regional anaesthesia is a useful way of avoiding systemic drug accumulation and can be used as the primary anaesthetic technique, depending on the type of surgery. Central neuraxial blockade is not absolutely contraindicated, but consequent hypotension must be attended to in order to prevent worsening of renal perfusion. The ability of the kidneys to excrete water and sodium is impaired, causing difficulty in handling large fluid loads. As glomerular filtration decreases, sodium and water are retained and because of the increased hydrostatic pressure, fluid moves into the extravascular space leading to the clinical manifestations of fluid overload such as generalised and pulmonary oedema. Loss of nephron function results in renal retention of potassium. As GFR declines, the remaining nephrons adapt by increasing renal excretion and this works as a compensatory mechanism when the GFR is >15 ml/min. The tendency towards hyperkalaemia is ameliorated by instituting a low potassium diet and avoiding drugs known to cause hyperkalaemia [10-22].

## 2. Results and Discussion

Renal ischemia/reperfusion (IR) injury is a leading cause of preoperative acute kidney injury (AKI), which frequently complicates major vascular, cardiac, transplant, and liver surgeries. AKI has been shown to occur after some major surgeries. Novel interventions that protect against IR injury are needed to improve early graft function after kidney transplantation. Available general and local anesthetics, including third generation inhaled anesthetics, propofol, and amide-class local anesthetics, are effective and safe with a low incidence of side effects. Propofol, a widely used anesthetic, has shown potential as a novel organ-protective agent through its efficient membrane-targeted and cytoprotective effects. This is not surprising since lipid emulsions are major components of propofol, which

accumulating data show provide organ protective effects against IR injury in many organs, such as the heart, kidney, liver, and intestines. Sevoflurane and enflurane, did not cause deterioration of postoperative renal function in patients with preexisting renal issues. The metabolism of enflurane to inorganic fluoride during and after surgery did not cause a clinically significant level of renal disease or dysfunction. There is significant clinical evidence for volatile anesthetic-mediated organ protection. Sevoflurane preconditioning significantly lessened the postoperative rise of transaminase levels in patients undergoing liver resection. A short period of sevoflurane preconditioning in patients undergoing coronary artery bypass graft surgery was shown to significantly decrease both the release of a myocardial contractile dysfunction marker and the levels of plasma cystatin C concentrations. Ketamine ameliorate the upregulation of inflammatory pathways and reduction of metabolism caused by hypoxia. Thiopental pretreatment reduced renal IR injury induced by free radicals. Opioids may be used, as they have no direct toxic effects on the kidney. They do, however, have an antidiuretic effect. Patients with Stage 5 CKD who have undergone renal transplantation require immunosuppression. Nonsteroidal anti-inflammatory drugs (NSAIDs) can contribute to a reduction of the residual renal function in CKD and should be avoided.

Chronic kidney disease may result in a wide array of neurological manifestations including myoclonus, asterixis, chorea, uraemic encephalopathy and seizures. Autonomic neuropathy is common in patients with CKD because of decreased baroreceptor sensitivity, sympathetic overactivity and parasympathetic dysfunction. In haemodialysis, dialysate is pumped in a counter-current direction to blood flow and solutes equilibrate after diffusion across a semipermeable membrane. This too has associated complications, which include those related to vascular access, hypotension, arrhythmias and dysequilibration syndrome. Disease-specific treatment should be continued throughout the perioperative period where it is safe and practicable to do so. Certain medications such as ACE inhibitors remain controversial in the preoperative setting with no consensus regarding whether preoperative ACE inhibitor therapy is beneficial or harmful. Therefore, preoperative antihypertensive management should be considered on an individual basis and remains at the discretion of the anaesthetist. Should a patient require dialysis before surgery, liaise with the patient's specialist team.

---

### 3. Conclusion

The majority of research studies have focused on the non-anesthetic effects of propofol, such as its antioxidant, immunomodulatory, analgesic, and neuroprotective effects. It was found that propofol and ketamine were the 2 anesthetic agents that produced the least renal oxidative stress. AKI incidence, morbidity, and ICU stay were all found to be significantly lower for patients who received dexmedetomidine. Bupivacaine's ability to decreased cell damage suggests that it may have some protective effects against renal IR injury. Further clinical studies are required to address the optimal medication regimen that can be used for postoperative pain management in the more severe stages of CKD, including hemodialysis.

---

### References

- [1] Livio M, Gotti E, Marchesi D. Uraemic bleeding: role of anaemia and beneficial effect of red cell transfusions. *Lancet*. 1982; 2:1013-1015.
- [2] Ocak G, Lijfering W.M, Verduijn M. Risk of venous thrombosis in patients with chronic kidney disease: identification of high-risk groups *J Thromb Haemost*. 2013; 11:627-633.
- [3] Títóff V, Moury H.N, Títóff I.B. Seizures, antiepileptic drugs, and CKDAm *J Kid Dis*. 2019; 73:90-101.
- [4] Craig R.G, Hunter J.M. Recent developments in the perioperative management of adult patients with chronic kidney disease *Br J Anaesth*. 2008; 101:296-310.
- [5] National Institute for Health and Care Excellence. Chronic kidney disease in adults: assessment and management, clinical guideline CG182 NICE, London, 2015.
- [6] Wright M, Southcott E, MacLaughlin H. Clinical practice guideline on undernutrition in chronic kidney disease. *BMC Nephrol*. 2019; 1620:370.
- [7] Gemmell L, Docking R.D, Black E. Renal replacement therapy in critical care *BJA Educ*. 2017; 17:88-93.
- [8] Bradley T, Teare T, Milner Q. Anaesthetic management of patients requiring vascular access surgery for renal dialysis *BJA Educ*. 2017; 17:269-274.
- [9] Lea-Henry T.N, Carland J.E, Stocker S.L. Clinical pharmacokinetics in kidney disease *CJASN*. 2018; 13:1085-1095.

- [10] Fukazawa Kyota, Lee H. Thomas. Volatile anesthetics and AKI: risks, mechanisms, and a potential therapeutic window JASN. 2014; 25:884-892.
- [11] R. G. Craig, J. M. Hunter. Recent developments in the perioperative management of adult patients with chronic kidney disease, BJA: British Journal of Anaesthesia, Volume 101, Issue 3, September 2008, Pages 296–310, <https://doi.org/10.1093/bja/aen203>.
- [12] Centers for Disease Control and Prevention. National Chronic Kidney Disease Fact Sheet, 2017. National Chronic Kidney Disease Fact Sheet, 2017.
- [13] Krishnan M. Preoperative care of patients with kidney disease. Am Fam Physician. 2002 Oct 15;66(8):1471-6, 1379. PubMed PMID: 12408421.
- [14] Domi R, Huti G, Sula H, Baftiu N, Kaci M, et al. From Pre-Existing Renal Failure to Perioperative Renal Protection: The Anesthesiologist's Dilemmas, Anesth Pain Med. 2016; 6(3):e61545. doi: 10.5812/aapm.32386.
- [15] Sellbrant I, Brattwall M, Jildenstål P, Warren-Stomberg M, Forsberg S, Jakobsson JG. Anaesthetics and analgesics; neurocognitive effects, organ protection and cancer reoccurrence an update. Int J Surg 2016; 34: 41–46.
- [16] Andrews DT, Royse C, Royse AG. The mitochondrial permeability transition pore and its role in anaesthesia-triggered cellular protection during ischaemia-reperfusion injury. Anaesth Intensive Care 2012; 40: 46–70.
- [17] L, Ruffenach G, Kararigas G, et al. Intralipid protects the heart in late pregnancy against ischemia/reperfusion injury via Caveolin2/STAT3/GSK-3 $\beta$  pathway. J Mol Cell Cardiol 2017; 102: 108–116.
- [18] Li J, Motayagheni N, Barakati N, Eghbali M. Intralipid protects the heart in late pregnancy against ischemia/reperfusion injury by reducing cardiomyocyte apoptosis via Mir122 induction. Circ Res 2017; 119:A442.
- [19] Motayagheni N, Eghbali M. Complete reversal of xylazine-induced bradycardia with intralipid in female mice. Circ Res 2016; 119:A253.
- [20] Motayagheni N, Phan S, Eshraghi C, Eghbali M. Inhibition of leptin receptor abolishes intralipid-induced cardioprotection against ischemia-reperfusion injury. Cardiology 2016; 134: 241.
- [21] Motayagheni N, Eghbali M. Reversal of xylazine-induced bradycardia with intralipid. Cardiology 2016; 134: 431.
- [22] Motayagheni N. From bupivacaine to intralipid: Leading edge. J Anesth Clin Res 2016; 4: 00164.



## Protocol of Pain

Maria Dalamagka

### ► To cite this version:

Maria Dalamagka. Protocol of Pain. Novel Research in Sciences, 2021, 6, 10.31031/nrs.2021.06.000627 . hal-05433483

**HAL Id: hal-05433483**

**<https://hal.science/view/index/docid/5433483>**

Submitted on 28 Dec 2025

**HAL** is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



Distributed under a Creative Commons Attribution 4.0 International License

# Protocol of Pain

**Dalamagka M\***

Anesthesia Department, General Hospital of Larisa, Greece

## Mini Review



**\*Corresponding author:** Dalamagka M,  
Anesthesia Department, General Hospital  
of Larisa, Greece

**Submission:**  February 01, 2021

**Published:**  March 02, 2021

Volume 6 - Issue 1

**How to cite this article:** Dalamagka M.  
Protocol of Pain. Nov Res Sci. 6(1). NRS.  
000627. 2021.  
DOI: [10.31031/NRS.2021.06.000627](https://doi.org/10.31031/NRS.2021.06.000627)

**Copyright@** Dalamagka M, This article is  
distributed under the terms of the Creative  
Commons Attribution 4.0 International  
License, which permits unrestricted use  
and redistribution provided that the  
original author and source are credited.

When acute pain progresses to chronic, then it is not a symptom of a condition, but is in itself a condition. It lasts longer than the expected course of the disease or injury. Usual period of 3-6 months. In the past the answer to chronic pain was: "everything is in your mind". A useful definition by Margo McCaffrey is: "pain is what describes the person who experiences it and exists when he says he exists [1]." The International Union for the Study of Pain says it is "an unpleasant aesthetic and emotional experience, combined with actual or potential tissue damage, or described in terms of such damage". Pain is transmitted through the body through the nervous system when nerve endings detect damage to a part of the body. Today, pain specialists can understand how pain is created: the way the nervous system, including the spinal cord, interacts with the brain to create the sensation of pain [2]. Knowledge of the neurotransmitter system, the chemical messengers that transmit nerve signals, has paved the way for important new methods of treating pain. In recent years, scientists have learned how to manage these chemical messengers to change the way they interact with brain signals. This has led to the use of antidepressants and other drugs, which work with certain chemicals in the brain, such as which affect emotions and help in perceiving pain. There are now drugs that are very effective. And with advances in MRI, researchers can prove that the changes are very real in the brain. We can see exactly where the sensation of pain in the brain is created, when it is activated by a stimulus. We can see the effects of pain on a person's emotional state. There is also a new concept, a process called central awareness. If the initial pain from an injury is not treated properly, then these pain signals are sent repeatedly, resulting in changes in the central nervous system, which make it increasingly vulnerable. So over time, even normal stimuli can be perceived as painful. With this knowledge, pain specialists are now prescribing medications that attack moderate to severe chronic pain from different perspectives: innovative drugs, nerve-targeting techniques, and drug delivery pumps that offer strong nerve analgesia [3].

Doctors also approve of the use of psychotherapy, relaxation techniques, hypnosis and alternative therapies, such as acupuncture, which rely on the growing evidence of a brain-body connection to relieve chronic pain. There is still much to learn, but research has shown evidence of the development of even newer treatment options. Pain specialists now prescribe medications that attack moderate to severe chronic pain from different angles: innovative drugs, nerve-targeting techniques, and drug pumps, which offer strong nerve root analgesia. Doctors also approve of the use of psychotherapy, relaxation techniques, hypnosis and alternative therapies, such as acupuncture, which rely on the growing evidence of a brain-body connection to relieve chronic pain. There is still much to learn, but research has shown evidence of the development of even newer treatment options. Pain specialists now prescribe medications that attack moderate to severe chronic pain from different angles: innovative drugs, nerve-targeting techniques, and drug pumps, which offer strong nerve root analgesia.

Doctors also approve of the use of psychotherapy, relaxation techniques, hypnosis and alternative therapies, such as acupuncture, which rely on the growing evidence of a brain-body connection to relieve chronic pain. There is still much to learn, but research has shown evidence of the development of even newer treatment options. and drug delivery pumps, which offer a strong analgesia to the nerve roots. Doctors also approve of the use

of psychotherapy, relaxation techniques, hypnosis and alternative therapies, such as acupuncture, which rely on the growing evidence of a brain-body connection to relieve chronic pain. There is still much to learn, but research has shown evidence of the development of even newer treatment options. and drug delivery pumps, which offer a strong analgesia to the nerve roots. Doctors also approve of the use of psychotherapy, relaxation techniques, hypnosis and alternative therapies, such as acupuncture, which rely on the growing evidence of a brain-body connection to relieve chronic pain. There is still much to learn, but research has shown evidence of the development of even newer treatment options [4,5]. Many patients come for treatment in the final stages of chronic pain, when it is more difficult to treat. The sooner treatment is started, the better the chances of successfully treating the pain. When the pain is severe then doctors turn to stronger drugs to treat it:

a. Antiepileptics: Medications used to treat seizures are effective in treating chronic pain. It is unclear how they control pain, but it is thought to lead to milder effects of neuropathic pain, such as post-herpes zoster neuralgia. These include Pregabalin (Lyrica), gabapentin (Neurontin) and Carbamazepine (Tegretol). A new generation of antiepileptic drugs seems to be promising, with fewer side effects [6].

b. Antidepressants: Low doses of common antidepressants are prescribed for many chronic pain problems. These drugs regulate the levels of chemicals in the brain, and this is thought to be their mechanism in controlling pain. Antidepressants often help when other treatments do not lead to complete pain control. They lead to pain relief, whether the person suffers from depression or not. The doses used to treat pain are usually lower than those used to treat depression. Amitriptyline (Elavil), Nortriptyline (Pamelor) and Norpramin are tricyclic antidepressants, prescribed for chronic pain, especially cancer pain, neuropathic pain from diabetic neuropathy and postherpetic neuralgia from shingles. They affect the levels of chemicals in the brain, such as norepinephrine and serotonin. Duloxetine (Cymbalta) is a serotonin and norepinephrine reuptake inhibitor, which increases the availability of brain chemicals serotonin and norepinephrine. Duloxetine has been approved for the treatment of diabetic neuropathy, fibromyalgia and musculoskeletal pain, such as osteoporosis and chronic back pain [7].

c. Pain relief creams: Topical analgesics, such as capsaicin-containing Zostrix, are often helpful. Capsaicin works by reducing the transmission of a pain chemical called substance P to the brain. Products with these ingredients also have a similar effect: salicylate (found in products such as Aspercreme and Bengay), a substance that reduces inflammation and provides pain relief, and anti-irritants such as camphor, eucalyptus oil and the menthol, which lead to pain relief by causing cold or heat at the site of the pain [8].

d. Skin patches: A transdermal patch containing lidocaine can provide relief from chronic pain. Patches have been approved for neuropathic pain from shingles, a condition known as post-herpes neuralgia. Lidoderm and Lidopain are two skin patches of

lidocaine. Capsaicin is also available in a patch and is placed by the doctor himself and is called Qutenza. It can be used every three months [9].

e. Opioids: When the pain is severe then the next stage is opioids. Opioids such as codeine, fentanyl, morphine, oxycodone act on pain receptors at the level of nerve cells and are very effective in controlling severe chronic pain. But opioid use has always been controversial. There is a perception among doctors that they will get into legal trouble if they undergo treatment or show excessive zeal in treating opioid pain. It is a factor associated with inadequate training on these drugs. They are very effective for the right patients. They should be used carefully, but they can be used in the long run. There is a small risk of addiction. But studies show that the risk is small when used properly. When prescribed because, pain specialists often use combinations of medications, such as new prolonged-release antidepressants. The combination of drugs allows us to reduce the amount of opioids and leads to better pain control, because the mechanism of action of opioids is different from other drugs, such as antidepressants and antiepileptics. This approach is critical to treating neuropathic pain, such as diabetic neuropathy. Tramadol (Ultram ER) is a non-opioid drug that acts on opioid receptors. It is indicated in moderate to moderate pain, when continuous pain management is required. Synthetic opioids do not appear to be addictive. They are effective in treating many different pain syndromes. Many doctors prefer them before moving on to opioids [10].

f. Among the newest opioids for pain control are: The Duragesic transdermal patch for the treatment of moderate to severe pain. Provides continuous supply of opioid fentanyl for 72 hours. More options for pain flares: There are two fast-acting drugs that contain fentanyl. They were developed for cancer patients who have sudden pain and are already taking opioids for cancer pain. Fentanyl citrate (Actiq) comes in the form of a lollipop and Fentora is a soluble tray in the mouth.

#### **The following procedures can also help control pain:**

i. Nerve blockages: When a group of nerves causes pain in a specific organ or area of the body, the pain can be ruled out by injecting a local anesthetic. This is a nervous breakdown. Injections and nerve blocks are more effective in treating acute pain. But also, in patients with a depressed nerve, nerve blockages can alleviate the pain so that the patient can function and start physical therapy. And if treatment is started early, the development of chronic pain will be prevented. Radiofrequency ablation: A small area of nerve tissue is heated to reduce pain signals from that area. The procedure is performed under the guidance of CT imaging. A needle is inserted into the affected nerve area and an electric current is used to thermally destroy the target. A new technique, the application of high frequency pulses, offers only neurotransformation, without leading to nerve damage, as the temperature does not exceed 45 oC. The control of chronic pain lasts from three to six months. This is a great advantage, because it is a very localized and very specific treatment for pain. It's not a panacea, but it can really make

a difference in some cases. TENS : Percutaneous electrical nerve stimulation. The treatment is useful for short-term pain relief. It includes a small device, which distributes low level electricity and helps to exclude pain. It is very useful in the treatment of various types of muscle pain and is often used with infusions at T trigger point (myoperitoneal trigger points, which are alginate in pressure, parts of the body).

ii. Trigger point injection: These are sore spots on muscle or connective tissue. They can sensitize the nerves around them and cause pain in other parts of the body. Particular sensitivity can also develop in nearby muscles or areas of the body. An injection of a local anesthetic (sometimes with a steroid) is given into the trigger point to relieve the pain. It usually requires only a few treatments to resolve the trigger point and the pain that arises from it. It is a relatively simple and safe process.

iii. Pain pacemakers: The technique is called electrical spinal cord stimulation and involves a pacemaker (neurostimulator) implanted in the body. The neurostimulator provides low-level electrical signals to the spinal cord or to specific nerves and prevents the transmission of pain to the brain (electrodes are placed in the epidural space and connected to the neurostimulator). The patient can adjust the on/off switch and adjust the intensity of the electrical signals. Electrical stimulation of the spinal cord is applied when other techniques have failed, as well as when a cancerous pain has infiltrated a nerve root. Implantable drug delivery pumps. These are also called intrathecal pumps because they send analgesic drugs to the spinal cord. Local anesthetics, Opioids and other analgesics can be given through these implantable pumps. At the touch of a button, it is injected analgesic and nerve block, so as not to transmit pain to the spinal cord. These pumps are often used in cancer patients, but also in patients who have tried drugs but developed side effects. The dosage is much lower than that of the oral one, so the side effects are less. There is also a psychological benefit to pumps, as controlling pain can help in the prevention of post-traumatic stress.

iv. Surgery: Surgery can help in some cases. Removing a tumor can offer pain relief, as can shrinking a tumor with radiotherapy. In neurosurgery, a cross-section of the nerves is made to control the pain.

### Advice for better

Dealing with chronic pain: It is very difficult to live with a chronic pain, which leads to depression, anxiety, anger and can make the pain worse. Negative emotions reduce the body's endogenous opioids and increase its sensitivity to pain. When chronic pain settles, the person's life shrinks to give way to pain. Activities are limited and this perpetuates the vicious cycle of pain as the perception of pain becomes worse. Health, work and interpersonal

relationships "bleed". Sleep and mood disorders perpetuate the sensation of pain. With counseling, patients gain skills in managing chronic pain. They can also find solutions to everyday problems that cause them stress and depression [11].

### Alternative approaches to

**Chronic pain:** Stress aggravates the pain, so a relaxation technique is great useful in all types of pain. When the patient is upset with something, his pain will rise several points on the pain scale. Biofeedback, for example, helps people train their minds to control bodily functions such as muscle tension, respiration and heart rate, leading to a reduction in stress and stress responses. Relaxation techniques are an important part of treating pain. Deep breathing, meditation, guided mental imagery, and hypnosis allow the mind to help the body. Regular exercise helps reduce stress and promote relaxation, which helps relieve chronic pain. Acupuncture, a traditional Chinese technique, has earned the respect of Western medicine. The National Institutes of Health recognizes acupuncture as an effective way to treat pain, especially in headaches and back pain, and suggests that the technique can help with other chronic pain conditions, such as arthritis, fibromyalgia and muscle aches. Acupuncture is extremely useful with pain and more and more insurance companies abroad are covering acupuncture treatments.

### References

1. Margo McCaffery, Chris Pasero (1999) Pain: Clinical manual. 2<sup>nd</sup> (edn.), Mosby publishers, US, pp. 1-795.
2. French JA, Gazzola DM (2011) New generation antiepileptic drugs: What do they offer in terms of improved tolerability and safety? *Ther Adv Drug Saf* 2(4): 141-158.
3. Breivik H, Collett B, Ventafridda V, Cohen R, Gallacher D (2006) Survey of chronic pain in Europe: prevalence, impact on daily life, and treatment. *Eur J Pain* 10(4): 287-333.
4. Morley S, Eccleston C, Williams AC (1999) Systematic review and meta-analysis of randomized controlled trials of cognitive behavioral therapy for chronic pain in adults, excluding headache. *Pain* 80(1-2): 1-13.
5. (2003) Pain management services: Good practice. RCoA and BPS, UK.
6. (2006) Desirable criteria for pain treatment facilities. IASP Seattle, USA.
7. (2004) A practical guide to the provision of chronic pain services for adults in primary care. BPS, UK, pp. 1-90.
8. Maniadakis N, Gray A (2000) The economic burden of back pain in the UK. *Pain* 84(1): 95-103.
9. Jensen MP, Chodroff MJ, Dworkin RH (2007) The impact of neuropathic pain on health-related quality of life: review and implications. *Neurology* 68(15): 1178-1782.
10. (2003) Acupuncture: Review and analysis of reports on controlled clinical trials. WHO Health Systems Library, pp. 1-87.
11. Dimitris Vasilakos, Pain & its treatment.

For possible submissions Click below:

Submit Article



## Dimorphic Behavior and cognition

Maria I Dalamagka

### ► To cite this version:

Maria I Dalamagka. Dimorphic Behavior and cognition. Clinical Research and Clinical Trials. ISSN: 2693-4779 , 2024, 10.31579/2693-4779/235 . hal-05433482

**HAL Id: hal-05433482**

**<https://hal.science/view/index/docid/5433482>**

Submitted on 28 Dec 2025

**HAL** is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



Distributed under a Creative Commons Attribution 4.0 International License

# Dimorphic Behavior and cognition

Maria I. Dalamagka\*

Anesthesiology department, General Hospital of Larisa, Greece

**\*Corresponding Author:** Maria I. Dalamagka, Anesthesiology department, General Hospital of Larisa, Greece.

**Received date:** September 21, 2024; **Accepted date:** October 16, 2024; **Published date:** November 29, 2024

**Citation:** Maria I. Dalamagka, (2024), Dimorphic Behavior and cognition, *Clinical Research and Clinical Trials*, 11(2); DOI:10.31579/2693-4779/235

**Copyright:** © 2024, Maria I. Dalamagka. This is an open access article distributed under the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited.

## Abstract:

Sexual differences in the structure and functions of the human brain have been the subject of much speculation ever since the time of Greek antiquity. Aristotle, designated the moment at which the male fetus receives its soul at the 40th day of gestation, whereas the female fetus was supposed to become animated only six weeks later, around the 80th day of pregnancy. In the course of 19th century, the interest in the sexual dimorphism of the human brain grew rapidly. The first studies reported that the male brains were larger and more asymmetrical than the female brains and that men had relatively more brain substance in front of the central sulcus than behind. The existence of these comparatively minor and seemingly random morphological sex differences in the human brain were often used in support of the biological view of that era, that men were intellectually superior to women and that white upperclass people were superior to the other races and lower classes.

**Keywords:** sex hormones; behavior; human brain; X-chromosome

## Introduction

Sex differences in behavior are the result of natural and sexual selection. The dimorphic classes of behavior described here, courtship, copulatory, and parental behaviors, reflect both kinds of evolutionary selective pressures. The term dimorphism refers to the existence of two distinct forms within a single species. The term sexually dimorphic behavior, by extension, implies two different forms of behavior exhibited by the male and the female. To describe these behavioral differences between the sexes as sexual dimorphisms does not violate common usage of the term by the morphological sciences. Among mammalian forms, both sexes have a pelvis. The difference between the sexes is not in the presence or absence of a pelvis or pelvic outlet, but in its size, girth, or other quantitative measure. Countless examples exist in morphology of sexual dimorphisms based only on quantitative differences, differences in intensity, or in response of a specific structure to hormonal stimulation. Current concepts of morphogenesis hold that the genetic sex of all vertebrates determines whether the embryonic genital ridge develops into a testis or an ovary. The means of action by which chromosomes direct the differentiation of the embryonic gonad are unknown; but it is known that the type of gonad differentiated determines by its secretory products whether male or female secondary reproductive organs develop. According to the organizational hypothesis (Phoenix et al., 1959), not only the reproductive organs but also the neural processes mediating sexual behavior in mammals have the intrinsic tendency to develop according to a female pattern of body structure and behavior. Sex steroids can be regarded as master regulators of sex-specific behaviors (Morris et al., 2004; Baum, 2003). The developmental influence (organizational role) of sex hormones can lead to enduring effects on brain and behavior. By contrast, in adults sex steroids elicit reversible changes (activational role) in neural circuits and behavior. Gonadal hormones bind to distinct nuclear hormone receptors that are essential for sex-typical displays (Scordalakes and Rissman, 2003; Raskin et al., 2009; Kudwa and Rissman, 2003;

Wersinger et al., 1997; Juntti et al., 2010; Ogawa et al., 2000; Lydon et al., 1995). These receptors directly regulate gene expression by binding DNA (Mangelsdorf et al., 1995), and they can initiate nontranscriptional signaling via mechanisms such as interactions with intracellular kinases and transmembrane receptors (Foradori et al., 2008; Lishko et al., 2011; Micevych and Dominguez, 2009; Revankar et al., 2005; Vasudevan and Pfaff, 2008; McDevitt et al., 2008). Sex hormones or their metabolites can also bind to neurotransmitter receptors to gate their activity (Henderson, 2007). Such nontranscriptional signaling can control neural function at time scales that allow real time modulation of behavior. Prior work has identified genes downstream of sex hormones that regulate sexually dimorphic behaviors (Kayasuga et al., 2007; Wersinger et al., 2002; Nelson et al., 1995; Winslow and Insel, 2002). The relative paucity of such genes is in contrast to the diversity of these behaviors, and suggests that the underlying neural circuits may be regulated largely by nontranscriptional hormone signaling [1-24].

## Results and Discussion

Sexually dimorphic influences on human cognition and behaviour may affect the phenotypic expression of 'disorders' and 'traits'. 'Disorders' are sporadic/heritable abnormalities due to non-functional or otherwise mutated genes. 'Traits' represent normal variation in sexually dimorphic characteristics. X-linked disorders, such as fragile X syndrome or Rett syndrome, are sexually dimorphic in their expression but they represent extreme cases – dysfunction of a critical gene, even though expression of that gene might be neither dominant nor recessive in the conventional Mendelian sense. X-linked behavioural traits, quantitative variants, include male aggression and parental behaviour. Sexually dimorphic cognitive traits include spatial orientation; in rodents, male spatial learning advantages observed in the radial or water maze are caused by male-female differences

in strategy selection. Females (rats and humans) navigate preferentially using landmarks, but males rely on a broader set of spatial representations. These traits are probably influenced by a Y-linked locus, although an X-linked locus may play a contributory role. During evolution, could X-linked genes for specific cognitive abilities, and a female preference for males who demonstrate those traits, have become closely linked, and hence jointly inherited? Owing to the obligatory expression of all X-linked genes in males, any X-linked trait that is advantageous to males (or to females) would spread rapidly in the population. If higher cognitive abilities were a critical step in our own evolution, it makes sense that you might find those functions on the X-chromosome.

## Conclusion

Studies on human facial sexual dimorphism have yielded intriguing insights into perceptions of other characteristics, such as attractiveness or trustworthiness, based on facial masculinity or femininity. Specifically, by manipulating the degree of sexually dimorphic facial traits in computerized faces, scientists have been able to study what judgmental differences arise as a result of physical alterations. What is significant about these findings is their evolutionary implications, in particular the facial cues that trigger innate judgments in reaction to a masculine or feminine face. Multiple studies have confirmed the correlations between sexually dimorphic faces and ratings of attractiveness, and many psychologists theorize that such judgments are evolutionarily based. Sexual selection in humans is largely based upon facial cues and their reflection of an individual's reproductive quality. In a study of Little et al. (2008) found that symmetry and sexual dimorphism in faces are both judged as more attractive to the opposite sex, leading the researchers to conclude that both qualities are reflective of biological quality, and that such judgments are likely to be the result of sexual selective pressures and mate choice preferences. Much research has been conducted on external judgments of personality as they relate to facial asymmetry and neuroticism, facial symmetry and personality, facial attractiveness and narcissism and judgment accuracy differences between the sexes.

## References

1. Blaustein, J.D. (2008). Neuroendocrine regulation of feminine sexual behavior: lessons from rodent models and thoughts about humans. *Annu. Rev. Psychol.* 59, 93-118.
2. de Bono, M., and Bargmann, C.I. (1998). Natural variation in a neuropeptide Y receptor homolog modifies social behavior and food response in *C. elegans*. *Cell* 94, 679-689.
3. Cahill, L. (2006). Why sex matters for neuroscience. *Nat. Rev. Neurosci.* 7, 477-484.
4. Canteras, N.S., Simerly, R.B., and Swanson, L.W. (1995). Organization of projections from the medial nucleus of the amygdala: a PHAL study in the rat. *J. Comp. Neurol.* 360, 213-245.
5. Carroll, J.S., Meyer, C.A., Song, J., Li, W., Geistlinger, T.R., Eeckhoute, J., Brodsky, A.S., Keeton, E.K., Fertuck, K.C., Hall, G.F., et al. (2006). Genome-wide analysis of estrogen receptor binding sites. *Nat. Genet.* 38, 1289-1297.
6. Clodfelter, K.H., Holloway, M.G., Hodor, P., Park, S.-H., Ray, W.J., and Waxman, D.J. (2006). Sex-dependent liver gene expression is extensive and largely dependent upon signal transducer and activator of transcription 5b (STAT5b): STAT5b-dependent activation of male genes and repression of female genes revealed by microarray analysis. *Mol. Endocrinol.* 20, 1333-1351.
7. Cooke, B.M. (2006). Steroid-dependent plasticity in the medial amygdala. *Neuroscience* 138, 997-1005.
8. Cooke, B., Hegstrom, C.D., Villeneuve, L.S., and Breedlove, S.M. (1998). Sexual differentiation of the vertebrate brain: principles and mechanisms. *Front. Neuroendocrinol.* 19, 323-362.
9. Dugger, B.N., Morris, J.A., Jordan, C.L., and Breedlove, S.M. (2007). Androgen receptors are required for full masculinization of the ventromedial hypothalamus (VMH) in rats. *Horm. Behav.* 51, 195-201.
10. Dulac, C., and Wagner, S. (2006). Genetic analysis of brain circuits underlying pheromone signaling. *Annu. Rev. Genet.* 40, 449-467.
11. Enoch, M.-A., Hodgkinson, C.A., Yuan, Q., Albaugh, B., Virkkunen, M., and Goldman, D. (2009). GABRG1 and GABRA2 as independent predictors for alcoholism in two populations. *Neuropsychopharmacology* 34, 1245-1254.
12. Foradori, C.D., Weiser, M.J., and Handa, R.J. (2008). Non-genomic actions of androgens. *Front. Neuroendocrinol.* 29, 169-181.
13. Gagnidze, K., Pfaff, D.W., and Mong, J.A. (2010). Gene expression in neuroendocrine cells during the critical period for sexual differentiation of the brain. *Prog. Brain Res.* 186, 97-111.
14. Garcia, L.R., Mehta, P., and Sternberg, P.W. (2001). Regulation of distinct muscle behaviors controls the *C. elegans* male's copulatory spicules during mating. *Cell* 107, 777-788.
15. Henderson, L.P. (2007). Steroid modulation of GABAA receptor-mediated transmission in the hypothalamus: effects on reproductive function. *Neuropharmacology* 52, 1439-1453.
16. Barrett, L. F., & Pietromonaco, P. R. (1997). Accuracy of the five-factor model in predicting perceptions of daily social interactions. *Personality and Social Psychology Bulletin*, 23, 1173-1187.
17. Bronstad, M., Langlois, J., Russell, R. (2008). Computational models of facial attractiveness judgments. *Perception*, 37, 126-142.
18. Carré, J., McCormick, C., Mondloch, C. (2009). Facial Structure Is a Reliable Cue of Aggressive Behavior. *Psychological Science*, 20, 1194-1198.
19. Fink, B., Neave N., Manning J., Grammer K. (2006). Facial symmetry and judgements of attractiveness, health and personality. *Personality and Individual Differences*, 41, 491-499.
20. Fink, B., Neave N., Manning J., Grammer K. (2005). Facial symmetry and the 'big-five' personality factors. *Personality and Individual Differences*, 39, 523-529.
21. Holtzman, N.S., Strube, M.J. (2009). Holtzman, N.S., Strube, M.J., Narcissism and Attractiveness, *Journal of Research in Personality*.
22. Noor, F., Evans, D.C. (2003). The effect of facial symmetry on perceptions of personality and attractiveness. *Journal of Research in Personality*, 37, 339-347.
23. Penton-Voak, I.S., Pound, N., Little, A.C., & Perrett, D.I. (2006). Personality judgments from natural and composite facial images: More evidence for a "kernel of truth" in social perception. *Social Cognition*, 24, 490-524.
24. Smith, F., Jones, B., DeBruine, L., Little, A. (2008). Interactions between masculinity-femininity and apparent health in face preferences. *Behavioral Ecology*, 20.

25. Khil PP & Camerini-Otero RD. (2005). Molecular features and functional constraints in the evolution of the mammalian X

chromosome. Critical Reviews in Biochemistry and Molecular Biology 40 313-330.



This work is licensed under Creative Commons Attribution 4.0 License

To Submit Your Article Click Here:

**Submit Manuscript**

DOI:[10.31579/2693-4779/235](https://doi.org/10.31579/2693-4779/235)

**Ready to submit your research? Choose Auctores and benefit from:**

- fast, convenient online submission
- rigorous peer review by experienced research in your field
- rapid publication on acceptance
- authors retain copyrights
- unique DOI for all articles
- immediate, unrestricted online access

At Auctores, research is always in progress.

Learn more <https://auctoresonline.org/journals/clinical-research-and-clinical-trials>



# Pediatric neurodegenerative disorders and anesthesia considerations

Maria I Dalamagka

## ► To cite this version:

Maria I Dalamagka. Pediatric neurodegenerative disorders and anesthesia considerations. GSC Advanced Research and Reviews , 2025, 22, pp.077 - 083. 10.30574/gscarr.2025.22.1.0006 . hal-05434058

**HAL Id: hal-05434058**

**<https://hal.science/view/index/docid/5434058>**

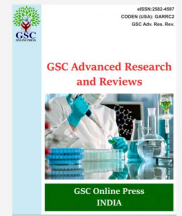
Submitted on 29 Dec 2025

**HAL** is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



Distributed under a Creative Commons Attribution 4.0 International License



## Pediatric neurodegenerative disorders and anesthesia considerations

Maria I. Dalamagka \*

*Department of Anesthesiology, General Hospital of Larisa, Greece.*

GSC Advanced Research and Reviews, 2025, 22(01), 077-083

Publication history: Received on 27 November 2024; revised on 06 January 2025; accepted on 09 January 2025

Article DOI: <https://doi.org/10.30574/gscarr.2025.22.1.0006>

### Abstract

Pediatric neurodegenerative disorders encompass a wide range of conditions that result from progressive damage to cells and nervous system connections that are essential for mobility, coordination, strength, sensation, and cognition. Neurodegenerative diseases affect millions of people worldwide. Neurological disorders are disorders that involve the brain, spinal cord or nerve and muscle. Children with neurological disorders may have disorders such as epilepsy, developmental delay, cerebral palsy, meningitis, genetic/metabolic diseases or diseases of muscle or nerve such as muscular dystrophy or peripheral neuropathy. These illnesses are frequently severe and potentially have long lasting consequences for the affected children. The pathologies most frequently responsible for psycho-physical disorders can be summarized into three groups: collaboration difficulties (autism spectrum disorders, intellectual impairment, phobia); motor dysfunction (cerebral palsy, epilepsy, other brain pathologies, neuromuscular disorders), and craniofacial anomalies (Down syndrome, other genetic syndromes). Due to their clinical history, and the lack of collaboration, in pediatric non-cooperative patients with Special Needs (SN), the anesthesiological risk may result difficult to assess, despite the fact that they meet the criteria of the outpatient management applied for the patient without disabilities. The perioperative management of pediatric patients with psycho-physical disorders with related relational and cognitive problems must be carefully planned, in order to make the entire hospitalization process as comfortable and as less traumatic as possible. General anesthesia is the most suitable type of anesthesia in pediatric patients with Special Needs (SN), although anesthetic complications are more frequent in these patients, mainly due to comorbidities, taken drugs and anatomical peculiarities. Some studies observed no correlation between multiple anesthesia exposures and neurodevelopmental deficits, while others reported that even a single exposure could increase the risk of deficits. An increased risk of developmental or behavioral disorders as the consequence of surgeries requiring general anesthesia was observed. Based on current studies, it is necessary to endeavor to limit the duration and numbers of anesthesia and the dose of anesthetic agents. The evaluation must include history and physical examination pertaining to the conditions requiring special anaesthetic considerations. Early diagnosis and intervention are crucial in managing these neurological disorders in pediatric patients to improve overall quality of life and long-term outcomes.

**Keywords:** Neurological; Disorder; Pediatric; Anesthesia; Seizures; Complications

### 1. Introduction

Pediatric functional neurological disorder (FND) is a common condition that can affect multiple types of movement or sensation. It is caused by abnormalities in how the brain responds to sensory information. The brain “misfires” and sends signals to the body by mistake. These abnormalities are present at birth and do not come from or cause injury to the structure of the brain. This makes the recovery of FND better, on average, than many brain-based disorders. FND is sometimes confused with other conditions. Doctors used to call it “conversion disorder” or “psychogenic,” but these terms are not up to date with the latest science. You may also hear people refer to “functional movement disorder” or “non-epileptic seizures.” These are subtypes of FND. Importantly, people with FND are not “faking it.” They are not

\* Corresponding author: Maria I. Dalamagka

consciously controlling their symptoms or using the symptoms to get attention. Neurological disorders commonly seen in pediatric patients include attention deficit hyperactivity disorder (ADHD), developmental delays, epilepsy, migraines, and neuromuscular diseases. These conditions are commonly found in pediatric patients due to various factors such as genetic predisposition, environmental influences, and abnormalities in brain development. ADHD is characterized by inattention, hyperactivity, and impulsivity, and is often diagnosed in childhood. Developmental delays encompass a wide range of conditions that affect a child's physical, cognitive, communication, and social skills. Epilepsy is a neurological disorder characterized by recurrent seizures and is one of the most common neurological conditions in children. Migraines, while often associated with adults, can also occur in pediatric patients and are characterized by severe headaches, often accompanied by nausea, vomiting, and sensitivity to light and sound. Neuromuscular diseases, such as muscular dystrophy, can also manifest in childhood and affect the muscles and their control over movements. There are four types of FND. Each type is named for the effect it has on the body, but all are thought to come from the same brain changes. A person can have more than one type of FND at a time, and the forms they have can change over time. Functional movement disorders are one of the most common types of FND. It affects how a child moves or walks. It can be episodic (come and go) or be constant. Functional seizures (non-epileptic seizures) are another common type. It causes a child to have jerky, shaky episodes. Functional seizures are sometimes mistaken for epilepsy, but they have different causes. Epilepsy is caused by uncontrolled electrical signals in the brain, and functional seizures are caused by the brain handling information incorrectly. Functional disorders of consciousness involve blackouts (periods of unresponsiveness) or episodes of confusion. Functional sensory disorders may result in tingling, numbness or other unusual sensations in parts of the body. These sensations follow patterns that are different than when they have a brain or nerve injury. Symptoms of pediatric functional neurological disorders (FND): weakness, tremor, falls, events that look like seizures, abnormal walking, changes to vision or hearing. Epilepsy is one of the most prevalent neurological disorders in children, characterized by recurrent seizures. Seizures can vary in severity and frequency, affecting a child's daily activities and quality of life. Causes of epilepsy in children may include genetic factors, brain injury, or developmental disorders. Treatment often involves medications to control seizures, lifestyle modifications, and sometimes surgical interventions. A set of conditions known as cerebral palsy impact motor abilities, tone of muscles, and movement. Damage to the developing brain, which frequently happens prior to birth or throughout infancy, is the cause of it. Children with cerebral palsy may experience difficulties with walking, coordination, and fine motor skills. Treatment aims to improve mobility and function through physical therapy, occupational therapy, and sometimes surgical interventions. Autism Spectrum Disorder (ASD) is a kind of developmental disability that impacts behavior, social interaction, and communication. While the exact cause of ASD is unknown, it is believed to involve a combination of genetic and environmental factors. Children with ASD may exhibit repetitive behaviors, difficulties with verbal and non-verbal communication, and challenges in social situations. Early intervention with behavioral therapy, speech therapy, and educational support can help improve outcomes for children with ASD. Attention-Deficit/Hyperactivity Disorder (ADHD) is a neurodevelopmental disease marked by hyperactivity, impulsivity, and trouble focusing. It can significantly impact a child's academic performance, social relationships, and overall functioning. While the exact cause of ADHD is not fully understood, it is thought to involve a combination of genetic, environmental, and neurological factors. Treatment often involves a combination of behavioral therapy, medication, and support services. Developmental delay refers to a child's failure to reach developmental milestones within the expected time frame. It can encompass delays in areas such as language, motor skills, cognitive abilities, and social development. Developmental delay can result from various factors, including genetic conditions, prenatal exposure to toxins, or environmental factors. Early intervention with therapies such as speech therapy, occupational therapy, and educational support can help address developmental delays and promote optimal development [1-7].

---

## 2. Neurological Disorders and drug interactions

Neurological disorders in children occur when something is abnormal in the brain, the nervous system or the muscle cells. These disorders can vary from epilepsy to migraine headaches to tic or movement disorders and more. International scientific literature defines pediatric patients with "special needs" (SN) as children suffering from psychophysical disorders with related relational and cognitive problems. The perioperative management of these pediatric patients must be carefully planned, in order to make the entire hospitalization process as comfortable and as least traumatic as possible. The standard perioperative path is inapplicable, mainly due to the surgical and anesthetic criticalities. These patients have a much-increased perioperative risk due to their anatomical features, underlying disease and the coexistence of complex comorbidities. Down Syndrome (DS) represents 9% of all interventions in patients with SN and 1.25% of all anesthetic procedures. Patients with DS have peculiar anatomical features such as a low set of ears, small teeth, flat nose (flat nasal bridge), stature delay, abnormal fingerprints and hypotonia. They also present microbrachycephaly, short neck, macroglossia and protruding tongue, hypertrophic tonsils and adenoids, narrow subglottic area and prolapsed epiglottis. Frequently, these patients are obese and obesity can add further anesthetic problems: omega epiglottis and tracheomalacia. Patients with DS are at risk of atlantoaxial instability (AAI). This condition is also referred to as an atlantoaxial subluxation and it has been reported to occur in 6.8% to 27% of the

DS population. Cri du Chat Syndrome is a very rare congenital genetic syndrome with peculiar clinical characteristics such as microcephaly, enlarged nose root, epicanthus, hypertelorism, low-set ears, micrognathia and typical acute crying which gives the name of the syndrome. The acute crying pitch depends on laryngeal anatomical abnormalities, including laryngomalacia with vocal cord paralysis, small and/or narrow larynx with diamond-shaped vocal cords, and epiglottitis, which may be elongated, curved and fluctuating, or short and flaccid, hypoplastic and hypotonic. Autism Spectrum Disorders (ASD) cause impairment in social interaction, communication and isolation problems, behavioral disturbances, disorders of the sensory sphere, as well as significantly altered motor skills. The problems of communication and social interaction are linked to a limited vocabulary, language deficits (deficit of acquisition, echolalia, occasional mutism), the inability to understand figurative language and the meaning of conversations. Some patients exhibit behavioral disturbances, such as self-mutilation, aggression, psychotic disturbances, resistance to change, gesture repetition, mechanical memory, excessive emotional reactions, and motor awkwardness, stereotyped patterns of movement, language or manipulation of objects, excessive adherence to a daily routine made up of motor or verbal rituals. Infantile cerebral palsy (CP) includes several clinical manifestations, from monoplegia with normal cognitive function to spastic quadriplegia with mental retardation. The causes are multiple and are responsible for damage to the central nervous system that can occur in the prenatal, perinatal or postnatal period. The Spastic Form, associated with intellectual disability and epilepsy, presents an altered control of the buccal muscles, tongue and pharynx resulting in various degrees of difficulty in feeding (nasogastric feeding or gastrostomy are often necessary). The dyskinetic form can be associated with dystonia, athetosis and chorea; epilepsy often concomitates, while the IQ is usually normal. Balance disorders, cerebellar tremors, language difficulties, epilepsy and cognitive deficits are associated with the ataxic form. Epileptic patients usually take long-term anti-epileptic therapy; monotherapy antiepileptic drugs are often effective in seizure control and a significant number of these patients require polypharmacological anti-comital therapy. It can be associated with the treatment of concomitant pathologies. Epileptic patients are exposed to an increased risk of clinically significant drug interactions, especially with traditional anti-epileptic drugs. With regards to the interactions examined between new generation anti-epileptic drugs and drugs used for the treatment of related non-epilepsy disorders, pharmacokinetic interactions have always referred to hepatic enzyme induction or inhibition. Furthermore, pharmacological dosages of anesthetic drugs, such as hypno-inducers was highlighted. The required dose of Propofol is significantly lower and the recovery time was longer. Oral benzodiazepines (midazolam) may be beneficial for small children as they provide sedation without respiratory depression and should be administered under supervision. An intravenous (IV) induction with propofol and neuromuscular block to facilitate endotracheal intubation is ideal in children. Children at risk for aspiration should undergo rapid-sequence anaesthetic induction. Anaesthesia is maintained either with low end-tidal volatile agents (minimum alveolar concentration (MAC) 1) or with total IV anaesthesia (TIVA), along with short-acting opioids (fentanyl or remifentanyl), inhaled nitrous oxide and controlled ventilation. Sevoflurane has almost replaced halothane as an agent for induction; isoflurane, sevoflurane or desflurane is used for maintenance of anaesthesia. Sevoflurane provides smooth induction followed by a rapid recovery. Neuromuscular blockade with non-depolarising muscle relaxants is given to prevent patient movement and minimise the amount of anaesthetic agent required. Fentanyl is the most commonly used opioids, but its half-life increases with repeated dosing. It requires hepatic metabolism which is immature in premature infants. The sedative and respiratory depressive effects of fentanyl may be prolonged in these children. Remifentanyl is unique in the sense it is cleared rapidly by the plasma esterases, but the associated rapid recovery may be accompanied by delirium and inadequate analgesia. However, it requires supplementation of other analgesics during postoperative period. The goal is "rapid awakening" in order to help early neurological assessment, haemodynamic stability, and minimal coughing and straining in the ETT. Sevoflurane provides better recovery profile, as compared to isoflurane in children. The anaesthetic concerns in epilepsy surgery, include perioperative seizures and the effects of anticonvulsants. Chronic use of anticonvulsants induces rapid metabolism and clears anaesthetic agents such as neuromuscular blockers and opioids, thereby increasing the anaesthetic requirements. Anaesthesia for children undergoing neurosurgery requires knowledge on cerebral pathophysiology and the interaction of anaesthetic agents on brain [8-19].

### 3. Neurodevelopmental disorders

More than 600 neurological disorders can affect people of all ages. Some conditions that can affect newborns, infants, children and teens include: Cerebral palsy: This group of disorders affects muscle control, causing difficulty with muscle movements. Cerebral palsy results from an injury or illness that affects brain development before birth. Seizures and epilepsy: Seizures are surges of electrical activity in the brain that affect behavior, sensations, movements or consciousness. Headache and migraine: Headaches and migraine can make learning difficult and affect a child's daily activities and quality of life. Sleep disorders: Children, especially teens, can have sleep disorders, which can lead to other problems such as daytime sleepiness and poor performance in school. Neurodegenerative diseases: These conditions occur when brain cells become damaged or begin to die. Neurological disorders in newborns: These conditions occur when an injury or other problem affects brain development before, during or shortly after birth. Neurodevelopmental disorders affect a child's development, causing physical, behavioral, learning and other difficulties. Attention-

deficit/hyperactivity disorder (ADHD): This condition causes behavior issues, such as self-control and ability to focus and concentrate. Autism spectrum disorder: This disorder affects a child's ability to develop healthy social and communication skills. Learning disabilities: This group of disorders causes difficulty with written and spoken language, math and other subjects. Rett syndrome: This genetic disorder occurs mostly in girls, affecting learning, social behaviors, speaking, balance and movement. Speech, language and swallowing disorders: Aphasia is a type of neurological disorder that affects the ability to read, write, speak and understand speech. Dysphagia, or difficulty swallowing, affects your ability to eat and drink. Tourette syndrome: This neurological disorder involves tics, which are sudden, repeated, involuntary (unintentional) movements or sounds. The disease manifest as a set of three of impaired social interaction, impairment of verbal (spoken) and non-verbal (unspoken) communication, alongside confined, repetitive patterns of behavior with a lot of adherence to routines and limited interests. Epilepsy is a common problem tracked down in individuals of all ages. This is a chronic condition characterized by the occurrence of multiple seizures. A seizure is caused as a result of a brief disturbance in the brain cells. Epilepsy might be caused by hereditary factors, developmental problems, infectious illnesses and injury caused to the brain preceding birth. Most headaches in aren't symptoms of neurological problems, nonetheless, at times they might be caused by a serious underlying issue. Headaches can be caused by changes in the brain or in the body which send pain messages to the brain and result in headaches. It is important to screen headaches especially when accompanied by other symptoms like reduced alertness, vomiting, clumsiness. Attention Deficit Hyperactivity Disorder (ADHD), one of the most widely known neurological problem than might in fact go on until adolescents in ADHD; frequently left undiagnosed and marked as naughty given the symptoms they experience like forgetfulness and fantasizing. Dyspraxia is a condition which makes it difficult to coordinate physical movements. Children with dyslexia can read to upto average levels, whenever diagnosed early. Cerebral Palsy influences motor skills and is brought about by brain damage caused before or during birth or childhood life. Pediatric Stroke, one in around 4000 new borns are affected by this condition of pediatric stroke. New born infants or children with sickle cell anemia, immune disorders or blood clotting troubles might be prone to this condition [19-24].

---

#### 4. Anesthesiology complications

The risk of postoperative anesthetic complications in patients with SN depends on the patient's ASA classification, clinical condition, type of anesthetic used and type of surgical procedure performed. About 4.2% of cases have moderate complications, such as hypotension. Airway obstruction is the most common complication, followed by nausea and vomiting. All drugs used may produce sedation and respiratory depression and should always be administered under supervision and monitoring. Anti-epileptic drugs cause a prolonged awakening time from anesthesia, but these drugs should not be discontinued during the preoperative in order to reduce perioperative seizures. Monoamine oxidase inhibitors (MAOIs) or selective serotonin reuptake inhibitors (SSRIs) have an increased risk of hypo/hypertension and prolonged awakening after general anesthesia. Risperidone can cause hypotension under general anesthesia and can lead arrhythmias. It may be appropriate to discontinue long-acting antipsychotics and switch to short-acting or lower dose antipsychotics after consultation with the psychiatrist. During anesthesia, Clozapine can cause agranulocytosis, hyperthermia, cardiac conduction problems and hypotension. Psychostimulants can: increase the dose of sedative needed during anesthesia and the risk of hypertension and arrhythmias, lower the seizure threshold and interact with vasopressor. Congenital heart disease affects approximately 50% of patients with DS and this can increase the risk of complications during anesthesia, such as bradycardia. The incidence of bradycardia in DS patients is approximately 3.7%, significantly higher rate than the approximately 0.36% found in healthy patients. Congenital heart disease can also lead to other complications such as pulmonary hypertension. Airway management can be difficult for patients with cerebral palsy due to excessive secretions and the risk of aspiration during anesthesia due to gastroesophageal reflux. These patients also have a greater risk of hypoxia during general anesthesia. In addition, about 30% of patients with cerebral palsy take home anti-comital therapy with an increased risk of drug interactions. General anesthetics mitigate distress and exaggerated hemodynamic responses to pain and stressful stimulation, allowing surgery and diagnostic procedures to be performed worldwide in millions of children every year. The safety of pediatric anesthesia has increased over the past several decades with improved monitoring, equipment, medications, and growing subspecialization and regionalization of pediatric care. Despite these advances, anesthetic-related complications occur more often in children compared to adults [25-43].

---

#### 5. Discussion and Outcomes

The nervous system is the brain, spinal cord and nerves, which work together to control the body. Neurological diseases and disorders are conditions that affect the nervous system. They can range from minor issues, such as a headache, to life-threatening conditions, such as stroke or seizure. Causes of these conditions include genetic disorders, developmental issues, injuries and infections. Depending on which areas of the nervous system are affected, neurological disorders can cause problems with: muscle movement, thinking and learning, speaking and swallowing,

mood, behavior and sleep. Children are either born with the disorder, such as spina bifida or hydrocephalus (fluid in the brain), or they acquire the disorder later in life; the result of a traumatic injury or serious infection. Most studies induced or maintained a surgical plane of anesthesia with the application of multiple types of anesthetic drugs in combination. Despite their limitations, extant studies suggest that single brief anesthetic exposures do not appear to produce a measurable effect whereas repeated exposures consistently demonstrate associations between exposure and subsequent deficits in learning and behavior. It is still difficult to draw a definitive conclusion about the relationship between anesthesia and neurodevelopment based on the available studies. In the aspect of outcome measure, we need further exploration into a sensitive and stable outcome to estimate the neurocognitive harm in children. It's consistently certified in most studies covering prolonged exposure that a longer duration contributes to a higher risk of long-term diminution of neurocognitive function. However, longer cumulative duration and multiple exposures may be closely linked. Most studies leaving a lack of strong evidence for the validity of measuring neurodevelopmental changes due to anesthesia, probably because considering the complexity of its design, the difficulty of its implementation, the large human and financial resources required, and the associated ethical issues. To provide optimal neuroanaesthesia care, the anaesthesiologist must have the knowledge of basic neurophysiology of developing brain and effects of various drugs on cerebral haemodynamics. Some studies observed no correlation between multiple anesthesia exposures and neurodevelopmental deficits, while others reported that even a single exposure could increase the risk of deficits.

The anesthetic technique can play a significant role in altering the state of the brain during neurosurgical procedures through effects on the cerebral circulation and metabolism. Pre-existing neuromuscular disease may also have specific anesthetic implications such as cardiorespiratory involvement (eg, myotonia dystrophica), the potential for drug interactions (eg, myasthenia gravis) or abnormal responses to commonly used drugs (eg, malignant hyperthermia).

---

## 6. Conclusion

A neurologic disorder is caused by a dysfunction in the brain or nervous system (i.e. spinal cord and nerves). This dysfunction can result in physical and psychological symptoms. Due to their clinical history, and the lack of collaboration, in pediatric non-cooperative patients with Special Needs (SN), the anesthesiological risk may result difficult to assess, despite the fact that they meet the criteria of the outpatient management applied for the patient without disabilities. Reasons for this increased anesthetic risk in young children are multifactorial and include limited cardiopulmonary reserve, multiorgan immaturity, altered total body water composition relative to adults, limited pharmacokinetic and pharmacodynamics data on commonly used medications in children, altered sensitivity to drugs relative to older children, temperature lability, care team experience, and monitoring difficulties. Millions of children experienced surgery procedures requiring general anesthesia. Any potential neurodevelopmental risks of pediatric anesthesia can be a serious public health issue. Based on current studies, it is necessary to endeavor to limit the duration and numbers of anesthesia and the dose of anesthetic agents.

---

## Compliance with ethical standards

### *Disclosure of conflict of interest*

The author declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

---

## References

- [1] Mallineni, S.K.; Yiu, C.K.Y. Dental Treatment under General Anesthesia for Special-Needs Patients: Analysis of the Literature. *J. Investig. Clin. Dent.* 2016, 7, 325–331.
- [2] Stanková, M.; Buček, A.; Dostálová, T.; Ginzlová, K.; Pacáková, Z.; Seydlová, M. Patients with Special Needs within Treatment under General Anesthesia-Meta-Analysis. *Prague Med. Rep.* 2011, 112, 216–225.
- [3] Rathi, M.D.; Kashani, R.; Chinn, C.H.; Nandi, S.S. Compliance of Special Health Care Needs and Healthy Pediatric Patients with Preventive Visits after Dental Treatment under General Anesthesia. *J. Dent. Child.* 2021, 88, 74–79.
- [4] Chaudhary, K.; Bagharwal, P.; Wadhawan, S. Anesthesia for Intellectually Disabled. *J. Anaesthesiol. Clin. Pharmacol.* 2017, 33, 432–440.
- [5] Graham, R.J.; Wachendorf, M.T.; Burns, J.P.; Mancuso, T.J. Successful and Safe Delivery of Anesthesia and Perioperative Care for Children with Complex Special Health Care Needs. *J. Clin. Anesth.* 2009, 21, 165–172.

- [6] Roizen, N.J.; Patterson, D. Down's Syndrome. *Lancet* 2003, 361, 1281–1289.
- [7] Mitchell, V.; Howard, R.; Facer, E. Down's Syndrome and Anaesthesia. *Paediatr. Anaesth.* 1995, 5, 379–384.
- [8] Lollar, D.J.; Hartzell, M.S.; Evans, M.A. Functional Difficulties and Health Conditions among Children with Special Health Needs. *Pediatrics* 2012, 129, e714–e722.
- [9] Bouchard, M.; Bauer, J.M.; Bompadre, V.; Krengel, W.F. An Updated Algorithm for Radiographic Screening of Upper Cervical Instability in Patients With Down Syndrome. *Spine Deform.* 2019, 7, 950–956.
- [10] Hickey, F.; Hickey, E.; Summar, K.L. Medical Update for Children with Down Syndrome for the Pediatrician and Family Practitioner. *Adv. Pediatr.* 2012, 59, 137–157.
- [11] Comprehensive Language and Speech Treatment for Down Syndrome. Available online: <https://www.ds-health.com/speech.htm> (accessed on 31 July 2022).
- [12] Chorney, J.M.; Kain, Z.N. Behavioral Analysis of Children's Response to Induction of Anesthesia. *Anesth. Analg.* 2009, 109, 1434–1440.
- [13] Meyers, E.F.; Muravchick, S. Anesthesia Induction Technics in Pediatric Patients: A Controlled Study of Behavioral Consequences. *Anesth. Analg.* 1977, 56, 538–542.
- [14] Barbic, D.; Andolfatto, G.; Grunau, B.; Scheuermeyer, F.X.; Macewan, B.; Qian, H.; Wong, H.; Barbic, S.P.; Honer, W.G. Rapid Agitation Control With Ketamine in the Emergency Department: A Blinded, Randomized Controlled Trial. *Ann. Emerg. Med.* 2021, 78, 788–795.
- [15] Mihaljević, S.; Pavlović, M.; Reiner, K.; Ćaćić, M. Therapeutic Mechanisms of Ketamine. *Psychiatr. Danub.* 2020, 32, 325–333.
- [16] McCormick, A.S.M.; Thomas, V.L.; Berry, D.; Thomas, P.W. Plasma Concentrations and Sedation Scores after Nebulized and Intranasal Midazolam in Healthy Volunteers. *Br. J. Anaesth.* 2008, 100, 631–636.
- [17] Nishina, K.; Mikawa, K. Clonidine in Paediatric Anaesthesia. *Curr. Opin. Anaesthesiol.* 2002, 15, 309–316.
- [18] Dahmani, S.; Stany, I.; Brasher, C.; Lejeune, C.; Bruneau, B.; Wood, C.; Nivoche, Y.; Constant, I.; Murat, I. Pharmacological Prevention of Sevoflurane- and Desflurane-Related Emergence Agitation in Children: A Meta-Analysis of Published Studies. *Br. J. Anaesth.* 2010, 104, 216–223.
- [19] Jun, J.H.; Kim, K.N.; Kim, J.Y.; Song, S.M. The Effects of Intranasal Dexmedetomidine Premedication in Children: A Systematic Review and Meta-Analysis. *Can. J. Anaesth.* 2017, 64, 947–961.
- [20] Simonini, A.; Vittori, A.; Cascella, M.; Calevo, M.G.; Marinangeli, F. The Impact of Emergence Delirium on Hospital Length of Stay for Children Who Underwent Tonsillectomy/Adenotonsillectomy: An Observational Retrospective Study. *Braz. J. Anesthesiol.* 2021. online ahead of print.
- [21] Lewis, J.; Bailey, C.R. Intranasal Dexmedetomidine for Sedation in Children; a Review. *J. Perioper. Pract.* 2020, 30, 170–175.
- [22] Natoli, S. The Multiple Faces of Ketamine in Anaesthesia and Analgesia. *Drugs Context* 2021, 10, 2020-12-18.
- [23] Lee, J.-H.; Jung, H.-K.; Lee, G.-G.; Kim, H.-Y.; Park, S.-G.; Woo, S.-C. Effect of Behavioral Intervention Using Smartphone Application for Preoperative Anxiety in Pediatric Patients. *Korean J. Anesthesiol.* 2013, 65, 508–518.
- [24] Okur, S.; Arıkan, M.; Temel, G.; Temel, V. BIS-Guided Total Intravenous Anesthesia for Orchiopexy and Circumcision in a Child with Severe Autism: A Case Report. *Case Rep. Anesthesiol.* 2012, 2012, 718594.
- [25] Taghizadeh, N.; Davidson, A.; Williams, K.; Story, D. Autism Spectrum Disorder (ASD) and Its Perioperative Management. *Paediatr. Anaesth.* 2015, 25, 1076–1084.
- [26] Frost, E.A.M. Differential Diagnosis of Delayed Awakening from General Anesthesia: A Review. *Middle East J. Anaesthesiol.* 2014, 22, 537–548.
- [27] Choi, J.; Kim, S. Delayed Awakening Time from General Anesthesia for Dental Treatment of Patients with Disabilities. *J. Dent. Anesth. Pain Med.* 2021, 21, 219–226.
- [28] Sedaghat, A.R.; Flax-Goldenberg, R.B.; Gayler, B.W.; Capone, G.T.; Ishman, S.L. A Case-Control Comparison of Lingual Tonsillar Size in Children with and without Down Syndrome. *Laryngoscope* 2012, 122, 1165–1169.
- [29] Nolan, J.; Chalkiadis, G.A.; Low, J.; Olesch, C.A.; Brown, T.C. Anaesthesia and Pain Management in Cerebral Palsy. *Anaesthesia* 2000, 55, 32–41.

- [30] Wass, C.T.; Warner, M.E.; Worrell, G.A.; Castagno, J.A.; Howe, M.; Kerber, K.A.; Palzkill, J.M.; Schroeder, D.R.; Cascino, G.D. Effect of General Anesthesia in Patients with Cerebral Palsy at the Turn of the New Millennium: A Population-Based Study Evaluating Perioperative Outcome and Brief Overview of Anesthetic Implications of This Coexisting Disease. *J. Child Neurol.* 2012, 27, 859–866.
- [31] Maeda, S.; Tomoyasu, Y.; Higuchi, H.; Ishii-Maruhama, M.; Egusa, M.; Miyawaki, T. Independent Predictors of Delay in Emergence from General Anesthesia. *Anesth. Prog.* 2015, 62, 8–13.
- [32] Association American Psychiatric (2013). *Diagnostic and Statistical Manual of Mental Disorders*. Arlington: American Psychiatric Publishing.
- [33] Backeljauw B., Holland S. K., Altaye M., Loepke A. W. (2015). Cognition and brain structure following early childhood surgery with anesthesia. *Pediatrics* 136, e1–e12. 10.1542/peds.2014-3526.
- [34] Bakri M. H., Ismail E. A., Ali M. S., Elsedfy G. O., Sayed T. A., Ibrahim A. (2015). Behavioral and emotional effects of repeated general anesthesia in young children. *Saudi J. Anaesth.* 9, 161–166. 10.4103/1658-354X.152843.
- [35] Bartels M., Althoff R. R., Boomsma D. I. (2009). Anesthesia and cognitive performance in children: no evidence for a causal relationship. *Win Res Hum Genet.* 12, 246–253. 10.1375/twin.12.3.246.
- [36] Barton K., Dielman T. E., Cattell R. B. (1972). Personality and IQ measures as predictors of school achievement. *J. Educ. Psychol.* 63, 398–404. 10.1037/h0033573.
- [37] Glatz P., Sandin R. H., Pedersen N. L., Bonamy A. K., Eriksson L. I., Granath F. (2017). Association of anesthesia and surgery during childhood with long-term academic performance. *JAMA Pediatr.* 171, e163470. 10.1001/jamapediatrics.2016.3470.
- [38] Ing C., DiMaggio C., Whitehouse A., Hegarty M. K., Brady J., von Ungern-Sternberg B. S., et al. (2012). Long-term differences in language and cognitive function after childhood exposure to anesthesia. *Pediatrics* 130, e476–e485. 10.1542/peds.2011-3822.
- [39] Ing C., Hegarty M. K., Perkins J. W., Whitehouse A., DiMaggio C. J., Sun M., et al. (2017a). Duration of general anaesthetic exposure in early childhood and long-term language and cognitive ability. *Br. J. Anaesth.* 119, 532–540. 10.1093/bja/aew413.
- [40] Ing C., Jackson W. M., Zaccariello M. J., Goldberg T. E., McCann M. E., Grobler A., et al. (2021). Prospectively assessed neurodevelopmental outcomes in studies of anaesthetic neurotoxicity in children: a systematic review and meta-analysis. *Br. J. Anaesth.* 126, 433–444. 10.1016/j.bja.2020.10.022.
- [41] Kayaalp L., Bozkurt P., Odabasi G., Dogangun B., Cavusoglu P., Bolat N., et al. (2006). Psychological effects of repeated general anesthesia in children. *Paediatr. Anaesth.* 16, 822–827. 10.1111/j.1460-9592.2006.01867.
- [42] Ko W. R., Huang J. Y., Chiang Y. C., Nfor O. N., Ko P. C., Jan S. R., et al. (2015). Risk of autistic disorder after exposure to general anaesthesia and surgery: a nationwide, retrospective matched cohort study. *Eur. J. Anaesthesiol.* 32, 303–310. 10.1097/EJA.000000000000130.
- [43] Ko W. R., Liaw Y. P., Huang J. Y., Zhao D. H., Chang H. C., Ko P. C., et al. (2014). Exposure to general anesthesia in early life and the risk of attention-deficit/hyperactivity disorder development: a nationwide, retrospective matched-cohort study. *Paediatr. Anaesth.* 24, 741–748. 10.1111/pan.12371.



# Autoimmune diseases in children and anesthesia

Maria I Dalamagka

## ► To cite this version:

Maria I Dalamagka. Autoimmune diseases in children and anesthesia. Magna Scientia Advanced Biology and Pharmacy, 2025, 14, pp.045 - 052. 10.30574/msabp.2025.14.1.0022 . hal-05434063

**HAL Id: hal-05434063**

**<https://hal.science/view/index/docid/5434063>**

Submitted on 29 Dec 2025

**HAL** is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



Distributed under a Creative Commons Attribution 4.0 International License

## Autoimmune diseases in children and anesthesia

Maria I Dalamagka \*

*Department of Anesthesiology, General Hospital of Larisa, Greece.*

Magna Scientia Advanced Biology and Pharmacy, 2025, 14(01), 045-052

Publication history: Received on 26 December 2024; revised on 13 February 2025; accepted on 16 February 2025

Article DOI: <https://doi.org/10.30574/msabp.2025.14.1.0022>

### Abstract

Autoimmune diseases in children may be linked to genetic, environmental, and hormonal factors. It's estimated that 5% of children worldwide have at least one autoimmune condition, with many experiencing multiple autoimmune challenges. Around 80 autoimmune diseases are present. However, autoimmune diseases in children are rare. When they occur they can be challenging to diagnose and difficult to treat. Advances in our knowledge of the immune system are uncovering connections between inflammation and many different diseases. All autoimmune diseases have relapsing and remitting tendencies. Autoimmune diseases can affect almost any part of the body, though they often target connective tissues (skin, muscle and joints). Symptoms can range from fatigue and mild rashes to rare, serious side effects, like seizures. Diagnosis can be difficult, because many symptoms tend to come and go and are frequently nonspecific. They occur in different kinds of autoimmune diseases as well as other types of illnesses, like infection and cancer. Autoimmune diseases occur most often in females by a 3-to-1 margin over males. Organ-specific disorders (also called localized) focus on one organ or a specific type of tissue: Addison's disease affects the adrenal glands; Autoimmune hepatitis affects the liver; Crohn's disease affects the gastrointestinal tract; Multiple sclerosis (MS) affects the central nervous system; Type 1 diabetes affects the pancreas; Ulcerative colitis affects the gastrointestinal tract. Non-organ-specific disorders (also called systemic) cause problems throughout the body: Juvenile dermatomyositis affects the skin and muscles; Juvenile idiopathic arthritis affects the joints and sometimes the skin and lungs; Lupus affects the joints, skin, liver, kidneys, heart, brain, and other organs; Scleroderma affects the skin, joints, intestine, and sometimes the lungs. Pediatric autoimmune neuropsychiatric disorders associated with streptococcal infections (PANDAS) are rare neurobehavioral disorders in children. Affected children may have a diverse array of perioperative manifestations including compulsive behavior, agitation, and abnormal movements. Intravenous anesthetics have anti-inflammatory properties, which in most septic cases are useful for patients. The anti-inflammatory effects of ketamine may be related to the suppression of TNF production by macrophage in the presence of bacteria. Propofol inhibits the phagocytosis and chemotaxis of human monocytes through GABAA receptors. Inhalational anesthetics, in a dose-dependent manner, suppress cytokine release, reduces lymphocyte proliferation, induce apoptosis of the lymphocytes, and inhibits the function of neutrophils. Inhalational anesthetics influence the endocrine response from the hypothalamus-pituitary-adrenal axis and indirectly through the secretions of hormones, such as glucocorticoids and catecholamines. Long-term administration of general anesthesia drugs, due to their effects on cytokines, can lead to disease progression in patients with immune deficiency.

**Keywords:** Autoimmune diseases; General anesthesia; Children; Immune system; PANDAS; OCD

### 1. Introduction

The production of antibodies to streptococcal antigens associated with the M-protein of Group A beta hemolytic streptococcus (GABHS) has been shown to cross-react with epitopes on neuronal tissue. It has been postulated that the pathogenesis of Sydenham's chorea results from immune complex disease produced by nondestructive anti-streptococcal antibodies that localize to the basal ganglia and striatal areas of the brain. Structural and functional

\* Corresponding author: Maria I Dalamagka

neuroimaging studies have demonstrated abnormalities of the basal ganglia structures and their related corticostriato-thalamocortical circuitry in the pathobiology of obsessive compulsive disorder (OCD). The PANDAS subgroup is identified by five clinical criteria: presence of OCD and/or tic disorder; prepubertal symptom onset; episodic course characterized by acute, severe onset and dramatic symptom exacerbations; neurological abnormalities (e.g., choreiform movements) present during symptom exacerbations; temporal relation between GABHS infections and symptom exacerbations. The major distinguishing feature of the PANDAS subgroup is the temporal association between neuropsychiatric symptom exacerbations and GABHS infections. Anesthetic Implications: PANDAS are rare and affected children have significant anesthetic implications. The abnormal behavior and movement of children with PANDAS are a real challenge to anesthesiologists. We have described the first case in anesthesia literature of a patient with PANDAS who required anesthesia for a series of therapeutic plasmapheresis. The children usually have motor or vocal tics, obsessions, and/or compulsions. Often communication with them is a major problem. In addition to these symptoms, children may also become moody, irritable or show concerns about separating from parents. In our patient, premedication with oral ketamine and midazolam worked well to separate the patient from his parents and gain an intravenous access. Propofol infusion at usual doses provided optimum sedation during plasmapheresis and the recovery from propofol was favorable. With frequent plasmapheresis, there is a potential for coagulopathy from washout/dilution of plasma clotting factors. Evaluating coagulation status, especially after the first few plasmaphereses may identify patients at risk of bleeding. In conclusion, we described our anesthetic management of a case of PANDAS in a young boy who underwent a series of therapeutic plasmapheresis. Anesthesiologists should be aware of the abnormal behavior and movements with this disorder and potential anesthetic implications. Atopic dermatitis is one of the most common skin disorders in childhood, and general anaesthesia has a risk of developing potential complications in various organs and anaphylaxis and allergies. However, the relationship between exposure to general anaesthesia and atopic dermatitis remains unknown. General anaesthesia (GA) has a risk of developing potential complications in various organs and anaphylaxis and allergies. General anaesthesia could affect various immune responses, including Th1 and Th2 immunity, which might also affect cells that play an important role in the pathogenesis of atopic dermatitis. The number of patients undergoing GA in the paediatric population is increasing and thus interest in whether GA could be a risk factor for various diseases has increased. In particular, previous studies revealed that exposure to GA in early childhood and multiple exposures increased the risk of developmental delay and attention deficit hyperactivity disorder. Conversely, autistic disorders or learning disabilities were not associated with a single exposure to GA. Also, it has been reported that exposure to GA in early life reduced the risk of asthma, allergic rhinitis, and AD. Congenital myasthenic syndromes are genetically, mainly recessive, transmitted disorders. This group is clinically very heterogeneous, children present shortly after birth with a feeding problem and muscular hypotonia. As a severe symptom is respiratory insufficiency is well described and children often need artificial ventilation. A few children respond to therapy with acetylcholinesterase-blockers. The defect could be located presynaptic (eg. CholinacetylTransferase Defect CHAT), synaptic (eg. mutations in the gene encoding for Collagen Q COLQ) and postsynaptic (eg. disturbances in fast or slow channels in acetylcolinrezeptors or mutations in the gene encoding for the Rapsyn gene). A special feature in CHAT defects are recurrent apnea in children suffering from simple infectious diseases. These children require fast intubation and artificial ventilation. Transient Neonatal Myasthenia gravis occurs in approximately 12% of the newborns of mothers with MG and improves within the first weeks of life as the antibody concentration decreases. The first signs are often muscular hypotonia, feeding problems and respiratory [1-24].

---

## 2. Autoimmune diseases

The exact causes of autoimmune conditions in children are not fully understood. Research suggests there may be a complex interplay between environmental factors, genetics, and increased intestinal permeability. A disease is defined as autoimmune if the tissue injury is caused by an adaptive immune response to self-antigen. Development of the disease by transferring the active component of the immune response to an appropriate recipient (usually an animal model, or in some instances materno-fetal transfer of autoantibody) is the best proof that a disease is caused by autoimmunity. Autoimmune diseases can be considered on a continuum between those which are organ specific, and those which are non-organ specific. There are over 100 identified autoimmune conditions. Some commonly known in children include: Type 1 diabetes, the immune system affects the beta cells of the pancreas, impacting insulin production; Celiac disease, the immune system reacts to gliadin, a protein in gluten-containing foods, affecting the small intestine lining; Juvenile idiopathic arthritis, the immune system produces markers that primarily affect the joints; Pediatric lupus, the immune system targets various tissues, including the lungs, skin, and kidneys. Autoantibodies are commonly regarded as being intrinsically involved in the tissue injury associated with autoimmune diseases. Whilst this may be true in a number of important examples (e.g. systemic lupus erythematosus (SLE), Goodpasture's disease, and myasthenia gravis), this is not true of all autoimmune diseases. For example, autoantibodies may form secondary to the tissue damage associated with the underlying disease (e.g. cardiac antibodies produced after myocardial infarction). Corticosteroids, cytotoxic drugs and other immunosuppressant agents are an important part of the therapeutic approach to many non-malignant autoimmune disorders. Thus, treatment protocols for these diseases in pediatrics are often based on adult practice, but

despite the similarities in disease processes, the most widely used treatments have different effects in children. Some of the side effects of chronic steroid use, including linear growth deceleration, bone demineralization, and chronic weight issues, are more consequential in children than in adults. Although steroids remain a cornerstone of therapy in JDM and are useful in many cases of CIDP and JMG, other immunomodulatory therapies with similar efficacy may be used more frequently. Early treatment with intravenous immunoglobulin (IVIG) or plasma exchange (PE) is recommended for any child with Guillain-Barré syndrome who has difficulty ambulating. Azathioprine is the most widely used steroid-sparing therapy in childhood CIDP (Chronic inflammatory demyelinating polyradiculoneuropathy). Patients with juvenile myasthenia gravis (JMG) can be offered three main categories of disease-modifying therapy: acetylcholinesterase inhibitors (AChE-I), medical immunomodulation or immunosuppression, and surgical intervention (thymectomy). Corticosteroids (prednisone, prednisolone, methylprednisolone) are effective immunosuppressants with good evidence for their efficacy in MG. Methotrexate is often used as an alternative immunosuppressant agent in adults with MG who are unable to tolerate azathioprine. Cyclophosphamide therapy has been used for severe, treatment-resistant adult MG. Rituximab has been used with promising results in adults with severe, refractory MG. Plasma exchange (PE) is an alternative to IVIG for treating acute exacerbations. The immune system relies on a large number of genes for its normal development. In children affected by combined immunodeficiency (CID), a defect in one or more of these genes results in the absence or malfunction of a protein necessary for normal functioning of the immune system. Different types of CID are named based on the particular protein or gene that is affected. Some of the better-known types include Wiskott-Aldrich syndrome, DOCK8 deficiency and recombination activating gene (RAG) deficiency. Some live vaccines might need to be avoided, particularly BCG, rotavirus, measles/mumps/rubella (MMR) and chickenpox (VZV). Haematopoietic stem cell transplantation In some cases, haematopoietic stem cell transplant (HSCT) (including bone marrow transplant, BMT) offers the potential for long-term cure of CID. This treatment is most well-established for defined types of CID that are known to be life-limiting because of predictably severe complications. Gene therapy aims to correct the underlying genetic abnormality by replacing the faulty gene in immune cells with a normal copy. The commonest condition affecting thymus development is called 22q11 deletion syndrome (also known as DiGeorge syndrome). In the small fraction (<1%) of DiGeorge patients whose thymus is completely missing, thymic transplant may need to be considered [25-63].

### 3. Anesthesia - Anesthetic Drugs and immune system

General anesthesia, which is a drug-induced deliberated state, is the most commonly used strategy in anesthesia care. There are various drugs to ensure unconsciousness, analgesia, amnesia, and loss of reflexes of the autonomic nervous system. General anesthesia usually induces using intravenous anesthetics, inhalational (volatile) anesthetics or a combination of both. Also, opioids and benzodiazepines are often used as adjuvants in general anesthesia. In some cases, ventilation of the patients should be controlled. Anesthesia and drugs that administer to induce the anesthesia, affect the immune system, mainly immune cells, during the perioperative period. The effects of anesthesia on B-lymphocytes, T-lymphocytes, NK cells, macrophages, erythrocytes, and leukocytes is well documented. Anesthesia affects the pro-inflammatory and anti-inflammatory cytokines' secretion. The main Pro-inflammatory cytokines include IL-6, IL-8, IL-1, and tumor necrosis factor- $\alpha$  (TNF- $\alpha$ ). The main anti-inflammatory cytokine includes IL-10. The levels of interferon Gamma (IFN- $\gamma$ ) and TNF are changed by general anesthesia. Evaluation of the effects of general anesthesia agents on the immune system helps to improve the management of anesthesia. The most common method to produce general anesthesia is the intravenous anesthesia (IV anesthesia). Sodium thiopental, propofol, and ketamine are examples of drugs which commonly use in IV anesthesia. Ketamine and sodium thiopental reduce the number of T-helper cells and NK cell activities and increase the T-inhibiting cells. Sodium thiopental and ketamine inhibit the release of IL-1, IL-6, TNF- $\alpha$ , and IL-8. It is well-documented that low dose ketamine, as an N methyl-D-aspartate (NMDA) receptors antagonist, reduces the lifetime of IL-6. In addition, these drugs increase the level of IL-10. However, according to the results reported by Song et al., ketamine inhibits the production of anti-inflammatory cytokines. Sodium thiopental is a rapid induction intravenous anesthetic agent which affects the GABA- A receptors that their presence in immune cells is confirmed. In a study, the effect of propofol and sodium thiopental on Th1/Th2 balance is compared by measuring the levels of IFN- $\gamma$ , IL-4, and IL-2. Compared to propofol, sodium thiopental reduce the concentration of IFN- $\gamma$  and IL-4 without affecting the concentration of IL-2. Propofol (2,6-diisopropylphenol) is a commonly used intravenous anesthetic agent. Besides, it uses for maintenance of general anesthesia and sedation in the intensive care unit. Propofol causes rapid induction of and recovery from anesthesia. This drug has anti-oxidant and anti-inflammatory characteristics. Information about whether anesthesia with propofol increases or decreases the IL-6 levels is controversial. Propofol in vitro inhibits IL-6 production by stimulated lipoproteins. The study conducted by Gonzalez-Correa et al. reported that propofol decreased the concentration of IL-1, TNF- $\alpha$ , and IL-6 cytokines. Propofol increases the concentration of IL-10. A higher concentration of anti-inflammatory cytokines in patients who received propofol is the reason for its anti-inflammatory properties. Jin et al. reported that Injection of high dose propofol increased the ratio of IFN- $\gamma$ /IL-4, while low dose propofol did not change the concentration of the cytokine. Volatile anesthetics can inhibit the release of IFN in animals, reduce the number of NK cells, and the production of cytokines in the human body.

Desflurane increases the number of neutrophil cell and pro-inflammatory cytokines expression in alveolar macrophages. Desflurane produces more pro-inflammatory responses compared to sevoflurane. Evidence suggests that opioids affect the immune system, although the results are contradictory. Both the suppressive and stimulant effects of the opioids on the immune system are reported by various studies. Opioids cause a significant reduction in the concentration of TNF- $\alpha$ , IL-1, and IL-6. Morphine produces TGF- $\beta$  by stimulating lymphocytes, monocytes, and macrophages. These effects are mainly the result of long-term treatment, which leads to an increase in the concentration of IL-4 and IL-5 as well as the reduction of IL-2 and IFN- $\gamma$ . Fentanyl exhibited cytotoxicity against NK cells. It is found that morphine, remifentanyl, fentanyl, methadone, and codeine alter the immune system more than oxycodone, hydrocodone, and tramadol. Remifentanyl decreases IFN- $\gamma$  concentrations. IL-6, TNF, IL-10, and IL-2 concentrations do not change during treatments with remifentanyl or fentanyl. Midazolam is more common than other benzodiazepines, because if administered intravenously causes less pain, and it's a short-acting drug compared to others. Midazolam can inhibit the production of IL-2 and IL-8 and produces an immunosuppression effect. Midazolam can inhibit the activity of TNF- $\alpha$  [64-81].

---

#### 4. Discussion

One of the ventilation methods during general anesthesia is controlled ventilation. Controlled ventilation, through damaging alveolar cells and mechanical stress, suppresses the stability of the ventilation/perfusion ratio, which results in the release of inflammatory mediators and worsening of the gas exchange during the postoperative period. The other factor that causes lung tissue damage and increases the release of cytokines is hypoxia during surgeries. Sevoflurane reduces the inflammatory response during one-lung ventilation in chest surgery and may be preferable in patients who the level of their pro-inflammatory cytokines is more than expected. Treatment of childhood autoimmune disorders is based upon published prospective and retrospective cohort studies, expert opinion, pediatric randomized controlled trials (particularly for Guillain-Barré syndrome and dermatomyositis), and extrapolation of results from adult studies. Early diagnosis and initiation of treatment can significantly reduce long-term morbidity for these diseases. In addition to affecting quality of life, pediatric autoimmune conditions may increase the risk of various health issues, including cardiometabolic concerns, cancer, and mental health challenges. Conventional treatments can provide some relief but may not address all underlying factors of autoimmunity.

---

#### 5. Conclusion

Autoimmune neuromuscular disorders in childhood include Guillain-Barré syndrome and its variants, chronic inflammatory demyelinating polyradiculoneuropathy (CIDP), juvenile myasthenia gravis (JMG), and juvenile dermatomyositis (JDM), along with other disorders rarely seen in childhood. Intravenous anesthetics have anti-inflammatory properties, which in most septic cases are useful for patients. The anti-inflammatory effects of ketamine may be related to the suppression of TNF production by macrophage in the presence of bacteria. Propofol inhibits the phagocytosis and chemotaxis of human monocytes through GABAA receptors. Inhalational anesthetics, in a dose-dependent manner, suppress cytokine release, reduces lymphocyte proliferation, induce apoptosis of the lymphocytes, and inhibits the function of neutrophils. Inhalational anesthetics influence the endocrine response from the hypothalamus-pituitary-adrenal axis and indirectly through the secretions of hormones, such as glucocorticoids and catecholamines. Long-term administration of general anesthesia drugs, due to their effects on cytokines, can lead to disease progression in patients with immune deficiency. Many autoimmune diseases share common pathogenic mechanisms. However, there are limited studies quantifying the coexistence of autoimmune diseases.

---

#### References

- [1] Yang EJ, Beck KM, Sekhon S, Bhutani T, Koo J. The impact of pediatric atopic dermatitis on families: a review. *Pediatr Dermatol* 2019; 36: 66–71.
- [2] Kantor R, Thyssen J, Paller A, Silverberg J. Atopic dermatitis, atopic eczema, or eczema? A systematic review, meta-analysis, and recommendation for uniform use of 'atopic dermatitis'. *Allergy* 2016; 71: 1480–1485.
- [3] Çetinkaya PG, Şahiner ÜM. Childhood atopic dermatitis: current developments, treatment approaches, and future expectations. *Turk J Med Sci* 2019; 49: 963–984.
- [4] Ha J, Lee SW, Yon DK. Ten-Year trends and prevalence of asthma, allergic rhinitis, and atopic dermatitis among the Korean population, 2008–2017. *Clin Exp Pediatr* 2020; 63: 278.
- [5] Pols DH, Bohnen AM, Nielen MM, Korevaar JC, Bindels PJ. Risks for comorbidity in children with atopic disorders: an observational study in Dutch general practices. *BMJ open* 2017; 7: e018091.

- [6] Gilaberte Y, Pérez-Gilaberte JB, Poblador-Plou B, Bliiek-Bueno K, Gimeno-Miguel A, Prados-Torres A. Prevalence and comorbidity of atopic dermatitis in children: a large-scale population study based on real-world data. *J Clin Med* 2020; 9: 1632.
- [7] Huang AH, Roh YS, Sutaria N, Choi J, Williams KA, Canner JK, et al. Real-world comorbidities of atopic dermatitis in the pediatric ambulatory population in the United States. *J Am Acad Dermatol* 2021; 85: 893–900.
- [8] Dharmage SC, Lowe A, Matheson MC, Burgess J, Allen K, Abramson MJ. Atopic dermatitis and the atopic march revisited. *Allergy* 2014; 69: 17–27.
- [9] Harris M, Chung F. Complications of general anesthesia. *Clin Plast Surg* 2013; 40: 503–513.
- [10] Stepanovic B, Sommerfield D, Lucas M, von Ungern-Sternberg BS. An update on allergy and anaphylaxis in pediatric anesthesia. *Paediatr Anaesth* 2019; 29: 892–900.
- [11] Kim JC, Kim DC, Choi YW, Lee ES, Choi JW. Association of chronic spontaneous urticaria with the first exposure to general anaesthesia. *Clin Exp Allergy* 2022; 52: 990–993.
- [12] Kuo H-C, Yang Y-L, Ho S-C, Guo MM-H, Jiang J-H, Huang Y-H. General anesthesia exposure in early life reduces the risk of allergic diseases: a nationwide population-based cohort study. *Medicine* 2016; 95: e4269.
- [13] Palacios AS, Ponce MO, Pérez AR, Medina FS, Marrero JG. Modification of mediators of immune reaction after general anaesthesia. *Allergol Immunopathol (Madr)* 2004; 32: 352–360.
- [14] Lee J, Lee JS, Park SH, Shin SA, Kim K. Cohort profile: The National Health Insurance Service-National Sample Cohort (NHIS-NSC), South Korea. *Int J Epidemiol* 2017; 46: e15.
- [15] Yoon J, Choi JW. Primary cicatricial alopecia in a single-race Asian population: a 10-year nationwide population-based study in South Korea. *J Dermatol* 2018; 45: 1306–1311.
- [16] Soh BW, Kim SM, Kim YC, Choi GS, Choi JW. Increasing prevalence of alopecia areata in South Korea. *J Dermatol* 2019; 46: e331–e332.
- [17] Kim JC, Lee ES, Choi JW. Impact of alopecia areata on psychiatric disorders: a retrospective cohort study. *J Am Acad Dermatol* 2020; 82: 484–486.
- [18] 18. Kim JC, Choi JW. Impact of alopecia areata on subsequent pregnancy rate: a retrospective cohort study. *Australas J Dermatol* 2021; 62: e121–e123.
- [19] Lee H, Kim YC, Choi JW. Alopecia areata is not a risk factor for heart diseases: a 10-year retrospective cohort study. *PLoS One* 2021; 16: e0250216.
- [20] Sundararajan V, Quan H, Halfon P, Fushimi K, Luthi JC, Burnand B, et al. Cross-national comparative performance of three versions of the ICD-10 Charlson index. *Med Care* 2007; 45: 1210–1215.
- [21] Tzong KY, Han S, Roh A, Ing C. Epidemiology of pediatric surgical admissions in US children: data from the HCUP kids inpatient database. *J Neurosurg Anesthesiol* 2012; 24: 391–395.
- [22] von Ungern-Sternberg BS, Boda K, Chambers NA, Rebmann C, Johnson C, Sly PD, et al. Risk assessment for respiratory complications in paediatric anaesthesia: a prospective cohort study. *The Lancet* 2010; 376: 773–783.
- [23] Ko W-R, Huang J-Y, Chiang Y-C, Nfor ON, Ko P-C, Jan S-R, et al. Risk of autistic disorder after exposure to general anaesthesia and surgery: a nationwide, retrospective matched cohort study. *Eur J Anaesthesiol* 2015; 32: 303–310.
- [24] Tsai C-J, Lee CT-C, Liang SH-Y, Tsai P-J, Chen VC-H, Gossop M. Risk of ADHD after multiple exposures to general anesthesia: a nationwide retrospective cohort study. *J Atten Disord* 2018; 22: 229–239.
- [25] Feng Y-P, Yang T-S, Chung C-H, Chien W-C, Wong C-S. Early childhood general anesthesia exposure associated with later developmental delay: a national population-based cohort study. *PLoS One* 2020; 15: e0238289.
- [26] Ng YT, Chew FT. A systematic review and meta-analysis of risk factors associated with atopic dermatitis in Asia. *World Allergy Organ J* 2020; 13: 100477.
- [27] Mortz C, Andersen K, Dellgren C, Barington T, Bindslev-Jensen C. Atopic dermatitis from adolescence to adulthood in the TOACS cohort: prevalence, persistence and comorbidities. *Allergy* 2015; 70: 836–845.
- [28] Czarnowicki T, He H, Krueger JG, Guttman-Yassky E. Atopic dermatitis endotypes and implications for targeted therapeutics. *J Allergy Clin Immunol* 2019; 143: 1–11.

- [29] Prescott SL, Macaubas C, Smallacombe T, Holt BJ, Sly PD, Holt PG. Development of allergen-specific T-cell memory in atopic and normal children. *The Lancet* 1999; 353: 196–200.
- [30] Kawamoto N, Kaneko H, Takemura M, Seishima M, Sakurai S, Fukao T, et al. Age-related changes in intracellular cytokine profiles and Th2 dominance in allergic children. *Pediatr Allergy Immunol* 2006; 17: 125–133.
- [31] Jafarzadeh A, Hadavi M, Hassanshahi G, Rezaeian M, Vazirinejad R. General anesthetics on immune system cytokines: a narrative review article. *Anesth Pain Med* 2020; 10: e103033.
- [32] Lee J, Lozano-Ruiz B, Yang FM, Fan DD, Shen L, González-Navajas JM. The multifaceted role of Th1, Th9, and Th17 cells in immune checkpoint inhibition therapy. *Front Immunol* 2021; 12: 625667.
- [33] Jin Z, Mendu SK, Birnir B. GABA is an effective immunomodulatory molecule. *Amino Acids* 2013; 45: 87–94.
- [34] Snider LA, Seligman LD, Ketchen BR et al. Tics and problem behaviors in schoolchildren: prevalence, characterization, and associations. *Pediatrics* 2002; 110: 331–336.
- [35] National Institute of Mental Health. Pediatric and Developmental Neuro-psychiatry Branch. <http://intramural.nimh.nih.gov/pdn/web.htm>, Posted January 26, 2005.
- [36] Swedo SE. Sydenham's chorea. A model for childhood autoimmune neuropsychiatric disorders. *JAMA* 1994; 272: 1788–1791.
- [37] Zai G, Bezchlibnyk YB, Richter MA et al. Myelin oligodendrocyte glycoprotein (MOG) gene is associated with obsessive-compulsive disorder. *Am J Med Genet B Neuropsychiatr Genet* 2004; 129B: 64–68.
- [38] Snider LA, Swedo SE. PANDAS: current status and directions for research. *Mol Psychiatry* 2004; 9: 900–907.
- [39] Lougee L, Perlmutter SJ, Nicolson R et al. Psychiatric disorders in first-degree relatives of children with pediatric autoimmune neuropsychiatric disorders associated with streptococcal infections (PANDAS). *J Am Acad Child Adolesc Psychiatry* 2000; 39: 1120–1126.
- [40] Giedd JN, Rapoport JL, Leonard HL et al. Case study: acute basal ganglia enlargement and obsessive-compulsive symptoms in an adolescent boy. *J Am Acad Child Adolesc Psychiatry* 1996; 35: 913–915.
- [41] Citak EC, Gucuyener K, Karabacak NI et al. Functional brain imaging in Sydenham's chorea and streptococcal tic disorders. *J Child Neurol* 2004; 19: 387–390.
- [42] Swedo SE, Leonard HL, Garvey M et al. Pediatric autoimmune neuropsychiatric disorders associated with streptococcal infections: clinical description of the first 50 Cases. *Am J Psychiatry* 1998; 155: 265–271.
- [43] van Toorn R, Weyers HH, Schoeman JF. Distinguishing PANDAS from Sydenham's chorea: case report and review of the literature. *Eur J Paediatr Neurol* 2004; 8: 211–216.
- [44] Snider LA, Sachdev V, MaCkaronis JE et al. Echocardiographic findings in the PANDAS subgroup. *Pediatrics* 2004; 114: 748–751.
- [45] Swedo SE, Grant PJ. Annotation: PANDAS: a model for human autoimmune disease. *J Child Psychol Psychiatry* 2005; 46: 227–234.
- [46] Storch EA, Gerdes AC, Adkins JW et al. Behavioral treatment of a child with PANDAS. *J Am Acad Child Adolesc Psychiatry* 2004; 43: 510–511.
- [47] Naguib M, Koorn R. Interactions between psychotropics, anaesthetics and electroconvulsive therapy: implications for drug choice and patient management. *CNS Drugs* 2002; 16: 229–247.
- [48] Kudoh A, Katagai H, Takase H et al. Effect of preoperative discontinuation of antipsychotics in schizophrenic patients on outcome during and after anaesthesia. *Eur J Anaesthesiol* 2004; 21: 414–416.
- [49] Coulter DM, Bate A, Meyboom RH et al. Antipsychotic drugs and heart muscle disorder in international pharmacovigilance: data mining study. *BMJ* 2001; 322: 1207–1209.
- [50] Bookheimer S, Schrader LM, Rausch R et al. Reduced anesthetization during the intracarotid amobarbital (Wada) test in patients taking carbonic anhydrase-inhibiting medications. *Epilepsia* 2005; 46: 236–243.
- [51] Perlmutter SJ, Leitman SF, Garvey MA et al. Therapeutic plasma exchange and intravenous immunoglobulin for obsessive-compulsive disorder and tic disorders in childhood. *The Lancet* 1999; 354: 1153–1159.
- [52] Orvidas LJ, Slattey MJ. Pediatric autoimmune neuropsychiatric disorders and streptococcal infections: role of otolaryngologist. *Laryngoscope* 2001; 111: 1515–1519.

- [53] Murphy ML, Pichichero ME. Prospective identification and treatment of children with pediatric autoimmune neuropsychiatric disorder associated with group A streptococcal infection (PANDAS). *Arch Pediatr Adolesc Med* 2002; 156: 356–361.
- [54] Heubi C, Shott SR. PANDAS: pediatric autoimmune neuropsychiatric disorders associated with streptococcal infections – an uncommon, but important indication for tonsillectomy. *Int J Pediatr Otorhinolaryngol* 2003; 67: 837–840.
- [55] Wood L, Jacobs P. The effect of serial therapeutic plasmapheresis on platelet count, coagulation factors, plasma immunoglobulin, and complement levels. *J Clin Apheresis* 1986; 3: 124–128.
- [56] Qudah AA, Shahar E, Logan WJ, Murphy EG. Neonatal Guillain-Barre syndrome. *Pediatr Neurol.* 1988;4(4):255–256. doi: 10.1016/0887-8994(88)90042-2.
- [57] Vajsar J, Fehlings D, Stephens D. Long-term outcome in children with Guillain-Barre syndrome. *J Pediatr.* 2003;142(3):305–309. doi: 10.1067/mpd.2003.115.
- [58] Bradshaw DY, Jones HR., Jr Guillain-Barre syndrome in children: clinical course, electrodiagnosis, and prognosis. *Muscle Nerve.* 1992;15(4):500–506. doi: 10.1002/mus.880150415.
- [59] Nguyen DK, Agenarioti-Belanger S, Vanasse M. Pain and the Guillain-Barre syndrome in children under 6 years old. *J Pediatr.* 1999;134(6):773–776. doi: 10.1016/S0022-3476(99)70297-0.
- [60] Roodbol J, de Wit MC, Walgaard C, de Hoog M, Catsman-Berrevoets CE, Jacobs BC. Recognizing Guillain-Barre syndrome in preschool children. *Neurology.* 2011;76(9):807–810. doi: 10.1212/WNL.0b013e31820e7b62.
- [61] Asbury AK, Cornblath DR. Assessment of current diagnostic criteria for Guillain-Barre syndrome. *Ann Neurol.* 1990;27(Suppl):S21–24. doi: 10.1002/ana.410270707.
- [62] Yikilmaz A, Doganay S, Gumus H, Per H, Kumandas S, Coskun A. Magnetic resonance imaging of childhood Guillain-Barre syndrome. *Childs Nerv Syst.* 2010;26(8):1103–1108. doi: 10.1007/s00381-010-1197-8.
- [63] Nagasawa K, Kuwabara S, Misawa S, Fujii K, Tanabe Y, Yuki N, Hattori T, Kohno Y. Electrophysiological subtypes and prognosis of childhood Guillain-Barre syndrome in Japan. *Muscle Nerve.* 2006;33(6):766–770. doi: 10.1002/mus.20520.
- [64] Chen L, Deng H, Cui H, Fang J, Zuo Z, Deng J, et al. Inflammatory responses and inflammation-associated diseases in organs. *Oncotarget.* 2018;9(6):7204–18. doi: 10.18632/oncotarget.23208.
- [65] Akdis CA. Therapies for allergic inflammation: Refining strategies to induce tolerance. *Nat Med.* 2012;18(5):736–49. doi: 10.1038/nm.2754.
- [66] Abbas AK, Lichtman AH, Pillai S. Cellular and molecular immunology. 8th ed. US: Elsevier Sanders; 2014.
- [67] Mahmoud K, Ammar A. Immunomodulatory effects of anesthetics during thoracic surgery. *Anesthesiol Res Pract.* 2011;2011:317410. doi: 10.1155/2011/317410.
- [68] Jafarzadeh A, Larussa T, Nemati M, Jalapour S. T cell subsets play an important role in the determination of the clinical outcome of *Helicobacter pylori* infection. *Microb Pathog.* 2018;116:227–36. doi: 10.1016/j.micpath.2018.01.040.
- [69] Nemati M, Malla N, Yadav M, Khorramdelazad H, Jafarzadeh A. Humoral and T cell-mediated immune response against trichomoniasis. *Parasite Immunol.* 2018;40(3) doi: 10.1111/pim.12510.
- [70] Kalayci D, Dikmen B, Kacmaz M, Taspinar V, Ornek D, Turan O. Plasma levels of interleukin-10 and nitric oxide in response to two different desflurane anesthesia flow rates. *Braz J Anesthesiol.* 2014;64(4):292–8. doi: 10.1016/j.bjane.2013.06.008.
- [71] Jafarzadeh A, Shokri F. The antibody response to HBs antigen is regulated by coordinated Th1 and Th2 cytokine production in healthy neonates. *Clin Exp Immunol.* 2003;131(3):451–6. doi: 10.1046/j.1365-2249.2003.02093.
- [72] Lenz A, Franklin GA, Cheadle WG. Systemic inflammation after trauma. *Injury.* 2007;38(12):1336–45. doi: 10.1016/j.injury.2007.10.003.
- [73] Pirbudak Cocelli L, Ugur MG, Karadasli H. Comparison of effects of low-flow sevoflurane and desflurane anesthesia on neutrophil and T-cell populations. *Curr Ther Res Clin Exp.* 2012;73(1-2):41–51. doi: 10.1016/j.curtheres.2012.02.005.

- [74] Gong L, Qin Q, Zhou L, Ouyang W, Li Y, Wu Y, et al. Effects of fentanyl anesthesia and sufentanil anesthesia on regulatory T cells frequencies. *Int J Clin Exp Pathol*. 2014;7(11):7708–16.
- [75] Alsina E, Matute E, Ruiz-Huerta AD, Gilsanz F. The effects of sevoflurane or remifentanil on the stress response to surgical stimulus. *Curr Pharm Des*. 2014;20(34):5449–68. doi: 10.2174/1381612820666140325105723.
- [76] Brown EN, Lydic R, Schiff ND. General anesthesia, sleep, and coma. *N Engl J Med*. 2010;363(27):2638–50. doi: 10.1056/NEJMr0808281.
- [77] Miller RD. *Miller's anesthesia*. Philadelphia: Elsevier; 2015.
- [78] Melamed R, Bar-Yosef S, Shakhar G, Shakhar K, Ben-Eliyahu S. Suppression of natural killer cell activity and promotion of tumor metastasis by ketamine, thiopental, and halothane, but not by propofol: mediating mechanisms and prophylactic measures. *Anesth Analg*. 2003;97(5):1331–9. doi: 10.1213/01.ane.0000082995.44040.07.
- [79] Schneemilch CE, Hachenberg T, Ansorge S, Ittenson A, Bank U. Effects of different anaesthetic agents on immune cell function in vitro. *Eur J Anaesthesiol*. 2005;22(8):616–23. doi: 10.1017/s0265021505001031.
- [80] Snyder GL, Greenberg S. Effect of anaesthetic technique and other perioperative factors on cancer recurrence. *Br J Anaesth*. 2010;105(2):106–15. doi: 10.1093/bja/aeq164.
- [81] Welters ID, Hafer G, Menzebach A, Muhling J, Neuhauser C, Browning P, et al. Ketamine inhibits transcription factors activator protein 1 and nuclear factor-kappaB, interleukin-8 production, as well as CD11b and CD16 expression: studies in human leukocytes and leukocytic cell lines. *Anesth Analg*. 2010;110(3):934–41. doi: 10.1213/ANE.0b013e3181c95cfa.



# Autistic Spectrum disorder and Anaesthesia

Maria I Dalamagka

## ► To cite this version:

Maria I Dalamagka. Autistic Spectrum disorder and Anaesthesia. World Journal of Advanced Research and Reviews, 2024, 21, pp.1656 - 1658. 10.30574/wjarr.2024.21.3.0918 . hal-05434068

**HAL Id: hal-05434068**

**<https://hal.science/view/index/docid/5434068>**

Submitted on 29 Dec 2025

**HAL** is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



Distributed under a Creative Commons Attribution 4.0 International License

## Autistic Spectrum disorder and Anaesthesia

Maria I Dalamagka \*

*Anaesthesia Department, General Hospital of Larisa, Greece*

World Journal of Advanced Research and Reviews, 2024, 21(03), 1656–1658

Publication history: Received on 12 February 2024; revised on 19 March 2024; accepted on 21 March 2024

Article DOI: <https://doi.org/10.30574/wjarr.2024.21.3.0918>

### Abstract

One of the groups of patients which may require the greatest flexibility of approach is those with autistic spectrum disorder (ASD). ASD is a lifelong developmental disability, affecting four times as many males than females, which affects how a person communicates with, and relates to, other people and the world around them. Challenging behaviors may be triggered by the unfamiliar perioperative environment and may make anesthesia related procedures difficult or impossible. Children with more severe ASD may exhibit verbal or physical aggression, antisocial or disruptive behavior, temper tantrums, screaming, panic attacks, and self-injurious behavior, and may have aberrant responses to sensory stimuli. Although anaesthesia and sedation do not present a problem for most children with ASD, unpredictable regression in skills and behaviour is noted in a small number of patients after general anaesthesia. A 10-year-old child, 53 kg, with autism under treatment, came for a dental abscess and related dental work under general anaesthesia. The pediatric neurological assessment indicated hyperactivity and lack of cooperation. Post-operatively, during resuscitation, the child showed agitation and an attempt to withdraw oxygen, with the physical presence of his family environment.

**Keywords:** Autism; Anaesthesia; ASD; Disability; Epilepsy; Behavior

### 1. Introduction

Autism spectrum disorder (ASD) is characterized by deficits in social communication and interaction; restrictive, repetitive patterns of behavior and interests that may include inflexible adherence to nonfunctional routines or rituals; and hyper- or hyposensitivity to sensory input. The prevalence of Autistic spectrum disorders (ASD) is estimated to be 1% worldwide. The incidence of autism is about 0.2%. One of the groups of patients which may require the greatest flexibility of approach is those with autistic spectrum disorder (ASD). ASD is a lifelong developmental disability, affecting four times as many males than females, which affects how a person communicates with, and relates to, other people and the world around them. This complex mental disability is about four times more prevalent in males but is more severe in females. It manifests during the first 3 years of life. It is important to recognize that for almost all cases, genetic testing is unlikely to establish a diagnosis of autism in the absence of careful clinical evaluation, since Fragile X Mental Retardation-1 gene, maternally inherited 15q11-q13 duplications, and other syndromes greatly increase the risk for autism, but do not lead to autism spectrum disorders in all cases. Epilepsy is an enduring predisposition for generating seizures which is active in approximately 0.4%–0.8% of the population. The majority of studies that examined sex differences in people with ASD found an increased risk in males, which is consistent with higher male prevalence for ASD in the general population. These results support the strong connection between intellectual disability, epilepsy and ASD [1-8].

\* Corresponding author: Maria I. Dalamagka

---

## 2. Case Study

A 10-year-old child, 53 kg, with autism under treatment, came for a dental abscess and related dental work. The pediatric neurological assessment indicated hyperactivity and lack of cooperation. Without cerebellar semiology, pyramidal and extrapyramidal involvement, as well as without neurodermal spots. His gait was normal, with no scoliosis present. Medication included Sir. Risperdal 0.75 ml, Melatonin spray 2 mg. Preanesthesia was done with 8 mg of Dormicum intranasally 20 minutes before induction into anesthesia. Anesthesia was induced with 0.1 mg Fentanyl, 180 mg Propofol and 50 mg Esmeron. An additional 0.1 mg of Fentanyl was given and anesthesia was maintained with 6% Desflurane in a mixture of 50% oxygen and 50% air. The child was mechanically ventilated on a Drager ventilator and volume control ventilation model. The duration of the surgery was 2.5 hours and the awakening, with 100% oxygen, took place in 15 minutes with the administration of Bridion 150 mg. Post-operatively, during resuscitation, the child showed agitation and an attempt to withdraw oxygen, with the physical presence of his family environment.

### 2.1. Management and outcome

Many children with special needs may attend hospital for investigations, medical management, or surgery. They may exhibit high levels of anxiety when faced with hospital treatment, and may have difficulty conforming to the usual pattern of care. In extreme cases, children may become so uncooperative that their procedure is postponed or abandoned, or can only be undertaken with the use of heavy sedative premedication or restraint. Anesthesia providers generally are aware of the prevalence of diagnosed ADHD and the various drugs these children might be on. Autistic children might also be on stimulant or antipsychotic drugs; therefore, they must consider specific, necessary anesthetic considerations. These drugs are combined with certain anesthetic drugs and an increase in central nervous system depression might result. Recommendations include good hydration, minimal fasting, care with Hartmann's solution (due to elevated blood lactate levels), maintenance of normal blood glucose, body temperature and acid–base balance, and avoidance of oxidative stress.

---

## 3. Discussion

Although there is a range of severity of ASD, most children with ASD are uncomfortable with or upset by unexpected or new situations. Challenging behaviors may be triggered by the unfamiliar perioperative environment and may make anesthesia related procedures (eg, application of a face mask for induction, intravenous catheter placement, application of monitors) difficult or impossible. Children with more severe ASD may exhibit verbal or physical aggression, antisocial or disruptive behavior, temper tantrums, screaming, panic attacks, and self-injurious behavior, and may have aberrant responses to sensory stimuli. A goal for perioperative care should be to avoid these difficulties, for patient comfort, and to avoid negative associations with the hospital environment.

---

## 4. Conclusion

Although anaesthesia and sedation do not present a problem for most children with ASD, unpredictable regression in skills and behaviour is noted in a small number of patients after general anaesthesia. It may be appropriate therefore to adopt an anaesthetic technique suitable for patients with mitochondrial disease.

---

## Compliance with ethical standards

### *Statement of informed consent*

Informed consent was obtained from all individual participants included in the study.

---

## References

- [1] Khan NZ, Gallo LA, Arghir A, et al. Autism and the grand challenges in global mental health. *Autism Res* 2012; 5: 156–59.
- [2] Bauman ML, Kemper TL. Neuroanatomic observations of the brain in autism: a review and future directions. *Int J Dev Neurosci* 2005; 23: 183–87.
- [3] American Psychiatric Association. *Diagnostic and Statistical Manual of Mental Disorders (DSM-5)*, 5th edn Washington, DC: American Psychiatric Association Publishing, 2013.

- [4] Lord C, Petkova E, Hus V, et al. A multisite study of the clinical diagnosis of different autism spectrum disorders. *Arch Gen Psychiatry* 2012; 69: 306–13.
- [5] Maenner MJ, Rice CE, Arneson CL, et al. Potential impact of DSM-5 criteria on autism spectrum disorder prevalence estimates. *JAMA Psychiatry* 2014; 71: 292–300.
- [6] Weitlauf AS, Gotham KO, Vehorn AC, Warren ZE. Brief report: DSM-5 “levels of support:” a comment on discrepant conceptualizations of severity in ASD. *J Autism Dev Disord* 2014; 44: 471–76.
- [7] Sicherman N, Loewenstein G, Tavassoli T, Buxbaum JD. Grandma knows best: Family structure and age of diagnosis of autism spectrum disorder. *Autism* 2018; 22: 368–76.
- [8] Weitlauf AS, McPheeters ML, Peters B, et al. Therapies for children with autism spectrum disorder: behavioral interventions update. Comparative Effectiveness Review No. 137. Rockville, MD: Agency for Healthcare Research and Quality, 2014



## Mechanism of Pain

Maria I Dalamagka

### ► To cite this version:

Maria I Dalamagka. Mechanism of Pain. World Journal of Biological and Pharmaceutical Research, 2023, 4, pp.001 - 005. 10.53346/wjbpr.2023.4.1.0047 . hal-05434336

**HAL Id: hal-05434336**

**<https://hal.science/view/index/docid/5434336>**

Submitted on 29 Dec 2025

**HAL** is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



Distributed under a Creative Commons Attribution 4.0 International License

## Mechanism of Pain

Maria I. Dalamagka \*

*Department of Anaesthesia, General Hospital of Larisa, Greece.*

World Journal of Biological and Pharmaceutical Research, 2023, 04(01), 001-005

Publication history: Received on 01 December 2022; revised on 14 January 2023; accepted on 17 January 2023

Article DOI: <https://doi.org/10.53346/wjbpr.2023.4.1.0047>

### Abstract

Word "pain" derives its origin from the Indo-European root *aleg* meaning to suffer. Word "pain" is later and comes from the Latin word "poena" meaning punishment. Since ancient times there has been disagreement regarding the perception of pain and its evaluation. Unlike sight, hearing and smell, pain does not seem to be a primary sensation, but rather an emotional experience. Most pain researchers view pain as a complex perception, induced by noxious stimuli. Although pain is the most frequent symptom in medicine and despite the enormous advances that have been made in the field of analgesia and anesthesia, the pathophysiological mechanisms involved in its generation and maintenance are not fully understood. Definition of pain was given in 1979 by the classification committee of the international association for the study of pain (IASP) "as an unpleasant aesthetic and emotional experience, associated with actual or potential tissue damage or described in terms of such damage". In other words, although physiology and anatomy determine a precise point of reference for the detection and transmission of messages interpreted as painful, what differentiates the experience of pain is the fact that there is always an emotional gradient to the experience of pain. The purpose of the review is to investigate the analgesic system. Pain signals can be blocked at their initial point of entry into the spinal cord. Analgesia system may also inhibit pain transmission elsewhere in the nociceptive pathway. Because most drugs that alter neuronal excitability act on synaptic receptors, it has been suggested that the "morphine receptors" of the analgesia system must actually be receptors for some morphine-like neurotransmitter that is secreted normally from the brain.

**Keywords:** Pain; Physiology; Enkephalins; Nociceptors; Nerves; Spinal

### 1. Introduction

Pain affects the person's daily life, the financial life and the relatives of the sufferer. In order to treat it as effectively as possible, it is necessary to have the perception of pain, as well as to be able to describe its characteristics (intensity, duration, etc.) by the patient himself. Various measurable biochemical changes are indicators of surgical stress and pain, which the anesthesiologist can take advantage of to observe in depth the effect of analgesia and its possible complications. Acupuncture is one of the methods of dealing with pain, it has been used for thousands of years in traditional medicine of Asian countries and in recent decades has joined the quiver of anesthesiologists of classical medicine. It is claimed that the application of acupuncture preoperatively and intraoperatively can reduce the need for co-administration of anesthetic substances, or even replace anesthesia. In such a case the improvement in pain management after surgery should be reflected both in the subjective testimonies of the patients and in the biochemical indicators. Pain acts as a protective mechanism of the body, as it forces the person to react in such a way as to move away from the painful stimulus. It is important not only for cases where there is marked tissue damage, but also for everyday simple activities. Thus, when a person sits on the hips for a long time, it is possible to cause tissue damage due to the inhibition of blood supply to the skin where the skin is compressed by the weight of the body. When the skin begins to hurt due to ischemia, the person completely unconsciously changes position. When the sensation of pain has

\* Corresponding author: Maria Dalamagka

been lost, as happens after damage to the spinal cord, the person cannot feel the pain and as a result does not change position. This condition very quickly leads to ulceration in the area where the pressure is applied (1).

## 2. Results

Feeling of pain is created by the stimulation of nerve receptors located in the skin and deep tissues. There are two different types of pain: fast pain and slow pain (fast and slow pain). Fast pain occurs within 0.1 second after the application of a painful stimulus, while slow pain begins to be felt after a second or more. Subsequently, its intensity increases slowly over several seconds, and in many cases even over several minutes. Fast pain is also described as sharp pain, dull pain, electric pain, etc. pain of this type is felt by inserting a needle into the skin, or by cutting the skin with a knife, as well as by the effect of electric discharge on the skin. Rapid pain is not felt by most deep tissues of the body. Slow pain is characterized as burning pain, deep pain, throbbing pain, chronic pain, etc. Pain of this type is usually associated with tissue destruction. It can become excruciating and can lead to long-term unbearable hopeless anguish. It can come from the skin as well as from any deep tissue or organ. Ways to treat these two types of pain are different. The centromere fibers that conduct fast pain are thin, myelinated type a-d, while the fibers for slow pain are type c and myelinated. Feeling of pain they transmit is characterized as "slow", prolonged, dull and diffuse pain. All nociceptors are free nerve endings. They abound in the superficial layers of the skin, as well as in some internal tissues, such as the periosteum, the walls of the arteries, the articular surfaces, as well as the sickle and tentacle of the skull. Most other deep tissues are not richly endowed with pain nerve endings. However, any extensive tissue damage can cause dull-type pain from these areas. Nociceptors are stimulated by mechanical, thermal and chemical stimuli. Although most pain nerve fibers are stimulated by multiple stimuli, some fibers are more responsive to excessive mechanical stretch, others to excessively high or low temperature, while others are more responsive to specific chemicals within the tissues. Nerve endings are respectively classified as mechanical, thermal and chemical nociceptors. Fast pain is triggered by mechanical and thermal receptors, while slow pain can be triggered by all three types of nociceptors. Some of the chemicals that stimulate nociceptors are bradykinin, serotonin, histamine, potassium ions, acids, acetylcholine, and proteolytic enzymes. In addition, prostaglandins enhance the sensitivity of pain nerve endings, without causing their direct stimulation. These chemicals are especially important in triggering the slow excruciating pain that occurs after tissue damage (2).

Although all pain nerve endings are free nerve endings, two separate nerve pathways are used to conduct these nerve impulses to the central nervous system. These two nerve pathways correspond to the two types of pain, i.e. one pathway for rapid, acute pain and another pathway for slow, chronic pain. Because of this dual nociceptive system, a sudden application of a painful stimulus produces a "double" pain sensation: a quick-sharp pain, followed after about a second by a slow, burning pain. Rapid pain informs the person about the effect of the harmful agent, therefore, pain of this type plays an important role for the immediate reaction of the person and his removal from the stimulus. Moreover, the slow burning sensation tends to become, over time, increasingly painful. Eventually, this feeling becomes unbearable constant pain (1, 2).

After entering the spinal cord with the posterior roots of the spinal nerves, the pain nerve fibers ascend or descend for one to three neurotomies in the tract of lissauer, which is located immediately behind the posterior horn of the spinal gray matter medulla, and then make synapses with neurons of the posterior horns. But also at this point there are two systems for processing the pain signals on their way to the brain. Skin nociceptors are called mechanoreceptors and are stimulated by the application of pressure, nudge or high temperature mechanoreceptors and  $\alpha$ - $\delta$  fibers have a "threshold", while c fibers lack this because they carry a variety of stimuli. However, the pain tolerance in the CNS determines the requirements for analgesics and not the "threshold" of the peripheral nerves. A-d and c fibers enter the gray matter of the posterior horns of the spinal cord. Fibers a-d terminate in zone 1, while fibers c in zone 2 of the rexed (gelatinous substance). Then they synapse through the ascending aorta with the dorsothalamic tract. The first and basic processing of painful stimuli takes place in the spinal cord through or with the help of inhibitory neurons or suppressive neurons of the formation cation. This mechanism is part of the "pain gate" as proposed by Melzack and Wall. A small number of the fibers of the neospinothalamic tract terminate in the reticular regions of the brainstem, but most fibers reach the optic thalamus and terminate in the ventromedial complex, along with the posterior fasciculi-internal pons tract. Also, a small number of these fibers terminate in the posterior nuclei of the optic chamber. From these areas signals are transmitted to other areas at the base of the brain, as well as to the somatosensory region of the cerebral cortex (1, 3).

Acute pain is caused by a painful stimulus, such as by superficial or deeper injuries, by muscle spasm of striated or smooth muscle fibers, by some disease of the viscera, or by pathopsychological mechanisms that can cause muscle spasm of striated and smooth muscle fibers. Acute pain causes alesthesia, i.e. a nervous response to painful stimuli. Receptors that convert painful stimuli into perceived stimuli are called nociceptors. Nociceptors are free nerve endings

that can perceive chemical, thermal and mechanical damage. Under normal conditions thermal, mechanical and chemical stimuli activate high pathway nociceptors. Nociceptors in turn activate the first order neuron in the spinal cord. Noxious stimuli are accompanied by tissue damage, which causes inflammation and leads to further stimulation of nociceptors. Pain is divided into two categories: normal and pathophysiological. Normal pain is protective and warns of further damage, is temporary and localized. The pathophysiological is accompanied by inflammation, neuropathy, and results from central or peripheral sensitization and extends to healthy areas (4).

The surgical trauma creates a body response known as an inflammatory response. algia in the inflammatory response is due to release of substance p, neurokinin  $\alpha$ , and calcitonin-related peptide (cgrp) at nociceptor nerve endings. These peptides act on sympathetic nerve fibers and lead to vasodilation, extravasation of plasma proteins, and release of chemical messengers from inflammatory cells. Then follows a cascade of inflammatory substances such as serotonin, histamine, bradykinin, potassium, cytokines, substance p, and nitric oxide, products of cyclooxygenase and lipoxygenase, which sensitize high pathway nociceptors, resulting in peripheral sensitization, where low-intensity stimuli are perceived as painfully. An area of primary hyperalgesia is created around the trauma, i.e. an enhanced perception of pain to a stimulus which is normally painful. In the region of the posterior horns of the spinal cord, the first-order centromeres terminate, and more specifically in lobes i, ii and v. in the posterior horns of the spinal cord, first-order centromeres connect to many second-order neurons. Some fibers ascend or descend several myelotomy within the bundle of lissauer before reaching synapses with neurons of higher or lower levels. In the posterior horns of the spinal cord the amino acids glutamate and aspartate play a dominant role in the transmission of painful stimuli and act on nmda receptors, non-nmda receptors and glutamate receptors. Other peptides released by first-order centrioles are substance p, neurokinin  $\alpha$  that act on neurokinin receptors, and cgrp (calcitonin-related peptide). Other receptors such as opioid,  $\gamma$ -aminobutyric acid (gaba), serotonin,  $\alpha$ -adrenergic and adenosine receptors, participate in the transmission and modulation of painful stimuli in the posterior horns of the spinal cord. After the injury, allodynia occurs, that is, an increased response to mechanical painless stimuli in an area around the wound zone (area of secondary hyperalgesia). This phenomenon is called central sensitization, in the context of the "plasticity" of the CNS, resulting in the modification of pain. After the transmission of the stimuli in the posterior horn of the spinal cord, inhibitory mechanisms are released, through local inhibitory neurons and descending fibers from the brain. Acute pain originates in epidermal, deep somatic or visceral areas. Acute superficial pain is characterized as cutting, burning, stabbing, throbbing, exfoliating and localized. Deep body pain is dull, deep, and less well localized. The visceral pain is diffuse, vaguely located, dull, deep, and when the pain worsens, nausea, vomiting, sweat appear. Also accompanied by hyperactivity of the sympathetic system, hyperalgesia and reflex muscle spasm(1,2,4).

Acute pain is often associated with neurotomy and supra-neurotomy reflex responses, which help homeostasis of the body's organs. Local Spinal Neurotomy reflexes, such as during stimulation of pre-ganglionic sympathetic neurons in the anterolateral horn of the spinal cord cause tachycardia, hypertension, increase in pulse volume, increase in cardiac work, metabolism, myocardial oxygen consumption and can lead to even in ileus, urinary retention and arrhythmias on worsening pain. Increased musculoskeletal tone may occur upon stimulation of somatomotor cells resulting in muscle spasm in the trunk and decreased compliance of the chest wall, increased intra-abdominal pressure and possible hypoxemia. Supra-Neurotomy reflexes occur in the premechanic centers of respiration, circulation and hypothalamus. They can influence hypothalamic sympathetic tone by releasing catecholamines resulting in an increase in cardiac output, blood pressure, peripheral resistance, oxygen consumption and an increase in the production of endocrine substances such as cortisol, glucagon, adrenocorticotrophic hormone (acth), growth hormone, of renin, angiotensin ii and aldosterone. After stress, the cascade of these endocrine substances leads to: hyperglycemia, glucose tolerance, insulin resistance, protein muscle catabolism, increased lipolysis and oxidation, water and sodium retention, as well as increased potassium excretion, due to increased aldosterone, cortisol and ADH. Through cortical stimulation, anxiety and fear increase the hypothalamic stress response. Cortisol and catecholamines released due to intense stress can affect the hypothalamus. Intense stress through the cortex can lead to increased blood viscosity, clotting time, fibrinolysis and platelet aggregation. Fear and anxiety are the main psychological concomitants of acute pain, resulting in a vicious circle, where pain and anxiety reinforce each other (5, 6).

Inhibition of the stress response, through analgesic treatment, can lead to an improvement in the outcome of surgical patients. Therapeutic goal is a VAS scale (visual analog scale) below 4, both at rest and during mobilization, stable hemodynamic status, satisfactory respiratory rate, absence of significant sedation, and control of adverse reactions such as nausea, vomiting, pruritus, urinary retention and bowel motility disorder. The pain gate theory was proposed in 1965 by Melzack and Wall in an attempt to explain the endogenous mechanisms of pain control. It has to do with how a stimulus can be differentiated at the level of the spinal cord. Melzack and wall hypothesized that the painful stimulus to reach the brain must pass through a "gate". The portal is located in the gel substance of the posterior horn of the spinal cord. In the posterior horns there are transmitting (efferent) neurons or "t" cells and inhibitory interneurons or "y" cells. Stimulation of t cells transmits pain to higher levels of the CNS. C cells when stimulated inhibit t cells. Ab fibers

stimulate c cells, while ad and c fibers inhibit them. A low-intensity stimulus excites the ab fibers and closes the gateway to the painful impulses carried by the ad and c fibers. If the stimulus continues, the ad and c fibers are activated, the portal of entry opens and the stimulus becomes painful. The portal is also controlled by impulses, which travel to the spinal cord from higher centers (stem). The existence of these mechanisms seems to explain how analgesia is achieved by acupuncture, transcutaneous electrical nerve stimulation, pressure and rubbing of the area. The goal is to activate the ab fibers and close the gate. The infiltration of the wound with local anesthetics aims to interrupt the activity of the ad and c fibers to give the avians the opportunity to close the gate (7).

Painful push releases excitatory neurotransmitters (substance p) from ad or c fibers and activates class b neurons, which then carry the information to higher centers. Interneurons of the substantia nigra can modulate neurotransmitter release by activating inhibitory presynaptic receptors. The inhibitory neurotransmitters are thought to be enkephalins. Endogenous opioids work in the same way. Serotonin and norepinephrine modulate the release of excitatory neurotransmitters in the substantia nigra following activation of descending inhibitory pathways. The activity of  $\alpha$ b fibers also suppresses the response to painful stimulation by release of gaba. The posterior horns are thought to be the gateway for completing and modifying the treatment of painful stimuli. Thus the notion that the posterior horns are a gate which can be closed by the processes of conversion, transference and modification forms the postulate which forms the basis for the effective treatment of pain. The proposed pain gate mechanism is part of the endogenous analgesia system, which the nervous system possesses and which explains, among other things, the differences in the response to pain from person to person. Analgesia system consists of three main parts: Gray matter, located around the aqueduct of silvius in the midbrain and upper pons; neurons from this area send nerve impulses to the Large Nucleus of the suture, a thin nucleus located in the midline, at the level of the lower part of the pons and the upper part of the longus. From this point nerve impulses are transmitted down the lateral posterior bundles of the spinal cord to a Nociceptive Complex located in the posterior horns of the spinal cord. In this position, the analgesia system blocks pain before it is channeled to the brain. By electrical stimulation of the gray matter around the aqueduct or large nucleus of the suture, numerous but very strong pain signals, which enter the nervous system with the posterior roots of the spinal cord, can be almost completely suppressed. Also, by stimulating areas located at even higher levels of the brain, which in turn stimulate the gray matter around the aqueduct and especially the periventricular nuclei of the hypothalamus, located next to the third ventricle and to a lesser extent the medial precerebral fasciculus, also located in the hypothalamus may suppress pain, although possibly to a lesser extent (8).

Among the various neurotransmitters involved in this analgesia system, enkephalin and serotonin stand out. Many of the nerve fibers originating in both the periventricular nuclei and the gray matter around the aqueduct secrete enkephalin from their endings. Thus, the endings of many nerve fibers within the nucleus accumbens secrete enkephalin. The nerve fibers that start from this nucleus and end in the posterior horns of the gray matter of the spinal cord secrete serotonin at their endings. Serotonin in turn acts on another group of neurons in the spinal cord, which secrete enkephalin. Enkephalin causes presynaptic inhibition of both type c pain afferents and type ad fibers at their synapse points within the posterior horns. Possibly this is done by blocking calcium channels in the cell membrane of the nerve endings. Since calcium ions are what because neurotransmitter release at the synapse, blocking calcium channels obviously results in presynaptic inhibition. In addition, this blockade appears to be long-lasting, because after activation of the analgesia system, analgesia is usually maintained for several minutes, or even hours. Thus, with the analgesia system, pain signals can be blocked at their initial point of entry into the spinal cord. It is possible that this analgesia system may also inhibit pain transmission elsewhere in the nociceptive pathway, and particularly in the reticular nuclei of the brainstem and the optic interlobular nuclei. It has been found that the injection of extremely small amounts of morphine into the periventricular nucleus around the third ventricle in the diencephalon, or in the gray matter around the aqueduct in the brainstem produces an extraordinary degree of analgesia. Morphine has also been found to exert actions at many other sites of the analgesia system, including the dorsal horns of the spinal cord gray matter. Because most drugs that alter neuronal excitability act on synaptic receptors, it has been suggested that the "morphine receptors" of the analgesia system must actually be receptors for some morphine-like neurotransmitter that is secreted normally from the brain. About a dozen opioid substances have been found to date in various parts of the nervous system, all of which are breakdown products of three major protein molecules: proopiomelanocortin, pro-enkephalin, and prodynorphin. In addition, the presence of receptors for opioids was demonstrated in many parts of the brain, and especially in the areas included in the analgesia system. Among the most important of the opioid substances are beta-endorphin, met-enkephalin, leu-enkephalin (leucine-enkephalin), and dynorphin. The two enkephalins are found in the parts of the analgesia system described above, while  $\beta$ -endorphin is found in both the hypothalamus and pituitary gland. dynorphin, although found in only small amounts in nervous tissue, is an important substance because it is an extremely powerful opioid, with an analgesic effect 200 times stronger than that of morphine, when injected directly into the analgesia system (5,9,10).

---

### 3. Conclusion

Thus, although all the details of the brain's opioid system are not yet fully understood, nevertheless activation of the analgesia system, either by nerve impulses carried to the gray matter around the aqueduct, or by morphinoid drugs, can suppress many signals pain, which enter the central nervous system with the peripheral nerves.

---

### Compliance with ethical standards

#### *Acknowledgments*

Acknowledged to Aristotle University.

---

### References

- [1] Guyton. Medical Physiology. 8th ed
- [2] Okuse K. Pain signalling pathways: from cytokines to ion channels, *Int J Biochem Cell Biol* 2007; 39(3):490-6.
- [3] Siddal P, Cousins M. Neurobiology of pain, *Int Anesthesiol Clin* 1997; 35: 1-26
- [4] Machelska H, Schopohl JK, Mousa SA, Labuz D, Schäfer M, Stein C. Different mechanisms of intrinsic pain inhibition in early and late inflammation. *J Neuroimmunol.* 2003 141(1-2):30-9.
- [5] Gustin SM, Peck CC, Wilcox SL, Nash PG, Murray GM, Henderson LA. Different pain, different brain: thalamic anatomy in neuropathic and nonneuropathic chronic pain syndromes, *J Neurosci* 2011; 31(16):5956-64.
- [6] Ossipov MH, Dussor GO, Porreca F. Central modulation of pain, *J Clin Invest* 2010; 120(11):3779-87.
- [7] Melzack R, Wall PD. Pain mechanisms: a new theory. *Science.* 1965; 150(3699):971–9.
- [8] Kunnumpurath S, Srinivasagopalan R, Vadivelu N: Spinal cord stimulation: principles of past, present and future practice: a review. *J Clin Monit Comput*; 2009 Oct;23(5):333-9
- [9] Schweinhardt P, Bushnell MC. Pain imaging in health and disease — how far have we come? *J Clin Invest.* 2010; 120(11):3788–3797. Basbaum AI, Fields HL. Endogenous pain control mechanisms: review and hypothesis. *Ann Neurol* 1978;iv: 451-62.
- [10] Lesniak A, Lipkowski AW. Opioid peptides in peripheral pain control, *Acta Neurobiol Exp (Wars)* 2011; 71(1):129-38.24. Olausson H, Wessberg J, Morrison I, McGlone F, Vallbo A. The neurophysiology of unmyelinated tactile afferents, *Neurosci Biobehav Rev* 2010; 34(2):185-91



# Anesthesia in the elderly with colon cancer and postoperative outcome

Maria I Dalamagka

## ► To cite this version:

Maria I Dalamagka. Anesthesia in the elderly with colon cancer and postoperative outcome. GSC Advanced Research and Reviews , 2022, 11, pp.176 - 179. 10.30574/gscarr.2022.11.1.0113 . hal-05434308

**HAL Id: hal-05434308**

**<https://hal.science/view/index/docid/5434308>**

Submitted on 29 Dec 2025

**HAL** is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



Distributed under a Creative Commons Attribution 4.0 International License

## Anesthesia in the elderly with colon cancer and postoperative outcome

Maria I Dalamagka \*

*Department of Anesthesia, General Hospital of Larisa, Greece.*

GSC Advanced Research and Reviews, 2022, 11(01), 176–179

Publication history: Received on 12 March 2022; revised on 19 April 2022; accepted on 21 April 2022

Article DOI: <https://doi.org/10.30574/gscarr.2022.11.1.0113>

### Abstract

Colorectal surgery is commonly performed for colorectal cancer but morbidity and mortality remain high and vary among surgical centers. Two elderly patients with Colorectal Cancer and a history of Chronic Obstructive Pulmonary Disease underwent epidural anesthesia plus general anesthesia for colectomy surgery. In addition, Bridium was used as an inhibitor of muscle relaxation to better awaken them. The aim of the study was to evaluate the postoperative outcome of elderly patients with colorectal cancer and session disease Chronic Obstructive Pulmonary Disease, when they undergo anesthesia to perform a major surgery such as colectomy. The combination of epidural and general anesthesia showed encouraging results in the postoperative outcome of these patients and Bridium's effect on winning was significant.

**Keywords:** Anesthesia; Cancer; COPD; Elderly; Surgery

### 1. Introduction

Colorectal surgery for cancer, diverticular, or inflammatory diseases is a high-risk surgery. Other indications for CR surgery include ischemic colitis, iatrogenic perforation or injury, and volvulus. Perioperative factors affecting colon blood flow and oxygenation, suppression of stress response, optimal fluid therapy, and multimodal pain management are essential for the successful anesthetic management and a favorable perioperative outcome. The goals of perioperative anesthetic management for surgical patients are minimizing stress and immune responses, maintaining systemic and colonic blood flow and oxygenation, meticulous fluid and electrolyte therapy, multimodal analgesia, and prevention of postoperative gut dysfunction. Chronic obstructive pulmonary disease (COPD) is a common disease in elderly patients. Although most COPD patients tolerate tracheal intubation under “smooth” anesthetic induction without serious adverse effects, regional anesthetic blockade and application of laryngeal masks or non-invasive positive pressure ventilation should be considered whenever possible, in order to provide optimal pain control and to prevent upper airway injuries as well as lung baro-volutrauma. Minimally-invasive procedures and modern multimodal analgesic regimen are helpful to minimize the surgical stress response, to speed up the physiological recovery process and to shorten the hospital stay. Reflex-induced bronchoconstriction and hyperdynamic inflation during mechanical ventilation could be prevented by using bronchodilating volatile anesthetics and adjusting the ventilatory settings with long expiration times. Intraoperatively, the depth of anesthesia, the circulatory volume and neuromuscular blockade should be assessed with modern physiological monitoring tools to titrate the administration of anesthetic agents, fluids and myorelaxant drugs [1-26].

\* Corresponding author: Maria I Dalamagka  
Department of Anesthesia, General Hospital of Larisa, Greece.

## 2. Case Study

Two patients aged 82 and 86 years, respectively, with a body weight of 70 kg and 75 kg, underwent colectomy with surgical etiology of colon cancer and concomitant chronic obstructive pulmonary disease with a relatively good response to bronchodilation. In both patients, an epidural catheter was placed at the lumbar level 2 - lumbar 3, with a test dose of 3 ml Lidocaine 2%, under continuous monitoring with NBP, ECG, SpO<sub>2</sub>. Introduction to anesthesia was performed with 1 mg Dormicum, Fentanyl 0.1 mg, Propofol 150 mg and Esmeron 80 mg and ligated after inhalation into Drager respirators using the Volume Control ventilation model (Tidal Volume: 420 ml, Frequency: 16, PEEP: 5). After vital signs were checked, 5 ml Naropain 0.75% was given through the epidural catheter. Maintenance of anesthesia was done with Sevoflurane 2.5% and O<sub>2</sub> / Air: 50/50%. After the start of the operation the patients were hydrated by slow infusion of 1L Plasmalite and epidurally given 10 ml Naropaine 0.375% plus 2 mg Morphine. The duration of the surgery was about two hours. Two hours after the start of the surgery, a Baxter continuous infusion pump, with an infusion rate of 5 ml / h, Naropaine 2%, was connected to the epidural catheter. The epidural infusion pump was maintained postoperatively.

After the completion of the surgery, the Sevoflurane faucet was closed and an O<sub>2</sub> / N<sub>2</sub>O: 50/50% mixture was placed with a Pressure Control ventilation model. When the minimal alveolar concentration of Sevoflurane dropped to 0.4, pure oxygen was given at a flow rate of 12 Lt and Bridium 300 mg at a slow infusion. The intubation was performed in full vigilance, after they had regained sufficient respiratory range. They were taken to the post-operative monitoring room of their vital signs, with a simple oxygen mask at 2 Lt for half an hour.

## 3. Management and Outcome

Rapid onset of epidural anesthesia with 3 ml test dose Lidocaine and 5 ml Naropaine 0.75% reduces the need for large doses of opioids. Then the second infusion of a lower concentration of Naropaine 0.375% and 2 mg Morphine creates a satisfactory level of analgesia. Thus epidural anesthesia reduces the required doses of opioids, offering satisfactory analgesia and helping to win these patients. In addition, Bridium for more effective reversal of muscle relaxation is necessary in elderly patients when they undergo major surgeries such as colectomy and especially when they have concomitant diseases such as chronic obstructive pulmonary disease. Sevoflurane is a good choice for respiratory patients as it does not irritate their respiratory system. Plasmalite was used to avoid sympatholysis. The continuous anesthetic pump in the epidural maintains a constant level of analgesia and helps in the better postoperative outcome of these patients.

## 4. Discussion

Elderly exiled patients are more difficult to release from the ventilator. The possibility of inability to release the ventilator increases when undergoing major surgery and especially abdominal surgery with large incisions as in a colectomy. When they present with concomitant diseases, very common in elderly patients and especially with respiratory diseases, such as chronic obstructive pulmonary disease, then the probability of hospitalization in the ICU and the failure of a good postoperative outcome of these patients greatly increases. The combination of epidural - general anesthesia and the use of Bridium in the awakening contributes to the faster release from the ventilator and the better postoperative course.

## 5. Conclusion

Epidural anesthesia plus general anesthesia, the use of Bridium in winning and the adequate analgesic action offered by the epidural local anesthetic infusion pump, reduce or eliminate ICU hospitalization days in elderly patients with chronic obstructive pulmonary disease and better obstructive pulmonary disease.

## Compliance with ethical standards

### *Acknowledgments*

General Hospital of Larisa.

*Statement of informed consent*

Written informed consent was obtained from the patient for publication of this case report. A copy of the written consent is available.

**References**

- [1] Beaussier M, El'Ayoubi H, Schiffer E, Rollin M, Parc Y, Mazoit JX, et al. Continuous preperitoneal infusion of ropivacaine provides effective analgesia and accelerates recovery after colorectal surgery: A randomized, double-blind, placebo-controlled study. *Anesthesiology*. 2007; 107: 461–8.
- [2] Liu SS, Richman JM, Thirlby RC, Wu CL. Efficacy of continuous wound catheters delivering local anesthetic for postoperative analgesia: A quantitative and qualitative systematic review of randomized controlled trials. *J Am Coll Surg*. 2006; 203: 914–32.
- [3] Polglase AL, McMurrick PJ, Simpson PJ, Wale RJ, Carne PW, Johnson W, et al. Continuous wound infusion of local anesthetic for the control of pain after elective abdominal colorectal surgery. *Dis Colon Rectum*. 2007; 50: 2158–67.
- [4] Karthikesalingam A, Walsh SR, Markar SR, Sadat U, Tang TY, Malata CM. Continuous wound infusion of local anaesthetic agents following colorectal surgery: Systematic review and meta-analysis. *World J Gastroenterol*. 2008; 14: 5301–5.
- [5] Beaussier M, El'ayoubi H, Rollin M, Parc Y, Atchabahian A, Chanques G, et al. Parietal analgesia decreases postoperative diaphragm dysfunction induced by abdominal surgery: A physiologic study. *Reg Anesth Pain Med*. 2009; 34: 393–7.
- [6] Partridge BL, Stabile BE. The effects of incisional bupivacaine on postoperative narcotic requirements, oxygen saturation and length of stay in the postanesthesia care unit. *Acta Anaesthesiol Scand*. 1990; 34: 486–91.
- [7] Conaghan P, Maxwell-Armstrong C, Bedford N, Gornall C, Baxendale B, Hong LL, et al. Efficacy of transversus abdominis plane blocks in laparoscopic colorectal resections. *Surg Endosc*. 2010; 24: 2480–4.
- [8] Beaussier M, Weickmans H, Parc Y, Delpierre E, Camus Y, Funck-Brentano C, et al. Postoperative analgesia and recovery course after major colorectal surgery in elderly patients: A randomized comparison between intrathecal morphine and intravenous PCA morphine. *Reg Anesth Pain Med*. 2006; 31: 531–8.
- [9] De Kock M, Lavand'homme P, Waterloos H. The short-lasting analgesia and long-term antihyperalgesic effect of intrathecal clonidine in patients undergoing colonic surgery. *Anesth Analg*. 2005; 101: 566–72.
- [10] Scott NB, James K, Murphy M, Kehlet H. Continuous thoracic epidural analgesia versus combined spinal/thoracic epidural analgesia on pain, pulmonary function and the metabolic response following colonic resection. *Acta Anaesthesiol Scand*. 1996; 40: 691–6.
- [11] Virlos I, Clements D, Beynon J, Ratnalikar V, Khot U. Short-term outcomes with intrathecal versus epidural analgesia in laparoscopic colorectal surgery. *Br J Surg*. 2010; 97: 1401–6.
- [12] Levy BF, Tilney HS, Dowson HM, Rockall TA. A systematic review of postoperative analgesia following laparoscopic colorectal surgery. *Colorectal Dis*. 2010; 12: 5–15.
- [13] Marret E, Rolin M, Beaussier M, Bonnet F. Meta-analysis of intravenous lidocaine and postoperative recovery after abdominal surgery. *Br J Surg*. 2008; 95: 1331–8.
- [14] Chen JY, Wu GJ, Mok MS, Chou YH, Sun WZ, Chen PL, et al. Effect of adding ketorolac to intravenous morphine patient-controlled analgesia on bowel function in colorectal surgery patients—a prospective, randomized, double-blind study. *Acta Anaesthesiol Scand*. 2005; 49: 546–51.
- [15] Schlachta CM, Burpee SE, Fernandez C, Chan B, Mamazza J, Poulin EC. Optimizing recovery after laparoscopic colon surgery (ORAL-CS): Effect of intravenous ketorolac on length of hospital stay. *Surg Endosc*. 2007; 21: 2212–9.
- [16] Chen JY, Ko TL, Wen YR, Wu SC, Chou YH, Yien HW, et al. Opioid-sparing effects of ketorolac and its correlation with the recovery of postoperative bowel function in colorectal surgery patients: A prospective randomized double-blinded study. *Clin J Pain*. 2009; 25: 485–9.
- [17] Abernethy AP, McDonald CF, Frith PA, et al. Effect of palliative oxygen versus room air in relief of breathlessness in patients with refractory dyspnoea: A double-blind, randomised controlled trial. *Lancet*. 2010; 376: 784–793.

- [18] Rafii R, Albertson TE, Louie S, Chan AL. Update on pharmaceutical and minimally invasive management strategies for chronic obstructive pulmonary disease. *Pulmonary Medicine*. 2011; 257496.
- [19] Fishman A, Martinez F, Naunheim K, et al. A randomized trial comparing lung-volume-reduction surgery with medical therapy for severe emphysema. *New Engl J Med*. 2003; 348: 2059–2073.
- [20] Berger RL, Decamp MM, Criner GJ, Celli BR. Lung volume reduction therapies for advanced emphysema: An update. *Chest*. 2010; 138: 407–417.
- [21] Elliott MW. Non-invasive ventilation: Established and expanding roles. *Clinical Medicine*. 2011; 11: 150–153. [PMC free article] [PubMed] [Google Scholar]
- [22] Smith TA, Davidson PM, Lam LT, Jenkins CR, Ingham JM. The use of non-invasive ventilation for the relief of dyspnoea in exacerbations of chronic obstructive pulmonary disease; a systematic review. *Respirology*. 2010; 17: 300–307.
- [23] Schwartzstein RM, Lahive K, Pope A, Weinberger SE, Weiss JW. Cold facial stimulation reduces breathlessness induced in normal subjects. *American Review of Respiratory Disease*. 1987; 136: 58–61.
- [24] Suzuki M, Muro S, Ando Y, et al. A randomized, placebo-controlled trial of acupuncture in patients with chronic obstructive pulmonary disease (copd): The copd-acupuncture trial (cat) *Arch Intern Med*. 2012; 172: 878–886.
- [25] Clini EM, Ambrosino N. Nonpharmacological treatment and relief of symptoms in copd. *Eur Respir J*. 2008; 32: 218–228.
- [26] Back AL, Arnold RM. Discussing prognosis: “How much do you want to know?” Talking to patients who are prepared for explicit information. *Journal of Clinical Oncology*. 2006; 24: 4209–4213.



## Emergence Delirium in children

Maria I Dalamagka

### ► To cite this version:

Maria I Dalamagka. Emergence Delirium in children. International Journal of Science and Research Archive, 2024, 12, pp.1577 - 1581. 10.30574/ijrsra.2024.12.2.1408 . hal-05434269

**HAL Id: hal-05434269**

**<https://hal.science/view/index/docid/5434269>**

Submitted on 29 Dec 2025

**HAL** is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



Distributed under a Creative Commons Attribution 4.0 International License

## Emergence Delirium in children

Maria I Dalamagka \*

*Anesthesiology Department, General Hospital of Larisa, Greece.*

International Journal of Science and Research Archive, 2024, 12(02), 1577–1581

Publication history: Received on 21 June 2024; revised on 02 August 2024; accepted on 04 August 2024

Article DOI: <https://doi.org/10.30574/ijrsra.2024.12.2.1408>

### Abstract

Delirium after anesthesia, also known as emergence delirium (ED) is a clinical condition in which patients have alterations to their attention, awareness, and perceptions. In children, this often results in behavioral disturbances such as crying, sobbing, thrashing and disorientation. Emergence Agitation (EA) and Emergence Delirium (ED) are commonly used interchangeably; they describe two distinct conditions with emergence delirium being described in the anesthesia literature as a state of mental confusion, agitation, and dis-inhibition marked by some degree of hyper-excitability during recovery from general anesthesia. The commonly reported incidence of emergence delirium is about 10% to 30% of paediatric patients. Risk factors associated with emergence delirium are age, preexisting behaviours, types of surgery and the use of volatile anaesthesia. Transient agitation - delirium from sevoflurane anesthesia can lead to a variety of adverse events, such as airway spasm, shedding or displaced tracheal tube, dehiscence, or bleeding. Volatile anaesthetics may affect brain activity by interfering with the balance between neuronal synaptic inhibition and excitation in the central nervous system. Elevated postoperative pain has been suggested to underlie ED. But given that ED is seen in patients undergoing MRI, pain cannot be the sole cause. Treatment options include the use of premedication, analgesic adjuvants, single dose of propofol at the conclusion of the case. Midazolam premedication, intraoperative dexmedetomidine and fentanyl were associated with lower incidence of ED. The incidence of ED in patients receiving propofol is markedly lower than those receiving sevoflurane, despite the similar rapid emergence profile of both agents. Paediatric Assessment of Emergence Delirium (PAED) scale, developed specifically for children, is a valid and reliable scale. Watcha score is a simpler, reliable tool to measure emergence behaviour. There has been considerable progress in the neuroscience of anaesthesia and the application of new pharmacological agents, but the mystery behind the exact mechanism of ED is elusive. ED is a diagnosis of exclusion once other causes have been dismissed. There is no strong evidence of long-term effects and outcomes in children who developed emergence delirium after anesthesia. Prevention may be the best treatment but no one medication is entirely effective.

**Keywords:** Emergence delirium; Agitation; Anesthesia; Volatile anesthesia

### 1. Introduction

Emergence delirium (ED) was first described in the literature in the early 1960s. Incidence of ED is two to three times more common in children than in adults. Current data would suggest that the incidence of ED varies from 20% to 80% of all paediatric anaesthetics with most of the literature suggesting it to be close to 20%. ED is a transient state of marked irritation and disassociation after the discontinuation of anaesthesia. Delirium and agitation can occur as a child awakens, or emerges, from anesthesia. Emergence agitation, emergence excitation, and emergence delirium are all terms utilized to describe the state of behavioral dysregulation that can happen during emergence from general anesthesia. The fact that ED is the most common in pre-schoolers does not necessarily mean that it cannot develop in different age categories. Although often used interchangeably with emergence agitation, the characteristics that make up ED include irritability, inconsolable crying, distress, uncontrolled movements and inability to cooperate. There are defined risk factors, and usually symptoms occur within 30 min of termination of anaesthesia and last for 15–30 min.

\* Corresponding author: Maria I Dalamagka

Sevoflurane anesthesia has advantages of rapid uptake and elimination and shows reduced opioid adverse events in pediatric ambulatory surgery. However, emergence agitation (EA) and emergence delirium (ED) are recognized as a significant complication after sevoflurane anesthesia in pediatric patients, with a reported prevalence between 10% and 80%. Emergence delirium may be distressing to the patient, parents, and caregivers, and can result in inadvertent removal of intravenous (IV) catheters, drains, and dressings, and rarely, self-harm. While pain is not the sole cause for ED, surgery associated with elevated postoperative pain has been thought to increase the risk of ED. The site of surgery has been proposed as a risk factor, as well. Otorhinolaryngological and ophthalmological procedures have been shown to increase the risk of ED in comparison with urological and general surgery procedures, pointing to the possibility that surgeries affecting the head and neck may be relatively more likely to induce ED. Children who are more emotional, more impulsive, less social, and less adaptable to environmental changes, has been identified as a risk factor for ED. Various causes of ED have been proposed, including rapid emergence after the use of short-acting volatile anaesthetics. The introduction of short-acting volatile anaesthetics has been historically associated with the increased incidence of ED. In non-surgical settings, with patients undergoing magnetic resonance imaging (MRI), sevoflurane has been shown to increase the risk of ED. Several perioperative factors may be associated with EA or ED. Volatile anesthesia exposure and depth are the major risk factors for EA/ED. Volatile agents induce EA/ED nearly four times more often than intravenous anesthetics, particular sevoflurane. The bispectral index (BIS) monitor is a common method for monitoring the depth of anesthesia. Previous studies have suggested low BIS value <40, affects postoperative recovery, including postoperative delirium, mortality. A definitive relationship between low BIS and EA- ED remains inconclusive. EA is commonly considered not always associated with significant changes in behavior or cognition. Intensity of EA can be scored using tools such as the Richmond Agitation-Sedation Scale (RASS), it is the version of typical sedation scales. Unlike EA, ED is acute brain dysfunction, which occurs in the setting of systemic disease or derangement. Due to the large range of behaviors in children, the Pediatric Anesthesia Emergence Delirium (PAED) scale was created to assess these behaviors in a reliable way. The Pediatric Anesthesia Emergence Delirium (PAED) is considered an effective tool for ED assessment and is widely used in the post-anesthesia care unit (PACU). One theory attributes ED to the unique neurodevelopmental characteristics in this age group and the effects of the newer volatile anesthetics (ie, sevoflurane, desflurane, isoflurane) on them. Another theory associates ED with the unique electroencephalographic (EEG) findings that occur with sevoflurane anesthesia. Sevoflurane but not propofol is associated with altered metabolism in some parts of the brain, which in turn is associated with a greater risk of ED. ED can be described with neurological symptoms and terms that amount to a discreet postoperative neurological state, characterised by core neurological and behavioural symptoms: motor agitation, confusion, and lack of recognition or appropriate interaction with their surrounding environment. It usually begins at the start of emergence, but the onset can be as long as 45 min after the termination of anaesthesia. ED occurs with a similar frequency after desflurane and isoflurane even though the EEG tracings during these inhaled anesthetics are quite dissimilar from sevoflurane. The EEG patterns that may predict ED have yet to be determined, and EEG monitoring cannot yet be used to predict which children might develop ED in the PACU. Postoperative pain can confound the identification of ED. It is often difficult, especially in the nonverbal child, to differentiate between pain and delirium. Patients who experience emergence delirium are at higher risk of developing postoperative maladaptive behaviours. Preoperative anxiety is recognised as a risk factor for ED. Nonpharmacologic, behavioural and distraction techniques to minimise preoperative anxiety are being studied. Delirium may also occur without signs of agitation restlessness or non-purposeful movement. This type of delirium is called hypoactive or quiet delirium. The clinical significance, cause and management of hypoactive delirium in the post-operative setting are unknown [1-8].

## 2. Prophylactic Measures

Sevoflurane and desflurane with their low blood–gas solubility coefficients are associated with rapid washout and emergence from anaesthesia. It remains unclear as to whether rapid emergence causes ED, or whether emergence in the presence of surgical pain and before the onset of effective analgesia is a major factor. The duration of deep anesthesia, bispectral index- BIS <45, does not correlate with the risk for developing ED. ED is as likely to occur after brief procedures, as it is after surgeries of greater duration. Children with purported agitation in the recovery room should first be assessed for potentially dangerous causes of agitation; hypoxia, hypotension, hypercarbia, hypoglycemia and for pain. If pain is a problem, treated appropriately with analgesics. The incidence of ED varies with the anesthetic agent used, age of the child, the procedure or surgery and in particular, with the criteria used to diagnose ED. Current practice routinely includes preemptive administration of analgesics, often sedatives and other adjunct agents to ensure a smooth emergence from anesthesia, which has reduced the clinical incidence of ED in the PACU. Elevated postoperative pain has been suggested to underlie ED. But given that ED is seen in patients undergoing MRI, pain cannot be the sole cause. It is worth noting that while postoperative pain is not the sole cause of ED, the degree of pain produced by a surgical procedure can change the characteristics of the observed ED and may affect the efficacy of non-analgesic adjuvants such as propofol used to treat or prevent ED. The presence of preexisting anxiety or maladaptive behaviours was found to pose the greatest risk for the development of ED, including agitated, uncooperative behaviour; exaggerated displays of

anger and low adaptability skills. The degree of preoperative patient anxiety has also been found to correlate with ED. In addition, a high level of parental anxiety contributes to a higher level of preoperative anxiety in the child. Patients aged 2 to 7 years have been found to be at higher risk of ED when compared with those older than 7 years. Rapid emergence has also been identified as another possible contributing factor. It is postulated that sudden awakening in an unfamiliar environment with strangers can worsen patients' underlying fear. In addition, preschool children have psychological immaturity and are less likely to cope with a rapid return of consciousness in a strange environment. Patient age, disposition (maladaptive personality type), degree of preoperative anxiety, surgical procedure and anaesthetic technique have all been identified as playing important roles in ED. Various narcotics, including fentanyl, remifentanyl, sufentanyl and alfentanil have been studied as prophylactic measures to prevent ED. The effectiveness of narcotics in preventing ED is unclear. Individual studies to date have had mixed results. Midazolam is the most commonly prescribed oral premedication in the preoperative setting. Its benefits include preoperative anxiolysis, amnesia, relatively rapid onset and short duration of action. Although most children have anxiolysis with midazolam, up to 29% may display a paradoxical agitation response. Oral melatonin doses up to 0.4 mg/kg (maximum 20 mg) is effective in reducing ED in children (age 3-7 years). Premedication with ketamine is shown to be more effective than midazolam in reducing the incidence of ED during the early period of recovery (10 minutes, 20 minutes) after sevoflurane anaesthesia in children. The addition of midazolam or an antiemetic enhanced the efficacy of other agents. Among single drugs, high dose melatonin (0.2 to 0.4 mg/kg) was the most effective preventive measure for ED. Regional anaesthesia has not been widely studied with respect to ED. Acupuncture may be an important technique in decreasing ED. Due to the rapid pharmacokinetics of propofol, a bolus of 1 mg/kg given at the end of the procedure or continuous infusion used during maintenance of anaesthesia results in increased concentrations during emergence resulting in a decreased incidence of ED. Several analgesics have been studied for the prevention of ED including: fentanyl (dose 1 µg/kg i.v. given 10 min before the end of a procedure), ketamine (0.25 mg/kg i.v. given at the end of procedure, or as a premedication 6 mg/kg orally), and α<sub>2</sub>-adrenoreceptor agonists such as clonidine (caudally 1–3 µg/kg; i.v. 2–3 µg/kg) and dexmedetomidine (0.15–0.3 µg/kg). These preventative strategies increase sedation and therefore should be balanced against the risk of prolonging emergence or delaying discharge from the post-anaesthesia care unit. Pain is a significant confounding factor. Propofol, ketamine, fentanyl, and preoperative analgesia were effective in reducing the risk of ED presumably by delaying emergence and reducing postoperative pain. Propofol and other adjuvants may reduce the risk but may prolong recovery. Total i.v. anaesthesia (TIVA) techniques have become popular in recent years with well-described techniques that can be adapted to most surgeries. Watching cartoons, video goggles, or hand-held video games have been shown to decrease anxiety in some patients as effectively as those who received oral midazolam as premedication. Many children with ED require no treatment other than support and prevention of harm. Approximately 95 percent of cases of ED in children resolve spontaneously within 20 minutes of onset without lasting sequelae. Although ED is a self-limited condition lasting most commonly from 5 to 15 min it can lead to several complications. Children with ED have been shown to have longer recovery periods, which is particularly undesirable in the ambulatory setting. Several prophylactic strategies have been used to prevent ED including perioperative behavioral management, perioperative analgesia, and various prophylactic adjuvant medications such as midazolam, propofol, ketamine, tramadol, and dexmedetomidine, with varying degrees of success. These drugs, however, have side effects; thus, their use should be carefully considered in the ambulatory care setting, especially since ED is a self-limited disorder [9-15].

---

### 3. Discussion

ED is an altered state of consciousness, which begins with emergence from anesthesia and continues through the early recovery period. ED is a disturbance of awareness of, or attention to, the child's environment, and manifests as disorientation, hyperactive behavior, and hypersensitivity in the immediate period after anesthesia. Particularly in the stressful pre- or postoperative period, any child may exhibit behaviours that do not reflect his usual behaviour. The child's preoperative psychological condition was also reported to affect the incidence of ED, with a higher incidence among children exhibiting preoperative anxiety as well as among those who were surrounded by parental anxiety. Emergence delirium is a problematic condition in children. Studies that reported an association between specific surgical procedures and ED (eg, tonsillectomy, urologic surgery without neuraxial analgesia, strabismus surgery) may have conflated postoperative pain-related behavior with ED. The cause for higher prevalence of ED among children may be related to brain maturation. Younger children are in general more prone to behavioral changes upon recovery from anesthesia, particularly the 2–5 age group. This may be due to the fact that they are more easily frightened and confused by unfamiliar experiences. In addition to their psychological immaturity, waking up in a strange environment can also be a factor that contributes to the higher incidence of ED among preschool children. Inhalation anesthetics, particularly desflurane and sevoflurane, were confirmed to be triggers of ED in children. Postoperative pain has been shown to cause agitation in children, and it is considered a potential confounder in all studies that investigate ED, however, during strabismus surgery pain is usually mild and easy to control. The wide range in incidence as per different studies may be due to a need for standardised measurement tools and confounding factors like pain. Some possible risk factors are age,

preoperative anxiety, shorter-acting inhalational agents, cultural differences and certain surgeries. ED is often a diagnosis of exclusion, and it is important to rule out other causes of altered mental status before ED is established. Elevated postoperative pain has been suggested to underlie ED. But given that ED is seen in patients undergoing MRI, pain cannot be the sole cause. It is worth noting that while postoperative pain is not the sole cause of ED, the degree of pain produced by a surgical procedure can change the characteristics of the observed ED and may affect the efficacy of non-analgesic adjuvants such as propofol used to treat or prevent ED.

---

#### 4. Conclusion

Although we know that there are some predisposing factors to emergence delirium, we still are unable to predict accurately those who are at greatest risk. Emergence delirium should be considered as a vital sign, which should be followed and documented in every child in the postanaesthesia recovery period. Sudden emergence from anaesthesia into a disordered state of consciousness or into an unfamiliar environment has been proposed as a cause of ED. However, the incidence of ED in patients receiving propofol is markedly lower than those receiving sevoflurane, despite the similar rapid emergence profile of both agents. The challenge lies in determining how to modify the perioperative experience in order to decrease the incidence of ED. Strategies for improving outcomes range from non-pharmacological techniques, such as behaviour management and distraction techniques, to modification of the delivery of anaesthesia. While emergence delirium remains poorly understood in terms of mechanism and causes, detecting risk factors preoperatively and using evidence-based prevention methods can significantly help to reduce the incidence of ED. The most effective measure for prevention of ED is to supplant potent inhalation anesthetics, during general anesthesia with total intravenous anesthesia -TIVA. Combination of preventive medications may be more effective than individual medications alone.

---

#### Compliance with ethical standards

##### *Disclosure of conflict of interest*

No conflict of interest to be disclosed.

---

#### References

- [1] Cravero J, Surgenor S, Whalen K. Emergence agitation in paediatric patients after sevoflurane anaesthesia and no surgery: a comparison with halothane, *Anaesth*, 2000, vol. 10 (pg. 419-24).
- [2] Bajwa SA, Costi D, Cyna AM. A comparison of emergence delirium scales following general anesthesia in children, *Paediatr Anaesth*, 2010, vol. 20 (pg. 704-11).
- [3] Pieters BJ, Penn E, Nicklaus P, et al. Emergence delirium and postoperative pain in children undergoing adenotonsillectomy: a comparison of propofol vs sevoflurane anesthesia, *Paediatr Anaesth*, 2010, vol. 20 (pg. 944-50).
- [4] Dahmani S, Stany I, Brasher C, et al. Pharmacological prevention of sevoflurane- and desflurane-related emergence agitation in children: a meta-analysis of published studies, *Br J Anaesth*, 2010, vol. 104 (pg. 216-23).
- [5] Voepel-Lewis T, Malviya S, Tait AR. A prospective cohort study of emergence agitation in the pediatric postanesthesia care unit, *Anesth Analg*, 2003, vol. 96 (pg. 1625-30).
- [6] Aono J, Ueda W, Mamiya K, et al. Greater incidence of delirium during recovery from sevoflurane in preschool boys, *Anesthesiology*, 1997, vol. 87 (pg. 1298-300).
- [7] Kain ZN, Caldwell-Andrews AA, Maranets I, et al. Preoperative anxiety and emergence delirium and postoperative maladaptive behaviors, *Anesth Analg*, 2004, vol. 99 (pg. 1648-54).
- [8] Cohen IT, Finkel JC, Hannallah RS, et al. The effect of fentanyl on the emergence characteristics after desflurane or sevoflurane anesthesia in children, *Anesth Analg*, 2002, vol. 94 (pg. 1178-81).
- [9] Mason K. Paediatric emergence delirium: a comprehensive review and interpretation of the literature. *Br J Anaesth*. 2017;118(3):335-343.
- [10] Voepel-Lewis T, Malviya S, Tait AR. A prospective cohort study of emergence agitation in the pediatric postanesthesia care unit. *Anesth Analg*. 2003;96(6):1625-1630.

- [11] Aono J, Ueda W, Mamiya K, Takimoto E, Manabe M. Greater incidence of delirium during recovery from sevoflurane anesthesia in preschool boys. *Anesthesiology*. 1997;87(6):1298-1300.
- [12] Somaini M, Engelhardt T, Fumagalli R, Ingelmo P. Emergence delirium or pain after anaesthesia—how to distinguish between the two in young children: a retrospective analysis of observational studies. *Br Jo Anaesth*. 2016;116(3):377-383.
- [13] Costi D, Ellwood J, Wallace A, Ahmed S, Waring L, Cyna A. Transition to propofol after sevoflurane anesthesia to prevent emergence agitation: a randomized controlled trial. *Pediatr Anesth*. 2015;25(5):517-523.
- [14] Das A, Kundu R, Mukherjee A, Basunia S, Bhattacharyya R, Chattopadhyay S. Emergence agitation prevention in paediatric ambulatory surgery: a comparison between intranasal dexmedetomidine and clonidine. *J Res Pharm Pract*. 2015;4(1):24.
- [15] Yao Y, Qian B, Lin Y, Wu W, Ye H, Chen Y. Intranasal dexmedetomidine premedication reduces minimum alveolar concentration of sevoflurane for laryngeal mask airway insertion and emergence delirium in children: a prospective, randomized, double-blind, placebo-controlled trial. *Pediatr Anesth*. 2014;25(5):492-498.



# Autism and general anesthesia for dental care

Maria I Dalamagka

## ► To cite this version:

Maria I Dalamagka. Autism and general anesthesia for dental care. GSC Advanced Research and Reviews , 2024, 18, pp.283 - 285. 10.30574/gscarr.2024.18.2.0065 . hal-05434110

**HAL Id: hal-05434110**

**<https://hal.science/view/index/docid/5434110>**

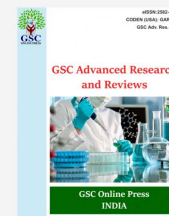
Submitted on 29 Dec 2025

**HAL** is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



Distributed under a Creative Commons Attribution 4.0 International License



## Autism and general anesthesia for dental care

Maria I. Dalamagka \*

*Department of Anaesthesia, General Hospital of Larisa, Greece.*

GSC Advanced Research and Reviews, 2024, 18(02), 283–285

Publication history: Received on 01 January 2024; revised on 11 February 2024; accepted on 14 February 2024

Article DOI: <https://doi.org/10.30574/gscarr.2024.18.2.0065>

### Abstract

Autism is the fastest growing serious developmental disability. Infantile autism is associated with a characteristic cognitive, language and behavioral features. Autism spectrum disorders (ASD) have lifelong effects on areas of individual daily life functioning such as learning, relationships and independence. The future of anesthetic management of ASD will depend on how well we understand the etiology, psycho-social and medical issues with the disease. A wide spectrum of medical and behavioral symptoms is exhibited by children with autism, which makes routine dental care very difficult in them. Autism is a neurobehavioral and cognitive disorder characterized by impaired development of interpersonal and communication skills, limited interests, and repetitive behaviors. American Academy of Paediatric Dentistry's guidelines on Caries Risk Assessment has categorized them under High risk for physicians and other nondental health care providers and Moderate risk for dental providers. Attitude and knowledge of the oral health care professionals are of utmost importance while rendering oral health care to such children. The treatment rendered currently provides a long-term benefit for the patient. More focus had to be given for the long-term maintenance of oral health in special children. A 24-year-old autistic child with a history of seizures came to the operating room where he received general anesthesia for dental work. The awakening was particularly slow, as the patient's condition was characterized as severe autism with the corresponding medication, which interacts with the anesthetic drugs and makes awakening difficult.

**Keywords:** Autism; Anaesthesia; Dental; Epilepsy; Disability

### 1. Introduction

The prevalence of Autistic spectrum disorders (ASD) is estimated to be 1% worldwide. The majority of studies that examined sex differences in people with ASD found an increased risk in males, which is consistent with higher male prevalence for ASD in the general population. These results support the strong connection between intellectual disability, epilepsy, and ASD. The signature characteristics of autism are atypical development of behavioral and social skills, and the inability to communicate. Symptoms of autism include, but are not limited to poor social relationships, underdeveloped communication skills, repetitive behaviors, unusual interests and activities, variable degree of intellectual disability, avoiding eye contact, avoiding social contact (preference toward being alone), not understanding other people's feelings and needs, physical aggressiveness, self destructive behavior, and tantrums when provoked. The different types of treatments available are generally broken down into the categories of behavior and communication approach. Medications approved by the FDA to treat aspects of ASD are the antipsychotics risperidone (risperidal) and aripiprazole (abilify). Risperidone is a novel atypical neuroleptic with favorable profile of side effects due to its unique pharmacological activity: it exhibits both potent dopamine and receptor blocking activity, as well as high affinity for alpha 1 and alpha 2 adrenergic receptors and histamine 5- HT3 receptor. These medications can help reduce irritability (aggression) and self harming acts [1-6].

\* Corresponding author: Maria Dalamagka

---

## 2. Case Report

A 24-year-old patient with a body weight of 70 kg, with autism and a history of seizures under treatment with Depakine 500 mg (therapeutic levels), Risperdal 4 mg, Seroquel 200 mg, Akineton 2 mg and Concor 5 mg was admitted to anesthesia for dental work. Patient was given intravenously: 2mg Dormicum, 0.1mg Fentanyl, 200 mg Propofol slowly intravenously and 100 mg Esmeron. After nasotracheal intubation with a 6.5 spiral tube, the patient was placed on mechanical ventilation with 50% oxygen and 50% air and sevoflurane 2.5% to maintain anesthesia. The ventilation model was volume control on Drager ventilators and positive end-expiratory pressure 4. The duration of anesthesia was 3 hours. After the end of the dental work, the patient was on 100% oxygen and the awakening was delayed to 40 minutes by administration of 500 mg Bridion and Anexate 0.5 mg. The patient maintained spontaneous breathing several minutes before extubation. In resuscitation he was drowsy for half an hour and then he started to communicate.

---

## 3. Management and outcomes

Anesthesia comes with significant adverse reactions such as: complete regression into autism, great difficulty coming out of the anesthesia, exceeding a normal time frame for “recovery”, developmental regressions. Risk factors for adverse reactions from anesthesia are: history of seizures, autism or other neurodevelopmental disorder such as ADD/ADHD, learning disabilities or Sensory Processing Disorder, preoperative respiratory problems, poor clinical condition of the patient prior to the procedure, undiagnosed or diagnosed mitochondrial dysfunction (disorder of energy production) such as mitochondrial oxidative phosphorylation (OXPHOS), MTHFR genetic defect. Many children with autism have a vitamin B12 deficiency, and nitrous oxide depletes levels of B12.

---

## 4. Discussion

Autistic children are considered difficult patients due to increased anxiety of patient and family members, uncooperative or combative behavior, or in extreme cases very violent behavior of patients. It is critical for the anesthesiologist to recognize these difficult cases and prepare for the necessary interventions beforehand. Following identification, there must be appropriate pre-anesthetic consultation and careful planning, which should include discussion with parents of the anesthetic plan and options.

---

## 5. Conclusion

Children with autism have multiple medical and behavioral problems, which make their dental treatment extremely difficult. Communication problems and poor mental capabilities are central concerns when treating children with autism; these children exhibit wide variations in their ability to cooperate during dental treatment. Children with autism who have mild to moderate mental retardation and an absence of severe behavioral problems can be treated successfully in the general practice setting. Nevertheless, behavioral problems like hyperactivity and quick frustration can hamper the provision of oral health care in patients with autism. Furthermore, the invasive nature of oral care may trigger violent and self-injurious behavior such as temper tantrums or headbanging.

---

## Compliance with ethical standards

### *Statement of informed consent*

Informed consent was obtained from all individual participants included in the study.

---

## References

- [1] Schreiber S1, Backer MM, Weizman R, Pick CG. Augmentation of opioid induced antinociception by the atypical antipsychotic drug risperidone in mice. *Neurosci Lett*. 1997; 228: 25-28.
- [2] Posey DJ, Stigler KA, Erickson CA, McDougale CJ. Antipsychotics in the treatment of autism. *J Clin Invest*. 2008; 118: 6-14.
- [3] Cook EH . Autism: review of neurochemical investigation. *Synapse*. 1990; 6: 292-308.
- [4] Bryan H. King, Nina de Lacy, Matthew Siegel. Psychiatric Assessment of Severe Presentations in Autism Spectrum Disorders and Intellectual Disability. *Child and Adolescent Psychiatric Clinics of North America*. 2014; 23:1-14.

- [5] ML1, Kemper TL. Neuroanatomic observations of the brain in autism: a review and future directions. *Int J Dev Neurosci*. 2005; 23: 183-187.
- [6] Munson J1, Dawson G, Abbott R, Faja S, Webb SJ. Amygdalar volume and behavioral development in autism. *Arch Gen Psychiatry*. 2006; 63: 686-693.



# Acupuncture & electrotherapy: An alternative and complementary treatment method

Maria I Dalamagka

## ► To cite this version:

Maria I Dalamagka. Acupuncture & electrotherapy: An alternative and complementary treatment method. World Journal of Advanced Research and Reviews, 2023, 17, pp.556 - 563. 10.30574/wjarr.2023.17.2.0293 . hal-05434105

**HAL Id: hal-05434105**

**<https://hal.science/view/index/docid/5434105>**

Submitted on 29 Dec 2025

**HAL** is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



Distributed under a Creative Commons Attribution 4.0 International License

# Acupuncture & electrotherapy: An alternative and complementary treatment method

Maria I. Dalamagka \*

*Department of Anaesthesia, General Hospital of Larisa, Greece.*

World Journal of Advanced Research and Reviews, 2023, 17(02), 556–563

Publication history: Received on 08 January 2023; revised on 18 February 2023; accepted on 20 February 2023

Article DOI: <https://doi.org/10.30574/wjarr.2023.17.2.0293>

## Abstract

Acupuncture involves activating specific points on the skin, usually by inserting needles. Acupuncture was based on the principles of traditional Chinese Medicine. Traditional acupuncturists understood health in terms of a vital force or energy called "Qi" that circulates between the organs along channels called meridians. The flow of 'Qi' energy must be of the right strength and quality in each of these meridians and organs so that health is maintained. Acupoints are located along the meridians and can alter the flow of energy and appear to correspond to nerve endings. There is a distinct difference between traditional and Western acupuncture, but the two approaches overlap significantly. In terms of Western theories, acupuncture induces signals in central nerves that modify the transmission through the spinal cord, as well as the perception of pain in the brain. Electrotherapy leads to stimulation of the sensory and motor nerves, and is especially used in the treatment of pain. Different set of parameters may lead to different physiological effects and treatment may be more specific. Electroacupuncture can be performed with or without needles, activating traditional acupuncture points or choosing points that have a neural background, such as trigger points or tender points. The purpose of this study is to verify that acupuncture works as an effective alternative and complementary treatment method. It is a safe, effective and natural approach, which helps to regain and maintain health and mental well-being.

**Keywords:** Acupuncture; Electrotherapy; Electroacupuncture; Pain; Endorphins

## 1. Introduction

In 1987 Pomeranz proposed the theory that acupuncture stimulation activates A-d and C fibers in the muscles, which leads to the transmission of signals to the spinal cord, which in turn leads to the local release of dynorphin and enkephalins. These local processes are transmitted via centromere pathways to the midbrain, where they activate a series of excitatory and inhibitory transmitters in the spinal cord. The eventual release of neurotransmitters such as serotonin, dopamine, and norepinephrine in the spinal cord results in pre- and postsynaptic inhibition and inhibition of pain transmission. When these signals reach the hypothalamus and pituitary gland, they trigger the release of cortico-adrenostimulating hormone and endorphins. Pomeranz's theory was confirmed by numerous experiments in his research laboratory, as well as by other researchers. This fundamental principle of acupuncture-induced analgesia has been explored over the past three decades and with a number of neurophysiological and imaging studies. Energy roads (King) are called meridians. Meridians are the imaginary lines on the body, joining specific acupuncture points. There are 12 meridians within which energy circulates at a certain rate non-stop. During two hours at each meridian there is a state of maximum energy supply. The 12 meridians are arranged in pairs, which means they are exactly symmetrical left and right, in relation to the midline of the body. Each meridian is responsible for an organ or a group of functions. These 12 meridians are then divided into 6 Yang and 6 Yin meridians, not meaning that each category has only Yang or possibly only Yin energy circulating. This name has to do with the operation of the particular organ for which they are responsible. Thus it is possible to distinguish between Yang and Yin conditions. In other words, these are diseases that have Yang hyperfunction or Yin deficiency or over (Yang) or under (Yin) function. These roads, the basic lines where the energy circulates, it must be emphasized that they have nothing to do with blood circulation (1).

\* Corresponding author: Maria I. Dalamagka

It has been shown that the transmission of information between two neurons is electrochemical and can be affected by acupuncture due to the metal construction of the needles. Chemical transmission of the stimulus simultaneously controls and influences the transmission of different information in the same neuron. In the central nervous system, chemical transmitters such as norepinephrine, dopamine and acetylcholine act stimulatory, while others such as serotonin act inhibitory. Impulses on axon terminals are concentrated from different regions and can release excitatory or inhibitory neurotransmitters, which have respective agonistic or antagonistic effects on the same postsynaptic neuron. Acupuncture can affect neurons, facilitating the transmission of the stimulus through the synapses (2).

In 1997 the US National Institutes of Health accepted the fact that there is sufficient evidence for acupuncture to expand its use in conventional medicine. Acupuncture has been used as a health care modality for over 3000 years. Practitioners of the medical practice of acupuncture have shown clinical success in a variety of health problems. Today acupuncture enjoys wide clinical acceptance. Most people have associated acupuncture with pain control. However, acupuncture has a proven track record in treating endocrine, circulatory and systemic diseases. Acupuncture and modern medicine have the potential to support and empower patients' health and well-being. What is known about the effects of acupuncture? In recent decades, research has attempted to explain how acupuncture works and what it is indicated for (3).

---

## 2. Results

In 1997 the National Institutes of Health (NIH) states that: "studies have shown that acupuncture can elicit biological responses mediated primarily by sensory neurons in many structures of the central nervous system. This can lead to activation of pathways that act on a variety of physiological systems in the brain, as well as in the periphery." The NIH also suggests that acupuncture may activate the hypothalamus and pituitary gland and lead to a wide range of systemic effects. Changes in neurotransmitter and hormone secretion and changes in blood flow regulation have been documented. There is also sufficient evidence for the observed changes in immune system function induced by acupuncture. The following are the current theories regarding the mechanism of action of acupuncture: The Neurotransmitter Theory: acupuncture acts on higher areas of the brain, activating the secretion of b-endorphins and enkephalins in the brain and spinal cord. The release of neurotransmitters affects the immune system and "anti-algesic" mechanisms to stop pain. The Autonomic Nervous System Theory: acupuncture stimulates the release of norepinephrine, acetylcholine as well as various types of opioids, leading to a balancing of the autonomic nervous system and a reduction in pain. The Gate Control Theory of pain: acupuncture activates "anti-algesic receptors", which prevent the transmission of analgesic signals to the dorsal horn and thus exclude noxious stimuli. Vascular-Interstitial Theory: acupuncture manages the body's electrical energy by creating a closed transport circuit in the tissues. This process facilitates healing by allowing the transfer of electrical energy between normal and injured tissue. The Blood Chemistry Theory: Acupuncture affects blood concentrations of triglycerides, cholesterol, and phospholipids, leading to the conclusion that acupuncture may increase or decrease peripheral blood components so that to regulate the body's homeostasis (4).

According to a study published in the Archives of Internal Medicine, 51% of physicians perceive the effectiveness of acupuncture, and most clinicians refer their patients to acupuncturists, more than any other alternative care provider (Astin1998). The NIH notes that clinical experience based on research data supports the use of acupuncture in a number of clinical conditions. There have also been positive clinical trials for addiction, stroke recovery, carpal tunnel syndrome, osteoarthritis and headache. It is also reported that acupuncture treatment can be useful in other conditions such as asthma, post-operative pain, back pain. Indeed, acupuncture has been used to treat many painful conditions, such as low back pain, chronic shoulder and neck pain, osteoarthritis of the knee, migraine, dysmenorrhea, and acute postoperative pain. Sun et al (2008) conducted a systematic review to assess the effectiveness of acupuncture and related techniques in the management of postoperative pain. They concluded that preoperative acupuncture can be a useful, additional tool in the treatment of postoperative pain. It is noted that under the "umbrella" of acupuncture, a number of different techniques are often placed and that the relevant studies are not always methodologically sound. As a consequence, although the results are encouraging, their generalization should be done with caution (5, 6).

Regarding the physiological mechanisms at the biochemical and genetic level, it has been found that acupuncture (traditional and electroacupuncture) activates the hypothalamus and nuclei of the brain, as well as the primary sensory cortex, while it deactivates part of the cortex of the afferent helix. At the gene level this translates into activation of endorphin production genes. One of the advantages of acupuncture is that the potential for side effects is significantly lower than that of many drugs and medical methods used for the same conditions. For example, musculoskeletal conditions such as fibromyalgia, epicondylitis, tennis elbow and fasciitis syndrome are conditions in which acupuncture can be helpful. These painful conditions are often treated with anti-inflammatory drugs (aspirin, ibuprofen) or steroid injections. Both medical methods have the potential for adverse side effects, but are widely used as acceptable treatments. The evidence supporting these treatments is no better than that for acupuncture. In summary, acupuncture

therapy has a direct relationship with the somatosensory nerves and the Autonomic Nervous System. Placing a needle in a strictly selected point and in a specific area inhibits the transmission of impulses to the somatosensory nerves. Acupuncture works by causing the pain gate to close, thereby blocking the transmission of impulses to the central nervous system. At the same time, it leads to the secretion of endogenous opioids (endorphins), which also explains the analgesic effect of acupuncture in surgery. Acupuncture works as an effective alternative and complementary treatment method. It is a safe, effective and natural approach, which helps to regain and maintain health and mental well-being (7,8).

"The phenomenon of electricity and the laws governing it are an admirable achievement, as they become evident in inorganic matter, but there can be no comparison at all when their application is extended to the nervous system and by extension to life itself" (Michael Faraday). Electroacupuncture has its roots in Western electrotherapy as well as Eastern traditions. After two and a half centuries, much remains unknown about how both direct current and low-frequency current affect us. In general, they have three actions: chemical (as in electrolysis), physical or stimulating and thermal action. These actions can affect the body at various levels such as: at the cellular, at the tissue, at the segmental and at the systemic level. At the cellular level, we may have effects on cellular chemistry (such as enzyme or protein synthesis), on fibroblasts, osteoblasts or mitochondria, changes in the permeability of cell membranes, as well as in the stimulation of peripheral nerves and muscle fibers and finally changes in microcirculation (arterial, venous or lymphatic). Actions on tissues include muscle contraction, tissue regeneration and remodeling, and changes in overall thermal and chemical balance. Segmental contraction of muscle groups can in turn produce a pump action in the larger arteries, veins and lymphatics. Direct current stimulation was the first form of electrotherapy, after a brief initial phase where static electrical discharges were used. Because of its electrolytic action, it is now of little use in nerve and muscle stimulation (although high-voltage monophasic stimulation with very short pulse duration may potentially have some use) (9).

Electrotherapy leads to stimulation of the sensory and motor nerves and is particularly used in the treatment of pain. Electroacupuncture mainly uses transdermal stimulation. Both transcutaneous electrical nerve stimulation (TENS) and electroacupuncture owe their action mainly to sensory stimulation, using low-frequency biphasic currents with low power (less than 5 W) and relatively short pulse duration (generally between 0.01 and 0.4 ms). With these parameters, any heat generated during the transfer of electrical energy to the tissues is easily dissipated and the oxidative and electrolytic action is minimal. There is of course a small chance that the ionic products released in one pulse are not necessarily recovered in the electrochemical reaction produced by the reverse pulse. This is impossible to happen if the interval between pulses is short. The electrical stimulation of a nerve above the threshold (minimum limit) of depolarization leads to the production of an energy potential, similar to that produced physiologically, only that its course along the nerves includes both directions, orthodromically (in the direction of conduction of the nerve) and in the opposite direction. The threshold for each fiber depends on the applied pulse, i.e. on its amplitude and duration. Once the stimulation threshold leads to the generation of the action potential then the application of stronger stimulation does not change the action potential, but activates more fibers. In a short-duration pulse, the current must be strong to result in an energy potential. The longer the pulse duration, the less current is required (pulse duration longer than about 5 ms for motor and sensory fibers). The initiation of stimulation of the algesthetic fibers takes place in a pulse duration of the order of 1 ms, both in healthy individuals and in patients with chronic pain. Therefore there is no advantage to using a longer pulse duration. By using trains of pulses, the amount of current per pulse required to activate the energy potential is less compared to that required for a single pulse. Thus pulse trains can be an effective and pleasant form of stimulation. Motor (Aa) nerves are deeper than cutaneous sensory (Ab) nerves, so require more stimulation to be activated transdermally (even though they are thicker and thus have a lower threshold and less electrical resistance to external stimulus). Noxious (A and C) skin fibers, which are thinner, need more stimulation to fire. The threshold differs in different parts of the body, so for example it is higher in transdermal stimulation for the lower extremities than for the forearm. There is a range of response for the different types of fibers, so the minimum pulse duration leading to the generation of action potentials for Ab, Ac and C fibers is 0.20 ms, 0.45 ms and 1.50 ms respectively. A pulse duration of 200  $\mu$ s is impossible to cause pain. More current can reach closer thin fibers than thicker more distant ones. To avoid inadvertent stimulation of C fibers, it is important that the electrodes are placed as close as possible to the target tissue. In addition to pulse width and duration, rise time and pulse shape are equally important. Emil DuBois-Reymond observed that adaptation to a stimulus exists only when the change is gradual. If a pulse increases gradually, the firing threshold increases automatically. Because of the larger capacitance of the contact electrodes, this effect is particularly pronounced with transdermal stimulation, rather than with needle stimulation. To trigger fast depolarization in a nerve fiber, a rise time of less than 60  $\mu$ s is required, with rectangular or spike-shaped waveforms, but not triangular. Thicker nerve fibers have a shorter inexcitable period, as well as a lower threshold, so an energy potential is generated in response to each pulse, with current characteristics such as high frequency, low intensity, and short pulse duration. Minimal current is required at frequencies of 60-100 Hz for medium and large myelinated fibers. In the finer fibers with the longest non-excitation period, low frequency stimulation with high

intensity and longer pulse duration is required. Electrical stimulation at the motor level activates motor and sensory neurons, leads to muscle contraction, detectable by palpation and then visible by the movement of muscle groups. However, the response to external stimulation is diametrically opposed to the normal one and is characterized by rapid, easy fatigue of the motor units and their thicker neurons firing rapidly. The electrical capacitance of muscle fibers is ten times greater than that of nerve fibers, and for this reason it takes longer to reach the firing pathway (about 35 ms). The result is that for muscles that are normally innervated, the electrical stimulation is always via the motor neuron. In denervated muscles, where the distribution of the nerve has been interrupted, the muscle fiber itself is activated. As for the effect on the muscles of continuous pulses and bursts of pulses (pulses interrupted by a few milliseconds), in the latter case the intermittent current enables the muscles to relax between contractions and thus less fatigue occurs of the muscles. Ratio (start/stop) of the order of 1:3 and 1:5 lead to less fatigue than a ratio of 1:1 or 1:2. Stimulation at the motor level can lead to muscle strengthening, facilitation of muscle control, reduction of spasticity, beneficial effect on microcirculation and faster tissue healing. Muscle mobilization, whether natural or artificial, exerts a pump action on blood and lymph circulation, enhancing oxygenation and removing lactic acid and other metabolic products. The skin is like an electrolyte conduit rich in extracellular and intracellular ions. The existence of ions on the outer cell surface (surface potential) causes the appearance of an electromagnetic field. Factors affecting the passive electrical activities of the skin are: skin perfusion, skin thickness, hair growth in the area, electrode difference on the skin, sweat, psyche, intense visual and auditory phenomena and the contact surface of the electrodes. The tissue electrical resistance to the current passage depends on several factors. Body fluids have electrolytic properties. The path that the electric current moves within the body after leaving the needles is related to the ion current, which follows the points of the body with the least electrical resistance. Internally the body is a good conductor of electricity. By applying two electrodes into the skin with an electrical voltage, the current tends to flow into the extracellular space because cell membranes have a high resistance. The ions respectively reach the electrodes and depending on their degree of ionization, donate or take electrons. The cell membrane constitutes a biological protective barrier to currents from external sources. The high electrical resistance (low conductivity) of the outer layer of the skin is an electrical barrier protecting the body from external electrical reactions. High-intensity electric currents, if they can cross the cell membrane, cause permanent or non-permanent damage. The low intensity electric currents that penetrate the distance between the two electrodes, cross the road with the least electrical resistance. The passage of small electrical currents in the skin does not correspond to known anatomical pathways. It is, however, the same as about 75% of the superficial course of the Chinese meridians (10).

Acupuncture points have lower electrical resistances than their surroundings and therefore current passes much easier through these points. Meridians are channels of energy because the current produced by the natural electrical activity of the cells, as it travels within the tissues, follows the specific course of least electrical resistance. The application of two electrodes at different points on the skin causes an electric current between them, which is formed by the ions arranged in the extracellular fluid and a part of it crosses the cells, which are stimulated. The response is modulated by the type of cell stimulated, visual image (iris cells), muscle contraction (muscle cell), auditory signal (auditory cells). Human tissue is a good conductor of electricity. Conductivity is different from person to person, but even within the same person there are differences between areas of the body. Acupuncture points are not only found in the skin, but also in the underlying tissues. The resistance of the current is affected by the distance between the acupuncture points, as well as by the fat content of the skin. The movement of electric charge ( $Q$ ) in a certain direction is called electric current. If the electric current has the same direction it is called direct, while if it constantly changes direction then it is called alternating. The surface of the skin carries positive surface charges and the glands carry negative charges. The basal layer of the skin carries positive charges. From the constant exchange of ions between the outer and inner layers of the skin, a change in the polarity of the skin layers is created, with the possible eventual manifestation of some pathology. The electrical activity of the skin is controlled by the postganglionic fibers of the sympathetic system, the subcutaneous glands, the vascular perfusion and the myoepithelial fibers of the glands. By stimulating the points of acupuncture with electric current, we increase the results of acupuncture. With the help of electroacupuncture we have intense local tissue stimulation, resulting in the improvement of local blood and lymph circulation, an increase in tissue metabolism, faster tissue healing and special effect on muscle and nerve level (11).

The electric generator of the electroacupuncture machine creates the appropriate voltage ( $V$ ), which it sends out in a specific form of pulses. Then, with the help of two electrodes, the electric current is channeled onto the corresponding needles, which have previously been placed at specific acupuncture points. The superficial and deep tissues of the region exhibit a resistance ( $R$ ) to the passage of current. According to the well-known Ohm's law ( $I = V/R$ ), which states the relationship between intensity ( $I$ ), voltage ( $V$ ) and resistance ( $R$ ) of the electric current, when the intensity increases then the voltage also increases, as these two parameters are directly proportional. Thus, correspondingly, the current delivered to the patient will be more intense, when the intensity of the electric current increases. The resistance ( $R$ ) depends on the distance between the electrodes. When the distance increases, then the resistance increases, resulting in a decrease in intensity. Electro-acupuncture machines are usually low voltage (0.5-1.5 V) and can cause appropriate

electrical stimulation to muscles and nerves. This is particularly helped by placing the needles in strategic points of muscle and nerve stimulation, very close to the corresponding area of action, so that only a low-voltage current is necessary (12).

In direct contact with the muscle, muscle contraction is observed at a voltage of 1-3 V, while in direct contact with the nerve, muscle contraction is observed at a voltage of 0.4-1.5 Volts. The patient generally feels the current at a voltage of 0.3-1.2V. By gradually increasing the current, visible muscle contractions appear, as the threshold potential for muscle contraction ranges from 0.4-2.5V. We should be especially careful when adjusting the machine, because a continuous increase in voltage causes pain. It should be noted here that the threshold potential for the sensation of pain is 7-14 Volts, as long as the other parameters of the current are constant. The electrical current in electroacupuncture is usually alternating current with an asymmetric waveform. Usually the direction of electron flow reverses steadily and periodically. The magnitude of the current flow is constantly changing, but the duration of the flow can be changed. There are many forms of pulses from electroacupuncture machines. The sharp or square pulse usually predominates. With the use of the sharp pulse, stimulation occurs, while with the application of the square pulse, decay occurs (13).

The intensity of the current administered during electroacupuncture is at particularly low levels. When electroacupuncture is applied to the acupuncture point at a depth of 1 cm with a sharp waveform with a frequency of 20 Hz, then at 50-80  $\mu$ A the sensation of current begins, at 500-650  $\mu$ A the patient feels pain, and at 900  $\mu$ A he cannot tolerate the current of electroacupuncture. So the ideal intensity when applying electroacupuncture is 100-500 mA which is comfortably tolerated by the patient. After a few minutes of applying the current, the electroacupuncture sensation usually diminishes and then we increase the intensity in approximately 50 mA increments. This is where the phenomenon of addiction or adaptation occurs, in which the sensory and muscle fibers become addicted to stimulation by the current because there is an increase in the valvular stimulus. Valvular stimulus in physiology is called the smallest possible value of a stimulus required to excite a fiber. That is, inhibition of the stimulation of the corresponding fibers is observed, because they become addicted to the specific intensity of the electroacupuncture current. Applying current to acupoints or muscle motor points when the intensity rises to the valvular stimulus level of 40  $\mu$ A results in muscle stimulation, which is usually visible. It is essential that the intensity of the current should be increased by the potentiometer of the electroacupuncture machine slowly and gradually, so that the patient tolerates the increasing intensity well. In the case of a sudden increase in the intensity of the electroacupuncture machine, the patient will report a sudden sharp pain. When the intensity of the current increases to the point where muscle spasms are visible and felt, then we have stimulation (14).

The operating frequencies of electroacupuncture machines are among the main electrical parameters, because the beneficial effects of acupuncture depend on them. Frequency is measured in Hz, i.e. cycles per 1 sec. The frequency categories are divided into: low frequencies 0.1-10 Hz which are the best frequencies for stimulating the sympathetic system, medium frequencies 10-100 Hz which are the best frequencies for stimulating the parasympathetic system, high frequencies 100-1000 Hz, very high frequencies greater than 1000 Hz, which are not needed in daily practice except to achieve analgesia in surgical procedures. To induce analgesia during electroacupuncture anesthesia, high frequencies of the order of 2000 are used Hz close to the area where the operation is performed, while at the same time distant points are stimulated with low frequencies of 5-15 Hz. In general, frequencies of 1-35 Hz are usually applied to stimulate motor nerve fibers. The low frequencies of electroacupuncture, 2-4 Hz with medium intensity current are the most common in daily practice and cause analgesia with the release of endorphins. This is referred to as the opiate-dependent electroacupuncture pain relief system. The higher frequencies of 100-200 Hz have a similar effect by releasing serotonin and dynorphin. This is referred to as the non-opioid electroacupuncture pain relief system. When we apply an electric current with a frequency of up to 10 Hz on needles, which are close to motor points of muscles, then we have visible muscle contractions. If the frequency rises to 20 Hz, then muscle contractions are no longer visible, but muscle tetany occurs (15).

The current of electroacupuncture devices is also characterized by its emission rate. It is divided into continuous or intermittent rhythm. The rhythm of the current, when it is continuous but constantly interrupted, with a frequency of less than 4 Hz lasting 4 sec, while its inactive period is 4 sec, produces stimulation. However, in the event that the frequency is greater than 30 Hz lasting 4 sec, while the inactive period is 4 sec, it causes dispersion. The most successful rate of electroacupuncture is that of scanning (Dens-disperse). In this case, the rhythm is alternating-intermittent, with two different frequencies, a low 1-8 Hz and a high 30-100 Hz with intermediate pauses of 1-4 sec. Its main indication is the achievement of analgesia. With the scan it achieves, the tolerance effect, which is often observed in electroacupuncture, can be avoided. Electro-acupuncture pulse width Pulse width is the pulse duration of each waveform, emitted by the electro-acupuncture machines. The pulse range ranges from 100-500  $\mu$ sec and can activate nerve and motor fibers, while at the same time it is well tolerated by patients. When we decrease the frequency of the electroacupuncture machine, then the pulse width automatically increases. That is, the lower the frequency of the

electroacupuncture machine, the greater the pulse width. When the pulse width increases, greater muscle wasting occurs due to its particularly intense effect on the muscles. On the contrary, when we reduce the pulse width, a greater effect is achieved on the sensory nerves, resulting in analgesia (16).

The traditional form of acupuncture involves selecting acupoints, according to the principles of Chinese Medicine, inserting fine needles into the selected points and manipulating the needles with certain techniques, such as twisting and lifting, to achieve "De Qi" sensation (arrival of vital energy Chi), which includes a feeling of heaviness, numbness, distension or warmth. "Chi" is one of the key concepts in Chinese Medicine and is considered to be the metaphysical understanding of the human body and body functions. According to traditional Chinese Medicine, Qi sustains human life and the movement of Qi enables physiological and psychological functions. Normal functions depend on the smooth movement of Chi. Acupuncture points are where Chi is in and out of the body to connect with nature. Acupoints are also where the body's Chi can be regulated by applying treatment such as acupuncture, moxa and massage to these points. Traditional Chinese Medicine books mention 365 points on the human body. Each point seems to have specific functions. With the introduction of modern technology and neurophysiological studies on acupuncture, some acupuncture techniques have been challenged and modified. Electroacupuncture is a typical example and was first applied in Japan and France between 1800. Instead of handling the needles by hand, they apply electrodes to the acupuncture points either directly or output it to an electronic device with the handle of a needle, thereby allowing the electric current to pass through these selected points. Obviously, there are many advantages to using Electroacupuncture. First, unlike manual needle manipulation, electrical stimulation can be easily controlled and reliably repeated. Second, an electric current has a number of parameters, which we can easily change. Different set of parameters may lead to different physiological effects and treatment may be more specific. For example, Electroacupuncture of different frequencies (2 Hz, 15 Hz, 200 Hz) has been shown to increase the release of the three types of opioid peptides (endorphins, enkephalins, dynorphins) respectively. In clinical practice, the use of alternating frequencies, such as 2/15 Hz, or 2/100 Hz, has been proposed to reduce opioid tolerance and enhance analgesia. Finally, it has been reported that electroacupuncture leads to a better analgesic effect, compared to classical acupuncture. Electroacupuncture can be performed with or without needles, activating traditional acupuncture points or choosing points that have a neural background, such as trigger points or tender points. The type of electroacupuncture can also vary depending on electrical parameters such as frequency, intensity, pulse duration and electrode size (17, 18-20).

Electrotherapy leads to stimulation of the sensory and motor nerves, and is especially used in the treatment of pain. Electroacupuncture mainly uses transdermal stimulation. Its action is mainly due to sensory stimulation, using low-frequency biphasic currents with low power (less than 5 W) and relatively short pulse duration (generally between 0.01 and 0.4 ms). With these parameters, any heat generated during the transfer of electrical energy to the tissues is easily dissipated and the oxidative and electrolytic action is minimal. Increased opioid release is thought to be the underlying mechanism for the analgesic effect of acupuncture and transcutaneous electrical nerve stimulation (TENS). Endogenous opioids are divided into endorphins, enkephalins and dynorphins. These three types of opioids have different affinities for the three types of receptors (21).

Electroacupuncture can increase the release of all three types of endogenous opioids, and the type of endogenous opioid peptide released depends on the frequency of the electrical stimulation. In rats, the analgesia elicited after electroacupuncture with a frequency of 2Hz at points C 6 and Sp 6 was abolished after the administration of naloxone at a dose of 1mg/Kg, while the analgesia elicited by the frequency of 100 Hz is abolished by the administration of ten times the aforementioned dose of naloxone. The correlated frequency of opioid release has also been confirmed in humans. The level of immunoreactive met-enkephalin and immunoreactive dynorphin A increases by 36% and 49% after applying TENS to the selected points with frequencies of 2 Hz and 100 Hz respectively. One study looked at the frequency of electroacupuncture at the clinical level. Subcutaneous injection of formalin in rats elicits a biphasic response, an early response due to direct stimulation of nociceptors lasting 5 minutes and a late inflammatory response occurring 20 minutes after injection and lasting 20-30 minutes. Sciatic nerve stimulation with 2 or 15 Hz produces an analgesic effect in both phases. Conversely stimulation with 100 Hz produces a strong analgesic effect only in the late phase. The results of this study confirm that the frequency-dependent analgesic effect of electrical stimulation can also be used at a clinical level to treat pain. Furthermore, in a comparison made between the postoperative application of electroacupuncture alone and that combined with preoperative application of electroacupuncture with 2 Hz, the latter led to a better analgesic effect in the late phase (22,23).

Pulse duration is another parameter of electroacupuncture analgesia. In a study Romita et al (1997) examined the effects of parameters of electroacupuncture, more specifically the withdrawal reflex after noxious thermal radiation in rats. Withdrawal latency is considered to be the pain threshold, with an increased latency indicating analgesia. Given a specific frequency (4Hz) and intensity (20 times threshold, 20T), electrical stimulation with 0.2 ms, 2 ms, and 5 ms reduced the latency. However, only electrical stimulation with 2 ms or 5 ms produces analgesia, lasting up to an hour.

The underlying mechanism linking analgesia to pulse duration may depend on the activation of different fiber centromeres. Electrical stimulation with a pulse duration of 2-50  $\mu$ s makes the sensation of stimulation perceptible, but does not cause pain (sensory level). Electrical stimulation above 0.15 ms can cause muscle contraction and further activation of Ab fiber centromeres in the muscle (motor level). Electrical stimulation for more than 1.0 ms activates C fibers and leads to painful sensation (level of noxious stimulus). Analgesia induced by stimulation at the sensory level does not last longer than electrical stimulation, while analgesia induced at the motor and noxious levels lasts for hours. Thus Romita et al. (1997) with electrical stimulation of 2 or 5 ms apparently stimulate C afferent fibers and further activate endogenous nociceptive systems and produce this prolonged analgesic effect. Conversely electrical stimulation with 0.2 ms can activate Ad fibers. However, the author does not report whether this type of stimulation caused muscle contraction. In addition, the stimulation lasted only 20 minutes, much less than the required stimulation time, which is 30-45 minutes for the motor level stimulation. Thus transient analgesia can be induced through segmental pain control (24).

Intensity is perhaps the most important parameter in electroacupuncture analgesia. High-intensity electroacupuncture that produces a "strong but tolerable" or "strong but not unpleasant" sensation results in more satisfactory analgesia than low-intensity electroacupuncture. In rats, electrical stimulation at 4 Hz, and 10 times above the threshold that can cause muscle contraction (10 T), causes only a brief increase in the withdrawal latency, whereas electrical stimulation 20 times above the threshold causes analgesia lasting up to an hour. The use of electroacupuncture in postoperative pain control showed that high-intensity electroacupuncture (9-12 mA) reduced hydromorphone PCA (patient-controlled analgesia) requirements, as well as opioid adverse effects after surgery. In contrast, the effect of PCA and electroacupuncture with low intensity (4-5 mA) was not statistically significant, compared to that caused by PCA and "sham" electroacupuncture (the "sham" electroacupuncture is the application of needles at different points from those defined as the strictly selected acupuncture points) (25,26).

---

### 3. Conclusion

Electrotherapy leads to stimulation of the sensory and motor nerves, and is especially used in the treatment of pain. Electroacupuncture of different frequencies (2 Hz, 15 Hz, 200 Hz) has been shown to increase the release of the three types of opioid peptides (endorphins, enkephalins, dynorphins) respectively. In clinical practice, the use of alternating frequencies, such as 2/15 Hz, or 2/100 Hz, has been proposed to reduce opioid tolerance and enhance analgesia. Finally, it has been reported that electroacupuncture leads to a better analgesic effect, compared to classical acupuncture.

---

### Compliance with ethical standards

#### *Acknowledgments*

Aristotle University.

---

### References

- [1] Liem MS, van der Graaf Y, van Steensel CJ, Boelhouwer RU, Clevers GJ, Meijer WS, et al. Comparison of conventional anterior surgery and laparoscopic surgery for inguinal-hernia repair. *N Engl J Med* 1997;336(22):1541–7.
- [2] Stark E, Oestreich K, Wendl K, Rumstadt B, Hagmuller E. Nerve irritation after laparoscopic hernia repair. *Surg Endosc* 1999;13(9):878–81. 3. Olmi S, Scaini A, Erba L, Guaglio M, Croce E. Quantification of pain in laparoscopic transabdominal preperitoneal (TAPP) inguinal hernioplasty identifies marked differences between prosthesis fixation systems. *Surgery* 2007;142(1):40–6.
- [3] Lovisetto F, Zonta S, Rota E, Mazzilli M, Bardone M, Bottero L, et al. Use of human fibrin glue (Tissucol) versus staples for mesh fixation in laparoscopic transabdominal preperitoneal hernioplasty: a prospective, randomized study. *Ann Surg* 2007;245(2):222–3.
- [4] Chu DW, Lee DT, Chan TT et al. Acupuncture anaesthesia in inguinal hernia repair. *ANZ J Surg* 2003; 73: 125–127. 6. Melzack R, Wall PD. Pain mechanisms: a new theory. *Science*. 1965;150(3699):971–9.
- [5] Kunnumpurath S, Srinivasagopalan R, Vadivelu N: Spinal cord stimulation: principles of past, present and future practice: a review. *J Clin Monit Comput*; 2009 Oct;23(5):333-9.

- [6] Schweinhardt P, Bushnell MC. Pain imaging in health and disease — how far have we come? *J Clin Invest.* 2010;120(11):3788–3797. Basbaum AI, Fields HL. Endogenous pain control mechanisms: review and hypothesis. *Ann Neurol* 1978;iv:451-62.
- [7] Lesniak A, Lipkowski AW. Opioid peptides in peripheral pain control, *Acta Neurobiol Exp (Wars)* 2011;71(1):129-38.24. Olausson H, Wessberg J, Morrison I, McGlone F, Vallbo A. The neurophysiology of unmyelinated tactile afferents, *Neurosci Biobehav Rev* 2010; 34(2):185-91.
- [8] Lee D, Xu H, Lin JG, Watson K, Wu RS, Chen KB. Needle-free electroacupuncture for postoperative pain management, *Evid Based Complement Alternat Med* 2011; 2011:696754.
- [9] Sim CK, Xu PC, Pua HL, Zhang G, Lee TL. Effects of electroacupuncture on intraoperative and postoperative analgesic requirement, *Acupunct Med* 2002; 20: 56–65.
- [10] Pomeranz B. Acupuncture analgesia for chronic pain: brief survey of clinical trials. In: Pomeranz B, Stux G, eds. *Scientific bases of acupuncture*. Berlin/Heidelberg, Springer-Verlag, 1989: 197-199.
- [11] Han JS (1987): *The Neurochemical Basis of Pain Relief by Acupuncture*. Beijing: Chinese Medical Science and Technology Press, 1987.
- [12] Fukazawa Y, Maeda T, Kishioka S. The pharmacological mechanisms of electroacupuncture, *Curr Opin Investig Drugs* 2009;10(1):62-9.
- [13] Astin JA, Marie A, Pelletier KR, Hansen E, Haskell WL. A review of the incorporation of complementary and alternative medicine by mainstream physicians, *Arch Intern Med* 1998;158(21):2303-10.
- [14] Sun Y, Gan TJ, Dubose JW, Habib AS. Acupuncture and related techniques for postoperative pain: a systematic review of randomized controlled trials, *Br J Anaesth* 2008; 101(2):151-60.
- [15] Zhao ZQ. Neural mechanism underlying acupuncture analgesia, *Prog Neurobiol* 2008; 85(4):355-75.33. Ulett GA, Han S, Han JS. Electroacupuncture: mechanisms and clinical application, *Biol Psychiatry* 1998; 44:129 –38.
- [16] *Electroacupuncture: A Practical Manual and Resource* David F Mayor (editor) Elsevier Churchill Livingstone, 2007 ISBN 0443063699, Chapter: 3, 4, 6, 9.10, 9.13, 9.14.
- [17] Ulett G, Nichols J (1996): *The Endorphin Connection*. Gleve, Australia: Wild and Woolley, 1996.
- [18] Han JS. Central neurotransmitters and acupuncture analgesia. In: Pomerranz B, Stux G (Eds). *Scientific Bases of Acupuncture*. Springer-Verlag, 1989, pp 7–34.
- [19] Wang JQ, Mao L, Han JS (1992b): Comparison of the antinociceptive effects induced by electroacupuncture and transcutaneous electrical nerve stimulation in the rat. *Int J Neurosci* 65:117–129.
- [20] Thorén P, Floras JS, Hoffmann P, Seals DR. Endorphins and exercise: physiological mechanisms and clinical implications, *Med Sci Sports Exerc* 1990; 22(4):417-28.
- [21] Han, J.S., Chen, X.H., Shun, S.L., Xu, X.J., Yuan, Y., Yan, S.C., Hao, J.X. and Terenius, L., Effect of low- and high-frequency TENS on met-enkephalinagr-phe and dynorphin A immunoreactivity in human lumbar CSF, *Pain*, 47 (1991) 295-298.
- [22] Hsieh, C.L., C.C.Kuo, Y.S. Chen, T.C. Li, C.T.Hsieh, C.J. Lao, C.J. Lee and J.G. Lin. Analgesic effect of electric stimulation of peripheral nerves with different electric frequencis using the formalin test. *Am. J. Chin. Med.* 28: 291–299, 2000.
- [23] Romita, V.V. and Henry, J.L., Parametric studies on electroacupuncture-like stimulation in a rat model: effects of intensity, frequency, and duration of stimulation on evoked antinociception, *Brain Research Bulletin*, 42 (1997) 289-296.
- [24] Wang B, Tang J, White PF, Naruse R, Sloninsky A, Kariger R, Gold J, Wender RH. Effect of the intensity of transcutaneous acupoint electrical stimulation on the postoperative analgesic requirement. *Anesth Analg* 1997;85:406–413.